PROGRAM BOOK

ASLO 2019 AQUATIC SCIENCES MEETING

PLANETWATER

CHALLENGES AND SUCCESSES

23 FEBRUARY-2 MARCH 2019 · SAN JUAN, PUERTO RICO

WWW.ASLO.ORG · FACEBOOK.COM/ASLO.ORG · TWITTER.COM/ASLO_ORG · #ASLO19

ASLO thanks the following organizations for supporting the 2019 Aquatic Sciences Meeting:

AAAS Caribbean Division · Associação Brasileira de Limnologia · SAM · Sea Grant Puerto Rico · SCL Sociedad Chilena de Limnología · ISOR Caribe

Sponsored by:

ASLO Association for the Sciences of Limnology and Oceanography

TABLE OF CONTENTS

This program is produced for reference on site at the meeting. Changes received after the printing of the program can be found on the conference web site.

Planet Water - Challenges and Successes 1
Association for the Sciences of Limnology and Oceanography 1
Supporting Organizations 1
Web Site and Social Media 1
ASLO 2019 Aquatic Sciences Meeting Committee 2
ASLO Board of Directors 2
ASLO Staff 2
Plenary Sessions 3-4
2019 ASLO Award Presentations 5-6
At The Puerto Rico Conference Center 6
- Internet Access 6
- Family Room 6
- Nursing Mothers Room 6
- Lactation Room 6
- Press Room 6
- Emergencies and First Aid at the Puerto Rico Convention Center 7
- Special Needs 7
- Lost and Found 7
- Receipts and Letters of Participation 7
- Conference Registration and Information 7
- Water Bottles 7
- Message Boards 7
- Coffee Breaks 7
- Lunch, Daily Entertainment, and Workshops at the PRCC 7
- Important Safety Reminders 8

ASLO Events Code of Conduct 8
Exhibitors 8-16
Meeting Agenda 10-14
Education Activities and Public Discussion 16
- ASLO 2019 Education Fair 16
- High School Poster Session 16
- Public Discussion-What Can Aquatic Scientists Do for You? 17

ASLO StoryTellers 17
Special Events 17
- Local Artisans of Puerto Rico - Artesanos de Puerto Rico 17

Art and Science Workshops 17
- WATER MARKS: Where Art Meets Science 17

Workshops, Auxiliary Events, and Town Halls 18
- Aquatic Science Education and Outreach Workshop 18
- LOREX Orientation Workshop 18
- The Science of Science Communication Workshop 18
- Make Your Science Communication More Effective Workshop 18
- Teaching Introductory Aquatic Sciences Courses Workshop 18
- Improv to Improve International Collaborations Workshop 19
- FlowCam Software Workshop 19
- Interdisciplinary Presentations Workshop 19
- Elements of a Teaching Philosophy Statement Workshop 19
- Introducing an Ocean Protein Portal Town Hall 19
- Real-time QC of pH Data Workshop 19
- Research Collaboration Across Borders Panel Discussion 20
- Limnology & Oceanography WikiWorkshop 20
- Project Redefining Recognition Town Hall 20
- Update and Status of the Arctic-COLORS Science Program Town Hall 20
- National Science Foundation Ocean Sciences Town Hall 20
- Mixotrophy Workshop 21
- Strategies for Transboundary HABs Management Town Hall 21
- Applying to Graduate School Workshop: Tips from current students 21
- Teaching and Mentoring Under the Threat of Climate Change 21
- Writing Effective Abstracts and Summaries Workshop 21
- Teaching Aquatic Biology with Aquaponics Workshop 22

Meetings and Working Groups 22
- Deep Search Annual Project Meeting 22
- Society for Women in Marine Science (SWMS) Meeting 22
- JPR Editorial Board Get Together and Meeting 22
- Blue Carbon Discussion Group 22

Society Events 22
- Opening Mixer Reception 22
- ASLO Fellows and Membership Reception 22
- ASLO Business Meeting 23

Student Events 23
- Student Worker Training Session 23
- Sunday Evening Pub Crawl 23
- ASLO Student Mixer 23
- ASLO Student Workshop - I Need the Degree AND the Money 23
- ASLO Student Beach Cleanup 23

Early Career Events 23
- Early Career Social Mixer 23
- ASLO Early Career Workshop - How to Successfully Write Proposals and Receive Funding 23

ASLO Meeting Mentor Program 23
ASLO Multicultural Program 24
Social and Evening Events 24
- Jam Session 24
- Fiesta! 24

Outreach Activities 24
Presenter Information 24
- On-Site Submission of Oral Presentations 24
- Poster Sessions and Receptions 25
- Poster Set Up and Teardown 25
- Poster Printing in San Juan 25

Schedule-at-a-Glance 26-30
Puerto Rico Convention Center Maps 31-33
Poster/Exhibit Hall Map 34
Monday Orals 35-42
Tuesday Orals 43-50
Tuesday Posters 51-58
Wednesday Orals 59-66
Thursday Orals 67-74
Thursday Posters 75-82
Friday Orals 83-90
Author Index 91-104

PLANET WATER - CHALLENGES AND SUCCESSES

A wealth of topics will be discussed at the meeting, many stemming from key areas of importance to the area itself at this crucial time-the role of science at the center of all economic, societal and environmental recovery and development efforts, studies on renewable energy, environmental sustainability, clean water, and rebuilding and maintaining terrestrial ecosystems.

Science and innovative problem-solving are keys to creating a better and more resilient world, and this speaks truth to the residents of Puerto Rico. While ASLO's meeting in San Juan will benefit Puerto Rico for the short-term, the outcomes of the ASLO 2019 Aquatic Sciences Meeting can impact the island for the long-term. This presents an excellent opportunity for ASLO to showcase how they are doing their part to train scientists in communication to the public and to teach effective research skills to future generations.

The scientific program will take place Sunday, 24 February, through Friday, 1 March. The full meeting dates are set Saturday, 23 February, to Saturday, 2 March, to allow meeting participants to take part in educational activities, volunteer opportunities, and culturally relevant events that will focus on environmental and ecosystem restoration as well as the resilience of the land.

Abstracts of papers presented during the meeting will be available on the meeting website. Abstracts also will be archived following the meeting. ASLO no longer publishes an abstract book.

ASLO RECORDING POLICY

Please! ASLO policy does not allow recording of individual talks, poster images, or sessions. This includes the opening plenary session as well as the poster and oral presentations that will take place throughout the week. Additionally, audio taping, videotaping, or photographing presentations, including slides and posters, is not allowed at the meeting. Thank you for your cooperation!

ASSOCIATION FOR THE SCIENCES OF LIMNOLOGY AND OCEANOGRAPHY

The purpose of ASLO is to foster a diverse, international scientific community that creates, integrates and communicates knowledge across the full spectrum of aquatic sciences, advances public awareness and education about aquatic resources and research, and promotes scientific stewardship of aquatic resources for the public interest. Its products and activities are directed toward these ends.

For more than 50 years, ASLO has been the leading professional organization for researchers and educators in the field of aquatic science. ASLO traces its roots to the Limnological Society of America (LSA), which was established in 1936 to further interest and research in limnological science. While the LSA had members working in both freshwater and marine systems, the name did not reflect this diversity until 1948 when the Oceanographic Society of the Pacific merged with the LSA to become the American Society of Limnology and Oceanography. ASLO is incorporated as a non-stock (non-profit) corporation in the State of Wisconsin. Membership in the society is presently more than 3,800 members. Members are drawn from 58 countries including the United States, and more than a quarter of the members reside outside the U.S. In 2011, ASLO members voted to change its name to the Association for the Sciences of Limnology and Oceanography, reflecting the increasingly international nature of the society.

SUPPORTING ORGANIZATIONS

ASLO thanks the following organizations for supporting the 2019 Aquatic Sciences Meeting:

- American Association for the Advancement of Science / AAAS Caribbean
- Brazilian Association of Limnology / Associação Brasileira de Limnologia
- Chilean Society of Limnology / Sociedad Chilena de Limnología (SCL)
- Institute for Socio-Ecological Research, Inc. (ISER)
- Marine Environment Society / Sociedad Ambiente Marino (SAM)
- Sea Grant Puerto Rico

INCLUDED IN THIS PROGRAM

This program is produced for reference on site at the meeting. It contains program information through 1 February 2019. Additional information including changes received after the printing of the program can be found on the conference web site.

WEB SITE AND SOCIAL MEDIA

We encourage you to use the meeting web site for all current information and to navigate the meeting.
Facebook: facebook.com/ASLO.org or Twitter: #ASLO19

OVERVIEW OF THE PROGRAM SCHEDULE

The opening reception will be held on Sunday evening from 18:30 to 20:30 to welcome attendees to Puerto Rico. The meeting begins Monday morning, as it will each day of the week, with concurrent sessions in various rooms at the Puerto Rico Convention Center. There will be a break following the concurrent sessions that will allow attendees time to network with one another before moving into the awards presentations and plenary session in the ballroom on the third floor. Poster sessions and receptions are scheduled Tuesday and Thursday in the exhibit hall following the conclusion of the concurrent oral sessions on those days. Posters will be up beginning Tuesday and will be in place through Thursday evening. Friday's schedule will include a last talk lottery drawing featuring prizes for those who are able to stay through the end of the meeting.

ASLO 2019 AQUATIC SCIENCES MEETING COMMITTEE

MEETING CO-CHAIRS

Deborah Bronk, Bigelow Laboratory for Ocean Sciences, dbronk@bigelow.org

John Downing, Minnesota Sea Grant College Program & LLO, University of Minnesota, downing@d.umn.edu

SCIENTIFIC COMMITTEE

Carmen Aguilar, University of Wisconsin-Milwaukee, aguilar@uwm.edu

Javier Alcocer Durand, UNAM FES Iztacala, jalcocer@unam.mx

Lisa Campbell, Texas A&M University, lisacampbell@tamu.edu

Ruperto Chaparro Seranno, University of Puerto Rico Mayagüez, ruperto.chaparro@upr.edu

Anne Deininger, University of Agder and Norwegian Institute of Water Research, anne.deininger@posteo.de

Lucinda Johnson, University of Minnesota Duluth, ljohnson@d.umn.edu

Jean-François Lapierre, University of Montréal, jfrancoislapierre@gmail.com

Paty Matrai, Bigelow Laboratory for Ocean Sciences, pmatrai@bigelow.org

Bill McDowell, University of New Hampshire, bill.mcdowell@unh.edu

Tiara Moore, ASLO Student Board Member, UCLA, tiaranmoore@gmail.com

Áurea Rodriguez, Taller Ecológico de Puerto Rico, auryro@gmail.com

Fabio Roland, Universida de Federal de Juiz de For a, fabio.roland@ufjf.edu.br

Rachel Sipler, Memorial University of Newfoundland, resipler@mun.ca

Conference Management for the 2019 Aquatic Sciences Meeting is provided by sg Meeting and Marketing Services, Waco, Texas

Helen Schneider Lemay, ASLO Business Manager, helens@sgmeet.com

Chris Schneider, Vice President, MIS and Communications, The Schneider Group, Inc., chriss@sgmeet.com

Lynda West, ASLO Meeting Project Manager, lyndaw@sgmeet.com

ASLO BOARD OF DIRECTORS

OFFICERS/EXECUTIVE COMMITTEE

Michael Pace, President

Linda Duguay, Past-President

Roxane Maranger, President-Elect

Dianne Greenfield, Secretary

Lisa Campbell, Past-Secretary

Phillip Taylor, Treasurer

MEMBERS-AT-LARGE

Robert Chen

Hans-Peter Grossart

Amina Pollard

Ajit Subramaniam

Diana Varela

Kim Wickland

STUDENT BOARD MEMBERS

Camille Gaynus

Ngozi M. Oguguah

ASLO STAFF

Teresa Curto, Executive Director

Chris Filstrup, Deputy Editor, *Limnology and Oceanography Bulletin*

Dave Hambright, Editor-in-Chief, *Limnology and Oceanography*

Paul Kemp, Web Editor and Editor-in-Chief, *Limnology and Oceanography: Methods*

Helen Schneider Lemay, Business Manager

Patricia Soranno, Editor-in-Chief, *Limnology and Oceanography Letters*

Adrienne Sponberg, Co-Editor, *Limnology and Oceanography Bulletin* and Director of Communications and Science

Maggie Xenopoulos, Deputy Editor, *Limnology and Oceanography*

THANK YOU TO THE GORDON AND BETTY MOORE FOUNDATION

GORDON AND BETTY MOORE FOUNDATION

ASLO thanks the Gordon and Betty Moore Foundation for supporting the ASLO 2019 Meeting. For more information about the foundation, please go to: https://www.moore.org/

PLENARY SESSIONS

MONDAY PLENARY SESSION

Monday, 25 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

WELCOME BY ASLO PRESIDENT

Michael L. Pace, Professor and Department Chair, University of Virginia, Department of Environmental Sciences, Charlottesville, Virginia

WELCOME BY MEETING CO-CHAIRS

Deborah A. Bronk, President and CEO, Bigelow Laboratory for Ocean Sciences, East Boothbay, Maine

John A. Downing, Director, Minnesota Sea Grant, and Professor of Biology and Large Lakes Observatory, University of Minnesota, Duluth, Minnesota

WELCOME TO SAN JUAN

Carmen Yulín Cruz, Mayor of San Juan, Puerto Rico

U.S. SENATOR SHELDON WHITEHOUSE (D-RI)

Newport, Rhode Island (Via Video)

Plenary Presentation: The Importance of the Science-Policy Interface for Marine and Coastal Resources

Biographical Information: Sheldon Whitehouse, a member of the Senate Environment and Public Works Committee (EPW), plays a key role in crafting policies addressing environmental protection and climate change. In 2011, he joined with Democrats and Republicans to form the Senate Oceans Caucus to increase awareness of and find common ground on issues facing the oceans and coasts. The Caucus helped gain Senate approval of four international fisheries treaties and passage of the IUU Fishing Enforcement Act that will prevent illegal, unreported, and unregulated fishing. Whitehouse has worked to boost federal support for fisheries science and cooperative fisheries research as well as efforts to improve transparency and efficiency in the commercial and recreational fisheries management process.

Senator Whitehouse has long advocated for a dedicated fund to support ocean and coastal research, restoration, and conservation. In 2015, the National Oceans and Coastal Security Fund was created to provide grants that support work for the oceans, coasts, and Great Lakes. It received funding for the first time as part of the FY2018 spending bill. He also led the successful bipartisan effort to reauthorize the Environmental Protection Agency's National Estuaries Program to protect and study coastal habitats. Whitehouse was a lead sponsor of the Save Our Seas Act, a bipartisan bill driven by the Caucus's leadership to reauthorize NOAA's marine debris program and strengthen the U.S.'s role in combatting the global marine debris crisis. The SOS Act became law in October.

A graduate of Yale University and the University of Virginia School of Law, Whitehouse served as United States Attorney and Attorney General of Rhode Island before being elected to the Senate in 2006. In addition to EPW, he is a member of the Budget Committee; the Judiciary Committee; and the Finance Committee. He and his wife Sandra, a marine biologist and environmental advocate, live in Newport, Rhode Island.

TUESDAY PLENARY SESSION

Tuesday, 26 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

MARÍA FALCÓN

Television Producer, Director, Presenter, and Journalist, Puerto Rico

Plenary Presentation: Television + Fun + Education... Total Immersion

Biographical Information: A University of Puerto Rico graduate with more than 30 years of experience in mass media, María Falcón has distinguished herself as a journalist, producer and director of educational TV programs and documentaries focused on nature, environment and cultural topics. She is most known for her work in GeoAmbiente.

Her credits contain hundreds of titles that have resulted in countless recognitions: (10) Emmy Suncoast Awards, the Conservation Filmmaker of the Year Award [Filmmakers for Conservation - Bristol, UK 2008], (3) Excellence in Ecological Journalism Awards [Overseas Press Club], the Accolade Award [California], the Environmental Hero Award [National Oceanographic & Atmospheric Administration - US 2005], the Centennial Award [US Forest Service - 2005], (2) Excellence Awards [Environmental Protection Agency], as well as other achievements at festivals in England, Brazil, Spain, Portugal and the United States. In addition, the Universidad Metropolitana (AGMUS) awarded Falcón an Honorary Doctorate Degree in Sciences - Environmental Management in 2017.

Currently, she is a member of the Puerto Rico Climate Change Council, the Sea Grant Program UPR - Advisory Council, the El Yunque Management Plan Review Committee and the international organization Filmmakers for Conservation. Also, as part of her commitment to the conservation of our natural resources and the environmental health, she offers talks and workshops to students of our local schools and universities.

WEDNESDAY PLENARY SESSION

Wednesday, 27 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

ADA MONZÓN

Chief Meteorologist, WIPR-TV, Univision Radio WKAQ 580 am, and Noticel; Founder and President, EcoExploratorio, Puerto Rico

Plenary Presentation: Building Resilience through Innovation in Education: Case Study: Hurricane Maria

Biographical Information: Ada Monzón is the commonwealth's chief meteorologist for WIPR-TV and Univision Radio WKAQ 580 am and Noticel. She is also the founder and president of EcoExploratorio: Science Museum of Puerto Rico. After earning her M.S. in Meteorology at the Florida State University, Ada joined the U.S. NWS Forecast Office in San Juan, where she became a forecaster and the warning and preparedness meteorologist. Ada is the first female American Meteorological Society (AMS) Fellow and Certified Broadcast Meteorologist in Puerto Rico. She is a professional speaker and producer of STEM education programs and disaster mitigation educational campaigns, and she is widely recognized for her successful social media networks with over 1,000,000 followers. Her standout service and commitment during 2017 reassured the Commonwealth of Puerto Rico. She closely followed Irma as it passed near the Caribbean and then provided essential weather information during Hurricane Maria, helping the population prepare and remain calm throughout the most devastating disaster in Puerto Rico's record. Ada was recognized as the 2018 National Weatherperson of the Year and the 2019 AMS Award for Broadcast Meteorology. Her awards include the AMS Joanne Simpson Mentorship Award and the AMS Award for Excellence in Science Reporting by a Broadcast Meteorologist.

THURSDAY PLENARY SESSION

Thursday, 28 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

CHRISTOPHER M. VOLPE

Executive Director, ScienceCounts, Washington, DC

Plenary Presentation: Americans' Views of Science: Separating Fact from Fiction

Biographical Information: Christopher Volpe is the executive director and a founding board member of ScienceCounts. With an academic background in atmospheric chemistry and two decades of private sector experience in marketing and branding, Chris is strong proponent of using traditional marketing techniques to better understand public attitudes about science and its intersection with society. Through ScienceCounts, he is working to foster stronger connections between the scientific community and the general public. Previously, Chris was the president and co-founder of Prismatic Laser Programs LLC, the nation's leading provider of STEM-based assembly programs to elementary and middle schools. In his free time, Volpe is a demonstration pilot of various types of historically significant WW2 aircraft. Chris received his PhD from Scripps Institution of Oceanography.

FRIDAY PLENARY SESSION

Friday, 1 March 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

ERNESTO L. DÍAZ

Director, Coastal Management and Climate Change Office, Puerto Rico Department of Natural and Environmental Resources, San Juan, Puerto Rico

Plenary Presentation: The State of the Puerto Rico Climate 2018: An assessment of socio-ecological vulnerabilities for coastal and marine systems

Biographical Information: Ernesto Díaz is Director of the Puerto Rico Coastal Zone Management Program under the Department of Natural and Environmental Resources, National Oceanic and Atmospheric Administration. He also is Co-Incident Commander of Hurricane Maria's Sunken Vessel and Marine Debris Removal (DNER-USCG) Program and is the lead for coral reefs, beaches and dunes restoration and recovery efforts (DNER-FEMA). Díaz also is the current coordinator of the Puerto Rico Climate Change Council. He has served in various capacities within the DNER since 1995 and before that was Coordinator of Integrated Planning and Institutional Development under the United Nations Environment Program Regional Co-ordinating Unit in Kingston, Jamaica. Díaz is a prolific author and co-author on a number of papers and reports. He has formal training in many areas ranging from remote sensing, to strategic planning to climate change adaptation to national disaster preparedness. He holds degrees in oceanography, coastal and marine biology, engineering management, and has completed post-graduate studies in energy and environment.

2019 ASLO AWARD PRESENTATIONS

Society awards will be presented during the plenary sessions Monday through Friday. Following is a list of the 2019 award recipients and the day on which they will accept their award. Complete information on each award recipient including complete citations for each award are available on the ASLO web site: https://aslo.org/page/2019-award-recipients.

2019 RUTH PATRICK AWARD

Monday, 25 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

JENNIFER TANK
Department of Biological Sciences, University of Notre Dame, Notre Dame, Indiana

The Ruth Patrick Award honors scientists who have made outstanding contributions towards solving environmental problems. This year, ASLO recognizes Jennifer Tank with the 2019 Ruth Patrick Award for her significant contributions to aquatic biogeochemistry, applying those fundamental concepts to human-impacted agricultural landscapes, and informing how conservation practices influence stream management through community engagement. Tank is the Galla Professor of Biological Sciences at the University of Notre Dame and Director of the Notre Dame Environmental Change Initiative.

2019 A.C. REDFIELD LIFETIME ACHIEVEMENT AWARD

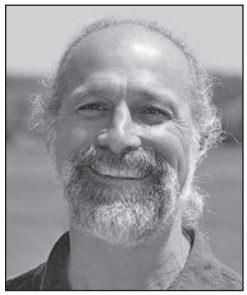
Tuesday, 26 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

STEPHEN CARPENTER
Center for Limnology, University of Wisconsin, Madison, Wisconsin

The A.C. Redfield Lifetime Achievement award honors major, long-term achievements in the fields of limnology and oceanography, including research, education, and service to the community and society. Stephen R. Carpenter is the 2019 recipient of the A.C. Redfield Award in recognition of his remarkable contributions to the field of limnology and lake management; including major advances in our understanding of trophic cascades, eutrophication, regime shifts, fisheries sustainability, and social ecological systems via approaches that confront theory with large-scale collaborative experimentation, long-term lake surveys, community outreach, and education.

2019 RAMÓN MARGALEF AWARD

Wednesday, 27 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center



DAVID FIELDS
Bigelow Laboratory for Ocean Sciences, East Boothbay, Maine

The Ramón Margalef Award is given to scientists and educators for excellence in teaching and mentoring in the fields of limnology and oceanography. David Fields of the Bigelow Laboratory for Ocean Sciences has been awarded this distinguished honor for his enthusiastic leadership and effusive commitment to developing transformative, hands-on marine science education programs, and for creating opportunities for authentic science experiences for high school and college students from all backgrounds.

2019 YENTSCH-SCHINDLER EARLY CAREER AWARD

Wednesday, 27 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

ROBERT SPENCER
Department of Earth, Ocean, and Atmospheric Science, Florida State University, Tallahassee, Florida

The Yentsch-Schindler Early Career Award honors an early-career scientist for outstanding and balanced contributions to research, education and society. Robert Spencer is the 2019 recipient of the Yentsch-Schindler Award for his contributions to our understanding of the biogeochemical processes involving the production, fate and transport of organic matter in terrestrial, freshwater and marine environments, and his projections on how anthropogenic impacts will affect these processes. Spencer is an associate professor at Florida State University.

2019 JOHN H. MARTIN AWARD

Thursday, 28 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

PHILIP BOYD
Institute for Marine and Antarctic Studies, University of Tasmania, Hobart, Australia

Accepted via video on behalf of co-authors.

ASLO presents the John H. Martin Award to one paper each year that has led to fundamental shifts in research

focus and interpretation of a large body of previous observations. The 2019 John H. Martin Award is for "A mesoscale phytoplankton bloom in the polar Southern Ocean stimulated by iron fertilization," a paper published in 2000 by Philip Boyd and co-authors that tested how phytoplankton respond to iron fertilization. (Lead author Philip W. Boyd and study co-authors Andrew J. Watson, Cliff S. Law, Edward R. Abraham, Thomas Trull, Rob Murdoch, Dorothee C. E. Bakker, Andrew R. Bowie, K. O. Buesseler, Hoe Chang, Matthew Charette, Peter Croot, Ken Downing, Russell Frew, Mark Gall, Mark Hadfield, Julie Hall, Mike Harvey, Greg Jameson, Julie LaRoche, Malcolm Liddicoat, Roger Ling , Maria T. Maldonado, R. Michael McKay, Scott Nodder, Stu Pickmere, Rick Pridmore, Steve Rintoul, Karl Safi, Philip Sutton, Robert Strzepek, Kim Tanneberger, Suzanne Turner, Anya Waite, and John Zeldis. Citation: Nature. 2000 Oct 12;407(6805): 695-702.

2019 RAYMOND L. LINDEMAN AWARD

Thursday, 28 February 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

ANDREA G. BRAVO

Institute of Marine Sciences, Barcelona, Spain

The Raymond L. Lindeman Award honors a young scientist for an outstanding peer-reviewed, English-language paper in the aquatic sciences. ASLO has awarded Andrea G. Bravo the 2019 Lindeman Award for her paper, "Molecular composition of organic matter controls methylmercury formation in boreal lakes." Bravo's research, published in Nature Communications in 2017, challenges the traditional mercury cycle by demonstrating the role of organic matter composition on bacterial activity, and ultimately, on mercury methylation rates in boreal lakes.

2019 G. EVELYN HUTCHINSON AWARD

Friday, 1 March 2019, 11:00 to 12:30
Ballroom A-Puerto Rico Convention Center

OSCAR SCHOFIELD

Department of Marine and Coastal Sciences, Rutgers University, New Brunswick, New Jersey

The G. Evelyn Hutchinson award honors a limnologist or oceanographer who has made considerable contributions to knowledge, and whose future work promises a continued legacy of scientific excellence. Dr. Oscar Schofield is the 2019 recipient of the G. Evelyn Hutchinson Award for transforming our understanding of the physical and chemical processes that govern marine phytoplankton physiology and ecology through the application of novel ocean observing tools, and for his skillful and enthusiastic leadership of the collaborative science necessary for addressing broad scale oceanographic challenges. Schofield is Distinguished Professor in the Department of Marine and Coastal Sciences at Rutgers University.

AT THE PUERTO RICO CONFERENCE CENTER

INTERNET ACCESS

Complimentary wireless Internet access is available at the Puerto Rico Convention Center. To connect to the Wi-Fi at the Convention Center you should enable your wireless access on your device and connect to Wi-Fi SSID PSAV WIFI. Once connected, open your favorite web browser and enter the password: ASLO2019 on the Access Code field. (The password is not case sensitive.) After you enter the access code, you will be asked to click on Connect. By doing so you agree with the terms. Once you log on, you will not be required to do so each day and will remain connected when you are on site at the Convention Center.

FAMILY ROOM

A family room will be open throughout the meeting in the Mezzanine Suite on the 2nd floor of the Puerto Rico Convention Center. This is a room where you may go to relax with your children if you bring them to the conference center. Please keep in mind that this is not a room for childcare, and no service is offered in this room. You may not leave children unattended.

NURSING MOTHERS ROOM

If you are a nursing mother and need a private place for you and your infant, a quiet room and a comfortable chair are available for you in the Mothers' Room on the 1st floor directly across from the Information Desk. Please go to the PRCC Information desk for instructions on how to access the room.

LACTATION ROOM

A private room for mothers to express breast milk is available on the 3rd floor. This room also includes a small refrigerator for breastmilk and offers storage for pumping necessities and supplies. It will be available throughout the meeting. Please go to the meeting registration desk for instructions on how to access the room.

PRESS ROOM

All members of the media who are attending the meeting must check in at the conference registration desk as soon as they arrive at the convention center to receive a media badge. A press room is set up exclusively for members of the media in Room 210 on the 2nd floor of the convention center. If a room is needed for a media interview, go to the conference registration desk, and conference management personnel can direct you to the Chrysler Conference Room. We remind the media that ASLO policy does not allow recording of individual talks, slides, or poster images during the sessions.

EMERGENCIES AND FIRST AID AT THE PUERTO RICO CONVENTION CENTER

In the event of a medical or other emergency situation, do not call 911 from your cell phone. Please find a member of the convention center staff, dial 0 from any house phone at the PRCC, or go to the conference registration desk on the 1st floor. Client services will call for emergency personnel, if needed.

Should you need minor first aid when you are the convention center, please come to the conference registration desk on Level 1. Staff at the desk will contact someone who can assist you and direct you to the first aid area.

SPECIAL NEEDS

If you have a disability or limitation that may require special consideration to ensure your full participation in this meeting, please see a staff person at the conference registration desk. You also may send an email to business@aslo.org prior to your arrival at the meeting.

LOST AND FOUND

Please come to the Registration Desk for inquiries concerning lost and found items.

RECEIPTS AND LETTERS OF PARTICIPATION

Your registration confirmation that was emailed to you when you registered for the meeting will serve as your receipt. In keeping with our conservation efforts, we will not provide printed receipts to attendees on site at the meeting. If you have misplaced your original receipt and need another copy, please send an email to the conference registrar, Jo Davis, at jdavis@sgmeet.com. Likewise, letters of participation only will be provided to those who are registered for the meeting. If you need a letter of participation, please contact Jo Davis at the email address given.

CONFERENCE REGISTRATION AND INFORMATION

Registration and check in for the meeting will be available all week in the 1st Floor Lobby Area of the Puerto Rico Convention Center. Please check in upon your arrival at the meeting in order to receive your name badge and other important materials and information.

Registration Desk Hours:

Sunday, 24 February 2019 .....15:00 to 20:00
Monday, 25 February 2019 .....08:00 to 17:30
Tuesday, 26 February 2019.....08:00 to 17:30
Wednesday, 27 February 2019.....08:00 to 17:30
Thursday, 28 February 2019.....08:30 to 17:30
Friday, 1 March 2019.....08:30 to 17:30

WATER BOTTLES

Please join ASLO's continuing sustainability efforts as we try to reduce our plastic footprint in Puerto Rico as much as possible. To help in this effort, we are asking all participants to bring their own refillable water bottle to the meeting! If you purchased a water bottle from the ASLO Student Committee, you will pick it up at registration. All proceeds after expenses will support ASLO student travel grants.

MESSAGE BOARDS

Message boards are on the 1st Floor Lobby Area near registration. This is where you may post announcements or check for messages throughout the conference.

COFFEE BREAKS

Morning coffee breaks are planned during the transition time between concurrent oral sessions and the plenaries Monday through Friday. Afternoon breaks are planned Monday through Friday as well. Complimentary coffee and tea will be served. Water will be provided in coolers, and attendees are encouraged to bring their own water bottles. Break locations vary throughout the week and are listed on the agenda page of this program.

LUNCH, DAILY ENTERTAINMENT, AND WORKSHOPS AT THE PRCC

To help you relax between mornings and afternoons full of science, multiple lunch options at reasonable prices will be available at the convention center. Rather than rush to find a place for lunch, make plans to stay at the convention center and participate in one of the workshops listed in this program. You also may fuel your creative side by joining the art and science activities planned as part of the WATER MARKS workshop. If you prefer to sit and visit with fellow attendees and colleagues, local musicians will play while you relax in comfortable places and take in the beautiful scenery. For those who want to get up and move, there will be Salsa dance lessons during the week (Monday, Tuesday and Thursday).

The following food options can be found in various locations throughout the convention center. Expect a variety of food from American-style lunches to Puerto Rican dishes like a Mofongo Bowl.

- Alto Grande Café-Located in the Main Lobby
  Opens at 07:00 and offers bakery and breakfast selections, sandwiches, wraps, soups, cookies, brownies and soft drinks throughout the day.
- Espresso Bar-Located on the 2nd and 3rd Floor Foyer Areas
  Opens at 07:00 and offers espresso, select coffees, latte, cappuccino and tea.
- Lobby Bar Lounge-Located in the Main Lobby
  Opens at 11:00 and features nachos, a variety of appetizers, sliders, cocktails, wine and Cava by the glass, domestic and imported beer, soft drinks and spring water.
- Mexican/Puerto Rican Movable Cafe-Located near the Exhibit Hall
  Open for lunch only. Themed selections will change daily. Soft drinks and water.

- Barbeque Movable Café-Located on the Third Floor Terrace Open for lunch only. Offering burgers, barbecued meats, and side items. Selections will change daily. Soft drinks and water.

IMPORTANT SAFETY REMINDERS

We want you to enjoy a fun, productive, and informative week in San Juan. As with all meetings in busy urban areas such as San Juan, taking measures to ensure your personal safety is very important. Here are some reminders:

- Make sure you have copies of your ID and passport. Bring one copy and store it separately from your original and leave another with a friend or relative.
- Streamline your wallet. Only carry the card(s) you plan to use on your trip and leave your checkbook and the rest of your cards at home.
- Always remove your meeting name badge when you leave the convention center.
- If you are walking between any destination and your hotel or the convention center, plan your travel route before you depart.
- Use caution when walking on sidewalks and crossing streets and be careful to cross only at designated crosswalks.
- Avoid visible use of cell phones while you are walking.
- Be aware of your surroundings.
- Make sure your belongings are secure.
- Walk with your colleagues.
- When transportation is needed, take or share a taxi or Uber.
- Check the local weather and pack the appropriate gear before you leave.

ASLO EVENTS CODE OF CONDUCT

A core goal of ASLO is to foster a scientific community that is safe, hospitable, and productive for all its members. Thus, ASLO seeks to provide a welcoming and productive environment for those attending our meetings, workshops, and events, regardless of gender, sexual orientation, gender identity, race, ethnicity, religion, disability, physical appearance, or career level. All participants, including, but not limited to, attendees, speakers, volunteers, exhibitors, ASLO staff, service providers, and others are expected to abide by this ASLO Events Code of Conduct. This Code of Conduct applies to all components of an ASLO event, including those sponsored by organizations other than ASLO but held in conjunction with ASLO events, in public or private facilities.

EXPECTED BEHAVIOR

- All participants, attendees, ASLO staff, and vendors are treated with respect and consideration, valuing a diversity of views and opinions.
- Be considerate, respectful, and collaborative.
- Communicate openly with respect for others, critiquing ideas rather than individuals.
- Avoid personal attacks directed toward other attendees, participants, ASLO staff, and suppliers/vendors.
- Be mindful of your surroundings and of your fellow participants. Alert staff if you notice a dangerous situation or someone in distress.
- Respect the rules and policies of the meeting venue, hotels, ASLO-contracted facility, or any other venue.

UNACCEPTABLE BEHAVIOR

- Harassment, sexual harassment, bullying, or discrimination in any form will not be tolerated.
- Physical or verbal abuse of any attendee, speaker, volunteer, exhibitor, ASLO staff member, service provider, or other event guest.
- Examples of unacceptable behavior include, but are not limited to, verbal comments related to gender, sexual orientation, disability, physical appearance, body size, race, religion, national origin, inappropriate use of nudity and/or sexual images in public spaces or in presentations, or threatening or stalking any attendee, speaker, volunteer, exhibitor, ASLO staff member, service provider, or other event guest.
- Disruption of talks at oral or poster sessions, in the exhibit hall, or at other events organized by ASLO at the event venue, hotels, or other ASLO-contracted facilities.

CONSEQUENCES

- Anyone requested to stop unacceptable behavior is expected to comply immediately.
- ASLO staff (or their designee) or security may take any action deemed necessary and appropriate, including immediate removal from the meeting without warning or refund.
- ASLO reserves the right to prohibit attendance at any future meeting.

REPORTING UNACCEPTABLE BEHAVIOR

- If you are the subject of unacceptable behavior or have witnessed any such behavior, please immediately notify an ASLO staff member or ASLO officer.

EXHIBITORS

Exhibits are in Exhibit Hall B. Exhibitors will set up on Monday, 25 February, and will be in place over the following days and times:

Tuesday, 26 February 2019.....09:00-19:00
Wednesday, 27 February 2019.....09:00-19:00
Thursday, 28 February 2019 .....09:00-19:00

Exhibitors will tear down beginning at 19:00 on Thursday following the end of the poster session and reception in the Exhibit Hall. Attendees will have access to the exhibits during the hours listed above. ASLO appreciates the support of the following organizations who are exhibiting at the 2019 Aquatic Sciences Meeting:

ASLO/WILEY

ASLO (#E-11 and #E-20)
1105 Wooded Acres, Suite 260
Waco, TX 76710 USA
Contact: Helen Schneider Lemay
Phone: 254-776-3550
Email: business@aslo.org
Website: http://aslo.org

John Wiley & Sons, Inc. (#E-11 and #E-20)
111 River Street-Mail Stop 8-02
Hoboken, NJ 07030 USA
Contact: Fiona Sarne
Phone: 201-748-7850
Email: fsarne@wiley.com
Website: www.wiley.com

EXHIBITORS:

BERMUDA INSTITUTE OF OCEAN SCIENCES (#E-19)

St. Georges GE 01 Bermuda
Contact: Matthew Hayden
Phone: 441-297-1880
Email: matthew.hayden@bios.edu
Website: www.bios.edu

CENTER FOR COASTAL AND MARINE ECOSYSTEMS (CCME) (#E-23)

1601 S. Martin L. King Jr. Blvd.
Tallahassee, FL 32307 USA
Contact: Sharmini Pitter
Phone: 850-561-2017
Email: sharmini.pitter@famu.edu
Website: www.ccme.famu.edu

COASTALOCEANVISION, INC. (#E-21)

10 Edgerton Drive, Suite 2
North Falmouth, MA 02556 USA
Contact: Scott Gallagher
Phone: 508-472-5520
Email: sgallager@coastaloceanvision.com
Website: www.coastaloceanvision.com

EUREKA WATER PROBES (#E-22)

2113 Wells Branch Parkway, Suite 4400
Austin, TX 78728 USA
Contact: Stuart Garner
Phone: 512-302-4333
Email: sgarner@waterprobes.com
Website: www.WaterProbes.com

FLORIDA INTERNATIONAL UNIVERSITY (#E-07)

Institute of Water Environment
11200 SW 8th St.
OE-148
Miami, FL 33199 USA
Contact: Bradley Schonhoff
Phone: 305-348-3095
Email: bschonho@fiu.edu
Website: www.inwe.fiu.edu

FLUID IMAGING TECHNOLOGIES (#E-15 AND #E-16)

200 Enterprise Drive
Scarborough, ME 04074 USA
Contact: Harry Nelson
Phone: 207-289-3242
Email: harry.nelson@fluidimaging.com
Website: www.fluidimaging.com

HYDROPTIC (#E-24)

33 route de Ségoufielle
L'isle Jourdain 32600 France
Contact: Jérôme Coindat
Phone: +33 (0)9 6324 8220
Mobile: +33 (0)6 7399 5790
Fax: +33 (0)5 6206 2334
Email: jerome.coindat@hydroptic.com
Website: www.hydroptic.com

JFE ADVANTECH CO., LTD (#E-18)

3-48, Takahata-cho
Nishinomiya, Hyogo 663-8202 Japan
Contact: Hua Li
Phone: +81-798-66-1783
Email: lihua@jfe-advantech.co.jp
Website: www.jfe-advanatech.co.jp/eng/

MCLANE RESEARCH LABORATORIES, INC. (#E-06)

121 Bernard St. Jean Drive
East Falmouth, MA 02536 USA
Contact: Jon Mogul
Phone: 1-508-495-4000
Email: mclane@mclanelabs.com
Website: www.mclanelabs.com

NOAA NATIONAL SEA GRANT COLLEGE PROGRAM (#E-08)

1315 East West Highway, SSMC3
Silver Spring, MD 20910 USA
Contact: Rebecca Briggs
Phone: 301-734-1084
Email: rebecca.briggs@noaa.gov
Website: www.seagrant.noaa.gov/

MEETING AGENDA

SATURDAY, 23 FEBRUARY 2019

08:00-17:00	ASLO Board Meeting	Sheraton Hotel
07:30-15:30	Outreach Activity: Volunteering with P.E.C.E.S.	PRCC: Check in at 07:00
07:30-13:00	Outreach Activity: Monitoring Dive at a Coral Reef Ecosystem at Escambron Beach and Condado Lagoon	PRCC: Check in at 07:15

SUNDAY, 24 FEBRUARY 2019

08:00-17:00	ASLO Board Meeting	Sheraton Hotel
08:00-17:00	The Science of Science Communication Workshop	Room 103 A
08:00-17:00	Deep Search Annual Project Meeting	Room 103 B
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
10:00-13:00	Make Your Science Communication More Effective Workshop	Room 101
13:00-17:00	Aquatic Science Education and Outreach Workshop	Room 208 C
14:00-18:00	LOREX Orientation Workshop (By Invitation Only)	Room 208 AB
15:00-20:00	Registration	PRCC Lobby-Main Level Prefunction Area
15:00-21:00	Presentation Room	Room 207
16:00-18:00	ASLO MP Student Program Training Session (Dr. Ben Cuker)	Ballroom B
16:30-17:30	Local Artisan Booth Set Up	3rd Floor Foyer
17:30-18:00	Student Worker Training Session	PRCC Lobby near Registration Desk
17:30-20:30	Local Artisans (Artesanos de Puerto Rico)	3rd Floor Foyer
18:30-20:30	Opening Mixer Reception	3rd Floor Terrace & Foyer
20:30	Sunday Evening Pub Crawl (Organized by ASLO Students)	Off-site: La Taberna Lupulo

MONDAY, 25 FEBRUARY 2019

07:00-19:00	Presentation Room	Room 207
08:00-17:30	Registration	PRCC Lobby-Main Level Prefunction Area
08:00-19:30	Literature Tables	PRCC-Main Level or 2nd Floor Prefunction Area
08:00-20:30	ASLO Mentor Program Room	Room 203
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
09:00-10:30	Concurrent Sessions	PRCC-Various Rooms
10:00-17:00	Exhibitor Set Up	Exhibit Hall B
10:30-11:00	Coffee Break	Ballroom A Foyer

11:00-12:30	Opening Plenary & Awards Session	Ballroom A
	Welcome by ASLO President: Michael L. Pace, University of Virginia, Charlottesville, VA	
	Welcome by Meeting Co-chairs: Deborah A. Bronk, Bigelow Lab for Ocean Sciences, East Boothbay, ME and John Downing, Minnesota Sea Grant and University of Minnesota, Duluth, MN	
	Welcome to San Juan: Carmen Yulín Cruz, Mayor, San Juan, PR	
	2019 Ruth Patrick Award Acceptance: Jennifer Tank, Univ. of Notre Dame, Notre Dame, IN	
	Plenary Presentation via Video: U.S. Senator Sheldon Whitehouse (D-RI), Newport, RI	
12:30-14:00	WATER MARKS Workshop-Where Science Meets Art	Room 206
12:30-14:00	Teaching Introductory Aquatic Sciences Courses Workshop	Room 101
12:30-14:00	Improv to Improve International Collaborations Workshop	Room 102
12:30-14:00	FlowCam Software Workshop	Room 103 A
12:30-14:00	Lunch Break	
12:30-17:00	Poster Set Up by Attendees	Exhibit Hall B
14:00-15:30	Concurrent Sessions	PRCC-Various Rooms
15:30-15:45	Coffee Break	Ballroom A Foyer
15:45-17:30	Concurrent Sessions	PRCC-Various Rooms
17:30-18:30	ASLO Fellows and Membership Reception	Ballroom B
18:30-19:30	ASLO Business and Membership Meeting	Ballroom A
19:30-20:30	Early Career Mixer	2nd Floor Prefunction Area
19:30-20:30	Student Mixer	3rd Floor Terrace & Lobby

TUESDAY, 26 FEBRUARY 2019

07:00-19:00	Presentation Room	Room 207
08:00-17:30	Registration	PRCC Lobby-Main Level Prefunction Area
08:00-19:30	Literature Tables	PRCC-Main Level or 2nd Floor Prefunction Area
08:00-20:30	ASLO Mentor Program Room	Room 203
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
08:00-08:45	Mentor Program Meeting (Includes Light Breakfast for Mentors and Mentees)	Ballroom B
09:00-19:00	Exhibits Open	Exhibit Hall B
09:00-19:00	Posters Open	Exhibit Hall B
09:00-10:30	Concurrent Sessions	PRCC-Various Rooms
10:30-11:00	Coffee Break	Exhibit Hall B

11:00-12:30	Plenary & Awards Session	Ballroom A
	2019 Redfield Award Acceptance: Stephen R. Carpenter, University of Wisconsin-Madison, Madison, WI	
	Plenary Presentation: Maria Falcon, Television Producer, Director, Presenter, and Journalist, PR	
12:30-14:00	WATER MARKS Workshop-Where Science Meets Art	Room 206
12:30-14:00	Interdisciplinary Presentations Workshop	Room 102
12:30-14:00	Elements of a Teaching Philosophy Statement Workshop	Room 208 C
12:30-14:00	Introducing an Ocean Protein Portal Town Hall	Room 204
12:30-14:00	Real-Time QC of pH Data Workshop	Room 209 C
12:30-14:00	Research Collaboration Across Borders Panel	Room 101
12:30-14:00	ASLO Student Workshop-I Need the Degree AND the Money	Room 104
12:30-14:00	Lunch Break and Salsa Lessons	3rd Floor Terrace
13:15-16:00	Outreach Activity: Water Quality Monitoring in Different Zones	PRCC: Check in at 13:00
14:00-15:30	Concurrent Sessions	PRCC-Various Rooms
15:30-15:45	Coffee Break	Exhibit Hall B
15:45-17:30	Concurrent Sessions	PRCC-Various Rooms
17:30-19:00	Poster Session and Reception	Exhibit Hall B
19:00-19:30	Society for Women in Marine Science Meeting	Room 204

WEDNESDAY, 27 FEBRUARY 2019

07:00-19:00	Presentation Room	Room 207
08:00-17:30	Registration	PRCC Lobby-Main Level Prefunction Area
08:00-19:00	Literature Tables	PRCC-Main Level or 2nd Floor Prefunction Area
08:00-20:30	ASLO Mentor Program Room	Room 203
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
08:30-12:30	Outreach Activity: Coral Reef Fragments Recovery Dive & Restoration Project	PRCC: Check in at 08:00
09:00-19:00	Exhibits Open	Exhibit Hall B
09:00-19:00	Posters Open	Exhibit Hall B
09:00-10:30	Concurrent Sessions	PRCC-Various Rooms
09:00-10:45	High School Poster Session	Exhibit Hall A
09:00-10:45	Education Fair	Exhibit Hall A
10:30-11:00	Coffee Break	Exhibit Hall B
11:00-12:30	Plenary & Awards Session	Ballroom A
	2019 Ramon Margalef Award for Excellence in Education Acceptance: David Fields, Bigelow Lab for Ocean Sciences, East Boothbay ME	
	2019 Yentsch-Schindler Early Career Award Acceptance: Robert Spencer, Florida State University, Tallahassee, FL	
	Plenary Presentation: Ada Monzón, Chief Meteorologist, WIPR-TV, Univision Radio WKAQ 580 am, and Noticel; Founder and President, EcoExploratorio, PR	
12:30-14:00	WATER MARKS Workshop-Where Science Meets Art	Room 206
12:30-14:00	JPR Editorial Board Get-together and Meeting (By Invitation Only)	Room 208 AB

12:30-14:00	Limnology and Oceanography Wiki Workshop	Room 204
12:30-14:00	Update and Status of the Arctic COLORS Science Program Town Hall	Room 209 C
12:30-14:00	Project Redefining Recognition Town Hall	Room 101
12:30-14:00	ASLO Early Career Workshop-How to Successfully Write Proposals and Receive Funding	Room 102
12:30-14:00	Lunch Break and Entertainment	3rd Floor Terrace
14:00-15:30	Concurrent Sessions	PRCC-Various Rooms
15:30-15:45	Coffee Break	Exhibit Hall B
15:45-17:30	Concurrent Sessions	PRCC-Various Rooms
18:00-19:30	Public Discussion-What Can Aquatic Scientists Do for You?	Ballroom A
18:00-21:00	National Science Foundation Ocean Sciences Town Hall	Room 204
18:00-21:00	Mixotrophy Workshop	Room 103 B
20:00-23:00	Jam Session	TBD

THURSDAY, 28 FEBRUARY 2019

07:00-19:00	Presentation Room	Room 207
08:00-17:30	Registration	PRCC Lobby-Main Level Prefunction Area
08:00-19:00	Literature Tables	PRCC-Main Level or 2nd Floor Prefunction Area
08:00-20:30	ASLO Mentor Program Room	Room 203
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
09:00-19:00	Exhibits Open	Exhibit Hall B
09:00-19:00	Posters Open	Exhibit Hall B
09:00-10:30	Concurrent Sessions	PRCC-Various Rooms
10:30-11:00	Coffee Break	Exhibit Hall B
11:00-12:30	Plenary & Awards Session	Ballroom A
	2019 John Martin Award Acceptance Via Video: Philip Boyd, Inst. for Marine and Antarctic Studies, University of Tasmania, Hobart, Tasmania, Australia	
	2019 Raymond Lindeman Award Acceptance: Andrea Bravo, IDAEA-CSIC, Barcelona, Spain	
	Plenary Presentation: Chris Volpe, Executive Director, ScienceCounts, Inc., Washington, DC	
12:30-14:00	WATER MARKS Workshop-Where Science Meets Art	Room 206
12:30-14:00	Teaching and Mentoring Under the Threat of Climate Change Workshop	Room 204
12:30-14:00	Strategies for Transboundary HABs Management Town Hall	Room 103 B
12:30-14:00	Blue Carbon Discussion Group	Room 208 C
12:30-14:00	Applying to Graduate School Workshop: Tips from Current Students	Room 103 A
12:30-14:00	Writing Effective Abstracts and Summaries Workshop	Room 209 C
12:30-14:00	ASLO Student Beach Cleanup	Location TBA
12:30-14:00	Lunch Break and Entertainment	3rd Floor Terrace
13:15-16:00	Outreach Activity: Water Quality Monitoring in Different Zones	PRCC: Check in at 13:00
14:00-15:30	Concurrent Sessions	PRCC-Various Rooms
15:30-15:45	Coffee Break	Exhibit Hall B

15:45-17:30	Concurrent Sessions	PRCC-Various Rooms
17:30-19:00	Poster Session and Reception	Exhibit Hall B
19:00-21:00	Exhibitor and Poster Teardown	Exhibit Hall B

FRIDAY, 1 MARCH 2019

07:00-17:00	Presentation Room	Room 207
08:00-17:30	Registration	PRCC Lobby-Main Level Prefunction Area
08:00-19:00	Literature Tables	PRCC-Main Level or 2nd Floor Prefunction Area
08:00-20:30	ASLO Mentor Program Room	Room 203
08:00-20:30	Family Room	Mezzanine Suite
08:00-20:30	Nursing Mothers' Room	Mothers' Room-1st Floor
08:00-20:30	Lactation Room	VIP Room-3rd Floor
08:00-20:30	Press Room	Room 210
08:00-12:00	Exhibitor Teardown	Exhibit Hall B
08:00-12:00	Poster Teardown	Exhibit Hall B
09:00-17:30	L&O Spotlight Session	209 AB
09:00-10:30	Concurrent Sessions	PRCC-Various Rooms
10:30-11:00	Coffee Break	Ballroom A Foyer
11:00-12:30	Plenary & Awards Session 2019 G. Evelyn Hutchinson Award Acceptance: Oscar Schofield, Rutgers Univ., New Brunswick, NJ Plenary Presentation: Ernesto Diaz, Director, Coastal Management and Climate Change Office, PR Dept. of Natural and Environmental Resources, San Juan, PR	Ballroom A
12:30-14:00	WATER MARKS Workshop-Where Science Meets Art	Room 206
12:30-14:00	Teaching Aquatic Biology with Aquaponics Workshop	Room 103 A
12:30-14:00	Lunch Break and Entertainment	3rd Floor Terrace
14:00-15:30	Concurrent Sessions	PRCC-Various Rooms
15:30-15:45	Coffee Break	Ballroom A Foyer
15:45-17:30	Concurrent Sessions	PRCC-Various Rooms
18:00	Outreach Activity: Coral Reef and Mangrove Keys Clean-up at La Parguera	PRCC: Check in at 17:30
18:00-21:00	Closing Fiesta! (Ticketed event)	PRCC-Terrace and 3rd Floor Prefunction Area

SATURDAY, 2 MARCH 2019

08:00-19:00	Outreach Activity: Coral Reefs and Mangrove Keys Clean-Up at La Parguera	On-site
07:30-13:30	Outreach Activity: Guakia Colectivo Agroecologico	PRCC: Check in at 07:00
07:30-13:00	Outreach Activity: Monitoring Dive at a Coral Reef Ecosystem at Escambrón Beach and Condado Lagoon	PRCC: Check in at 07:15
08:00-12:00	Outreach Activity: Educational Workshop About the Role and Importance of Aquatic Macro-Invertebrates in Fresh Water Quality at the Botanical Garden of Rio Pedras	PRCC: Check in at 07:30
08:30-13:30	Outreach Activity: Restoration, Management and Conservation Project of the Las Cucharillas Marsh	PRCC: Check in at 08:00
13:00-18:00	Outreach Activity: Educational Workshop About the Role and Importance of Aquatic Macro-Invertebrates in Fresh Water Quality (Quebrada Sonadora River at El Yunque National Forest)	PRCC: Check in at 12:30

OMNIBOND (#E-27)
220 Keowee Trail
Clemson, SC 29631 USA
Contact: Amy Cannon
Phone: 864-888-7451
Email: amycannon@omnibond.com
Website: www.cloudycluster.com

OXFORD UNIVERSITY PRESS (#E-14)
New York, NY 10016 USA
Contact: Ian Sherman
Phone: 919-677-0977
Email: gab.exhibitions.us@oup.com
Website: www.global.oup.com

PRECISION MEASUREMENT ENGINEERING (#E-09)
1487 Poinsettia Avenue
Vista, CA 92081 USA
Contact: Kristin Elliott
Phone: 760-727-0300
Email: kelliott@pme.com
Website: www.pme.com

PRESENS PRECISION SENSING GMBH (#E-05)
Am BioPark 11
Regensburg, Bavaria 93053 Germany
Contact: Christina Artinger
Phone: +49 941 942 72 132
Email: christina.artinger@presens.de
Website: www.presens.de

PYROSCIENCE GMBH (#E-03)
Hubertusstr. 35
D-52064 Aachen Germany
Contact: Dr. Andrea Wieland
Phone: +49 (0)241 518322-10
Fax: +49 (0)241 518322-99
Email: info@pyro-science.com
Website: http://www.pyro-science.com

ROYAL SOCIETY PUBLISHING (#E-29)
The Royal Society
6 Carlton House Terrace
London SW1Y 5AG United Kingdom
Contact: Surayya Johar
Phone: +44 20 7451 2250
Email: surayya.johar@royalsociety.org
Website: https://royalsociety.org/journals/

SIL-INTERNATIONAL SOCIETY OF LIMNOLOGY (#E-17)
Contact: Yves Prairie
Phone: +1-514-987-3000 ext 4870
Email: president@limnology.org
Website: https://limnology.org/

SONTEK - A XYLEM BRAND (#E-12)
9940 Summers Ridge Road
San Diego, CA 92121 USA
Contact: Christina Iarossi
Phone: 858-546-8327
Email: christina.iarossi@xyleminc.com
Website: www.sontek.com

SPRINGER NATURE (#E-30)
233 Spring Street
New York, NY 10013 USA
Contact: Alexandrine Cheronet
Phone: 212-451-8766
Email: Alexandrine.Cheronet@springernature.com
Website: www.springernature.com

STATE KEY LABORATORY OF MARINE ENVIRONMENTAL SCIENCE (#E-13)
Xiam University
4221 Xiang'an South Road
Xiam, Fujian 361102 China
Contact: Ying Huang
Phone: + 86-592-2186039
Fax: + 86-592-2184101
Email: hymel@xmu.edu.cn

TURNER DESIGNS (#E-02)
1995 N. 1st Street
San Jose, CA 95112 USA
Contact: Tom Brumett
Phone: 408-749-0994
Fax: 408-749-0998
Email: marketing@turnerdesigns.com
Website: www.turnerdesigns.com

UNISENSE A/S (#E-01 AND #E10)
Tueager 1
Aarhus 8200 Denmark
Contact: Thomas Rattenborg, PhD
Phone: +45 89-44-9500
Email: tr@unisense.com
Website: www.unisense.com

UNIVERSITY OF ALASKA FAIRBANKS (#E-04)

College of Fisheries and Ocean Sciences
2150 N Koyukuk Drive
Fairbanks, AK 99701 USA
Contact: Lauren Frisch
Phone: 907-474-5350
Email: lcfrisch@alaska.edu
Website: www.uaf.edu/cfos/

UNIVERSITY OF CONNECTICUT (#E-28)

Department of Marine Sciences
University of Connecticut
1080 Shennecossett Road
Groton, CT 06340 USA
Contact: Catherine Matassa
Phone: 860-405-9028
Email: catherine.matassa@uconn.edu
Website: www.Marinesciences.uconn.edu

UNIVERSITY OF DELAWARE (#E-26)

School of Marine Science and Policy
700 Pilottown Road
Lewes, DE 19958 USA
Contact: Jennifer Biddle
Phone: 302-645-4267
Email: jfbiddle@udel.edu
Website: www.ceoe.udel.edu

LOCAL NON-PROFIT ORGANIZATIONS:

INSTITUTE FOR SOCIO-ECOLOGICAL RESEARCH (TABLE #2)

PO Box 3151
Lajas, Puerto Rico 00667-3151
Contact: Stacey Williams, PhD
Phone: 787-702-5818
Email: iser@isercaribe.org
Website: www.isercaribe.org

SEA GRANT PUERTO RICO (BOOTH #E-31)

University of Puerto Rico
Call Box 9000
Mayaguez, PR 00681-9000
Contact: Ruperto Chaparro
Phone: 787-832-4040, ext. 2165
Email: ruperto.chaparro@upr.edu
Website: https://seagrantpr.org/

SOCIEDAD AMBIENTE MARINO (SAM) (BOOTH #E-32)

PO Box 22158
San Juan, PR 00931
Contact: Samuel E. Suleimán Ramos
Phone: 939-642-7264
Email: samuelsuleiman@gmail.com
Website: http://www.sampr.org

UNIVERSITY OF THE VIRGIN ISLANDS (UVI) (BOOTH #E-33)

Masters of Marine and Environmental
Science (MMES) Program
Contact: Lorraine Buckley
Phone: 340-693-1457
Email: lbuckle2@uvi.edu

ECOEXPLORATORIO: MUSEO DE CIENCIAS DE PUERTO RICO (TABLE #1)

Suite 103 Puerto Rico Science, Technology & Research Trust
Carr. # 21, Bo. Monacillos, San Juan PR 00927
Contact: Natalie K. Laboy Flores
Phone: 787-587-9616
Email: natalie.laboy@ecoexploratorio.org
Website: www.ecoexploratorio.org

EDUCATION ACTIVITIES AND PUBLIC DISCUSSION

ASLO 2019 EDUCATION FAIR

Wednesday, 27 February 2019, 09:00 to 10:45
Location: Exhibit Hall A

Local middle school and high school students and teachers will participate in an education fair. This will include approximately 20 stations with aquatic science demonstrations and materials from organizations related to the aquatic sciences and conservation efforts. Teachers will receive activity manuals and will be advised on how to incorporate these activities and materials into their curriculum.

HIGH SCHOOL POSTER SESSION

Wednesday, 27 February 2019, 09:00 to 10:45
Location: Exhibit Hall A

This is an opportunity for high school students to present their own research. This event is in conjunction with the ASLO 2019 Education Fair. During this poster session, high school scientists will have the opportunity to describe their work and to receive positive feedback from some of the scientists in attendance at the meeting.

PUBLIC DISCUSSION-WHAT CAN AQUATIC SCIENTISTS DO FOR YOU?

Wednesday, 27 February 2019, 18:00 to 19:30
Location: Ballroom A

A panel of aquatic scientists and science communicators will respond to suggestions, comments, and questions from the public audience and community stakeholders. While the aquatic science research community has had limited success in communicating scientific results to the general public, the goal of this discussion is to seek advice and input from the public. There are many environmental problems that diminish the value of the beautiful island of Puerto Rico. Aquatic scientists would like the greater Puerto Rican community to help improve science communication, so we can all understand and improve the use of these valuable resources.

We ask the audience to consider the following questions:

1. What can aquatic scientists do for you?
2. What knowledge, in what form, and at what time would be useful to you?
3. How can we help open a continuing dialog for a better future of Puerto Ricans thriving with the environment?

Moderator: Bob Chen

Panelists and Area of Expertise:

- Jon Sharp, Professor Emeritus, University of Delaware: Estuarine Biogeochemistry
- Brian Palermo, Actor, Comedian: Improvisation, Science Communication
- Mike Pace, President of ASLO, University of Virginia: Aquatic Ecology
- Marie Thoms, Minnesota SeaGrant: Communications and Public Relations Manager
- Maritza Barreto, University of Puerto Rico Rio Piedras: Marine Geology and Coastal Processes
- Jorge Ortiz, University of Puerto Rico Rio Piedras: Limnology, Water Management

Financial support for this event has been received from the Ocean Sciences Division of the U.S. National Science Foundation.

ASLO STORYTELLERS

This will take place at EcoExploratio (ecoexploratorio.org) on 26-28 February from 15:00-16:00. Scientists attending the meeting will engage eager young minds with storytelling from the ASLO book list. This activity is organized by the ASLO Education and Outreach Committee.

SPECIAL EVENTS

LOCAL ARTISANS OF PUERTO RICO - ARTESANOS DE PUERTO RICO

Sunday, 24 February, 17:30 to 20:30
Location: 3rd Floor Foyer Area-PRCC

Immerse yourself in the culture of Puerto Rico by visiting artisans during the Opening Reception on Sunday evening. This select group will be available at the PRCC from 17:30 to 20:30 in the 3rd floor lobby and terrace area, adjacent to the reception. Creations include custom jewelry, clay works representative of San Juan architecture, small wood-carved sculpture, leather work, small paintings, and handmade wooden pens, and others. By visiting these artists, you will preserve the artisanal work of Puerto Rico and take home unique, handmade mementos from San Juan and ASLO 2019.

ART AND SCIENCE WORKSHOPS

WATER MARKS: WHERE ART MEETS SCIENCE

Monday, 25 February - Friday, 1 March, 12:00 to 14:30
Location: Room 206

How do artists embrace scientific concerns and translate them across diverse art forms? How can both disciplines benefit from an increased exchange of practices? What new insights and methods might we discover together? Water Marks activities will be facilitated by six artists from across the Americas and Puerto Rico during the 2019 Aquatic Sciences Meeting. Together artists and scientists will be exploring the materials and methods that artists use to investigate scientific themes focused on water. You will also be introduced to the work of local artists, providing you with a personalized vision of environmental challenges on the island. Through a week-long series of workshops, attendees can choose from a wide variety of hands-on activities including: media documentation using cellphone cameras and the 360 camera; guided explorations of the environment through movement; sound composition; and the creation of ephemeral installations using materials sourced from the area. Our goal is to cultivate greater understanding and collaboration between artists and scientists regarding aquatic environments, both during the conference and into the future. Workshops will be scheduled during lunch hours to avoid competition with the science sessions. No prior skills are required. Enjoyment guaranteed. If you have questions about this event, please contact the workshop organizers, Lauren Elder (lelderoakland@gmail.com) or Elizabeth Miller (elizabeth.miller@concordia.ca).

WORKSHOPS, AUXILIARY EVENTS, AND TOWN HALLS

AQUATIC SCIENCE EDUCATION AND OUTREACH WORKSHOP

Sunday, 24 February 2019, 13:00 to 17:00
Location: Room 208 C

This workshop will focus on helping participants develop ideas for effective education and outreach activities based on their research. Featuring active, hands-on learning, small group discussions, and guided inquiry, this workshop will include short presentations on exemplary projects in formal and informal education designed for K-12, undergraduate, graduate, and public audiences to stimulate ideas. Discussions of how people learn, how to assess the effectiveness of outreach activities, and how to develop projects that meet specific goals will help support project development. Participants are welcome to bring ideas that they would like to develop and share, and for which they would like to receive feedback. Organized by Cynthia Hagley (University of Minnesota, Duluth; chagley@d.umn.edu) and Bob Chen (University of Massachusetts Boston; bob.chen@umb.edu) Workshop is open to all attendees; walk-ins are welcome. While there is no need to register for this workshop, if you have any questions, please do not hesitate to email the organizers.

LOREX ORIENTATION WORKSHOP

This is an invitation-only event.
Sunday, 24 February 2019, 14:00 to 18:00
Location: Room 208 AB

The Limnology and Oceanography Research Exchange (LOREX) program Orientation Workshop will create a space for the principal investigators to provide an overview of the LOREX program, provide time for participants to ask questions and express concerns, and create a space for LOREX participants to meet each other at the beginning of the meeting. Additionally, LOREX participants will learn how to integrate improv skills to facilitate international collaboration and communicate their science more effectively from improv actor, Brian Palermo. Attendance is limited to LOREX program participants and collaborators. For more information about this event, please contact: Adrienne Sponberg at sponberg@aslo.org.

Tentative agenda follows:

14:00-14:30	Welcome, Personnel Introductions, Program Expectations
14:30-16:00	Improv to Improve Collaboration with Brian Palermo
16:00-16:30	Q&A with full group including former IRES students
16:30-17:00	Next Steps for the Program
17:00-17:45	Breakout sessions by site to give hosts and students an opportunity to discuss site-specific concerns/issues
17:45-18:00	Evaluation questionnaire

THE SCIENCE OF SCIENCE COMMUNICATION WORKSHOP

Sunday, 24 February 2019, 08:00 to 17:00
Location: Room 103 A

Science communication is a diverse and transdisciplinary field that has been gaining traction in recent years. In tandem, the expectation that scientists become proficient at science communication to all audiences is rising. Early career scientists (here defined as undergraduate students, graduate students and those within a few years of graduation from a graduate degree program) are particularly in need of effective science communication skills as they are either beginning their science career at the undergraduate level or developing their scientific specialty at the graduate level. In order to bridge the gap between the scientific and public audiences, this workshop will focus on oral and visual communication skills. To register for this event, please sign up for free at http://signup.com/go/oQqqOpo. If you would like more information, you can contact: Hayley Schiebel at hayley.schiebel@gmail.com.

MAKE YOUR SCIENCE COMMUNICATION MORE EFFECTIVE WORKSHOP

Sunday, 24 February 2019, 10:00 to 13:00
Location: Room 101

The Science Communication Lab in Puerto Rico will help you improve your communications skills so you can present your work more effectively. The Hollywood entertainment industry has traditionally been the source of both innovation and perfection of narrative elements. The same approach is applicable to the communication of science to all audiences, from the general public to fellow academics. Communications expert, Brian Palermo, will return to ASLO to lead the 2019 workshop. He is a professional actor and improv instructor (e.g., The Social Network, The Tonight Show with Jay Leno, the Groundlings Theatre, Los Angeles). It will build on the success of previous workshops by Palermo (2012, 2013, 2015, 2016, 2017, 2018) also organized by Jonathan Sharp (University of Delaware) and Adrienne Sponberg (ASLO). There will be a 3-hour workshop on Sunday (10:00-13:00) before the opening of the meeting. The workshop is free and is open to all, but participants must be pre-registered to participate. It will be a hands-on workshop with interaction between Palermo and audience members. Financial support for this workshop has been received from the Ocean Sciences Division of the US National Science Foundation. For more information about this event, please contact: Adrienne Sponberg at sponberg@aslo.org.

TEACHING INTRODUCTORY AQUATIC SCIENCES COURSES WORKSHOP

Monday, 25 February 2019, 12:30 to 14:00
Location: Room 101

Introductory aquatic and environmental science courses provide an excellent opportunity to prepare both majors and non-majors for thinking about some of the largest issues facing society such

as climate change and energy needs. Introductory courses can also serve to recruit students from highly diverse backgrounds into the field and/or to engage students of all majors to think deeply about environmental issues such as climate change. Often, these courses are large (>50 students). This workshop will provide strategies to overcome some of the challenges of these large introductory courses while making your teaching engaging, relevant, and effective for all students. A special emphasis is placed on a few big concepts that can help organize examples and knowledge for students to grapple with future ocean and environmental issues. Come ready to share ideas, to think actively about teaching and learning, and to discuss what works and why. For more information about this event, please contact: Robert Chen at bob.chen@umb.edu.

IMPROV TO IMPROVE INTERNATIONAL COLLABORATIONS WORKSHOP

Monday, 25 February 2019, 12:30 to 14:00
Location: Room 102

Initiating and maintaining collaborative research programs can be daunting. The additional challenges inherent to international research (e.g., language and cultural differences) can stop many potentially fruitful collaborations before they even begin. Join communication expert, actor, and improv comedy instructor, Brian Palermo, to learn how Hollywood narrative structure can help you initiate research collaborations and how Improv training can improve communication across language and cultural barriers. For more information about this event, please contact: Adrienne Sponberg at asponberg@aslo.org

FLOWCAM SOFTWARE WORKSHOP

Monday, 25 February 2019, 12:30 to 14:00
Location: Room 103 A

A demonstration of the newly released FlowCam software-VisualSpreadsheet 5.0. VisualSpreadsheet 5.0 allows the user to analyze multiple runs, enabling the comparison and contrast of data sets. "5.0" is a useful tool for longitudinal studies, time-series analyses, trend spotting, etc. The new FlowCam 8400 will be demonstrated as well. The workshop is geared towards existing users of the FlowCam.

For more information about this event, please contact: Harry Nelson at harry.nelson@fluidimaging.com or Frances Buerkens at frances.buerkens@fluidimaging.com.

INTERDISCIPLINARY PRESENTATIONS WORKSHOP

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 102

Limnology and Oceanography as multi-disciplinary sciences, combine aspects of physics, chemistry, biology, and geology; and often include socio-economics. At a meeting like this one, you could make a narrow-scope presentation, without setting up the reason for the project, using specialized terminology, and not explaining the relevance of your results. That may be fine for a small number of specialty peers. However, if you want to reach and appeal to a broader interdisciplinary audience, you need another approach. This workshop will address things to make presentations more engaging and appealing to those outside your specialty as well as those in your narrow research area. This workshop is open to all. It is organized by Jonathan Sharp (University of Delaware) and Adrienne Sponberg (ASLO); using the skills of communications expert, Brian Palermo (a professional actor and improv instructor; e.g., The Social Network, The Tonight Show with Jay Leno, the Groundlings Theatre, Los Angeles). We will attend a few presentations early in the 2019 Puerto Rico ASLO Aquatic Sciences Meeting. At the workshop, we will discuss specialty presentations in general, with some specific reference to what we saw in those talks. The main goal is to illustrate how to make improvements to transform a specialty presentation into one that is memorable and compelling to a broad interdisciplinary audience. Similar workshops have been conducted with help from Palermo in 2013, 2015, 2016, 2017 and 2018. For more information about this event, please contact: Jonathan Sharp at jsharp@udel.edu.

ELEMENTS OF A TEACHING PHILOSOPHY STATEMENT WORKSHOP

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 208 C

Seeking a career in academia-in teaching, research, or both? Regardless of your academic career path, chances are, you will need to include a teaching philosophy statement in your application package. In this workshop we will cover the elements of a good teaching philosophy statement and how to customize statements to match job descriptions and get your application noticed. Participants will leave the workshop prepared to write their own teaching philosophy statement. For more information about this event, please contact: Kylla Benes at kylla.benes@umontana.edu. Please register for the event online at https://www.surveymonkey.com/r/QWGZ237.

INTRODUCING AN OCEAN PROTEIN PORTAL TOWN HALL

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 204

We are announcing the release of the Ocean Protein Portal, an NSF EarthCube Prototype website that allows scientists to explore ocean metaproteomic datasets for research purposes. The event will include a description and demo of the portal, followed by a Q&A period. For more information about this event, please contact: Mak Saito at msaito@whoi.edu.

REAL-TIME QC OF PH DATA WORKSHOP

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 209 C

The U.S. IOOS Quality Assurance / Quality Control of Real-Time Oceanographic Data (QARTOD) will kick off the creation of a RT

QC manual for pH observations. All are invited to this workshop to discuss the methods and challenges involved, identify tests to employ, and determine the scope of this manual. For more information about this event, please contact: Mark Bushnell at mark.bushnell@noaa.gov.

RESEARCH COLLABORATION ACROSS BORDERS PANEL DISCUSSION

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 101

As challenges in the aquatic sciences become more global in nature, international research collaborations are increasingly prevalent, with over 35% of worldwide publications produced as a result of international partnerships. Join a panel of experts (TBA) that have successfully led international collaborations to discuss the unique set of challenges associated with conducting research internationally. Discussion topics will include identifying and contacting potential collaborators, strengthening ongoing collaborations, addressing language and cultural differences, and determining project goals, timelines, authorship, and logistics for fruitful collaborations. For more information about this event, please contact: Adrienne Sponberg at sponberg@aslo.org.

LIMNOLOGY & OCEANOGRAPHY WIKIWORKSHOP

Wednesday, 27 February 2019, 12:30 to 14:00
Location: Room 204

Wikipedia is among the most common ways people interact with information on the internet. When googling a search term, the Wikipedia article is generally the first hit and most read result, but many Limnology- and Oceanography-related Wikipedia articles are short and/or of low-quality. By improving the quality of Wikipedia articles in our research fields, we can improve general understanding and appreciation of Limnology and Oceanography. The Ecological Dissertations in the Aquatic Sciences (ECO-DAS) 2018 working group has launched an initiative to improve Wikipedia articles related to Limnology and Oceanography and we would like to expand this initiative to the larger ASLO community. The Limnology & Oceanography WikiWorkshop at ASLO 2019 will include a tutorial for editing Wikipedia pages and a short edit-a-thon. We will provide instructional materials for participants who want to organize similar workshops at their home institutions and we will advertise for future online edit-a-thons. For more information about this event, please contact: Margaret Brisbin at margaret.marsbrisbin@oist.jp.

PROJECT REDEFINING RECOGNITION TOWN HALL

Wednesday, 27 February 2019, 12:30 to 14:00
Location: Room 101

Project Redefining Recognition: Challenging Traditional Research Assessment—In our current system, research assessment and reward are weighted heavily towards scientific inputs and outputs, and research impact is measured in funding dollars awarded and research articles produced. The value of science, however, extends well beyond contributions to the research activity itself. Project Redefining Recognition is a cross-society initiative which seeks to develop a shared future vision and clear set of actions across the Earth, Space, and Environmental Sciences to expand research assessment to explicitly include, recognize, and reward actions that advance open science, broad communication and public engagement. As part of Project Redefining Recognition this session will look closely at the ways in which ASLO researchers are sharing and applying their science, and the challenges and opportunities in our current system to recognize and reward those efforts. ASLO Director of Communications and Science Adrienne Sponberg will moderate a panel discussion and audience Q&A with experts from funding agencies, academia and scientific journals about how key players are changing how they assess research impact. For more information about this event, please contact: Adrienne Sponberg at sponberg@aslo.org.

UPDATE AND STATUS OF THE ARCTIC-COLORS SCIENCE PROGRAM TOWN HALL

Wednesday, 27 February 2019, 12:30 to 14:00
Location: Room 209 C

Arctic-COLORS (AC) is a candidate NASA Ocean Biology and Biogeochemistry Program field campaign program. A NASA panel has reviewed and is endorsing the AC Science Plan. NASA will be moving forward on developing a plan for its implementation. AC proposes an integrative, interdisciplinary approach that combines detailed process studies, field surveys, advanced modeling tools, and enhanced remote-sensing retrievals from various platforms (ship-based, in-situ autonomous, airborne, and space-based). AC aims to quantify the response of the Arctic coastal environment to global change and anthropogenic disturbances-an imperative for developing mitigation and adaptation strategies for the region. The Arctic-COLORS field campaign represents the first attempt to study the nearshore coastal Arctic (from riverine deltas and estuaries out to the coastal sea) as an integrated land-ocean atmosphere-biosphere system, as is required to determine present and future impacts of terrigenous, atmospheric and oceanic fluxes on coastal ecology, biogeochemistry and ecosystem services in the context of environmental (short-term) and climate (long-term) changes. An update on the status of the Arctic-COLORS will be presented with Q&A and community input to follow. For more information about this event, please contact: Joe Salisbury at joe.salisbury@unh.edu.

NATIONAL SCIENCE FOUNDATION OCEAN SCIENCES TOWN HALL

Wednesday, 27 February 2019, 18:00 to 21:00
Location: Room 204

An update from the U.S. National Science Foundation, Division of Ocean Sciences, regarding recent developments in research funding, infrastructure, and education. A budgetary outlook and

discussion of the Division of Ocean Sciences initiatives throughout the Division portfolio will be presented. The Division Director will provide a brief presentation, to be followed by a question-and-answer session. National Science Foundation Program Officers will also be present to provide additional information as needed. For more information about this event, please contact: Terrence Quinn at terrymquinn@gmail.com.

MIXOTROPHY WORKSHOP

Wednesday, 27 February 2019, 18:00 to 21:00
Location: Room 103 B

During this workshop, participants will discuss ideas and future plans relating to SS062: Mixotrophic Protists: An Underrated Majority In Marine And Freshwater Ecosystems? For more information about this event, please contact: Stella Berger at berger@igb-berlin.de.

STRATEGIES FOR TRANSBOUNDARY HABS MANAGEMENT TOWN HALL

Thursday, 27 February 2019, 12:30 to 14:00
Location: Room 103 B

All ASLO participants are invited to participate in a town-hall style gathering to share strategies for transboundary HABs management. Many waters suffering from Harmful Algal Blooms are "transboundary"—shared by multiple political jurisdictions. This geo-political reality presents unique challenges and opportunities for HABS research and management. Facilitators will share perspectives from the binational Great Lakes region. Most of the time will be spent asking participants to share their perspectives or experiences in transboundary collaboration to synthesize, translate, and/or deliver science to HABs managers. Participants will also be asked to share experiences with engaging managers to understand their science needs. They will also be asked to share lessons learned about what hasn't worked and how we can we use those lessons to improve future activities. Finally, participants will document (for their keeping), and have the opportunity share with others, the strategies from this town hall discussion they plan to take back to their work. For more information about this event, please contact: Victoria Pebbles at vpebbles@glc.org.

APPLYING TO GRADUATE SCHOOL WORKSHOP: TIPS FROM CURRENT STUDENTS

Thursday, 27 February 2019, 12:30 to 14:00
Location: Room 103 A

Applying to graduate school is a long process and can feel overwhelming at first. In this workshop, future graduate students will hear from current students in a variety of fields and programs who have recently gone through this process. We will discuss many aspects of the application process, including deciding where to apply, writing a personal statement, asking for recommendation letters, and the increasingly important grad school interview. We will focus primarily on ASLO-related fields but hope that this workshop will be valuable to students considering any field of study. This will be a very active workshop-by students, for students. Our goal is that each attendant will leave with a concrete "game plan" for attacking application season and achieving their goals. All are welcome, but this event will be especially helpful to students planning to apply in the next 1-2 years. Advance registration is recommended by filling out the form at https://goo.gl/forms/8U6ALot4EbiGnLLg1 Contact Noelle Held (nheld@whoi.edu) for more information.

TEACHING AND MENTORING UNDER THE THREAT OF CLIMATE CHANGE

Thursday, 28 February 2019, 12:30 to 14:00
Location: Room 204

During this workshop, we will discuss strategies for effectively conveying information to students about the impacts of climate change on aquatic environments, as well as methods for combating the anxiety or apathy that may occur while living and working under this existential threat. Climate change is upon us and we are being bombarded with information about the negative effects of global warming. People from all walks of life-scientists, educators, politicians, journalists, and young and adult citizens-talk endlessly about this urgent issue, however, the relentless rhetoric about climate change impacts may result in fatigue and apathy from too much information about a seemingly insurmountable problem. Through guided discussions, we will answer questions, share ideas, and discuss what the best approach may be for educators and mentors to effectively reach students and mentees. But most importantly, we will need to hear from students and mentees about positive and effective ways to teach climate change awareness to new generations of students and scientists while increasing resiliency and optimism about the future for all. For more information about this event, please contact: Diana Varela at dvarela@uvic.ca.

WRITING EFFECTIVE ABSTRACTS AND SUMMARIES WORKSHOP

Thursday, 28 February 2019, 12:30 to 13:30
Location: Room 209 C

Journal articles are the key means of communicating scientific research. Unfortunately, academic texts are often characterized by a complicated writing style and abundance of jargon, which reduces clarity and effectiveness. Join ASLO Raelyn Cole Editorial Fellows Dr. Laura Falkenberg and Dr. Kelsey Poulson-Ellestad to explore these issues for critical components of your scientific articles, including the title, abstract, keywords, and summary. Our goal for this workshop is to enhance your awareness of these important issues, and increase your ability to write articles that are clear and accessible to a broad audience.For more information about this workshop, please contact Kelsey Poulson-Ellestad at kpoulsonellestad@roosevelt.edu.

TEACHING AQUATIC BIOLOGY WITH AQUAPONICS WORKSHOP

Friday, 1 March 2019, 12:30 to 14:00
Location: Room 103 A

We use fairly inexpensive catalogs bought and DIY aquaponic and hydroponic systems as the context and background for classes in introductory biology, microbiology and more. We have developed free digital content to support this pedagogy.The workshop will focus on digital content delivery as well as modeling systems we have developed for classroom and lab environments. For more information about this event, please contact: Christopher Perle at crperle@gmail.com.

MEETINGS AND WORKING GROUPS

DEEP SEARCH ANNUAL PROJECT MEETING

Sunday, 24 February 2019, 8:00 to 17:00
Location: Room 103 B

Deep Search is a large research program funded by BOEM, USGS, and NOAA OER under the NOPP umbrella. The program is two years into its 5-year duration and three research expeditions have been carried out so far. This meeting will bring together the PIs, their research labs, representatives of the funding agencies, the project's science review board, and other collaborators to discuss the project. This meeting will review the progress made to date, the plans for the upcoming field work, and discuss plans for synthesis and other potential projects that may not have been part of the original proposal. We welcome members of the community to join the meeting and learn more about the project and potential collaborations. For more information about this event, please contact: Erik Cordes at ecordes@temple.edu.

SOCIETY FOR WOMEN IN MARINE SCIENCE (SWMS) MEETING

Tuesday, 26 February 2019, 19:00 to 19:30
Location: Room 204

We welcome anyone who is currently a member of the Society for Women in Marine Science (SWMS) or anyone who would like to become a member of SWMS to come to this event. We will be providing some snacks and refreshments. Please RSVP here (https://goo.gl/mKQQPq) if you will be attending the meet up so that we can get an accurate headcount. Everyone is welcome to attend, even if they are not participating in the conference. For more information about this event, please contact: Sophie Chu at sochu@uw.edu.

JPR EDITORIAL BOARD GET TOGETHER AND MEETING

This is an invitation-only event.
Wednesday, 27 February 2019, 12:30 to 14:00
Location: Room 208 AB

A chance for all to get to know one another, learn who does what, and offer ideas for improving and strengthening the journal. The informal meeting will be run by the Editor-in-Chief of JPR (John Dolan) and Ian Sherman of Oxford University Press and open only to the Associate Editors and Editorial Board Members of JPR. For more information about this event, please contact: John Dolan at dolan@obs-vlfr.fr.

BLUE CARBON DISCUSSION GROUP

Thursday, 28 February 2019, 12:30 to 14:00
Location: Room 208 C

Following the celebration of the Special Session 19: "Blue Carbon, from the ecosystem to the markets," this workshop will serve to discuss in-depth some of the key issues presented during the session and to propose new ideas and strategies for this research field. For more information about this event, please contact: Miguel Mateo at mateo@ceab.csic.es.

SOCIETY EVENTS

OPENING MIXER RECEPTION

Sunday, 24 February 2019, 18:30 to 20:30
Location: 3rd Floor Terrace and Foyer

Be sure to arrive in time for the opening mixer on Sunday evening. We will celebrate the culture of Puerto Rico with local artisans displaying their crafts, entertainment to get us into the Latin spirit, and food and drinks with the flavors of the culture that created them. Rums of Puerto Rico will be hosting a rum bar so that attendees can taste the various rums available on the island. (Soft drinks also will be available.) This will be a great time to meet with colleagues and enjoy the beauty of San Juan from the terrace at the Puerto Rico Convention Center. Artesanos de Puerto Rico (Artisans of Puerto Rico) will be set up during the reception. Select unique creations include custom jewelry, clay works representative of San Juan architecture, small wood-carved sculpture, leather work, small paintings, and handmade wooden pens, and others. You will take home unique, handmade mementos from San Juan and ASLO 2019.

ASLO FELLOWS AND MEMBERSHIP RECEPTION

Monday, 25 February 2019, 17:30 to 18:30
Location: Ballroom B

A reception honoring ASLO fellows and sustaining fellows will precede the annual business meeting on Monday evening. This will be a great time to meet and to talk to ASLO officers and board members and to congratulate ASLO fellows. Reception food and drink will be served. Everyone is encouraged to attend the business meeting and the membership reception—especially new ASLO members and student members. The timing is planned so you can attend the business meeting and reception before you head out to the student or early career mixer.

ASLO BUSINESS MEETING

Monday, 25 February 2019, 18:30 to 19:30
Location: Ballroom A

The annual ASLO Business Meeting will be held during the conference on Monday evening. A reception honoring ASLO fellows and sustaining fellows will precede the business meeting, and food and drinks will be available during that time. The business meeting is open to all attendees—members and non-members. This will be a great opportunity to meet and talk to the ASLO officers, board members, and staff.

STUDENT EVENTS

STUDENT WORKER TRAINING SESSION

Students who have signed up to serve as student worker room monitors must attendee this training session. Meet at 17:30 near the registration desk area on the first floor (near the entrance to the exhibit hall). Please contact Sue Rulla at suer@sgmeet.com for more information or if you have a travel conflict and will not be available to attend the training session on Sunday.

SUNDAY EVENING PUB CRAWL

Grab an Uber to Old San Juan and join fellow students for the opening pub crawl following the opening mixer on Sunday evening. This is an opportunity to meet up with student participants before the meeting begins! Location: La Taberna Lúpulo at 151 Calle San Sebastián, San Juan, 00901.

ASLO STUDENT MIXER

Monday, 25 February 2019, 19:30 to 20:30
Location: 3rd Floor Terrace and Lobby

Come network with senior scientists and other students at the annual student mixer. This will follow the ASLO business and membership meeting on Monday evening. A limited number of drink tickets will be available!

ASLO STUDENT WORKSHOP - I NEED THE DEGREE AND THE MONEY

Tuesday, 26 February 2019, 12:30 to 14:00
Location: Room 104

A panel of experts will provide tips and tricks on the negotiations process of the job search. A limited number of box lunches will be available to those who participate in the workshop.

ASLO STUDENT BEACH CLEANUP

Thursday, 28 February 2019, 12:30 to 14:00
Exact location and departure location will be announced.

Join other students as they head over to a local beach to help beautify the area! A limited number of bag lunches will be provided by ASLO to those who participate. Please RSVP to tiaranmoore@gmail.com if you plan to participate in the beach cleanup.

EARLY CAREER EVENTS

EARLY CAREER SOCIAL MIXER

Monday, 25 February 2019, 19:30 to 20:30
Location: 2nd Floor Prefunction Area

A "meet and mix" reception is planned and organized by members of the ASLO early career (EC) committee to give early career members an opportunity to provide feedback on various topics relevant to them, including any concerns or expectations as an early career member. This is a social gathering for early career members to get to know each other and to network. Beverages and snacks will be served, and drink tickets will be available at the door (for soft drinks and limited alcoholic beverages). Come and meet the ASLO Board and members of the EC committee!

ASLO EARLY CAREER WORKSHOP - HOW TO SUCCESSFULLY WRITE PROPOSALS AND RECEIVE FUNDING

Wednesday, 27 February 2019, 12:30 to 14:00
Location: Room 102

A scientist's career depends mainly on research funds. Therefore, writing good applications is fundamental for career building. Based on a recent questionnaire sent to ASLO's early career researchers (ECRs), we offer a workshop on how to successfully write proposals and receive funding. We will invite both representatives of science foundations (e.g. NSF) and experienced scientists who have been successful in obtaining research funding from various agencies and organizations. There are many possibilities for funding such as the national research foundations for basic and applied sciences, local governments, private foundations, etc. To better navigate through the many funding options, the workshop aims to give you helpful advice. There will be time for questions related to individual cases. We also hope to provide helpful suggestions on how to write a good proposal based on practical aspects which will be put into practice before the writing process begins. The workshop also will allow for active exchange of knowledge and experience among senior scientists and those who participate. ASLO has high interest to provide solid advice pertaining to successful research career development of the community's young talents, i.e. you. For more information about this workshop, please contact: Hans-Peter Grossart at hgrossart@igb-berlin.de.

ASLO MEETING MENTOR PROGRAM

The ASLO Meeting Mentor Program is open to any participant looking for guidance on navigating the meeting and making new connections. Mentees are grouped with experienced scientists (mentors) who will provide guidance on navigating the meeting and introduce them to other scientists throughout the week. If you are participating as either a mentor or a mentee, please wear your badge ribbon that identifies you as a participant in this program. Ribbons will be available at registration when you pick up your badge and meeting materials. Room 203 on the 2nd floor of the convention

center is available to mentors and mentees throughout the week. This is set aside as a place to meet and discuss the week's activities.

MEETING FOR ASLO MEETING MENTORS AND MENTEES

Tuesday, 26 February, 08:00 to 08:45
Location: Ballroom B

This is a time for those who signed up to be ASLO meeting mentors and mentees and will take place first thing Tuesday morning prior to the first concurrent sessions. A light breakfast will be available to those who are participating.

ASLO MULTICULTURAL PROGRAM

Starting in 1990, the ASLO Multicultural Program has brought over 1,000 diverse undergraduate and graduate students to the annual ASLO meetings. The program features pre-conference dinner and field trip, meeting-mentors to help guide the students, a student-symposium, and various other activities. The goal of the program is to increase the human diversity of aquatic scientists and supports the full cost of participation in the meeting including travel, hotel, food and meeting registration. The program for 2019 will feature a special field trip taking advantage of the local environment around in Puerto Rico and a service learning activity. Mentors and students will go to some sessions together, and mentors will help guide their mentees through the meeting and introduce them to other scientists in the field. If you have questions about the program or the requirements, please contact Benjamin Cuker, benjamin.cuker@hamptonu.edu, ASLO Multicultural Program Director.

ASLO 2019 MULTICULTURAL PROGRAM MEETING OF MENTORS AND MENTEES

Sunday, 24 February , 16:00 to 18:00
Location: Ballroom B-3rd Floor

SOCIAL AND EVENING EVENTS

JAM SESSION

Wednesday, 27 February 2019, 20:00 to 23:00
Location: TBD

ASLO scientists-musicians will take the stage to jam for everyone! This event will be an opportunity to enjoy the musical talents of fellow scientists, mercifully supported by a great local band. Plan to experience an evening full of fun that will include dancing and making (or just enjoying) music. Having occurred during the 2010, 2012, 2014, and 2016 Ocean Sciences Meetings, we are continuing the tradition at this meeting. Musicians can bring their own musical instruments, or they can borrow one from the band on stage. For more information on this exciting event, contact Anne Deininger at anne.deininger@posteo.de or John Downing at downing@d.umn.edu. If you'd like to play or sing, please sign up by contacting Anne or John, add your name to the list near the meeting registration area, or just show up!

FIESTA!

Advance tickets are required.
Friday, 1 March 2019, 18:00 to 21:00
Location: 3rd Floor Terrace

An evening carefully planned to enhance the venue of San Juan. You can salsa dance to the sounds of one of San Juan's finest bands and enjoy a buffet with lots of local flair. You can salsa dance to the sounds of one of San Juan's finest bands and enjoy a buffet with lots of local flair. A ticket is required for this event, and space is limited. You must purchase tickets in advance, and they may not be available on site. Cost: $75.00 per person for professionals including those who are early career; $45.00 for students.

OUTREACH ACTIVITIES

San Juan's location makes it a natural host for an Aquatic Sciences Meeting with its freshwater ecosystems, extensive mangroves, seagrass beds, and the surrounding coral reefs. When Hurricane Maria swept across the island in September 2017, the meeting organizers and all of those in ASLO felt a strong attachment to the citizens of Puerto Rico and their struggles amid the devastation. Outreach activities were planned to allow the ASLO community to do its part by providing lots of hands and feet willing to work to help in the restoration efforts while they were here. During this week, meeting participants have signed up to take part in educational activities, volunteer opportunities, and culturally relevant events that will focus on environmental and ecosystem restoration as well as the resilience of the land. Almost of the outreach activities that are planned sold out in October. Participants who are participating in the outreach activities should have been contacted to receive information and instructions from the organizer. If you need more information, please visit the web site at: https://aslo.org/sanjuan2019/outreach. Advance registration is required for the outreach activities.

PRESENTER INFORMATION

ON-SITE SUBMISSION OF ORAL PRESENTATIONS

All oral presentations will need to be submitted in the Presentation Room, Room 207, located on the 2nd floor of the Puerto Rico Convention Center. This room will be staffed and run by audio visual technicians. Presenters may submit their presentations beginning at 15:00 on Sunday, 24 February. If you were assigned to give a talk, and you did not upload your slides in advance, you will need to upload your presentation preferably 24 hours in advance. Those who are presenting on Monday need to arrive in time to upload on Sunday.

All presenters are required to check in at the Presentation Room. This is the single most important action you will take to ensure your presentation is a success. Even if you submitted your presentation in advance via the online upload link, please check in at the Presentation Room to make sure that your presentation was

received. This will be your opportunity to review all fonts, images, and animations appear as expected and that all audio or video clips are working properly. You also can make minor changes to your presentation at this time. You must tell the A/V technician you have finalized your presentation file before you leave the Presentation Room. Be sure to bring a backup copy of your presentation with you to the meeting. USB/Flash drives are preferred. Please make sure you have all power, video, and networking adapters with you before you come to the Presentation Room.

Note: Personal laptops cannot be used in the session rooms. Dedicated internet access will not be available in the session rooms and cannot be used for presentations.

Hours for the presentation room are listed below:

Sunday	15:00 to 21:00
Monday	07:00 to 19:00
Tuesday	07:00 to 19:00
Wednesday	07:00 to 19:00
Thursday	07:00 to 19:00
Friday	07:00 to 17:00

POSTER SESSIONS AND RECEPTIONS

Poster sessions and receptions are planned Tuesday and Thursday from 17:30 to 19:00 in Exhibit Hall B. Posters will be organized in topical and session groupings for the entire meeting to maximize opportunities for viewing. Though poster presenters have been assigned a specific day for interaction with attendees during the poster session, you may be at your poster any time the area is open. The poster session times do not conflict with concurrent oral presentations.

POSTER SET UP AND TEARDOWN

Poster numbers are consecutive to placement on poster boards in the exhibit/poster hall. Poster numbers are included in this book under the scientific program schedule. Posters can go up Monday, 25 February, from 12:30 to 17:00 and will remain in place through 19:00 on Thursday, 28 February. They must be removed by 21:00 following the final poster session on Thursday or from 09:00 to 12:00 on Friday. Important note regarding poster presentations: The convention decorator may discard posters if the presenting author does not dismantle and remove them according to the posted times.

POSTER PRINTING IN SAN JUAN

If you had your poster printed locally by Go Print, you may pick it up on Monday from 08:00 to 17:00 near the meeting registration desk. Please call or contact Go Print to confirm that your order will be ready for pick up when you arrive: (787) 200-8085 or http://www.goprintpr.com/

MONDAY SCHEDULE-AT-A-GLANCE

Room	101	102	103 A	103 B	104	201	202	204	208 AB	208 C	209 AB	209 C
	SS036	**SS013**	**SS063**	**SS057**	**SS039**	**SS009**	**CS013**	**SS010**	**SS040**	**SS005**	**SS028**	**SS003**
09:00-10:30	Ecological Applications of Earth System Models and Regional Climate Models	Adaptation of Aquatic Biodiversity to Global Change	Ocean, Coastal, and Freshwater Acidification: Research and Education	Eastern Boundary Upwelling Systems in a Changing Ocean: Recent insights and future perspectives	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Hagase la luz- Light in aquatic ecosystems: Variability and ecological consequences	Dissolved Organic Matter-DOC, DON, DOP, fDOM, cDOM	Exploring What Makes Undergraduate Research Experiences Work: Evidence from students and mentors	Radionuclides in Aqueous Systems	ASLOMP Student Symposium	The Multiple Challenges of Rapidly Changing Tropical Freshwater Ecosystems	Adventures, Challenges, and Benefits of Conducting International Collaborative Research
10:30-11:00	COFFEE BREAK (Ballroom A Foyer-3rd Floor)											
11:00-12:30	PLENARY AND AWARD TALK SESSION (Ballroom A) 2019 Ruth Patrick Award Acceptance: Jennifer Tank, University of Notre Dame, Notre Dame, IN Plenary Presentation Via Video: U.S. Senator Sheldon Whitehouse (D-RI), Newport, RI											
12:30-14:00	LUNCH-WORKSHOPS AND ANCILLARY MEETINGS PLUS ENTERTAINMENT ON THE TERRACE (Salsa Dance Lessons)											
	SS036	**SS013**	**SS037**	**SS057**	**SS039**	**SS009**	**CS013**	**SS010**	**SS051**	**SS005**	**SS028**	**SS003**
14:00-15:30	Ecological Applications of Earth System Models and Regional Climate Models	Adaptation of Aquatic Biodiversity to Global Change	Terrestrial Materials in Planet Water: Tracking influences along the land-ocean continuum	Eastern Boundary Upwelling Systems in a Changing Ocean: Recent insights and future perspectives	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Hagase la luz- Light in aquatic ecosystems: Variability and ecological consequences	Dissolved Organic Matter-DOC, DON, DOP, fDOM, cDOM	Exploring What Makes Undergraduate Research Experiences Work: Evidence from students and mentors	New Views on the Biological Transformation of Metals in the Marine Environment	ASLOMP Student Symposium	The Multiple Challenges of Rapidly Changing Tropical Freshwater Ecosystems	Adventures, Challenges, and Benefits of Conducting International Collaborative Research
15:30-15:45	COFFEE BREAK (Ballroom A Foyer-3rd Floor)											
	AS002	**SS013**	**SS037**	**SS050**	**SS039**	**SS009**	**CS027**	**SS010**	**SS051**	**CS004**	**SS028**	**SS003**
15:45-17:30	Human and Social Dimensions-Research Meets Management: Where and How the Rubber Hits the Road (and Gets Traction)	Adaptation of Aquatic Biodiversity to Global Change	Terrestrial Materials in Planet Water: Tracking influences along the land-ocean continuum	Improving Coral Reef Resilience with Transformational Science	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Hagase la luz- Light in aquatic ecosystems: Variability and ecological consequences	Viruses	Exploring What Makes Undergraduate Research Experiences Work: Evidence from students and mentors	New Views on the Biological Transformation of Metals in the Marine Environment	Urban Ecosystems	The Multiple Challenges of Rapidly Changing Tropical Freshwater Ecosystems	Adventures, Challenges, and Benefits of Conducting International Collaborative Research
17:30-18:30	ASLO Fellows and Membership Reception (Ballroom B Foyer-3rd Floor)											
18:30-19:30	ASLO Business Meeting / Membership Meeting (Ballroom A)											
19:30-20:30	Student Mixer (Terrace and Foyer-3rd Floor) Early Career Mixer (Prefunction Area-2nd Floor)											

TUESDAY SCHEDULE-AT-A-GLANCE

Room	101	102	103 A	103 B	104	201	202	204	208 AB	208 C	209 AB	209 C
08:00-08:45	ASLO MENTOR PROGRAM MEETING (Includes light breakfast) (Ballroom B)											
	CS010	SS005	SS013	SS034	SS039	CS007	CS013	SS038	CS028	SS043	SS011	CS020
09:00-10:30	Nitrogen Biogeochemistry and Cycling	ASLOMP Student Symposium	Adaptation of Aquatic Biodiversity to Global Change	Forecasting is the Future: Advancing Methods and Applications of Forecasting in the Aquatic Sciences	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Hypoxia	Dissolved Organic Matter-DOC, DON, DOP, fDOM, cDOM	Real World Ecology Landscapes-Up-scaling Community Ecology Experiments in Aquatic Systems	Zooplankton Ecology and Physiology	Beyond the Numbers: Strategies for Inclusive Practices Across the Aquatic Sciences	Anatomy of a Bloom: Unraveling drivers of biomass change and carbon dynamics over the annual cycle	Fish and Fisheries
10:30-11:00	COFFEE BREAK (Exhibit Hall B)											
11:00-12:30	PLENARY AND AWARD TALK SESSION (Ballroom A) 2019 Redfield Award Acceptance: Stephen R. Carpenter, University of Wisconsin-Madison, Madison, WI Plenary Presentation: Maria Falcon, Television Producer and Show Host, San Juan, PR											
12:30-14:00	LUNCH-STUDENT WORKSHOP, OTHER WORKSHOPS AND AUXILIARY MEETINGS PLUS ENTERTAINMENT ON THE TERRACE (Salsa Dance Lessons)											
	CS010	SS005	SS063	SS034	SS039	CS023	CS013	SS038	CS028	SS043	SS011	SS059
14:00-15:30	Nitrogen Biogeochemistry and Cycling	ASLOMP Student Symposium	Ocean, Coastal, and Freshwater Acidification: Research and Education	Forecasting is the Future: Advancing Methods and Applications of Forecasting in the Aquatic Sciences	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Community Ecology	Dissolved Organic Matter-DOC, DON, DOP, fDOM, cDOM	Real World Ecology Landscapes-Up-scaling Community Ecology Experiments in Aquatic Systems	Zooplankton Ecology and Physiology	Beyond the Numbers: Strategies for Inclusive Practices Across the Aquatic Sciences	Anatomy of a Bloom: Unraveling drivers of biomass change and carbon dynamics over the annual cycle	Socio-ecological Research for Actionable Sustainable Solutions: examples, perspectives and challenges
15:30-15:45	COFFEE BREAK (Exhibit Hall B)											
	CS025	CS006	SS063	SS034	SS039	CS023	CS013	SS024	CS028	SS070	SS060	SS059
15:45-17:30	Phytoplankton Ecology and Physiology	Trace metals	Ocean, Coastal, and Freshwater Acidification: Research and Education	Forecasting is the Future: Advancing Methods and Applications of Forecasting in the Aquatic Sciences	Effects of Storm Events on Aquatic, Coastal, and Oceanic Environments and Ecosystems	Community Ecology	Dissolved Organic Matter-DOC, DON, DOP, fDOM, cDOM	Reconstructing Adaptive Responses in Aquatic Ecosystems Using Ancient DNA and Resurrection Ecology	Zooplankton Ecology and Physiology	Engaging Underrepresented Students in Ocean Science to Promote a Diverse Inclusive Workforce	The Advent of Sampling Biological Processes in Aquatic Systems Using Autonomous Platforms	Socio-ecological Research for Actionable Sustainable Solutions: examples, perspectives and challenges
17:30-19:00	POSTER SESSION and RECEPTION (Exhibit Hall B)											

WEDNESDAY SCHEDULE-AT-A-GLANCE

Room	101	102	103 A	103 B	104	201	202	204	208 AB	208 C	209 AB	209 C
	CS024	CS005	SS065	CS022	CS037	CS009	CS020	SS070	CS010	SS023	SS022	SS007
09:00-10:30	Microbial Ecology and Physiology	Acidification	Turning the Lights on for Deep-Sea Ecosystems in the Caribbean, Gulf of Mexico, and US SE Atlantic	Biodiversity	Coastal Ecosystems	Gas Fluxes	Fish and Fisheries	Engaging Underrep-resented Students in Ocean Science to Promote a Diverse Inclusive Workforce	Nitrogen Bio-geochemistry and Cycling	Carbon Cycling Across Gradients in the Land-Ocean-Continuum	Phytoplank-ton-bacteria Interactions: Molecular insights, chemical drivers, and behavioral dynamics	Halophila Stipulacea: Ecology and management of the globally invasive seagrass
9:00-10:45	EDUCATION FAIR AND POSTER SESSION FOR LOCAL HIGH SCHOOL STUDENTS (Exhibit Hall A)											
10:30-11:00	COFFEE BREAK (Exhibit Hall B)											
11:00-12:30	PLENARY AND AWARD TALK SESSION (Ballroom A) 2019 Ramon Margalef Award for Excellence in Education Acceptance: David Fields, Bigelow Lab for Ocean Sciences, East Boothbay ME 2019 Yentsch-Schindler Early Career Award Acceptance: Robert Spencer, Florida State University, Tallahassee, FL Plenary Presentation: Ada Monzon, Local Television Meteorologist, San Juan, PR											
12:30-14:00	LUNCH-EARLY CAREER WORKSHOP, OTHER WORKSHOPS AND AUXILIARY MEETINGS PLUS ENTERTAINMENT ON THE TERRACE (San Juan Music)											
	CS024	CS005	SS065	CS022	CS037	CS009	SS066	SS041	CS010	SS023	SS012	SS007
14:00-15:30	Microbial Ecology and Physiology	Acidification	Turning the Lights on for Deep-Sea Ecosystems in the Caribbean, Gulf of Mexico, and US SE Atlantic	Biodiversity	Coastal Ecosystems	Gas Fluxes	Large Rivers of the World as Pipes, Chimneys and Reactors	The Next Generation: Undergradu-ate Research in Puerto Rico and the US Virgin Islands	Nitrogen Bio-geochemistry and Cycling	Carbon Cycling Across Gradients in the Land-Ocean-Continuum	Tracing the Natural and Anthropo-genic Carbon Cycle Across Aquatic Environments	Halophila Stipulacea: Ecology and management of the globally invasive seagrass
15:30-15:45	COFFEE BREAK (Exhibit Hall B)											
	CS024	CS005	SS065	SS062	CS045	SS055	SS066	SS041	CS026	SS073	SS012	SS014
15:45-17:30	Microbial Ecology and Physiology	Acidification	Turning the Lights on for Deep-Sea Ecosystems in the Caribbean, Gulf of Mexico, and US SE Atlantic	Mixotrophic Protists: An underrated majority in marine and freshwater ecosystems?	Physical Dynamics	The 2017 Hurricane Season: Chal-lenges, Inno-vations, and Resiliency in Formal and Informal Education	Large Rivers of the World as Pipes, Chimneys and Reactors	The Next Generation: Undergradu-ate Research in Puerto Rico and the US Virgin Islands	Primary Production	The Challenge of Coral Reef Rehabilitation in the Context of Climate Change	Tracing the Natural and Anthropo-genic Carbon Cycle Across Aquatic Environments	Ecosystem Based Manage-ment: Holistic Approaches to Effective Management of Regional Ecosystem
18:30-20:00	Public Discussion-"What Can Aquatic Scientists Do For You?" (Ballroom A)											
20:00	JAM SESSION (Off-site)											

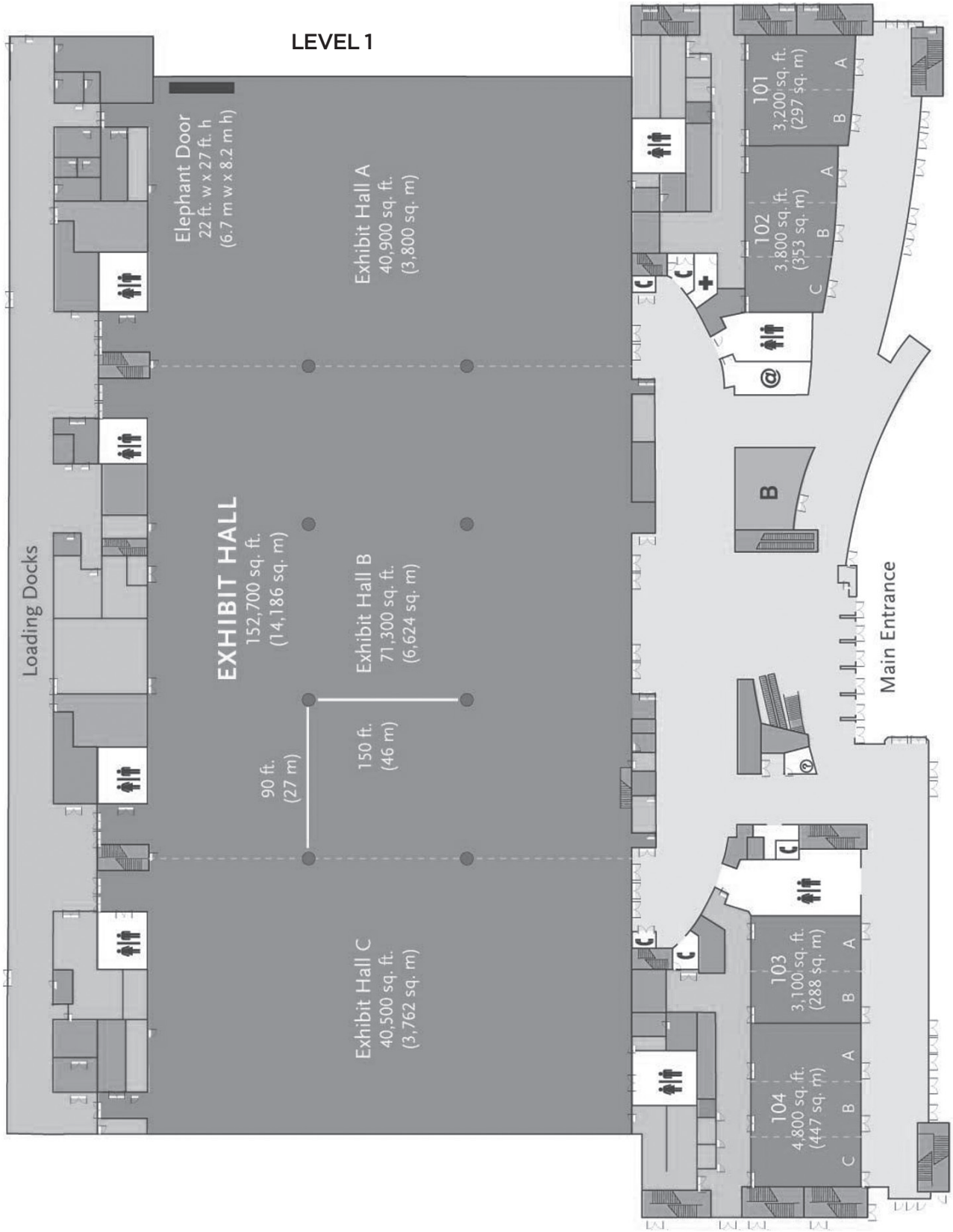
THURSDAY SCHEDULE-AT-A-GLANCE

Room	101	102	103 A	103 B	104	201	202	204	208 AB	208 C	209 AB	209 C
	SS072	CS017	SS052	SS027	SS023	SS018	SS048	AS001	SS035	SS020	SS045	SS067
09:00-10:30	Methods in Aquatic Science Education	Harmful Blooms	Carbon Cycling within Coastal Wetlands and Water Bodies	Small Scale Spatial and Temporal Patterns in Particles, Plankton, and Other Organisms	Carbon Cycling Across Gradients in the Land-Ocean-Continuum	Metabolic Diversity in Marine Biogeochemical Cycles in Present and Future Ocean	Basic and Applied Research for Resource Management and Environmental Compliance	Success through Science: Using Limnology and Oceanography to Tackle Difficult Management Questions	Exploring Microbial Interactions and Organic Matter Transformations within Oxygen Minimum Zones	Large Lakes of the World: Detecting changes using long-term monitoring	Drone Remote Sensing for Aquatic Sciences: Challenges and Successes	Microbial Phototrophy: Interactions and Impacts on Nutrient Cycling
10:30-11:00	COFFEE BREAK (Exhibit Hall B)											
11:00-12:30	PLENARY AND AWARD TALK SESSION (Ballroom A) 2019 John Martin Award Acceptance (Via Video): Philip Boyd, Inst. for Marine and Antarctic Studies, University of Tasmania, Hobart, Tasmania, Australia 2019 Raymond Lindeman Award Acceptance: Andrea Bravo, IDAEA-CSIC, Barcelona, Spain Plenary Presentation: Chris Volpe, Executive Director, ScienceCounts, Inc., Washington, DC											
12:30-14:00	LUNCH - STUDENT BEACH CLEAN UP, OTHER WORKSHOPS AND AUXILIARY MEETINGS PLUS ENTERTAINMENT ON THE TERRACE (Salsa Dance Lessons)											
	SS072	CS017	SS042	SS027	SS023	SS018	SS021	AS001	SS035	SS044	AS005	SS067
14:00-15:30	Methods in Aquatic Science Education	Harmful Blooms	Long-term ecological studies of freshwater ecosystems in the Greater Caribbean Biogeographic	Small Scale Spatial and Temporal Patterns in Particles, Plankton, and Other Organisms	Carbon Cycling Across Gradients in the Land-Ocean-Continuum	Metabolic Diversity in Marine Biogeochemical Cycles in Present and Future Ocean	Transitioning Scientific Research into Meaningful Applications	Success through Science: Using Limnology and Oceanography to Tackle Difficult Management Questions	Exploring Microbial Interactions and Organic Matter Transformations within Oxygen Minimum Zones	Connecting Watershed Characteristics to Fluvial Exports and In-stream Biogeochemical Processes	Extreme Events	Microbial Phototrophy: Interactions and Impacts on Nutrient Cycling
15:30-15:45	COFFEE BREAK (Exhibit Hall B)											
	SS072	CS017	CS011	SS027	CS033	SS018	SS021	CS019	SS069	SS044	AS005	CS008
15:45-17:30	Methods in Aquatic Science Education	Harmful Blooms	Phosphorus Biogeochemistry and Cycling	Small Scale Spatial and Temporal Patterns in Particles, Plankton, and Other Organisms	Coral Reef Ecosystems	Metabolic Diversity in Marine Biogeochemical Cycles in Present and Future Ocean	Transitioning Scientific Research into Meaningful Applications	Regime Shifts	Climate Adaptation in Aquatic and Human Systems	Connecting Watershed Characteristics to Fluvial Exports and In-stream Biogeochemical Processes	Extreme Events	Stable Isotopes
17:30-19:00	POSTER SESSION and RECEPTION (Exhibit Hall B)											

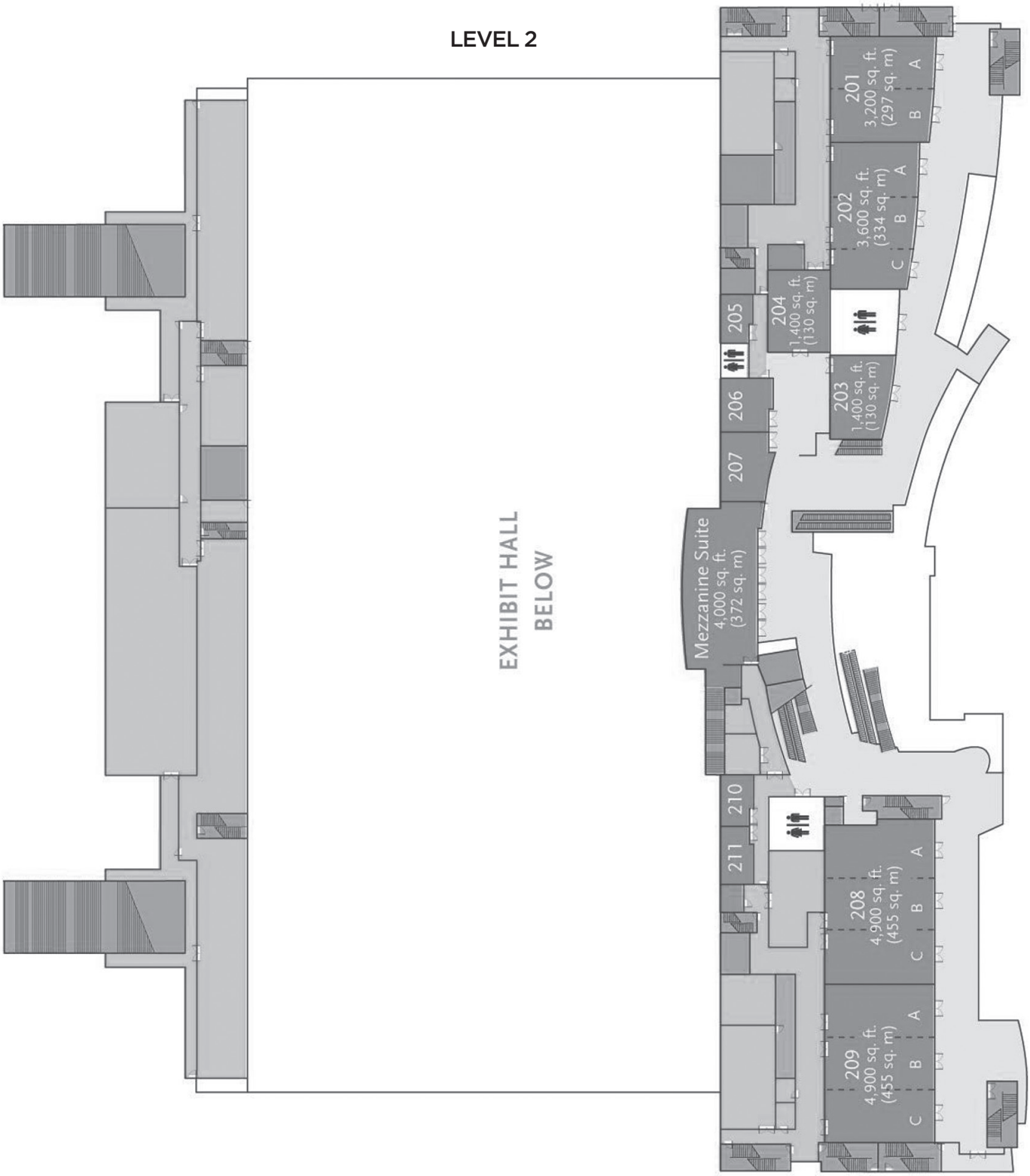
FRIDAY SCHEDULE-AT-A-GLANCE

Room	101	102	103 A	103 B	104	201	202	204	208 AB	208 C	209 AB	209 C
	SS030	CS025	SS002	CS021	CS033	SS006	CS035	SS019	SS058	CS043	SPOT001	SS031
09:00-10:30	Bioluminescent Bays of the Caribbean: Science, Management, Outreach, and Recovery	Phytoplankton Ecology and Physiology	Environmental Epigenetics: Key mechanisms of acclimatization and adaptation to global ocean change	Aquatic Food Webs	Coral Reef Ecosystems	Plastics in the Aquatic Environment	Changing Biogeochemistry and Ecology Across Polar Aquatic Systems in the 21st Century	Blue Carbon: From the ecosystem to the markets	Brave New World: The ecology of highly impacted waterbodies	Global Oceanography and Limnology	Publications Spotlight Session	Impacts of (Sub)Mesoscale Ocean Dynamics on Marine Ecosystems
10:30-11:00	COFFEE BREAK (Ballroom A Foyer-3rd Floor)											
11:00-12:30	PLENARY AND AWARD TALK SESSION (Ballroom A) 2019 G. Evelyn Hutchinson Award Acceptance: Oscar Schofield, Rutgers University, New Brunswick, NJ Plenary Presentation: Ernesto Diaz, Director, Coastal Management and Climate Change Office, PR Dept. of Natural and Environmental Resources, San Juan, Puerto Rico											
12:30-14:00	LUNCH - WORKSHOPS AND AUXILIARY MEETINGS PLUS ENTERTAINMENT ON THE TERRACE (Music of the Island)											
	CS012	CS025	SS002	CS021	CS033	SS006	CS035	SS019	SS058	CS043	SPOT001	SS031
14:00-15:30	Carbon Fluxes in FW and Marine Environment	Phytoplankton Ecology and Physiology	Environmental Epigenetics: Key mechanisms of acclimatization and adaptation to global ocean change	Aquatic Food Webs	Coral Reef Ecosystems	Plastics in the Aquatic Environment	Changing Biogeochemistry and Ecology Across Polar Aquatic Systems in the 21st Century	Blue Carbon: From the ecosystem to the markets	Brave New World: The ecology of highly impacted waterbodies	Global Oceanography and Limnology	Publications Spotlight Session	Impacts of (Sub)Mesoscale Ocean Dynamics on Marine Ecosystems
15:30-15:45	COFFEE BREAK (Ballroom A Foyer-3rd Floor)											
	CS012	CS025	CS015	CS021	CS033	SS006	SS054	SS019	SS058		SPOT001	SS033
15:45-17:30	Carbon Fluxes in FW and Marine Environment	Phytoplankton Ecology and Physiology	Environmental Change	Aquatic Food Webs	Coral Reef Ecosystems	Plastics in the Aquatic Environment	Remote Sensing of Water Quality and Quantity	Blue Carbon: From the ecosystem to the markets	Brave New World: The ecology of highly impacted waterbodies		Publications Spotlight Session	Going Deeper: Linkages across the oceanic water column
18:00-21:00	CLOSING FIESTA featuring La Bomba Show and Band (This is a ticketed event.) (Terrace and 3rd Floor Prefunction Area)											

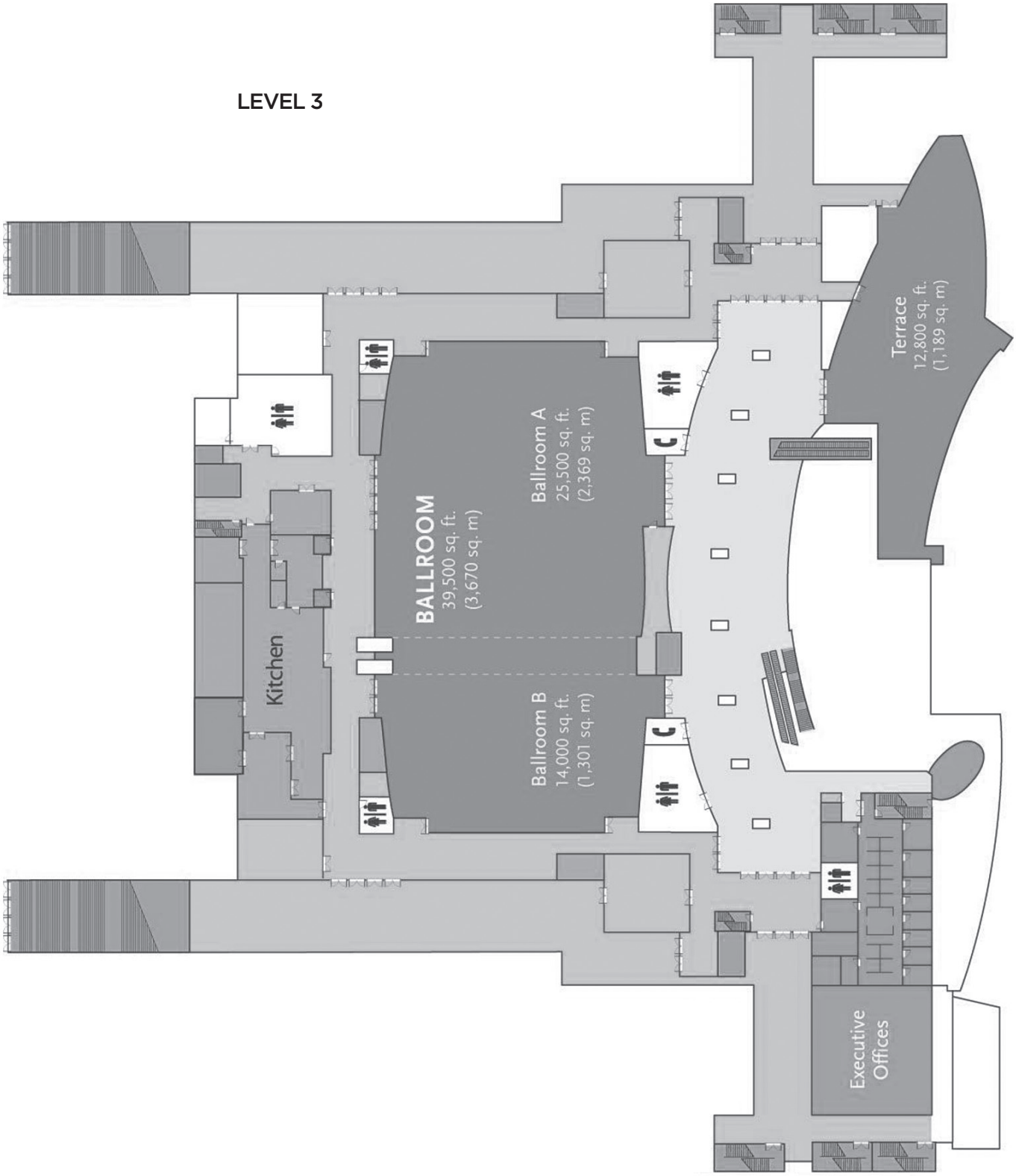
PUERTO RICO CONVENTION CENTER MAPS



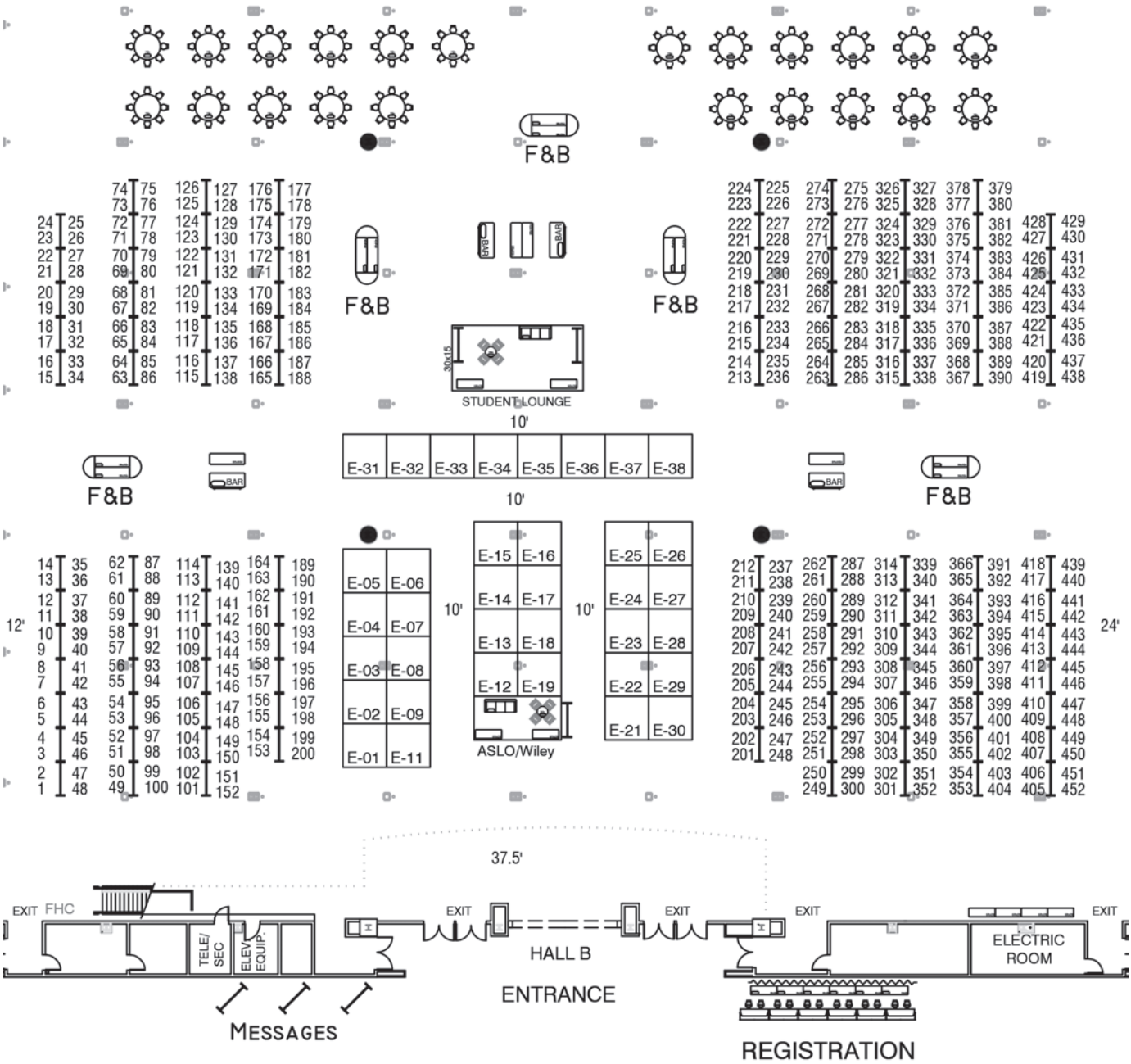
PUERTO RICO CONVENTION CENTER MAPS



PUERTO RICO CONVENTION CENTER MAPS



POSTER/EXHIBIT HALL MAP



MONDAY ORALS

AS002 HUMAN AND SOCIAL DIMENSIONS - RESEARCH MEETS MANAGEMENT: WHERE AND HOW THE RUBBER HITS THE ROAD (AND GETS TRACTION)

Chair(s): Lucinda Johnson, University of Minnesota Duluth (ljohnson@d.umn.edu)
Cynthia Hagley, University of Minnesota Duluth (chagley@d.umn.edu)

Location: Room 101 A/B

15:45 **Leon-Perez, M.**; Metz-Estrella, T.; Díaz-Velázquez, E.: LESSONS LEARNED FROM THE FIRST STEPS OF AN ADAPTIVE MANAGEMENT EXERCISE TOWARDS STRENGTHENING PUERTO RICO'S CORAL REEF MONITORING PROGRAM

16:00 **Suleiman Ramos, S.**; Hernandez Delgado, E.; Candelas, F.; Mercado, A.: INFORMAL EDUCATION FOR THE DEVELOPMENT OF COMMUNITY-BASED SCIENTIFIC CITIZEN AS STRATEGY TO MANAGE COASTAL RESOURCES.

16:15 **Hagley, C.**; Axler, R.; Schomberg, J.; Host, G.: HOW ENVIRONMENTAL DATA MADE ACCESSIBLE AND UNDERSTANDABLE AT THE LOCAL LEVEL CAN UNITE SCIENTISTS, MANAGERS, EDUCATORS, INDUSTRY, AND COMMUNITIES AROUND SHARED ENVIRONMENTAL GOALS

16:30 **Johnson, L.**; Bartsch, W.; Axler, R.; Hagley, C.; George Host Ce, D.; Hudak, G.; Peterson, D.: A DIGITAL NATURAL RESOURCE ATLAS FOR ENHANCING COMMUNICATION AND EFFECTIVE DECISION MAKING

16:45 **Pebbles, V.**; Evans, M.; Gibbons, K.: THE GREAT LAKES HABS COLLABORATORY: BUILDING A BOUNDARY SPACE FOR SCIENTISTS AND MANGERS

17:00 **Bradt, S.**; Snook, H.; Haney, J.; Hobbs, J.: THE CYANOBACTERIA MONITORING COLLABORATIVE – A SHINING EXAMPLE AND A CAUTIONARY TALE OF COLLABORATIVE EFFORTS TO MONITOR AND REPORT CYANOBACTERIA IN SURFACE WATERS

17:15 Bennett, M.; **Lee, S.;** Schofield, K.; Ridley, C.; Norton, S.; Webb, A.; Nichols, S.; Ogden, R.; Collins, A.: USING SYSTEMATIC REVIEW AND EVIDENCE BANKING TO INCREASE UPTAKE AND USE OF AQUATIC SCIENCE IN DECISION-MAKING

CS004 URBAN ECOSYSTEMS

Chair(s): Lisette de Senerpont Domis, Netherlands Institute of Ecology (L.deSenerpontDomis@nioo.knaw.nl)
Sven Teurlincx, Netherlands Institute of Ecology (S.Teurlincx@nioo.knaw.nl)

Location: Room 208 C

15:45 **de Senerpont Domis, L.**; Teurlincx, S.; Kuiper, J.; Lurling, M.: TOWARDS RESTORING URBAN WATERS: UNDERSTANDING THE MAIN PRESSURES

16:00 **Martinez, G.**; Pérez-Alegría, L.; Sotomayor, D.: SEARCHING FOR A NEEDLE IN A HAYSTACK: DEVELOPMENT OF A SYSTEMATIC STRATEGY FOR THE IDENTIFICATION OF CRITICAL SANITARY SPILLS ON A METROPOLITAN WATERSHED OF PUERTO RICO.

16:15 **Sowah, R.**; Snyder, B.; Georgacopoulos, O.; Molina, M.: ABUNDANCE AND CHARACTERISTICS OF SUL AND TET GENES AND FECAL INDICATORS IN AN URBAN WATERSHED

16:30 **Snyder, B.**; Molina, M.; Georgacopoulos, O.; Sowah, R.; Cyterski, M.: SEASONALITY OF FECAL INDICATOR BACTERIA, MICROBIAL SOURCE TRACKING MARKERS, AND PATHOGEN OCCURRENCE IN AN URBANIZED STREAM

16:45 **Lurling, M.**; Waajen, G.: IMPORTANCE OF A SYSTEM ANALYSIS IN URBAN LAKE AND POND RESTORATION

17:00 **Mucci, M.**; Teurlincx, S.; Lürling, M.; De Senerpont Domis, L.: MANAGING EUTROPHICATION IN A URBAN WATER THROUGH IRON-RICH SAND APPLICATION AND OXYGENATION

17:15 **Teurlincx, S.**; Mooij, W.; Seelen, L.; de Senerpont Domis, L.: FEEDING THE DUCKS: HOW A SEEMINGLY 'INNOCENT HABIT' CAN LIMIT ECOLOGICAL RESTORATION OF URBAN WATERS

CS013 DISSOLVED ORGANIC MATTER - DOC, DON, DOP, FDOM, CDOM

Chair(s): Rainer Amon, Texas A&M University at Galveston (amonr@tamug.edu)
Brittany Widner, Woods Hole Oceanographic Institution (bwidner@whoi.edu)
Rachel Sipler, Memorial University of Newfoundland (resipler@mun.ca)
Angela Knapp, Florida State University (anknapp@fsu.edu)
Robert Letscher, University of New Hampshire (robert.letscher@unh.edu)
Boris Koch, Alfred-Wegener-Institut (bkoch@awi.de)
Michael Gonsior, UMCES-CBL (gonsior@umces.edu)

Location: Room 202

09:00 **Herndl, G.**; Zhao, Z.: CHARACTERIZING SOURCES OF MARINE DISSOLVED ORGANIC MATTER USING –OMICS APPROACHES [1]

09:15 **Damashek, J.**; Okotie-Oyekan, A.; Wallsgrove, N.; Bayer, B.; Herndl, G.; Popp, B.; Hollibaugh, J.: FIELD RATES AND PHYSIOLOGICAL MECHANISMS OF POLYAMINE-N OXIDATION BY MARINE THAUMARCHAEOTA

09:30 **Kitzinger, K.**; Bristow, L.; Herbold, C.; Padilla, C.; Kidane, A.; Littmann, S.; Hach, P.; Daims, H.; Stewart, F.; Wagner, M.; Kuypers, M.; Marchant, H.: HIGHLY ACTIVE NITRITE OXIDIZERS UTILIZE UREA AND CYANATE AS N-SOURCES IN THE NORTHERN GULF OF MEXICO

09:45 **Avarachen, M.**; Ardelan, M.; Vadstein, O.; Gonzalez, S.; S. Vezhapparambu, V.; Olsen, Y.: BACTERIAL DEGRADATION OF DISSOLVED ORGANIC CARBON

10:00 **Yao, X.**; Sipler, R.; Stanley, B.; Roberts, Q.; Sanderson, M.; Bott, C.; Bronk, D.: QUANTIFYING EFFLUENT DISSOLVED ORGANIC NITROGEN (EDON) UPTAKE BY MICROBIAL COMMUNITIES ALONG A SALINITY GRADIENT IN THE YORK RIVER

10:15 **LaBrie, R.**; Fortin Saint-Gelais, N.; Péquin, B.; Guillemette, F.; Podgorski, D.; Tremblay, L.; Cherrier, J.; Spencer, R.; Maranger, R.: MIXING IT DOWN TO THE DEEP OCEAN ENHANCES THE PRODUCTION OF STABLE DISSOLVED ORGANIC CARBON

14:00 **Sipler, R.**; Isemonger, E.; Bornman, T.; Waterworth, S.; Venkatachalam, S.; Siwe-Noundou, X.; Dorrington, R.: UPTAKE OF ORGANIC AND INORGANIC NITROGEN BY SOUTH AFRICAN PERITIDAL STROMATOLITE MICROBIAL COMMUNITIES

14:15 **Fuessel, J.**; Rees, A.; Al-Moosawi, L.; Bibby, T.; Brown, I.; Skipp, P.; Johnson, R.; Bates, N.; Lam, P.: CONTRIBUTION OF SMALL ORGANICS TO NITRIFICATION IN THE MESOPELAGIC OCEAN

14:30 **Wymore, A.**; Krueger, S.; Van Der Hout, J.; McDowell, W.: EXPLORING THE ECOLOGICAL DUALITY OF DISSOLVED ORGANIC NITROGEN WITH FIELD BASED EXPERIMENTAL EVIDENCE

14:45 **Zhang, R.**; Wang, X.: DON CYCLING IN THE SURFACE SOUTH CHINA SEA AS INFERRED FROM ISOTOPE SIGNATURES

[1] REPRESENTS INVITED PRESENTATIONS

MONDAY

15:00 **Xu, M.**; Kao, S.: COUPLED EFFECT OF SUBSTRATE AND LIGHT ON ASSIMILATION AND OXIDATION OF REGENERATED NITROGEN IN THE EUPHOTIC OCEAN

15:15 **Knapp, A.**; Buck, K.; Boiteau, R.; Mirand, C.; McKenna, A.; Corilo, Y.; Caprara, S.; Chappell, P.: NOVEL OBSERVATIONS OF POSITIVELY CORRELATED DISSOLVED ORGANIC NITROGEN AND DISSOLVED IRON CONCENTRATIONS IN GULF OF MEXICO SURFACE WATERS

CS027 VIRUSES

Chair(s): Joaquin Martinez, Bigelow Lab for Ocean Sciences (jmartinez@bigelow.org)

Location: Room 202

15:45 **Veglia, A.**; Milford, C.; Schizas, N.: ISOLATING, CULTURING, AND GENOTYPING OF NOVEL CYANOPHAGES INHABITING CORAL REEF HOLOBIONTS

16:00 **Endo, H.**; Li, Y.; Blanc-Mathieu, R.; Ogata, H.: GLOBAL DISTRIBUTION OF NUCLEOCYTOPLASMIC LARGE DNA VIRUSES IN THE OCEAN

16:15 **Yakubovskaya, E.**; Zaliznyak, T.; Martinez Martinez, J.; Taylor, G.: PHYTOPLANKTON - VIRAL INTERACTIONS AT THE MOLECULAR LEVEL: ELEMENTAL TRANSFER STUDY BY RAMAN SPECTROSCOPY

16:30 **Rathwell, C.**; Liachko, I.; Rocap, G.: HI-C AT SEA: UNCOVERING VIRUS-HOST PAIRS IN A MARINE METAGENOME THROUGH PHYSICAL LINKAGES

16:45 **Coello-Camba, A.**: CLADE COMPOSITION DURING A NUTRIENT-INDUCED *SYNECHOCOCCUS* BLOOM AND COLLAPSE BY CYANOPHAGES

17:00 **Ignacio Espinoza, J.**; Alhgren, N.; Yeh, Y.; Fuhrman, J.: LONG-TERM VIRUS COMMUNITY STABILITY FACILITATED BY 'RED QUEEN' DYNAMICS

SS003 ADVENTURES, CHALLENGES, AND BENEFITS OF CONDUCTING INTERNATIONAL COLLABORATIVE RESEARCH

Chair(s): Adina Paytan, UCSC (apaytan@ucsc.edu)
Adrienne Sponberg, ASLO (sponberg@aslo.org)
Linda Duguay, USC (duguay@usc.edu)
Michael Pace, U. Virginia (mlp5fy@virginia.edu)

Location: Room 209 C

09:00 **Habtes, S.**; Montes, E.; Canonico, G.; Kavanaugh, M.; Marques, A.; Best, B.; Defeo, O.; Gutierrez, A.; Duffy, E.; Escobar-Briones, E.; Miloslavich, P.; Klein, E.; Bigatti, G.; Flores, A.; Cerderia, S.; Otis, D.; Chavez, F.; Muller-Karger, F.; Di Domenico, M.; Macaya, E.; Palomo, M.; Grimes, K.; Shah Esmaeili, Y.; Helmuth, B.; Rilov, G.; Londoño , E.; Mazzuco, A.; Lima, F.; Cooper, A.; Moiti, N.: MBON POLE TO POLE: BUILDING A REGIONAL COMMUNITY OF PRACTICE FOR UNDERSTANDING AND CONSERVING LIFE IN THE OCEAN

09:15 **Barber, P.**; Meyer, C.; Collins, C.; Cahyani, D.; Anggoro, A.; Mahardika, N.; Toha, H.: ADVANCING MARINE BIODIVERSITY RESEARCH IN DEVELOPING COUNTRIES THROUGH COLLABORATIVE RESEARCH AND EDUCATION: AN INDONESIAN CASE STUDY.

09:30 **Corman, J.**: TE AMO CUATRO CIENEGAS: LEARNING TO SAY GOOD-BYE

09:45 **Cisneros, A.**; Jiménez, F.; Elabd, I.; Miranda, A.; Brisson, S.; Cherepanova, O.: NEW FRONTIERS FOR INTEGRATED MARINE SCIENCES: DEVELOPING THE FIRST SOCIAL IMPACT ASSESSMENTS FOR OFFSHORE EXPLORATORY O&G PROJECTS IN THE GULF OF MEXICO

10:00 **Mantzouki, E.**; Ibelings, B.: SNAPSHOT SURVEYS FOR LAKE MONITORING, MORE THAN A SHOT IN THE DARK

10:15 **Demirel, N.**; Dursun, F.; Yilmaz, I.; Ediger, D.; Demir, V.; Yuksek, A.; Vardar, D.; Zeki, S.; Caner, H.; Altiok, H.: MARINE KNOWLEDGE SHARING PLATFORM FOR FEDERATING RESPONSIBLE RESEARCH AND INNOVATION COMMUNITIES – EU H2020: MARINA PROJECT

14:00 **Lecher, A.**: ABROAD AND AT HOME: UTILIZING THE COHORT MODEL TO HELP STUDENTS SUCCEED IN NOVEL RESEARCH EXPERIENCES[T]

14:30 **Killam, D.**; Al-Najjar, T.; Clapham, M.: STUDYING GIANT CLAMS IN A GEOPOLITICALLY AND GEOLOGICALLY COMPLEX SETTING: LESSONS LEARNED DURING A SUMMER COLLECTING FOSSIL AND MODERN SPECIMENS IN THE GULF OF AQABA

14:45 **Mescioglu, E.**; Paytan, A.: INTENSE REWARDING EXCITING SCIENCE (IRES) IN ISRAEL

15:00 **Bitterwolf, S.**: WHY YOU SHOULD PARTICIPATE IN AN INTERNATIONAL RESEARCH EXPERIENCE PROGRAM AND WHY YOU SHOULDN'T GO ALONE

15:15 **Lykkebo Petersen, K.**: EARLY-CAREER SCIENTISTS BENEFIT FROM INTERNATIONAL COLLABORATION

15:45 **Bitterwolf, K.**: BRIDGING BARRIERS WITH BARIUM (BA) ISOTOPES

16:00 **Martinez Fernandez, A.**: COORDINATING RESEARCH REMOTELY – LESSONS I LEARNT AFTER THINGS DID NOT WORK

16:15 **Chen, R.**; Ivanova, M.; Bender, J.; Petrova, M.: COASTS AND COMMUNITIES: GRADUATE STUDENTS COMPARE ENVIRONMENTAL ISSUES IN COASTAL MASSACHUSETTS AND RWANDA

16:30 **Sullivan, S.**: MINIMIZING RISK AND MAXIMIZING IMPACT IN INTERNATIONAL RESEARCH

16:45 **Broach, K.**; Torfstein, A.; Paytan, A.: MULTIPLE METHODS OF COMMUNICATION PROBABLY IMPROVE OUTCOMES OF INTERNATIONAL COLLABORATION

17:00 **Chien, C.**: WHAT I LEARNED FROM AN INTERNATIONAL COLLABORATIVE RESEARCH: A POINT OF VIEW FROM AN INTERNATIONAL STUDENT

SS005 ASLOMP STUDENT SYMPOSIUM

Chair(s): Benjamin Cuker, Hampton University (benjamin.cuker@hamptonu.edu)
Carolina Bonin, Hampton University (Carolina.Lewallen@hamptonu.edu)

Location: Room 208 C

09:00 **Layton, J.**: THE FEEDING ECOLOGY OF PACIFIC LAMPREYS ASSESSED BY GUT FULLNESS AND PREY IDENTIFICATION

09:15 **Stephens, D.**: SONIFEROUS SPECIES: AN INVENTORY FOR CHANNEL ISLANDS NATIONAL MARINE SANCTUARY (CINMS)

09:30 **McCarthy, J.**; Faulkner, P.; Petersen, L.: IMPACT OF SEX AND SALINITY ON THE SWIMMING PERFORMANCE AND METABOLISM OF SHEEPSHEAD MINNOWS (*CYPRINODON VARIEGATUS*)

09:45 **Ward, K.**; Carlisle, A.: A CHARACTERIZATION OF MIDWATER MICRONEKTON FOOD WEB DYNAMICS IN THE NORTH PACIFIC SUBTROPICAL GYRE AS DETERMINED THROUGH STABLE ISOTOPE ANALYSIS

10:00 **Min, M.**; Head, M.; Hastie, J.: LIMITATIONS AND APPLICATIONS OF MACROSCOPIC MATURITY ANALYSES: A COMPARISON OF HISTOLOGICAL AND VISUAL MATURITY STAGING IN MULTIPLE WEST COAST GROUNDFISH

14:00 **Williams-McLeod, S.**; Brill, R.; Horodysky, A.; Turner, C.: THE EFFECTS OF OCEAN ACIDIFICATION ON AUDITORY NEUROBIOLOGY IN A COASTAL FISH SPECIES.

[T] REPRESENTS TUTORIAL PRESENTATIONS

14:15 **Canava, K.**; Spore, P.; Alderman, D.; Costanza, C.; Davis, H.; Goehring, L.; Jaffery, R.; Leng, B.; Duke, J.; Scott, T.: DENITRIFICATION RATES OF FACULTATIVE ANAEROBES IN OXIC TURBID WATERS: A STUDY OF MICROHABITATS ON SUSPENDED SEDIMENTS

14:30 **Stojilovic, O.**; Goebel, M.; Krause, D.; Cruz, K.; Lewallen, E.; Bonin, C.: GENETIC DIVERSITY IN AN ANTARCTIC ECOSYSTEM TOP PREDATOR, THE LEOPARD SEAL HYDRURGA LEPTONIX.

14:45 **Eddy, T.**; Litvin, S.; Young, R.; Garza, C.: SEASONAL CHANGES IN DIET OF CALIFORNIA SPINY LOBSTER (PANULIRUS INTERRUPTUS) IN RELATION TO MARINE PROTECTED AREAS

SS009 HÁGASE LA LUZ- LIGHT IN AQUATIC ECOSYSTEMS: VARIABILITY AND ECOLOGICAL CONSEQUENCES

Chair(s): Rebecca North, University of Missouri (northr@missouri.edu)
Maren Striebel, University of Oldenburg (maren.striebel@uol.de)
Greg Silsbe, University of Maryland (gsilsbe@umces.edu)
Nur Ili Hamizah Mustaffa, University of Oldenburg (nur.ili.hamizah.mustaffa@uni-oldenburg.de)
Kevin Rose, Rensselaer Polytechnic Institute (rosek4@rpi.edu)
Jochen Wollschläger , University of Oldenburg (jochen.wollschlaeger@uni-oldenburg.de)
Robyn Smyth, Bard College (rsmyth@bard.edu)
Oliver Zielinski, University of Oldenburg (oliver.zielinski@uni-oldenburg.de)
Ruchi Bhattacharya, University of Missouri (bhattacharyar@missouri.edu)
Patrick Neale, Smithsonian Environmental Research Center (nealep@si.edu)

Location: Room 201

09:00 **Neale, P.**; Smyth, R.: ARE WARMER WATERS BRIGHTER WATERS?: THE IRRADIANCE ENVIRONMENT OF LAKES AND OCEANS IN A CHANGING CLIMATE

09:15 **Smyth, R.**; Neale, P.; Pilla, R.; Williamson, C.; Rose, K.: EFFECTS OF LAKE BROWNING AND MIXED LAYER DEPTH SHOALING ON LIGHT EXPOSURE AND PHOTOSYNTHESIS

09:30 **Köhler, J.**; Guislain, A.: INFLUENCE OF VERTICAL MIXING ON LIGHT DEPENDENCY OF PHOTOSYNTHESIS, RESPIRATION AND GROWTH OF PHYTOPLANKTON COMMUNITIES

09:45 **Trench, S.**; Fink, P.: EXTREME IRRADIANCE LEVELS IN A HIGH-ALTITUDE ECOSYSTEM AFFECT FUNCTIONAL GENE EXPRESSION PROFILES IN AQUATIC MICROEUKARYOTES

10:00 **Hellweger, F.**; van Sebille, E.; Sasaki, H.; Jabbur, L.; Johnson, C.: CIRCADIAN CLOCK HELPS CYANOBACTERIA MANAGE LIGHT ENERGY IN TURBID COASTAL AND HIGH LATITUDE OCEAN

10:15 **Sullivan, S.**; Hossler, K.; Meyer, L.; Rubinoff, B.: ARTIFICIAL LIGHTING AT NIGHT ALTERS FOOD WEBS AND ECOSYSTEM FUNCTIONS IN AQUATIC-RIPARIAN SYSTEMS

14:00 **Gorbunov, M.**; Park, J.; Kuzminov, F.; Bailleul, B.; Yang, E.; Nikonova, E.; Shirshin, E.; Falkowski, P.; Lee, S.: LIGHT AVAILABILITY RATHER THAN IRON IS THE MAIN CONTROL OF THE MAGNITUDE OF MASSIVE PHYTOPLANKTON BLOOM IN THE AMUNDSEN SEA, ANTARCTICA – LESSONS FROM A 5-YEAR SURVEY

14:15 **Gaskill, J.**; Harris, T.; deNoyelles , J.; Burgin , A.; Shields , A.; Baker, S.; Webb, L.; Klepikow , R.; Mash, H.; North, R.: EXPLORING PHYTOPLANKTON RESPONSE TO CHANGES IN LIGHT AND NUTRIENTS IN MESOCOSM TANKS: CAN GLACIAL ROCK FLOUR BE USED TO CONTROL CYANOBACTERIAL BLOOMS?

14:30 **Petty, E.**; Obrecht, D.; North, R.: PHYTOPLANKTON'S MOST WANTED: DOES LIGHT-NUTRIENT CO-DEFICIENCY LIMIT PHYTOPLANKTON COMMUNITIES IN TURBID MIDWESTERN USA RESERVOIRS?

14:45 **North, R.**; Graham, J.; Obrecht, D.; Baulch, H.; Hudson, J.; Dillon, P.; Smith, R.; Jones, J.: LIGHT IN COLD LAKES- IS IT LOW? IS IT CHANGING?

15:00 **Hrycik, A.**; Stockwell, J.: UNDER-ICE MESOCOSMS TO TEST INTERACTIONS OF LIGHT LIMITATION AND ZOOPLANKTON GRAZING ON PHYTOPLANKTON COMMUNITIES

15:15 **Bunnell, D.**; Ludsin, S.; Barbiero, R.; Collingsworth, P.; Hook, T.; Hug, J.; Knight, R.; Lesht, B.; Mayer, C.; Rudstam, L.; Rutherford, E.; Scofield, A.; Stewart, T.; Vanderploeg, H.; Weidel, B.: CONSEQUENCES OF CHANGING WATER CLARITY ON FISH AND FISHERIES IN LARGE LAKES: IMPLICATIONS FOR THE LAURENTIAN GREAT LAKES

15:45 **Zielinski, O.**; Mustaffa, N.; Pitarch, J.; Thewes, D.; Stanev, E.; Striebel, M.; Wollschlaeger, J.: DO WE WITNESS A COASTAL OCEAN DARKENING?

16:00 **Mustaffa, N.**; Kallajoki, L.; Wurl, O.; Streibel, M.: THE UPPER SEA SURFACE PHYTOPLANKTON COMMUNITIES' RESPONSES TO NUTRIENT AND LIGHT CHANGES

16:15 **Striebel, M.**; Mustaffa, N.; Biederbrick, J.; Binder, F.; Kallajoki, L.; Schlenker, A.: COASTAL OCEAN DARKENING EFFECTS ON PLANKTON: AN INDOOR MESOCOSM EXPERIMENT

16:30 **Wollschlaeger, J.**; Tietjen, B.; Zielinski, O.: AN EFFICIENT BIMODAL PARAMETRIZATION OF THE UNDERWATER LIGHT FIELD IN COASTAL WATERS

16:45 **Williamson, C.**; Overholt, E.; Pilla, R.; Knose, L.; Berry, N.; Wilkins, K.: ECOSYSTEM LEVEL CONSEQUENCES AND ZOOPLANKTON RESPONSES TO LONG-TERM BROWNING IN LAKES EXPERIENCING INCREASES IN DISSOLVED ORGANIC MATTER

17:00 **Gonçalves-Araujo, R.**; Markager, S.: LIGHT IN THE DARK: ESTIMATING LIGHT ATTENUATION FOR DANISH WATERS FROM IOP MEASUREMENTS

17:15 **Mirhakak Esfahani, M.**; Hedley, J.; LaJeunesse, T.; Dierssen, H.: ANALYSIS OF HYPERSPECTRAL REFLECTANCE OF CORALS IN RESPONSE TO THEIR HABITAT AND SYMBIONTS

SS010 EXPLORING WHAT MAKES UNDERGRADUATE RESEARCH EXPERIENCES WORK: EVIDENCE FROM STUDENTS AND MENTORS

Chair(s): Alan Wilson, Auburn University (wilson@auburn.edu)
Patrick Crumrine, Rowan University (crumrine@rowan.edu)
Gretchen Hofmann, University of California Santa Barbara (hofmann@ucsb.edu)
David Fields, Bigelow Laboratory for Ocean Sciences (dfields@bigelow.org)
Shane Rogers, Clarkson University (srogers@clarkson.edu)
Sarah Cohen, San Francisco State University (sarahcoh@sfsu.edu)

Location: Room 204

09:00 **Rogers, S.**; Twiss, M.: ATYPICAL REU SITE FEATURES LEAD TO PROGRAM SUCCESS

09:15 **Downs, E.**; Gordon, E.; Huang, J.; Ludlam, J.; O'Connor, A.; Welsh, D.: FITCHBURG STATE UNIVERSITY SUMMER RESEARCH COLLABORATIVE: AN INTERDISCIPLINARY RESEARCH EXPERIENCE TO IMPROVE STEM RETENTION AND GRADUATE SCHOOL ATTENDANCE

09:30 **Savoie, A.**; Newell, S.; McCarthy, M.; Hayes, C.; McClelland, J.: DISHES TO DEFENSE: ADVENTURES OF AN UNDERGRAD IN RESEARCH

[1] REPRESENTS INVITED PRESENTATIONS

MONDAY

09:45 **Cohen, S.**; Aleman, J.; Figueroa, J.; Flores, M.; Perez, C.; Hernandez, S.; Rohlfs, R.; Domingo, C.: INCREASING DIVERSITY IN ENVIRONMENTAL STUDIES AND EVOLUTIONARY DEVELOPMENTAL BIOLOGY

10:00 **Barber, P.**; Fong, C.; Habtes, S.; Fong, P.: THE DIVERSITY PROJECT: MAXIMIZING THE IMPACT OF UNDERGRADUATE RESEARCH EXPERIENCES THROUGH STUDENT DRIVEN RESEARCH INQUIRY

10:15 **Sloan, V.**; Haacker, R.; Morris, V.; Batchelor, R.; Manswell-Butty, J.: WHAT IS AND ISN'T WORKING IN REU DIVERSITY EFFORTS: TIPS FOR THE INTREPID

14:00 **Hofmann, G.**; Leach, T.: PROFESSIONAL DEVELOPMENT SERIES PROMOTES INCLUSION AND COMMUNITY: REPORT FROM THE SITE REU AT UC SANTA BARBARA

14:15 **Crumrine, P.**; Zwick, M.: FACTORS INFLUENCING MENTOR SELECTION IN UNDERGRADUATE RESEARCH EXPERIENCES

14:30 **Reed, M.**; McCarthy, M.; Newell, S.: UNDERGRADUATE RESEARCH: THE OPPORTUNITY FOR LEARNING AS A MENTOR AND MENTEE IN A RESEARCH LAB

14:45 **Coldsnow, K.**: UNDERGRADUATE RESEARCH CAN BE A CRUCIAL EXPERIENCE AND MENTORS AT ALL STAGES PLAY A HUGE ROLE.

15:00 **Ogurcak, D.**; Schonhoff, B.; Crowl, T.; Yanez Zapata, T.; Soares Quinete, N.; Dessu, S.; Teutonico, R.: COHORT BUILDING AND NEAR-PEER MENTORING AS INTEGRAL PARTS OF THE UNDERGRADUATE RESEARCH EXPERIENCE IN COASTAL ECOSYSTEM SCIENCE

15:15 **Walsh, E.**; Brown, P.; Pina, A.; Geller, H.; Hotchkin, P.; Lougheed, V.: USING AQUATIC SYSTEMS TO INTRODUCE COMMUNITY COLLEGE STUDENTS TO UNIVERSITY RESEARCH THROUGH MENTORING

15:45 **Greengrove, C.**; Masura, J.; Nuwer, M.; Kelley, D.: INTEGRATING OCEAN OBSERVATORIES INITIATIVE (OOI) DATA INTO UNDERGRADUATE INTRODUCTORY OCEANOGRAPHY COURSES

16:00 **Ramirez-Lugo, J.**; Velez Gonzalez, I.; Toledo Hernandez, C.; Ruiz Diaz, C.: A COURSE-BASED UNDERGRADUATE RESEARCH EXPERIENCE TO INVESTIGATE THE RESPONSES OF THE CORAL ACROPORA CERVICORNIS TO ENVIRONMENTAL STRESS

16:15 **Duke, J.**; Scott, T.: A COMPREHENSIVE INVESTIGATION-BASED AND COMMUNICATIONS-INTENSIVE UNDERGRADUATE COURSE IN AQUATIC SYSTEMS BIOLOGY

16:30 **Dzieweczynski, T.**: USING FISH ON DRUGS TO TRAIN THE NEXT GENERATION OF CRITICAL THINKERS

16:45 **Brown, M.**: LESSONS FROM THE LIBERAL ARTS ABOUT SUCCESSFUL ACADEMIC-YEAR RESEARCH WITH UNDERGRADUATE COLLEAGUES

17:00 Olden, J.; Lise, C.; Couto, T.; Koontz, E.; Kuehne, L.; **Messager, M.**; Rogosch, J.; Rubenson, E.; Stiling, R.: DISPATCHES FROM A GLOBAL SURVEY ON THE BENEFITS AND CHALLENGES OF COURSE-BASED UNDERGRADUATE RESEARCH FOR STUDENTS AND SCIENCE

17:15 **Wilson, A.**; Pollock, J.; Billick, I.; Domingo, C.; Fernandez-Figueroa, E.; Nagy, E.; Steury, T.; Summers, A.: ASSESSING SCIENCE TRAINING PROGRAMS: STRUCTURED UNDERGRADUATE RESEARCH PROGRAMS MAKE A DIFFERENCE

SS013 ADAPTATION OF AQUATIC BIODIVERSITY TO GLOBAL CHANGE

Chair(s): Vincent Fugere, University of Quebec at Montreal(vincent.fugere@mail.mcgill.ca)
Beatrix Beisner, University of Quebec at Montreal (beisner.beatrix@uqam.ca)

Location: Room 102 A/B/C

09:15 **Urban, M.**: ECO-EVOLUTIONARY RESILIENCE IN TEMPORARY PONDS

09:45 **Dam, H.**; Demayo, J.; Park, G.; Norton, L.; Brennan, R.; Finiguerra, M.; Baumann, H.; Pespeni, M.: RAPID, BUT LIMITED, ADAPTATION OF A MARINE COPEPOD TO A GREENHOUSE WORLD

10:00 **deMayo, J.**; Park, G.; Norton, L.; Finiguerra, M.; Dam, H.: ADAPTATION TO A GREENHOUSE WORLD COMES WITH A COST FOR THE COPEPOD, ACARTIA TONSA

10:15 **Fugere, V.**; Hendry, A.: HUMAN INFLUENCES ON THE STRENGTH OF PHENOTYPIC SELECTION

14:00 **Calosi, P.**; Thibault, C.; Massamba-N'SIala, G.; Noisette, F.; Vermandele, F.; Babin, M.: RARITY, TRANSGENERATIONAL RESPONSES AND THE GLOBAL CHANGE

14:15 **Baeza, M.**; Walsh, E.: MULTI-GENERATIONAL EXPOSURE OF AQUATIC MICROINVERTEBRATES TO ENHANCED ULTRAVIOLET RADIATION

14:30 **McLean, J.**; Rosa, P.; Diaz, H.: INVESTIGATING THE IMPACT OF TEMPERATURE AND PH ON BLUE CRAB JUVENILES

14:45 **Nyboer, E.**; Chrétien, E.; Kimera, S.; Chapman, L.: TROPICAL FISH IN A WARMING WORLD: EFFECTS OF ELEVATED TEMPERATURE ON METABOLIC PERFORMANCE OF COMMERCIALLY HARVESTED AFRICAN FRESHWATER FISHES

15:00 **Martinez, N.**; Roberson, D.; Toledo-Hernández, D.: THE IMPACT OF THERMAL STRESS ON THE PHYSIOLOGY AND GROWTH OF THREE DEVELOPMENTAL STAGES OF THE SCLERACTINIAN CORAL PORITES ASTREOIDES

15:15 **Bartley, T.**; McCann, K.; Guzzo, M.; McMeans, B.: BEHAVIORAL AND TROPHIC RESPONSES DRIVE WHOLE LAKE FOOD WEB REWIRING

15:45 **Post, D.**: DAMS, FISHWAYS, AND THE ECOLOGY AND EVOLUTION OF ALEWIFE DURING ISOLATION AND SECONDARY CONTACT

16:00 **Sasaki, M.**; Dam, H.: SOME LIKE IT HOT, SOME DON'T HAVE A CHOICE: GENE FLOW (SOMETIMES) CONSTRAINS THERMAL ADAPTATION AND AFFECTS VULNERABILITY TO WARMING IN A WIDESPREAD COPEPOD

16:15 **Oziolor, E.**; Reid, N.; Yair, S.; Lee, K.; VerPloeg, S.; Bruns, P.; Shaw, J.; Whitehead, A.; Matson, C.: ADAPTIVE INTROGRESSION ENABLES EVOLUTIONARY RESCUE FROM EXTREME ENVIRONMENTAL POLLUTION

16:30 **Schiebelhut, L.**; Dawson, M.: CONSEQUENCES OF AN EXTREME EVENT: (RE)SHAPING THE DISTRIBUTION OF CONTEMPORARY GENETIC VARIATION

16:45 **Fussmann, G.**; Fugère, V.; Hébert, M.; Costa, N.; Xu, C.; Barrett, R.; Beisner, B.; Bell, G.; Shapiro, J.; Yargeau, V.; Gonzalez, A.: COMMUNITY RESCUE: FROM THEORY TO EXPERIMENTAL FRESHWATER ECOSYSTEMS

17:00 **Beisner, B.**; Fugère, V.; Barrett, R.; Bell, G.; Cristescu, M.; Fussmann, G.; Shapiro, J.; Gonzalez, A.: EVOLUTIONARY RESCUE IN POND METACOMMUNITIES FACING SEVERE ACIDIFICATION

SS028 THE MULTIPLE CHALLENGES OF RAPIDLY CHANGING TROPICAL FRESHWATER ECOSYSTEMS

Chair(s): Ed Hall, Colorado State University (ed.hall@colostate.edu)
Flavia Tromboni, University of Nevada, Reno (flavia.tromboni@gmail.com)

Location: Room 209 A/B

09:00 **Benito-Granell, X.**; Schneider, T.; Fritz, S.; Ruhi, A.: A PALEOECOLOGICAL PERSPECTIVE OF RESILIENCE IN ANDEAN LAKES

09:15 **Seidel, M.**; Gomez-Saez, G.; Schnetger, B.; Rezende, C.; Dittmar, T.: MANGROVES AS THE MAIN SOURCE OF DISSOLVED ORGANIC SULFUR, PYROGENIC MATTER AND NUTRIENTS TO AN AMAZONIAN ESTUARY

09:30 **Melendez-Diaz, J.**; Melendez Diaz, J.; Garcia Rodriguez, B.; Flores Hernandez, M.; Barreto Velez, T.; Saez Uribe, R.; Figueroa, Y.; Perez Rivera, K.; De Jesus Roman, A.; Ramos Scharron, C.; Cabrera Rivera, L.; Castejon Brignoni, R.; Setegn, S.; Ortiz Zayas, J.: THE GUÁNICA BAY / RÍO LOCO WATERSHED: LINKING HISTORICAL LAND USES WITH WATER AND SEDIMENTS FLUXES AND HUMAN PERSPECTIVES OF THEIR IMPACTS TO COASTAL ECOSYSTEMS

09:45 **Flores-Hernandez, M.**; Flores-Hernandez, M.; Meléndez-Díaz, J.; Barreto-Vélez, T.; Castejón-Brignoni, R.; Cabrera-Rivera, L.; Setegn, S.; Ortiz-Zayas, J.: LINKING LAND USE CHANGE, HUMAN ACTIVITIES AND ECOSYSTEM SERVICES IN THE RIO GRANDE DE MANATI WATERSHED

10:00 **Barby, C.**; Fadum, J.; Hall, E.: CHANGING NUTRIENT DYNAMICS OF A LARGE TROPICAL WATERSHED

10:15 **Fadum, J.**; Barby, C.; Hall, E.: INVESTIGATING CHANGING NUTRIENT LIMITATION IN LARGE TROPICAL LAKES: A CASE STUDY OF LAKE YOJOA, HONDURAS

14:00 **Saltarelli, W.**; Finkler, N.; Cunha, D.: RESTORATION OF RIPARIAN FOREST IN TROPICAL STREAMS: EFFECTS ON AQUATIC METABOLISM AND MACRONUTRIENTS' RETENTION

14:15 **Bottino, F.**; dos Santos, A.; de Arruda, E.; Menezes, L.; Dergam, J.; Marques, R.; Pavan, D.: IMPRUDENT MANAGEMENT PRACTICES RESULTED IN A MASSIVE IMPACT ON A RIVER BASIN IN SOUTH AMERICA

14:30 **Flecker, A.**; García-Villacorta, R.; Gomes Selman, J.; Angarita, H.; Abad, J.; Almeida, R.; Forsberg, B.; Larson, E.; Sethi, S.; Steinschneider, S.; Gomes,C.; Amazon, D.: DAM PROLIFERATION IN THE AMAZON: A COMPUTATIONAL SUSTAINABILITY APPROACH FOR EXPLORING TRADEOFFS BETWEEN HYDROPOWER AND ECOSYSTEM SERVICES

14:45 **Cancel-Villamil, J.**; Locke, S.: COMPARISON OF FISH COMMUNITIES PRIOR TO REMOVAL OR MODIFICATION OF DAMS IN TROPICAL STREAMS

15:00 **Chappell, J.**; McKay, S.; Freeman, M.; Pringle, C.: LONG-TERM (37 YEARS) IMPACTS OF LOW-HEAD DAMS ON HABITAT CONNECTIVITY IN NORTHEASTERN PUERTO RICO: IMPLICATIONS FOR ISLAND-WIDE METAPOPULATION DYNAMICS FOR MIGRATORY SHRIMP

15:15 **Covich, A.**; Crowl, T.; Perez-Reyes, O.: CHANGING ECOSYSTEM SERVICES FOLLOWING SHORT-TERM AND LONG-TERM CUMULATIVE EFFECTS OF DROUGHT AND HURRICANE DISTURBANCES: THE ROLE OF FRESHWATER SHRIMP IN MAINTAINING CLEAR WATER IN LUQUILLO MOUNTAIN STREAMS, PUERTO RICO

15:45 **Moulton, T.**; Neres-Lima, V.; Aguiar, A.; Zandonà, E.; Silveira, R.: TROPHIC CASCADES AND STRONG INTERACTIONS IN TROPICAL STREAMS AND THEIR SENSITIVITY TO HYDROLOGIC DISCONTINUITY AND OTHER STRESSORS

16:00 **Pinceel, T.**; Thoré, E.; Philippe, C.; Brendonck, L.: AFRICAN KILLIFISH AS MODEL ORGANISMS IN AQUATIC SCIENCES

16:15 **Marazzi, L.**; Eppinga, M.; Angelini, C.; Castañeda-Moya, E.; Gaiser, E.; Sah, J.; Zhai, L.: WHY DO WE NEED TO DOCUMENT AND CONSERVE FOUNDATION SPECIES IN OLIGOTROPHIC WETLANDS?

16:30 **Santos, R.**; Barrato-Velez, T.; Santos-Crespo, M.; Perez-Reyes, O.; Covich, A.; Crowl, T.: INTEGRATING MOVEMENT ECOLOGY INTO LONG-TERM ECOLOGICAL MONITORING: A FEASIBILITY STUDY OF ACOUSTIC TELEMETRY TO MEASURE SHRIMP MOVEMENT IN TROPICAL STREAMS

16:45 **Torres-Pérez, W.**; Perez-Reyes, O.: EFFECT OF URBANISM ON WATER QUALITY AND MACROINVERTEBRATE COMMUNITIES IN NORTHEAST REGION OF PUERTO RICO

SS036 ECOLOGICAL APPLICATIONS OF EARTH SYSTEM MODELS AND REGIONAL CLIMATE MODELS

Chair(s): Rebecca Asch, East Carolina University (aschr16@ecu.edu)
Darren Pilcher, NOAA Pacific Marine Environmental Laboratory (darren.pilcher@noaa.gov)
Sara Rivero-Calle, University of North Carolina - Wilmington (riverocalles@uncw.edu)
Johnna Holding, Aarhus University (johnna@bios.au.dk)

Location: Room 101 A/B

09:00 **Dutkiewicz, S.**: MODELLING THE RESPONSES OF DIVERSE PHYTOPLANKTON COMMUNITIES TO CLIMATE CHANGE[T]

09:30 **Xue, P.**; Ye, X.; Huang, C.: IMPACT OF CLIMATE CHANGE ON THERMAL VARIABILITY AND ECOSYSTEM VULNERABILITY OF THE GREAT LAKES

09:45 **Chien, C.**; Pahlow, M.; Oschlies, A.: IMPROVED REPRESENTATION OF BIOGEOCHEMICAL TRACER DISTRIBUTIONS BY AN OPTIMALITY-BASED NON-REDFIELD ECOSYSTEM MODEL IN THE UVIC-ESCM

10:00 **Pilcher, D.**; Siedlecki, S.; Deutsch, C.; MacCready, P.; Klinger, T.; Newton, J.; Howard, E.; Sunday, J.: DYNAMICALLY DOWNSCALED CLIMATE CHANGE PROJECTIONS OF MULTIPLE MARINE ECOSYSTEM STRESSORS FOR THE CALFORNIA CURRENT SYSTEM

10:15 **Rose, K.**; Li, M.; Zhao, J.; Curchitser, E.; Fiechter, J.: PREDICTING CLIMATE CHANGE EFFECTS ON MARINE AND COASTAL FISH USING COUPLED BIO-PHYSICAL MODELS: CHALLENGES AND OPPORTUNITIES[I]

14:00 **Asch, R.**; Stock, C.; Sarmiento, J.: DO INCREASES IN TEMPERATURE AND SALINITY EXTREMES LEAD TO MORE FREQUENT AND LARGER MISMATHES BETWEEN THE PHENOLOGY OF FISH SPAWNING AND PLANKTON BLOOMS?

14:15 **Petrik, C.**; Stock, C.; Andersen, K.: DRIVERS OF FISHERIES PRODUCTION BY FUNCTIONAL TYPE AND THE EFFECTS OF CLIMATE CHANGE

14:30 **Drenkard, E.**; Miller, A.; McClatchie, S.; Webber, E.; Burnham, T.; Ramirez, S.; Shen, S.; Neilson, D.; Hovel, K.: MODELING CLIMATE CHANGE IMPACTS ON CALIFORNIA CURRENT SYSTEM FISHERIES

14:45 **Stock, C.**; Park, J.; Tommasi, D.; Dunne, J.; Pegion, K.; Alexander, M.; Checkley, D.; Methot, R.; Rosati, A.; Vecchi, G.; Yang, X.: SEASONAL TO MULTI-ANNUAL EARTH SYSTEM PREDICTIONS FOR LIVING MARINE RESOURCE RESILIENCE IN A CHANGING CLIMATE[I]

15:00 **Kearney, K.**; Hermann, A.; Aydin, K.: A SKILL ASSESSMENT OF BERING SEA COLD POOL SEASONAL PROJECTIONS FOR FISHERIES MANAGEMENT

[I] REPRESENTS INVITED PRESENTATIONS

MONDAY

15:15 **Krinos, A.**; Dixon, K.; Ross, A.; Stock, C.: UNDERSTANDING SPATIAL EFFECTS OF CLIMATE CHANGE ON CHESAPEAKE BAY BLUE CRAB USING STATISTICAL DOWNSCALING AND AGENT-BASED MODELING

SS037 TERRESTRIAL MATERIALS IN PLANET WATER: TRACKING INFLUENCES ALONG THE LAND-OCEAN CONTINUUM

Chair(s): Anne Deininger, University of Agder (anne.deininger@niva.no)
Allison Myers-Pigg, Memorial University of Newfoundland (amyerspigg@mun.ca)
Rachel Sipler, Memorial University of Newfoundland (resipler@mun.ca)
Amanda Poste, Norwegian Institute for Water Research (amanda.poste@niva.no)

Location: Room 103 A

14:00 **Liess, A.**; Rankinen, J.; Barnes, T.; Ndlovu, O.; Lannér, J.; Parkefelt, L.: BROWNING OF BOREAL LAKES CHANGES PHYTOPLANKTON PIGMENT COMPOSITION – CONCLUSIONS FROM A 6-WEEK MESOCOSM EXPERIMENT IN LAKE BOLMEN, SWEDEN

14:15 **Hutchins, R.**; Kuhn, M.; Tank, S.; Olefeldt, D.: COMPOSITIONAL SHIFTS IN DISSOLVED ORGANIC MATTER ALONG A LATITUDINAL PERMAFROST GRADIENT IN THE WESTERN CANADIAN BOREAL

14:30 **Attermeyer, K.**; Schelker, J.; Kainz, M.; Weigelhofer, G.: NUTRIENTS STIMULATE TERRESTRIAL CARBON DEGRADATION AND CAN COMPENSATE FOR THE LOW QUALITY OF TERRESTRIAL ORGANIC MATTER IN THE HYPORHEIC ZONE

14:45 **Deininger, A.**; Kaste, Ø.; Austnes, K.: RECENT CHANGES IN NUTRIENT STOICHIOMETRY IN NORWEGIAN RIVERS – SHIFTS FROM INORGANIC TO ORGANIC FRACTIONS

15:00 **Opdal, A.**; Lindemann, C.; Aksnes, D.: COULD LONG-TERM SHIFT IN COD PHENOLOGY BE LINKED TO TERRESTRIAL GREENING AND COASTAL BROWNING?

15:15 **Deering, N.**; Grinham, A.; Fisher, P.; Gibbes, B.; Cossu, R.; Linde, M.; Gale, D.; Albert, S.: EXTREME SUSPENDED SEDIMENT AND NUTRIENT LOADING FOLLOWING EX-TROPICAL CYCLONE IN SOUTHEAST QUEENSLAND

15:45 **Wilkinson, G.**; Walter, J.; Fleck, R.; Pace, M.: SYNCHRONIZATION AMONG TERRESTRIAL & AQUATIC ECOSYSTEMS

16:00 **Waters, M.**; Metz, A.; Smoak, J.: ALTERATIONS TO NUTRIENT DEPOSITION, SEDIMENT STOICHIOMETRY, AND PHYTOPLANKTON COMMUNITIES IN TWO SUBTROPICAL LAKES FROM REPEATED PRESCRIBED BURNING

16:15 **Xu, X.**; Wei, H.; Moffett, K.; McClelland, J.; Hardison, A.: TIDAL FRESHWATER ZONES AS HOTSPOTS FOR NITROGEN AND ORGANIC MATTER CYCLING IN TWO TEXAS RIVERS

16:30 **Polsenaere, P.**; Lannuzel, R.; Le Moine, O.; Soletchnik, P.: SPATIO-TEMPORAL VARIABILITY OF CARBON AND NUTRIENT CONCENTRATIONS AND FLUXES OVER A TEMPERATE TERRESTRIAL-AQUATIC CONTINUUM (FRENCH ATLANTIC COAST)

16:45 Erratt, K.; Creed, I.; **Trick, C.**: CONTEMPORARY UREA-BASED AGRICULTURAL FERTILIZERS ARE THE PREFERRED NITROGEN SOURCE FOR CYANOBACTERIA IN FRESHWATERS

17:00 **Brown, C.**; Zimmer-Faust, A.; Kaldy, J.; Mochon Collura, T.; Pacella, S.; Rugh, W.; Stecher, H.; Wise, D.: TRACKING LAND-BASED NUTRIENT AND BACTERIA INPUTS TO TILLAMOOK ESTUARY, OREGON (USA)

17:15 **Jiang, Y.**; Li, M.: AN INSIGHT INTO THE MECHANISMS OF TIO2 NANOPARTICLES TOXICITY ON PHOTOSYSTEM II (PS II) ACTIVITIES IN DUNALIELLA SALINA

SS039 EFFECTS OF STORM EVENTS ON AQUATIC, COASTAL, AND OCEANIC ENVIRONMENTS AND ECOSYSTEMS: FROM PHYSICAL PROCESSES TO FOOD WEBS

Chair(s): Beth Stauffer, University of Louisiana at Lafayette (stauffer@louisiana.edu)
Michael Martinez-Colon, Florida A&M University (michael.martinez@famu.edu)
Patricia Chardon-Maldonado, University of Puerto Rico at Mayaguez (patricia.chardon@upr.edu)
Jessica Fitzsimmons, Texas A&M University (jessfitz@tamu.edu)
Miguel F. Canals Silander, University of Puerto Rico at Mayaguez (miguelf.canals@upr.edu)
Lisa Campbell, Texas A&M University (lisacampbell@tamu.edu)
Simon Geist, Texas A&M University Corpus Christi (simon.geist@tamucc.edu)
Kam-Biu Liu, Louisiana State University (kliu1@lsu.edu)
Thomas Bianchette, University Of Michigan (tbianc@umich.edu)
Marianne Dietz, Louisiana State University (mdietz2@lsu.edu)

Location: Room 104

09:00 **DiMarco, S.**; Knap, A.; Whilden, K.; Potter, H.; Walpert, J.: PHYSICAL PROCESSES OF THE COASTAL OCEAN FROM SEVERE TROPICAL WEATHER: IMPACT OF HURRICANE HARVEY (2017) ON THE TEXAS COAST[T]

09:30 **Davis, K.**; Leichter, J.; Monismith, S.: TROPICAL CYCLONES INDUCE DEPTH-DEPENDENT TEMPERATURE CHANGES ON MARINE ECOSYSTEMS

09:45 **Browning, T.**; Sawyer, D.; Brooks, G.; Larson, R.; Ramos-Scharron, C.: WIDESPREAD DEPOSITION IN COASTAL ST. JOHN, USVI FOLLOWING THREE MAJOR 2017 HURRICANE EVENTS (IRMA, JOSE, AND MARIA)

10:00 **Casillas, P.**; Chardón Maldonado, P.; Canals, M.; Rodriguez, S.: NUMERICAL SIMULATIONS OF STORM-INDUCED NEARSHORE MORPHOLOGY CHANGE IN RINCON, PUERTO RICO

10:15 **Grafals-Soto, R.**: SPATIAL CHARACTERISTICS OF CARIBBEAN COASTAL DUNES: A PUERTO RICO CASE STUDY IN THE CONTEXT OF HURRICANE MARIA

14:00 **James, M.**; O'Donnell, J.: IMPROVING DRAG COEFFICIENTS IN SALT MARSH MODELS

14:15 **Rabalais, N.**; Justic, D.; Turner, R.: VARIABLE RESPONSES OF LOW-OXYGEN CONDITIONS TO TROPICAL STORMS AND HURRICANES IN THE NORTHERN GULF OF MEXICO WEST OF THE MISSISSIPPI RIVER DELTA

14:30 Liu, Y.; Pavia, A.; Kuo, L.; **Walsh, G.**; Chu, K.; Yang, S.; Yvon-Lewis, S.; Gold-Bouchot, G.; Sweet, S.; Wade, T.; Knap, A.; Tolic, N.; Chu, R.: DISTRIBUTION OF PER- AND POLYFLUORINATED ALKYL SUBSTANCES (PFAS) IN GALVESTON BAY, TX

14:45 **Zhou, Q.**; Wymore, A.; McDowell, W.: COMPARING THE IMPACTS OF TWO SIMULATED HURRICANES ON SOIL NITRATE CONCENTRATIONS: A MULTIPLE QUANTITATIVE APPROACH

15:00 **Fitzsimmons, J.**; Jensen, L.: TOXIC AND MICRONUTRIENT METALS IN WATERS OF GALVESTON BAY FOLLOWING HURRICANE HARVEY

15:15 **Steichen, J.**; Windham, R.; Kaiser, K.; Labonté, J.; Hala, D.; Bretherton, L.; Kamalanathan, M.; Setta, S.; Quigg, A.: CHANGES IN THE MICROBIAL COMMUNITY FOLLOWING AN EXTREME FLOODING EVENT IN THE WAKE OF HURRICANE HARVEY

[T] REPRESENTS TUTORIAL PRESENTATIONS

15:45 **Quigg, A.**; Windham, R.; McAmis, A.; Steichen, J.: EXTREME WEATHER EFFECTS ON PHYTOPLANKTON COMMUNITY COMPOSITION IN GALVESTON BAY, TEXAS: VISUALIZING ECOSYSTEM RESPONSES

16:00 **Hampel, J.**; McCarthy, M.; East, T.; Reed, M.; Newell, S.: THE EFFECTS OF HURRICANE IRMA ON NITRIFICATION AND HARMFUL ALGAL BLOOMS IN LAKE OKEECHOBEE AND THE ST. LUCIE ESTUARY (FLORIDA, USA)

16:15 **Buskey, E.**; Hyatt, C.; Scheef, L.: EFFECTS OF HURRICANE HARVEY ON WATER QUALITY AND THE PLANKTONIC FOOD WEB IN THE MISSION-ARANSAS ESTUARY, TEXAS

16:30 **Stauffer, B.**; Kurtay, G.; Pathare, M.; Schnetzer, A.: ENHANCED MICROZOOPLANKTON GRAZING IN SURFACE WATERS OF THE NORTHERN GULF OF MEXICO FOLLOWING HURRICANE HARVEY

16:45 **Topor, Z.**; Stauffer, B.; Robinson, K.: NEAR- TO OFFSHORE ZOOPLANKTON COMMUNITY STRUCTURE OF THE NORTHERN GULF OF MEXICO AFTER HURRICANE HARVEY

17:00 **Liu, H.**; Gilmartin, J.; Li, C.; Dziewit, M.: NEAR-TERM RESPONSES OF ESTUARINE ZOOPLANKTON TO EXTREME FLOODING AFTER A CATASTROPHIC HURRICANE

SS040 RADIONUCLIDES IN AQUEOUS SYSTEMS

Chair(s): Christopher G. Smith, U.S. Geological Survey (cgsmith@usgs.gov)
Ken Buesseler, Woods Hole Oceanographic Institution (kbuesseler@whoi.edu)
Mark Baskaran, Wayne State University (Baskaran@wayne.edu)
Sylvia Sander, IAEA Environment Laboratories (S.Sander@iaea.org)
Peter Swarzenski, IAEA Environment Laboratories (P.Swarzenski@iaea.org)
Kanchan Maiti, Louisiana State University (kmaiti@lsu.edu)
James Waples, University of Wisconsin-Milwaukee (jwaples@uwm.edu)
Erin Black, Dalhousie University (EBlack@dal.ca)

Location: Room 208 A/B

09:00 **Bam, W.**; Maiti, K.; Baskaran, M.; Krupp, K.: INFLUENCE OF PARTICLE CONCENTRATION AND COMPOSITION ON THE DISTRIBUTION OF 210-PO AND 210-PB ALONG US GEOTRACES ARCTIC TRANSECT

09:15 **Larson, R.**; Brooks, G.; Schwing, P.; Holmes, C.; Begin, C.; Browning, T.; Sawyer, D.; Diercks, A.: DETECTION OF SHORT-TERM SEDIMENTATION USING RADIOISOTOPES: FROM THE COAST TO THE DEEP SEA

09:30 **Charmasson, S.**; Wu, J.; Arnaud, M.; Cagnat, X.; Thomas, S.; Boulet, B.: OVERVIEW OF THE TRAPPING OF ARTIFICIAL RADIONUCLIDES OVER TIME IN THE SEDIMENTS OFF THE MOUTH OF THE RHONE RIVER (MEDITERRANEAN SEA)

09:45 **Savatier, M.**; Rocha, C.: SPATIAL AND SEASONAL CHANGES OF WATER AGE ALONG A FJORD IN IRELAND STUDIED WITH RA ISOTOPE RATIOS IN WATER TO GAIN FURTHER INSIGHT ON MIXING PROCESSES IN COASTAL SYSTEMS.

10:00 **McKenzie, T.**; Dulai, H.; Chang, J.: LINKING BASEFLOW AND SUBMARINE GROUNDWATER DISCHARGE: IMPACTS ON WATER QUALITY ALONG THE STREAM-COASTAL CONTINUUM IN KANEOHE BAY AND WATERSHED, HAWAII

10:15 **Douglas, A.**; Murgulet, D.; Peterson, R.: SPATIAL AND TEMPORAL VARIABILITY OF SUBMARINE GROUNDWATER DISCHARGE TO A DISTURBED SEMI-ARID ESTUARY

SS050 IMPROVING CORAL REEF RESILIENCE WITH TRANSFORMATIONAL SCIENCE

Chair(s): Emily Twigg, National Academies (etwigg@nas.edu)
Susan Roberts, National Academies (sroberts@nas.edu)
Stephen Palumbi, Stanford University (spalumbi@stanford.edu)
Tali Vardi, NOAA Fisheries (on contract from ECS) (tali.vardi@noaa.gov)

Location: Room 103 B

15:45 **Torres-Pulliza, D.**; Dornelas, M.; Madin, J.: DECONSTRUCTING REEF COMPLEXITY: RELATING CORAL DIVERSITY, RUGOSITY AND FRACTAL DIMENSION

16:00 **Williams, S.**: THE RESTOCKING DIADEMA ANTILLARUM POPULATIONS TO CONTROL THE ABUNDANCE OF RAMICRUSTA SPP. AND ALGAL COVER IN PUERTO RICO

16:15 **Martell, H.**; Hancock, H.; Zimmerman, R.: PREDICTING DYSBIOSIS FROM TEMPERATURE-DERIVED METABOLIC BUDGET MODELS

16:30 **Jaffe, M.**; Rodrigues, L.; Padilla-Gamino, J.: CARBON AND NITROGEN ACQUISITION AND TRANSLOCATION TO DEVELOPING EGGS IN NON-BLEACHED AND BLEACHED ADULT CORALS

16:45 **Vardi, T.**: THE CORAL RESTORATION CONSORTIUM: EFFORTS TO FUND, COORDINATE, AND TRANSFORM REEF CONSERVATION

17:00 **Palumbi, S.**; Anthony, K.; Baker, A.; Baskett, M.; Bhattacharya, D.; Bourne, D.; Knowlton, N.; Logan, C.; Naish, K.; Richmond, R.; Smith, T.; von Stackelberg, K.: A RESEARCH REVIEW OF INTERVENTIONS TO INCREASE THE PERSISTENCE AND RESILIENCE OF CORAL REEFS

SS051 NEW VIEWS ON THE BIOLOGICAL TRANSFORMATION OF METALS IN THE MARINE ENVIRONMENT

Chair(s): Randelle Bundy, University of Washington (rbundy@uw.edu)
Shane Hogle, Massachusetts Institute of Technology (shogle@mit.edu)
Katherine Heal, University of Washington (kheal@uw.edu)
Kristen Buck, University of South Florida (kristenbuck@usf.edu)
P. Dreux Chappell, Old Dominion University (pdchappe@odu.edu)

Location: Room 208 A/B

14:00 **Burns, S.**; Abdala, Z.; Sterling, A.; Bundy, R.; Chappell, P.; Jenkins, B.; Buck, K.: INVESTIGATING BIOGEOCHEMICAL FEEDBACKS BETWEEN TRACE METALS AND DIATOM GROWTH: INSIGHTS FROM SOUTHERN OCEAN PHYTOPLANKTON INCUBATION EXPERIMENTS

14:15 **Hawco, N.**; Fu, F.; Church, M.; Nelson, A.; Kelly, R.; Babcock-Adams, L.; Acker, M.; Ranieri, J.; Repeta, D.; Hutchins, D.; John, S.: NEW CONSTRAINTS ON IRON-LIGHT CO-LIMITATION IN THE DEEP CHLOROPHYLL MAXIMUM FROM CULTURE AND FIELD STUDIES

14:30 **Forsch, K.**; Fulton, K.; Manck, L.; Barbeau, K.: IRON LIMITATION OF A COASTAL FILAMENT IN THE SOUTHERN CALIFORNIA CURRENT ECOSYSTEM

14:45 **Geuer, J.**; Trimborn, S.; Brenneis, T.; Koch, F.; Krock, B.; Tebben, J.; Koch, B.: AN EXTREMIST'S STRATEGY - DOMOIC ACID AS IRON LIGAND OF THE ANTARCTIC DIATOM *PSEUDO-NITZSCHIA SUBCURVATA?*

15:00 **Manck, L.**; Dupont, C.; Barbeau, K.: FUNCTIONAL ROLE OF SIDEROPHORE BIOSYNTHESIS IN IRON ACQUISITION IN MARINE SYSTEMS

15:15 **Coale, T.**; Allen, A.: ASSIMILATION OF ORGANICALLY COMPLEXED IRON IN A MODEL PENNATE DIATOM

15:45 **Richon, C.**; Tagliabue, A.: DRIVERS OF THE COPPER DISTRIBUTION AND BIOGEOCHEMICAL CYCLING IN THE OCEAN

[1] REPRESENTS INVITED PRESENTATIONS

MONDAY

16:00 **Beghoura, H.**; Gorgues, T.; Aumont, O.; Planquette, H.; Tagliabue, A.: HOW ABIOTIC SEDIMENTARY PARTICULATE IRON MAY IMPACT THE DISSOLVED IRON DISTRIBUTION AND MAY CHANGE OUR UNDERSTANDING ON GLOBAL OCEAN BIOGEOCHEMICAL CYCLES?

16:15 **Kadko, D.**; Aguilar-Islas, A.; Bolt, C.; Buck, C.; Fitzsimmonsd , J.; Jensen, L.; Landing, W.; Marsay, C.; Rember, R.; Shiller, A.; Whitmore, L.; Anderson, R.: THE RESIDENCE TIMES OF TRACE ELEMENTS DETERMINED IN THE SURFACE ARCTIC OCEAN DURING THE 2015 US ARCTIC GEOTRACES EXPEDITION

16:30 **Black, E.**; Kienast, S.; Lam, P.; Buesseler, K.: UPPER OCEAN IRON RESIDENCE TIMES ACROSS A WIDE RANGE OF OCEAN REGIMES

16:45 **Held, N.**; McIlvin, M.; Webb, E.; Waterbury, J.; Saito, M.: NON-LIEBIG LIMITATION OF *TRICHODESMIUM* THROUGHOUT THE WORLD: INDIVIDUAL VARIABILITY IN METAL-MICROBE INTERACTIONS

17:00 **Hollister, A.**; Palmer, K.; Robert, M.; Tilney, C.; Breitbart, M.; Djurhuus, A.; Hubbart, K.; Buck, K.: REMINERALIZATION OF TRACE METALS AND MACRONUTRIENTS FROM NATURAL PHYTOPLANKTON ASSEMBLAGES, *PSEUDO-NITZSCHIA DOLOROSA*, AND *KARENIA BREVIS*

17:15 **Bundy, R.**; Buck, K.; Repeta, D.: THE ROLE OF SIDEROPHORES IN THE REGENERATION OF PARTICULATE IRON BY HETEROTROPHIC BACTERIA

SS057 EASTERN BOUNDARY UPWELLING SYSTEMS IN A CHANGING OCEAN: RECENT INSIGHTS AND FUTURE PERSPECTIVES

Chair(s): Francisco Chavez, Monterey Bay Aquarium Research Institute (MBARI) (chfr@mbari.org)
Carlos Henriquez, Millennium Institute of Oceanography (IMO) (chenriquez@lom.udec.cl)
Igor Fernandez, Millennium Institute of Oceanography (IMO) (igor.fernandez@imo-chile.cl)
Johana Medellín, Millennium Institute of Oceanography (IMO) (medellin.johana@gmail.com)
Natalia Osma, Millennium Institute of Oceanography (IMO) (natalia.osma@imo-chile.cl)
Paulina Aguayo, Millennium Institute of Oceanography (IMO) (paulina.aguayo@imo-chile.cl)

Location: Room 103 B

09:00 **Chavez, F.**: EASTERN BOUNDARY UPWELLING ECOSYSTEMS: PRESENT STATUS AND A VIEW TOWARDS THE FUTURE

09:15 **Morey, S.**: SPATIO-TEMPORAL VARIABILITY OF COASTAL UPWELLING FROM GLOBAL SATELLITE WIND COASTAL UPWELLING INDEX DATABASES

09:30 Flynn, R.; **Granger, J.**; Burger, J.; Veitch, J.; Siedlecki, S.; Pillay, K.; Fawcett, S.: ON-SHELF NUTRIENT TRAPPING ENHANCES THE FERTILITY OF THE SOUTHERN BENGUELA UPWELLING SYSTEM

09:45 **Grundle, D.**; Ji, Q.; Altabet, M.; Bange, H.; Graco, M.; Ma, X.; Arevalo-Martinez, D.: EL NIÑO EVENTS IN THE EASTERN TROPICAL SOUTH PACIFIC: IMPLICATIONS FOR N2O CYCLING AND DISTRIBUTIONS

10:00 **Henríquez, C.**; Garczarek, L.; Partensky, F.; van den Engh, G.; Ulloa, O.: GENETIC AND PIGMENT SHIFTS OF *SYNECHOCOCCUS* POPULATIONS ALONG THE CHILEAN EDGE OF THE HUMBOLDT CURRENT SYSTEM

10:15 **Pitz, K.**; Northcott, D.; Closek, C.; Michisaki, R.; Walz, K.; Boehm, A.; Chavez, F.: CHANGING BIODIVERSITY IN MONTEREY BAY, CALIFORNIA, FROM 2008-2016 AS MEASURED THROUGH EDNA METABARCODING

14:00 **Lilly, L.**; Ohman, M.: EL NIÑO -RELATED SPATIAL DISPLACEMENTS OF SUBTROPICAL AND COOL-WATER EUPHAUSIIDS IN THE SOUTHERN CALIFORNIA CURRENT SYSTEM

14:15 **Tapia, F.**; Sobarzo, M.; Moffat, C.; Gallardo-Escarate, C.: A MOSAIC OF LOCAL HYPOXIA REGIMES ON THE CENTRAL CHILE INNER SHELF – PHYSICAL FORCING AND IMPLICATIONS FOR BENTHIC ORGANISMS[T]

14:30 **Castelao, R.**; Luo, H.: UPWELLING JET SEPARATION IN THE CALIFORNIA CURRENT SYSTEM

14:45 **Auger, P.**; Gorgues, T.; Bento Netto Junior, J.; Berger Muñoz, T.; Morales, C.; Di Lorenzo, E.; Hormazabal Fritz, S.: WHAT DRIVES THE OFFSHORE EXTENSION OF COASTAL PRODUCTIVITY IN THE EASTERN SOUTH PACIFIC?

15:00 **von Appen, W.**; Strass, V.; Bracher, A.; Mathieu, L.: HIGH-RESOLUTION PHYSICAL-BIOGEOCHEMICAL STRUCTURE OF AN UPWELLING FILAMENT OFF NORTHWEST AFRICA

15:15 **Hill Cruz, M.**; Kriest, I.; Jose, Y.; Oschlies, A.: ZOOPLANKTON MORTALITY EFFECTS ON THE PLANKTONIC ECOSYSTEM OF THE UPWELLING SYSTEM OFF PERU: A REGIONAL BIOGEOCHEMICAL MODELLING APPROACH

SS063 OCEAN, COASTAL, AND FRESHWATER ACIDIFICATION: RESEARCH AND EDUCATION

Chair(s): Robert Chen, University of Massachusetts Boston (bob.chen@umb.edu)
Shannon Davis, University of Massachusetts Boston (Shannon.Davis002@umb.edu)
Joseph Salisbury, University of New Hampshire, Ocean Processes Analyses Lab (joe.salisbury@unh.edu)

Location: Room 103 A

09:00 **Seelmann, K.**; Aßmann, S.; Steinhoff, T.; Körtzinger, A.: PERFORMANCE EVALUATION OF THE FIRST COMMERCIAL AUTONOMOUS ANALYZER FOR TOTAL ALKALINITY (TA) IN SEAWATER

09:15 **Kulinski, K.**; Szymczycha, B.; Winogradow, A.; Stokowski, M.; Schneider, B.: PECULIARITIES OF THE ACID-BASE SYSTEM IN THE BALTIC SEA

09:30 **McCutcheon, M.**; Yao, H.; Staryk, C.; Hu, X.: TEMPORAL VARIABILITY AND DRIVING FACTORS OF THE CARBONATE SYSTEM IN THE TIDAL INLET OF A SEMIARID ESTUARY

09:45 **Tomasetti, S.**; Doall, M.; Gobler, C.: *CRASSOSTREA* CONTROL? CHARACTERIZATION OF THE TEMPORAL VARIABILITY IN DISSOLVED OXYGEN AND PH OF NEWLY CONSTRUCTED OYSTER REEFS.

10:00 Gassett, P.; **Turner, E.**; Stancioff, E.; Strong, A.; Grear, J.; Pimenta, A.; Bastidas, C.; O'Brien-Clayton, K.; Morrison, J.; Motyka, J.; DeRosia-Banick, K.: BROADENING CAPACITY FOR NORTHEAST OCEAN AND COASTAL ACIDIFICATION RESEARCH WITH COMMUNITY SCIENCE STAKEHOLDERS

10:15 **Lajaunie-Salla, K.**; Pinazo, C.; Diaz, F.: STUDY OF AIR-SEA CO2 FLUXES VARIATION AT THE REGIONAL SCALE OF THE BAY OF MARSEILLE, INSIGHTS FROM THE REGIONAL EARTH SYSTEM MODEL MASSILIA-CARBOX.

213 papers

[T] REPRESENTS TUTORIAL PRESENTATIONS

TUESDAY ORALS

CS006 TRACE METALS

Chair(s): Michael Twiss, Clarkson University (mtwiss@clarkson.edu)

Location: Room 102 A/B/C

15:45 **De Bonville, J.**; Amyot, M.; del Giorgio, P.; Tremblay, A.; Bilodeau, F.; Ponton, D.; Lapierre, J.: MOBILIZATION AND TRANSFORMATION OF MERCURY AND DISSOLVED ORGANIC MATTER ACROSS A DAMMED WATERSHED

16:00 **Brahmstedt, E.**; Zhou, H.; Eggleston, E.; Holsen, T.; Waller, M.; Windle, M.; Ridal, J.; Twiss, M.: SPATIAL VARIATION OF MERCURY UNDER CHANGING WATER LEVELS IN UPPER ST. LAWRENCE RIVER RIPARIAN WETLANDS

16:15 **Ayala Crespo, C.**; Brahmstedt, E.; Holsen, T.; Twiss, M.: MERCURY STORAGE IN HYDRIC SOILS AND BIOMASS OF A ST. LAWRENCE RIVER WETLAND

16:30 **Martinez-Colon, M.**; Alegria, H.; Huber, A.; Kurt, P.; Birgul, A.: BIOACCUMULATION AND BIOMAGNIFICATION OF HEAVY METALS IN BLACK MANGROVE AND FIDDLER CRABS FROM JOBOS BAY, PUERTO RICO

17:00 **Liu, Y.**; Liu, G.; Wu, Z.; Ge, M.; Liu, H.; Lam, P.: THE ACCUMULATION OF TRACE METALS IN AQUATIC ORGANISMS FROM A TYPICAL ESTUARINE-COASTAL ECOSYSTEM IN CHINA

17:15 **Mazzotta, M..**: CHARACTERIZATION OF THE METALLOPROTEOME OF THE MARINE BACTERIA, ALTEROMONAS

CS007 HYPOXIA

Chair(s): Steve DiMarco, Texas A&M University (sdimarco@tamu.edu)
Cory McDonald, Michigan Tech University (cpmcdona@mtu.edu)

Location: Room 201

09:00 **Seidel, L.**; Broman, E.; Ståhle, M.; Hylander, S.; Forsman, A.; Dopson, M.: TRACING BIOGEOCHEMICAL CHANGES IN AN ANOXIC BALTIC SEA COASTAL SEDIMENT DURING A TRANSITION TO OXIC CONDITONS

09:15 **Meyer, D.**; Prien, R.; Naumann, M.; Krüger, S.; Waniek, J.; Schulz-Bull, D.: SULFIDE OXIDATION EXPERIMENTS IN THE WORLD'S LARGEST HYPOXIC ZONE (BALTIC SEA) – IN SITU OBSERVATIONS WITH AN AUTOMATIC FLUID INJECTION SAMPLER

09:30 **LaBone, E.**; Justic, D.; Rose, K.; Wang, L.; Huang, H.: MODELING FISH MOVEMENT IN 3-D IN THE GULF OF MEXICO HYPOXIC ZONE

09:45 **DiMarco, S.**; Knap, A.; Chapman, P.; Potter, H.; Whilden, K.: DISSOLVED OXYGEN DYNAMICS OF THE TEXAS SHELF IN SUMMER 2017: RESPONSE TO SEASONAL UPWELLING AND HURRICANE HARVEY[IT]

10:00 **McDonald, C.**; Naziri-Saeed, M.; Robertson, D.; Prellwitz, S.; Siebers, B.: FORMATION OF A METALIMNETIC OXYGEN MINIMUM IN A DEEP LAKE

10:15 **Lorda, J.**; Data, B.; Rentería, M.; Molina, O.: INTER-ANNUAL DIFFERENCES IN THE VERTICAL DISTRIBUTION OF ZOOPLANKTON IN OMZ OF THE EASTERN TROPICAL MEXICAN PACIFIC

CS010 NITROGEN BIOGEOCHEMISTRY AND CYCLING

Chair(s): Julie Granger, University of Connecticut (julie.granger@uconn.edu)

Location: Room 101 A/B

09:00 **Montoya, J.**; Lee-Patterson, D.; Subramaniam, A.; Peterson, R.; Loick-Wilde, N.; Voss, M.: NITROGEN FIXATION AND NITROGEN DYNAMICS IN TROPICAL RIVER PLUMES

09:15 **Hallstrøm, S.**; Benavides, M.; Arístegui, J.; Riemann, L.: NITROGEN FIXATION AND DIAZOTROPH COMMUNITIES AT THE UPWELLING ECOSYSTEM OFF CAPE VERDE

09:30 **Marshall, T.**; Casciotti, K.; McIlvin, M.; Emeis, K.; Daehnke, K.; Fawcett, S.: THE ANGOLA GYRE, A HOTSPOT FOR N2 FIXATION IN THE TROPICAL SOUTH ATLANTIC OCEAN

09:45 Pedersen, J.; Bombar, D.; Paerl, R.; **Riemann, L.**: PARTICLE-ASSOCIATED N2-FIXATION AND DIAZOTROPH SUCCESSION ON ARTIFICIAL SURFACES IN COASTAL ESTUARINE WATERS

10:15 **Papacek, J.**; Phlips, E.; Lasi, M.; Strauss, S.; Inglett, P.: BIOLOGICAL NITROGEN FIXATION AS A POTENTIAL SOURCE OF NITROGEN FOR HARMFUL ALGAL BLOOMS IN A SUBTROPICAL, FLORIDA ESTUARY

14:00 **Trimmer, M.**; Fuessel, J.; Hemsley, V.; Duret, M.; Lam, P.: PARTICLE ASSOCIATED ORGANIC NITROGEN ESCAPES THE MARTIN CURVE IN THE SOUTH ATLANTIC

14:15 **Letscher, R.**; Villareal, T.: SEASONAL FORMATION RATES OF PREFORMED NITRATE ANOMALIES AT STATION ALOHA AND BATS: THE CASE FOR VERTICALLY MIGRATING PHYTOPLANKTON

14:30 **Shen, Y.**; Guilderson, T.; McCarthy, M.: DECIPHERING EXPORTED NITROGEN DYNAMICS IN THE CALIFORNIA MARGIN USING COMPOUND-SPECIFIC AMINO ACID D15N

15:00 **Fortin, S.**; Song, B.; Anderson, I.: SPATIAL AND TEMPORAL TRENDS OF MICROBIALLY MEDIATED REMOVAL AND RECYCLING OF FIXED NITROGEN IN THE YORK RIVER ESTUARY OF CHESAPEAKE BAY

15:15 **Zilius, M.**; Bartoli, M.; Bonaglia, S.; Griniene, E.; Liskow, I.; Overlinge, D.; Petkuviene, J.; Samuiloviene, A.; Stanislauskiene, R.; Vybernaite-Lubiene, I.; Voss, M.; Zemlys, P.; Bukaveckas, P.: THE INFLUENCE OF CYANOACTERIA BLOOMS ON THE ATTENUATION OF NITROGEN THROUGHPUTS IN A BALTIC COASTAL LAGOON

CS013 DISSOLVED ORGANIC MATTER - DOC, DON, DOP, FDOM, CDOM

Chair(s): Rainer Amon, Texas A&M University at Galveston (amonr@tamug.edu)
Brittany Widner, Woods Hole Oceanographic Institution (bwidner@whoi.edu)
Rachel Sipler, Memorial University of Newfoundland (resipler@mun.ca)
Angela Knapp, Florida State University (anknapp@fsu.edu)
Robert Letscher, University of New Hampshire (robert.letscher@unh.edu)
Boris Koch, Alfred-Wegener-Institut (bkoch@awi.de)
Michael Gonsior, UMCES-CBL (gonsior@umces.edu)

Location: Room 202

09:00 **Chanton, J.**; Spencer, R.; Drake, T.; Wilson, R.; Tfaily, M.; Cooper, W.; Griffiths, N.; Sebestyen, S.; Oleheiser, K.; Hanson, P.: TUTORIAL LECTURE ON DISSOLVED ORGANIC MATTER[T]

09:30 **Myers-Pigg, A.**; Bowering, K.; Prestegaard, K.; Skinner, A.; Spehn, N.; Ziegler, S.: SEASONALLY DISTINCT SOIL-STREAM FLOW PATHWAYS AND TERRESTRIAL ORGANIC MATTER (OM) SOURCES REGULATE DISSOLVED OM COMPOSITION IN A BOREAL FOREST HEADWATER STREAM

09:45 **Bowering, K.**; Myers-Pigg, A.; Edwards, K.; Ziegler, S.: SEASONAL VARIABILITY IN THE COMPOSITION OF MOBILIZED SOIL DISSOLVED ORGANIC MATTER IS SIMILAR IN TWO CONTRASTING BOREAL FOREST STAND TYPES

10:00 **Bhattacharya, R.**; Florea, K.; Petty, E.; North, R.: DISSOLVED ORGANIC MATTER CYCLING IN MID-WEST RESERVOIRS, USA

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

10:15 **Williams, C.**; Conrad, D.; Kothawala, D.; Baulch, H.: CHANGING AQUATIC DISSOLVED ORGANIC MATTER POOLS THREATEN DRINKING WATER RESOURCES

14:00 McKnight, D.; **Rue, G.**: ELECTRON-SHUTTLING BY HUMIC DISSOLVED ORGANIC MATTER IN A PERENNIALLY ICE-COVERED ANTARCTIC LAKE

14:15 **Lau, M.**; del Giorgio, P.: REACTIVITY AND FUNCTIONAL ROLES OF DISSOLVED ORGANIC MATTER IN THE ANOXIC HYPOLIMNION OF NORTH TEMPERATE LAKES

14:30 **Powers, L.**; Alvarez, J.; Del Vecchio, R.; Blough, N.; Gonsior, M.: DISSOLVED ORGANIC CARBON AND OPTICAL PROPERTY CHANGES IN LAGUNA GRANDE, PUERTO RICO, DURING THE 2018 SARGASSUM INUNDATION EVENT

14:45 **Miller, W.**; Powers, L.: PROBING THE ROLE OF CDOM/DOC SOURCE IN PHOTOCHEMICAL REDOX REACTIONS

15:00 **Stephens, B.**; Carlson, C.; Halewood, E.; Opalk, K.: MICROBIAL REMINERALIZATION OF DISSOLVED ORGANIC MATTER AT OCEAN STATION PAPA

15:15 **Calleja, M.**; Al-Otaibi, N.; Morán, X.: DISSOLVED ORGANIC CARBON TRANSPORT TO THE OCEAN INTERIOR: SEASONAL AND DIEL PROCESSES IN THE CENTRAL RED SEA

15:45 **Lechtenfeld, O.**; Koch, B.: ULTRA-HIGH RESOLUTION MASS SPECTROMETRY IN ORGANIC MATTER RESEARCH: ACTUAL AND PUTATIVE LIMITATIONS[I]

16:00 **Leefmann, T.**; Frickenhaus, S.; Koch, B.: ULTRAMASSEXPLORER – A NEW TOOL FOR THE EXPLORATION OF HIGH MASS RESOLUTION DATA IN DISSOLVED ORGANIC MATTER RESEARCH

16:15 **Gonsior, M.**; Powers, L.; Riemer, S.; Hemmler, D.; Harir, M.; Schmitt-Kopplin, P.: THE TIME-RESOLVED PHOTOCHEMISTRY OF CYANOBACTERIA-DERIVED PHYCOCYANOBILIN AND ITS RESULTING HIGHLY FLUORESCENT PHOTO-DEGRADATION PRODUCTS

16:30 **Li, Y.**; Benk, S.; Dittmar, T.; Gleixner, G.: MOLECULAR CHARACTERIZATION OF GROUNDWATER DISSOLVED ORGANIC MATTER IN THR CRITICAL ZONE

16:45 **Kamjunke, N.**; von Tümpling, W.; Hertkorn, N.; Harir, M.; Schmitt-Kopplin, P.; Griebler, C.; Brauns, M.; Weitere, M.; Herzsprung, P.: MOLECULAR CHANGE OF DISSOLVED ORGANIC MATTER AND PATTERNS OF BACTERIAL ACTIVITY IN A STREAM ALONG A LAND-USE GRADIENT

17:00 **Ianiri, H.**; Broek, T.; Bour, A.; McCarthy, M.: NEW D AMINO ACIDS PROVIDE INSIGHT TO MICROBIAL PRODUCTION OF REFRACTORY DISSOLVED ORGANIC NITROGEN IN THE OCEAN.

17:15 **McCarthy, M.**; Broek, T.; Vaughn, J.; Guilderson, T.: DOMINANT HETEROCYCLIC COMPOSITION OF DISSOLVED ORGANIC NITROGEN IN THE OCEAN

CS020 FISH AND FISHERIES

Chair(s): Brent A. Murry, U.S. Fish and Wildlife Service (brent.murry@gmail.com)

Location: Room 209 C

09:15 **Rubalcava, K.**; Chigbu, P.: RECRUITMENT AND DENSITY-DEPENDENT GROWTH OF SPOT (LEIOSTOMUS XANTHURUS) IN THE MARYLAND COASTAL BAYS

09:30 **Turley, B.**; Rykaczewski, R.: INFLUENCE OF WIND EVENTS ON LARVAL FISH MORTALITY RATES IN THE SOUTHERN CALIFORNIA CURRENT ECOSYSTEM

09:45 **Jarvis, L.**; McMeans, B.; Chu, C.: CONSIDERING THE CUMULATIVE EFFECTS OF ENVIRONMENTAL, BIOLOGICAL AND ANTHROPOGENIC STRESSORS ON WALLEYE PRODUCTIVITY

10:00 **Holbrook, S.**; Rassweiler, A.; Lauer, M.; Lester, S.; Schmitt, R.: LINKING SOCIAL AND ECOLOGICAL DATA TO UNDERSTAND HOW PACIFIC ISLAND FISHERS NAVIGATE CHANGING CORAL REEFS

10:15 **Garcia-Bermudez, M.**; Engman, A.; Llerandi-Roman, I.; Galindo, A.; Murry, B.; Olmeda, M.: A WATERSCAPE APPROACH TO FRESHWATER ARTISANAL FISHERIES

CS023 COMMUNITY ECOLOGY

Chair(s): Beatrix Beisner, University of Quebec at Montreal (beisner.beatrix@uqam.ca)
Nicolas Fortin-St-Gelais, University of Quebec at Montreal (nicolas.fstgelais@gmail.com)

Location: Room 201

14:00 **Winder, M.**; Zamora-Terol, S.; Novotny, A.: DRIVERS OF BIOTIC INTERACTIONS IN PLANKTON COMMUNITIES IDENTIFIED BY DNA METABARCODING[T]

14:15 **Kunzmann, A.**; Yohannes, E.; Straile, D.; Rothhaupt, K.: PREDATOR-PREY INTERACTION IN AN OLIGOTROPHIC LAKE: THE DIETARY SPECTRUM OF A KEYSTONE FRESHWATER COPEPOD

14:30 **Brown, A.**; Wares, J.; Osenberg, C.: EXTENDED PHENOTYPES ON CORAL REEFS: CRYPTIC VARIATION AND SPECIES INTERACTIONS

14:45 **Boyd, A.**; Zhang, S.; Silliman, B.; Riley, K.: A FOOD-WEB APPROACH TO SEAGRASS CONSERVATION: UTILIZING PINFISH (LAGODON RHOMBOIDES) TO INCREASE SEAGRASS GROWTH AND PRODUCTIVITY

15:15 **Herren, C.**: EXPECTED DIFFERENCES IN DIVERSITY AND RARITY BETWEEN COMMUNITIES CONTAINING ASEXUALLY VERSUS SEXUALLY REPRODUCING TAXA

15:45 **Keller, A.**; Cordes, E.: COLD SEEP HABITAT MAPPING OF COSTA RICA'S PACIFIC CONTINENTAL MARGIN

16:00 **Katkov, E.**; Low-Décarie, É.; Fussmann, G.: RESOURCE FLUCTUATIONS IN RELATIVE ABUNDANCE OF NITROGEN, PHOSPHORUS, CO2 UNLIKELY DRIVERS OF SEASONAL PHYTOPLANKTON SUCCESSION IN A TEMPERATE MESOTROPHIC LAKE

16:15 **Ares, A.**; Sato, K.; Mars Brisbin, M.; Diaz, J.; Ripken, C.; Mitarai, S.: PLANKTON COMMUNITY STRUCTURE DYNAMICS UNDER RED SOIL POLLUTION WITHIN CORAL REEF ECOSYSTEMS

16:30 **Cohen, R.**; Gray, D.: RESPONSES OF MACROINVERTEBRATE COMMUNITIES TO WATER QUALITY VARIABLES IN SMALL ARCTIC LAKES

16:45 **Gomez-Carrasquillo, J.**; Gomez, J.; Gutiérrez-Fonseca, P.; Ramírez, A.: HURRICANE EFFECTS ON AQUATIC INSECT ASSEMBLAGES

17:00 **Matassa, C.**; Ewanchuk, P.: DRIVERS OF SALT MARSH DECLINE IN A SOUTHERN NEW ENGLAND ESTUARY

17:15 **Gopalakrishnan, K.**; Pedersen, A.; Kashian, D.: EFFECT OF TEMPERATURE AND CALCIUM ON QUAGGA MUSSELS' FILTRATION RATES

[T] REPRESENTS TUTORIAL PRESENTATIONS

CS025 PHYTOPLANKTON ECOLOGY AND PHYSIOLOGY

Chair(s): Michael Behrenfeld, Oregon State University (mjb@science.oregonstate.edu)

Location: Room 101 A/B

15:45 **Prasil, O.**; Bernát, G.; Bonomi Barufi, J.; Campbell, D.; Claquin, P.; Cerveny, J.; Eichner, M.; Berman-Frank, I.; Giordano, M.; Halsey, K.; Kotabova, E.; Kromkamp, J.; Lombardi, A.; Lukeš, M.; Masuda, T.; Polerecky, L.; Rabouille, S.; Sediva, B.; Suggett, D.; Vancova, M.; Zavrel, T.: NITROGEN SOURCE AND CARBON LIMITATION INTERACT TO REGULATE DIEL PATTERNS OF ELECTRON FLOW AND BIOMASS IN CYANOTHECE

16:00 **Wang, J.**; Wagner, N.; Osburn, F.; Boedecker, A.; Scott, T.: VARIATION IN LIGHT CLIMATE AND NITROGEN AVAILABILITY AFFECT RESOURCE ALLOCATION TO PHOTOSYNTHETIC APPARATI IN CYANOBACTERIA

16:15 **Pombrol, M.**; Fisher, N.; Penta, W.; Halsey, K.: STRATEGIES FOR ALTERING ENERGY FLUX IN THALASSIOSIRA PSEUDONANA ELUCIDATED BY TRANSCRIPTOMICS

16:30 Bonisteel, E.; Turner, B.; Murphy, C.; Melanson, J.; Beardsall, B.; Duff, N.; Xu, K.; Campbell, D.; **Cockshutt, A.**: PHOTOSYSTEM II REPAIR CORRELATES WITH FTSH CONTENT AND ISOFORM EXPRESSION IN PICOCYANOBACTERIA

16:45 **Penta, W.**; Halsey, K.: CONTRASTING STRATEGIES FOR PHYTOPLANKTON PHYSIOLOGICAL ACCLIMATION DURING MIDYEAR MIXING EVENTS

17:00 **Graff, J.**; Behrenfeld, M.; Morison, F.; Harvey, E.; Menden-Deuer, S.; VerWey, B.: PHOTOACCLIMATION RESPONSES OF NATURAL PHYTOPLANKTON COMMUNITIES TO NATURAL AND ARTIFICIAL SHIFTS IN GROWTH IRRADIANCE

17:15 **D'souza, N.**; Passow, U.: THE COMBINED IMPACT OF MULTIPLE STRESSORS (CO2, TEMPERATURE, AND LIGHT) ON GROWTH AND PRODUCTIVITY OF THE MARINE DIATOM THALASSIOSIRA PSEUDONANA

CS028 ZOOPLANKTON ECOLOGY AND PHYSIOLOGY

Chair(s): John Dolan, Laboratoire d'Océanographie de Villefranche (dolan@obs-vlfr.fr)
George McManus, University of Connecticut (george.mcmanus@uconn.edu)

Location: Room 208 A/B

09:00 **Niemisto, M.**; Wahle, R.; Waller, J.; Clark, F.; Greenwood, S.; Fields, D.: GENE REGULATORY RESPONSE TO END-CENTURY TEMPERATURE AND PCO2 IN POSTLARVAL AMERICAN LOBSTER

09:15 **Djeghri, N.**; Pondaven, P.; Stibor, H.; Stockenreiter, M.: CAN δ13C, δ15N, AND C:N BE USED AS INDICATORS OF THE NUTRITION OF ZOOXANTHELLATE JELLYFISHES?

09:30 **Mendoza Islas, H.**; Hopcroft, R.: SCYPHOMEDUSAE ABUNDANCE AND DISTRIBUTIONS OF GELATINOUS ZOOPLANKTON IN THE NORTHERN GULF OF ALASKA

09:45 **Clardy, T.**; Heinle, M.; Al-Nuwairah, M.; Thomas, B.; Qurban, M.; Hikmawan, T.; Prihartato, P.; Abdulkader, K.: SPATIAL DIVERSITY AND DISTRIBUTION OF ZOOPLANKTON IN THE WESTERN ARABIAN GULF

10:00 **Dubickas, K.**; Daly, K.; Ferguson, M.; Zapfe, G.: ZOOPLANKTON COMMUNITY STRUCTURE IN THE NE GULF OF MEXICO BEFORE AND AFTER THE DEEPWATER HORIZON OIL SPILL

10:15 **Poje, A.**; Smoot, C.; Hopcroft, R.: GROWTH RATES OF CALANOID COPEPODS IN THE NORTHERN BERING AND SOUTHERN CHUKCHI SEAS

14:00 **Hopcroft, R.**; Smoot, C.: RESPIRATORY RATES OF COPEPODS IN THE PACIFIC ARCTIC

14:15 **Tyrell, A.**; Jiang, H.; Fisher, N.: COPEPOD RESPONSE TO TEMPERATURE CHANGE: SEPARATING THERMAL AND VISCOUS EFFECTS

14:30 **Dolan, J.**; Ciobanu, M.; Marro, S.; Coppola, L.: THE SURPRISING MICROZOOPLANKTON OF THE MESOPELAGIC MEDITERRANEAN SEA

14:45 **Anderson, S.**; Harvey, E.: COASTAL COMPLEXITIES: UNRAVELING TEMPORAL PHYTOPLANKTON RATE DYNAMICS IN THE SKIDAWAY RIVER ESTUARY

15:00 **Smith, S.**; Santoferrara, L.; Maurer-Alcalá, X.; Yan, Y.; Grattepanche, J.; Katz, L.; McManus, G.: CILIATE DIVERSITY: ADDRESSING THE SPECIES PROBLEM WITH "OMICS"

15:15 **Jarczak, J.**; Martin-Creuzburg, D.: NUTRIENTS MATTER MOST: EVOLUTION OF GRAZER RESISTANCE TO CYANOBACTERIA

15:45 **Otake, Y.**; Ohtsuki, H.; Urabe, J.; Kimura, S.; Yamada, K.; Yoshida, T.: LONG-TERM DYNAMICS OF CLADOCERAN COMMUNITY IN RESPONSE TO EUTROPHICATION PROCESSES SINCE LAKE FORMATION: RETROSPECTIVE ANALYSIS USING VARVED SEDIMENTS AND SUBFOSSILS

16:15 **Hahn, M.**; Effertz, C.; Bigler, L.; von Elert, E.: DIEL VERTICAL MIGRATION OF ZOOPLANKTON - IDENTIFICATION OF THE FISH KAIROMONE

16:30 Starke, C.; **Frost, P.**: INTERACTIVE EFFECTS OF TEMPERATURE AND FOOD QUALITY ON THE LIFE-HISTORY OF DAPHNIA PULICARIA

16:45 **Watkins, J.**; Weidel, B.; Luckey, F.; Barbiero, R.; Rudstam, L.; Lantry, B.: LAKE ONTARIO- IS THERE ENOUGH ZOOPLANKTON TO SUSTAIN ALL OF THE FISH?

SS005 ASLOMP STUDENT SYMPOSIUM

Chair(s): Benjamin Cuker, Hampton University (benjamin.cuker@hamptonu.edu)
Carolina Bonin, Hampton University (Carolina.Lewallen@hamptonu.edu)

Location: Room 102 A/B/C

09:00 **Das, N.**; Mayor, E.; Chigbu, P.: POPULATION DYNAMICS OF BLUE CRABS IN THE MARYLAND COASTAL BAYS

09:15 **Arzola, N.**; Ross, M.; Johnson, B.; Volpitta, A.; Schott, E.; O'Neil, J.: TRACKING MICROBIAL CONTAMINANTS IN BALTIMORE HARBOR: ARE CURRENT TECHNIQUES SUFFICIENT FOR ASSESSING HUMAN RISK?

09:30 **Selig, G.**; Netburn, A.; Malik, M.: INVESTIGATION OF LATITUDINAL AND VERTICAL DISTRIBUTIONS OF PELAGOTHURIA IN THE CENTRAL PACIFIC OCEAN USING NOAA EXPLORATORY DATA

09:45 **Cruz Reyes, R.**; Vohsen, S.; Girard, F.; Bullock, L.; Glickman, O.; Albertson, T.; DiNicola, G.; Fisher, C.: DEEP-SEA CALLOGORGIA DELTA OCTOCORALS CLOSELY ASSOCIATED WITH ACTIVE HYDROCARBON SEEPS ARE IN SIGNIFICANTLY BETTER CONDITION THAN COLONIES FOUND FAR FROM SEEPS

10:00 **Murarka, M.**; Prokopenko, M.; Robinson, L.; Li, T.; Burke, A.; Chen, T.: AN EXPLORATION OF DEEP-SEA CORALS AS ARCHIVES OF OCEAN RARE EARTH ELEMENT CONCENTRATIONS VIA LASER ABLATION-INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY

14:00 **Gade, A.**; Vohsen, S.; Fisher, C.; Baums, I.: EVOLUTIONARY HISTORY OF APICOMPLEXAN-RELATED SYMBIONTS IN CORALS: COMPARING NUCLEAR AND PLATID PHYLOGENY

TUESDAY

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

14:15 **Haggett, E.**; Cretoiu, S.; Bhattacharjee, A.; Cavicchioli, R.; Martínez Martínez, J.: SEASONAL VARIATION IN DSDNA EUKARYOTIC VIRUSES IN ANTARCTIC ACE LAKE

14:30 **Groves, S.**; Honisch, B.; Price, N.: PHYSIOLOGICAL IMPACTS OF CULTIVATING MUSSELS ON A KELP FARM: A SOLUTION TO OCEAN ACIDIFICATION?

14:45 **Adams, H.**; Fitzsimmons, J.: NATURAL AND TOXIC HEAVY METALS IN SEDIMENTS OF GALVESTON BAY, TEXAS, FOLLOWING HURRICANE HARVEY

SS011 ANATOMY OF A BLOOM: UNRAVELING DRIVERS OF BIOMASS CHANGE AND CARBON DYNAMICS OVER THE ANNUAL CYCLE

Chair(s): Jason Graff, Oregon State University (jrgraff@science.oregonstate.edu)
Peter Gaube, University of Washington - Applied Physics Lab (pgaube@apl.washington.edu)
Kay Bidle, Rutgers University (kbidle@marine.rutgers.edu)
Susanne Menden-Deuer, University of Rhode Island (smenden@uri.edu)
Craig Carlson, University of California Santa Barbara (craig_carlson@ucsb.edu)
Scott Doney, University of Virginia (sdoney@virginia.edu)

Location: Room 209 A/B

09:00 **Behrenfeld, M.**: DEVELOPING THOUGHTS ON PHYTOPLANKTON BLOOMS IN THE CONTEXT OF ANNUAL BIOMASS CYCLES[T]

09:30 **Eveleth, R.**; Doney, S.; Lima, I.; Glover, D.; Long, M.: GEOSTATISTICAL ANALYSIS OF NORTH ATLANTIC MESOSCALE BIOPHYSICAL VARIABILITY AND ITS DRIVERS IN AN EDDY RESOLVING CESM RUN

09:45 **Morison, F.**; Franzé, G.; Harvey, E.; Menden-Deuer, S.: THE INFLUENCE OF MIXED-LAYER LIGHT VARIABILITY ON PLANKTON POPULATION DYNAMICS.

10:00 **Bolanos, L.**; Menden-Deuer, S.; Karp-Boss, L.; Morrison, F.; Choi, C.; Worden, A.; Graff, j.; Carbonell-Moore, C.; Westberry, T.; Chase, A.; Haëntjens, N.; Gaube, P.; Boss, E.; Behrenfeld, M.; Giovannoni, S.: DIVERSE PHYTOPLANKTON FORM THE WESTERN NORTH ATLANTIC SPRING BLOOM

14:00 **Mayot, N.**; Matrai, P.; Arjona, A.; Bélanger, S.; Marchese, C.; Jaegler, T.: OPEN-OCEAN PHYTOPLANKTON PRODUCTION PATTERNS ASSOCIATED WITH SEA ICE: A GREENLAND SEA ANALYSIS

14:15 **Nicholson, D.**; Palevsky, H.: ANNUAL DYNAMICS OF NET COMMUNITY PRODUCTION IN THE IRMINGER SEA

14:30 **Hinson, A.**; Talmy, D.: WHAT CONTROLS ALGAL-VIRUS POPULATION DYNAMICS IN DIVERSE SYSTEMS?

14:45 **Schatz, D.**; Schleyer, G.; Saltvedt, M.; Rosenwasser, S.; Feldmesser, E.; Lawrence, J.; Sandaa, R.; Vardi, A.: COMMUNICATION VIA EXTRACELLULAR VESICLES ENHANCES VIRAL INFECTION OF A COSMOPOLITAN ALGA – EVIDENCE FROM THE LAB AND THE FIELD

15:00 Goode, A.; Fields, D.; Archer, S.; **Martinez, J.**: PHYSIOLOGICAL RESPONSES OF OXYRRHIS MARINA TO A DIET OF VIRALLY INFECTED EMILIANIA HUXLEYI

15:15 **Knowles, B.**; Bidle, K.: UNEXPECTED TEMPERATE VIRAL INFECTION NICHES IN THE LABORATORY AND IN NATURE

SS013 ADAPTATION OF AQUATIC BIODIVERSITY TO GLOBAL CHANGE

Chair(s): Vincent Fugere, University of Quebec at Montreal (vincent.fugere@mail.mcgill.ca)
Beatrix Beisner, University of Quebec at Montreal (beisner.beatrix@uqam.ca)

Location: Room 103 A

09:00 **Tan, E.**; Zou, W.; Zheng, Z.; Du, M.; Hsu, T.; Xu, J.; Tian, L.; Kao, S.: TEMPERATURE DEPENDENCE OF SEDIMENTARY NITROGEN REMOVAL PROCESSES AND ASSOCIATED N2O RELEASE

09:15 **Garcia Rodriguez, B.**: A METAGENOMICS APPROACH TO STUDY CYANOBACTERIAS AS BIOINDICATORS OF GLOBAL CLIMATE CHANGE

09:30 **Jin, P.**; Agustí, S.: FAST ADAPTATION OF TROPICAL DIATOMS TO INCREASED WARMING WITH TRADE-OFFS

09:45 Jin, P.; **Agusti, S.**: FAST ADAPTATION OF TROPICAL DIATOMS TO INCREASED WARMING

10:00 **Severin, T.**; Kremer, C.; Waters, J.; Hutchins, D.; Rynearson, T.: SEASONAL DIFFERENCES IN THERMAL TRAITS OF PHYTOPLANKTON FROM A TEMPERATE ESTUARY SUGGEST LOCAL ADAPTATION TO TEMPERATURE

10:15 **Gaedke, U.**; Ehrlich, E.; Kath, N.; Ceulemans, R.; Klauschies, T.; Guill, C.: THE ROLE OF TRADE-OFFS AND MUTUAL TRAIT ADAPTATION ACROSS THREE TROPHIC LEVELS OF A NATURAL PLANKTON FOOD WEB IN A CHANGING ENVIRONMENT

SS024 RECONSTRUCTING ADAPTIVE RESPONSES IN AQUATIC ECOSYSTEMS USING ANCIENT DNA AND RESURRECTION ECOLOGY

Chair(s): Raffaele Siano, Ifremer (raffaele.siano@ifremer.fr)
Marianne Ellegaard, University of Copenhagen (me@plen.ku.dk)
Isabelle Domaizon, INRA (isabelle.domaizon@inra.fr)
John Anderson, Loughborough University (N.J.Anderson@lboro.ac.uk)
Delphine Latour, University of Clermont-Ferrand (delphine.latour@uca.fr)

Location: Room 204

15:45 **Gauthier, J.**; Walsh, D.; Selbie, D.; Bourgeois, A.; Griffiths, K.; Domaizon, I.; Gregory-Eaves, I.: EVALUATING THE POTENTIAL OF DNA-BASED METHODS IN PALEOLIMNOLOGY USING A 3-YEAR SEDIMENT TRAP TIME-SERIES FROM A TEMPERATE MONOMICTIC LAKE

16:00 **Lassudrie, M.**; Cuzin, P.; Mertens, K.; Quintric, L.; Durand, P.; Schmidt, S.; Siano, R.: LONG-TERM DYNAMICS OF PROTIST PALEOCOMMUNITIES IN A COASTAL MARINE ECOSYSTEM (BAY OF BREST, NW FRANCE) BY DNA METABARCODING

16:15 **Domaizon, I.**: TRACKING LONG-TERM CHANGES IN MICROBIAL EUKARYOTIC DIVERSITY IN LAKES FACING NUTRIENT ENRICHMENT AND CLIMATE WARMING

16:30 **Zimmermann, H.**; Stoof-Leichsenring, K.; Kruse, S.; Müller, J.; Stein, R.; Tiedemann, R.; Herzschuh, U.: ANCIENT DIATOM DNA IN LATE PLEISTOCENE SEDIMENT CORES FROM THE NORTH ATLANTIC AND NORTH PACIFIC OCEANS

16:45 **Siano, R.**; Delebecq, G.: REVIVIFICATION OF ANCIENT MARINE DINOFLAGELLATE USING MOLECULAR BIOSTIMULATION

17:00 **Brosnahan, M.**; Fischer, A.; Solow, A.; Kulis, D.; Anderson, D.: COMPARING CHILLING RESPONSES OF RED TIDE DINOFLAGELLATE CYSTS FORMED DURING MODERN AND PAST DECADES

[T] REPRESENTS TUTORIAL PRESENTATIONS

17:15 **Walsh, E.**; Samaniego, S.; Mohl, J.: ROTIFER COMMUNITY PROFILING IN TEMPORARY WATERS THROUGH RESURRECTION ECOLOGY AND NEXT-GENERATION SEQUENCING

SS034 FORECASTING IS THE FUTURE: ADVANCING METHODS AND APPLICATIONS OF FORECASTING IN THE AQUATIC SCIENCES

Chair(s): Cayelan Carey, Virginia Tech (cayelan@vt.edu)
Rafael Marce, Catalan Institute for Water Research (rmarce@icra.cat)
Paul Hanson, University of Wisconsin-Madison (pchanson@wisc.edu)

Location: Room 103 B

09:15 **LaDeau, S.**; Weathers, K.; Dietze, M.: A FLEXIBLE, STATE-SPACE APPROACH FOR EVALUATING THE SCALE AND SCOPE OF PREDICTABILITY IN NATURE[IT]

09:45 **Baracchini, T.**; Chu, P.; Bouffard, D.; Wüest, A.: 3D HYDRODYNAMIC AND BIO-PHYSICAL LAKE MODELS WITH DATA ASSIMILATION – APPLICATION OF AN OPERATIONAL SYSTEM FOR SWISS LAKES

10:00 **Mercado-Bettín, D.**; Iturbe, M.; Marcé, R.; Frías, M.; Herrera, S.: FROM SEASONAL FORECASTING TO LAKE WATER QUALITY: A PRACTICAL TOOL

10:15 **Carey, C.**; Thomas, R.; Figueiredo, R.; Daneshmand, V.: REAL-TIME ECOLOGICAL FORECASTING ENABLES ADAPTIVE WATER QUALITY MANAGEMENT IN A DRINKING WATER RESERVOIR

14:00 **Pomati, F.**; Isles, P.: FORECASTING PHYTOPLANKTON COMMUNITY CHANGE USING IN SITU MONITORING AND MACHINE LEARNING

14:15 **Jones, I.**; Woolway, I.; Mackay, E.; Maberly, S.; Elliott, A.: PREDICTING FUTURE CHANGES TO PHYTOPLANKTON COMMUNITIES AROUND THE WORLD

14:30 **Gronchi, E.**; Joehnk, K.; Straile, D.; Peeters, F.: PROXY-BASED MODELLING OF PHYTOPLANKTON SPRING BLOOM

14:45 **Slavin, E.**; Perkins, R.; Wain, D.; Blenkinsopp, C.: CAN IN-RESERVOIR MONITORING BE USED TO PREDICT TASTE AND ODOUR EVENTS IN DRINKING WATER RESERVOIRS?

15:00 **Han, Y.**; Smithheart, J.; Smyth, R.; Obenour, D.; Aziz, T.: ASSESSING VERTICAL MIXING AS A POTENTIAL CONTROL ON CYANOBACTERIA DOMINANCE IN SHALLOW TURBID RESERVOIRS

15:15 **Justic, D.**; Wang, L.: THE FUTURE OF COASTAL HYPOXIA UNDER SCENARIOS OF RIVER MANAGEMENT

15:45 **Obenour, D.**; Del Giudice, D.; Katin, A.: EXPLORING OXYGEN DEMANDS AND DYNAMICS IN COASTAL SYSTEMS USING A BAYESIAN MECHANISTIC APPROACH

16:00 **Harvey, J.**: FORECASTING THE RESTORATION OF A FREE-FLOWING EVERGLADES BASED ON LARGE-SCALE HIGH-FLOW EXPERIMENTS AND WHOLE-SYSTEM MODELING

16:15 **Hanson, P.**; Stillman, A.; Carey, C.; Jia, X.; Kumar, V.: IMPROVED UNDERSTANDING AND PREDICTION OF 37 YEARS OF SURFACE WATER PHOSPHORUS DYNAMICS USING THEORY GUIDED MACHINE LEARNING

SS038 REAL WORLD ECOLOGY LANDSCAPES – UP-SCALING COMMUNITY ECOLOGY EXPERIMENTS IN AQUATIC SYSTEMS

Chair(s): Maria Stockenreiter, Ludwig-Maximilians-University (LMU) Munich (stockenreiter@bio.lmu.de)
Patrick Fink, University of Cologne (patrick.fink@uni-koeln.de)
Jens Christian Nejstgaard, Leibniz-Institute of Freshwater Ecology and Inland (nejstgaard@igb-berlin.de)

Location: Room 204

09:00 **Meunier, C.**; Groß, E.; Köring, M.; Duarte Moreno, H.; Tremblay, N.; Boersma, M.: IMPACT OF GLOBAL CHANGE ON MARINE PLANKTON, A MULTIPLE STRESSOR APPROACH

09:15 **Almeda, R.**; Christensen, A.; Dinh, K.; Frková, Z.; Antoniou, E.; Kalogerakis, N.; Pavloudi, C.; Parinos, C.; Hatzianestis, I.; Franzellitti, S.; Fabbri, E.; Rosato, A.; Magiopoulos, I.; Pitta, P.; Zanaroli, G.: IMPACT OF CRUDE OIL BURNING BY-PRODUCTS ON ZOOPLANKTON: A MICRO- AND MESOCOSM APPROACH.

09:30 **Huynh, M.**; Gray, D.: DISPERSAL AS A BUFFER AGAINST ZOOPLANKTON COMMUNITY CHANGE IN RESPONSE TO INCREASING SALINITY LEVELS ON THE GREAT PLAINS

09:45 **Berger, S.**; Wollrab, S.; Dilewski, C.; Beisner, B.; Grossart, H.; Hansson, T.; Penske, A.; Nejstgaard, J.: EXPERIMENTAL WINTER LIMNOLOGY IN SITU – FIRST TEST OF LAKE ECOSYSTEM RESPONSES TO CONTROLLED ICE COVER DURATION IN LARGE-SCALE ENCLOSURES

10:00 **Gollnisch, R.**; Stockenreiter, M.; Rengefors, K.: ENVIRONMENTAL FACTORS AFFECTING THE EXPANSION OF THE INVASIVE NUISANCE MICROALGA GONYOSTOMUM SEMEN

10:15 **Stockenreiter, M.**; Jarczak, J.: CYANOBACTERIA DOMINANCE INDUCED BY REDUCED MIXING DEPTH IN A EUTROPHIC LAKE

14:00 **Urrutia Cordero, P.**; Striebel, M.; Peter, E.; Bertilsson, S.; Colom, W.; Kelpsiene, E.; Laudon, H.; Lundgren, M.; Pierson, D.; Travik, L.; Langenheder, S.; Hillebrand, H.: LAKE ECOSYSTEM FUNCTIONING IN RESPONSE TO MULTIPLE ENVIRONMENTAL DISTURBANCES - IDIOSYNCRATIC AND GENERAL MECHANISTIC UNDERSTANDING FROM A MODULARIZED EXPERIMENT ACROSS LAKES

14:15 **Chaguaceda Borjabad, F.**; Scharnweber, K.; Dalman, E.; Tranvik, L.; Eklöv, P.: TOP-DOWN PREDATION DRIVE SHIFTS IN APPARENT COMPETITION ALONG A PRODUCTIVITY GRADIENT CAUSING CROSS-BOUNDARY SUBSIDY EFFECTS

14:30 Benitez-Requena, L.; **Titocci, J.**; Stockenreiter, M.; Stibor, H.; Fink, P.: LOSS OF FUNCTIONAL TRAITS AFFECTS PHYTOPLANKTON-ZOOPLANKTON DYNAMICS IN A LAKE MESOCOSM EXPERIMENT

14:45 **Ilic, M.**; Stockenreiter, M.; Stibor, H.; Fink, P.: EFFECTS OF PHYTOPLANKTON DIVERSITY ON INTRASPECIFIC COMPETITION IN *DAPHNIA* - A COMMON GARDEN EXPERIMENT

15:00 **Fink, P.**; Norf, H.; Weitere, M.; Borchardt, D.: MOBILE AQUATIC MESOCOSMS (MOBICOS) - A MODULAR EXPERIMENTAL INFRASTRUCTURE FOR PROCESS BASED HYDRO-ECOLOGICAL RESEARCH

15:15 Walles, T.; Sed'a, J.; Greer, A.; **Nejstgaard, J.**: DOUBLE-DEEP: ZOOPLANKTON ABUNDANCE AND FINE-SCALE MIGRATION PATTERNS QUANTIFIED WITH DEEP-FOCUS IMAGING AND DEEP-LEARNING ALGORITHMS

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

SS039 EFFECTS OF STORM EVENTS ON AQUATIC, COASTAL, AND OCEANIC ENVIRONMENTS AND ECOSYSTEMS: FROM PHYSICAL PROCESSES TO FOOD WEBS

Chair(s): Beth Stauffer, University of Louisiana at Lafayette (stauffer@louisiana.edu)
Michael Martinez-Colon, Florida A&M University (michael.martinez@famu.edu)
Patricia Chardon-Maldonado, University of Puerto Rico at Mayaguez (patricia.chardon@upr.edu)
Jessica Fitzsimmons, Texas A&M University (jessfitz@tamu.edu)
Miguel F. Canals Silander, University of Puerto Rico at Mayaguez (miguelf.canals@upr.edu)
Lisa Campbell, Texas A&M University (lisacampbell@tamu.edu)
Simon Geist, Texas A&M University Corpus Christi (simon.geist@tamucc.edu)
Kam-Biu Liu, Louisiana State University (kliu1@lsu.edu)
Thomas Bianchette, University Of Michigan (tbianc@umich.edu)
Marianne Dietz, Louisiana State University (mdietz2@lsu.edu)

Location: Room 104

09:00 **Fiorendino, J.**; Campbell, L.; Henrichs, D.: SHIFTS IN PLANKTON COMMUNITIES ON THE TEXAS COAST FOLLOWING HURRICANE HARVEY

09:15 **Henrichs, D.**; Fiorendino, J.; Campbell, L.: PEEKING AT THE PLANKTON: USING METATRANSCRIPTOMICS TO OBSERVE THE PLANKTON COMMUNITY RESPONSE TO HURRICANE HARVEY

09:30 **Campbell, L.**; Henrichs, D.; Anglès, S.: CHANGING CHAINS: IN SITU OBSERVATIONS OF DIATOM CHAIN LENGTH IN RESPONSE TO ENVIRONMENTAL FORCING FROM HURRICANES AND UPWELLING EVENTS

09:45 **Smith, T.**; Mukherjee, S.; Ennis, R.; Brandtneris, V.; Brandt, M.; Canals, M.: HURRICANE GENERATED WAVE THRESHOLDS FOR DAMAGE TO CORAL REEFS IN THE US VIRGIN ISLANDS

10:15 **Brandt, M.**; Olson, J.; Chaves-Fonnegra, A.; Smith, T.; Ennis, R.; Gochfeld, D.: IMPACTS OF HURRICANES IRMA AND MARIA ON CORAL REEF SPONGE COMMUNITIES IN ST. THOMAS, U.S. VIRGIN ISLANDS

14:00 **Rogers, C.**: HURRICANE DAMAGE TO A UNIQUE CORAL/MANGROVE ECOSYSTEM AND POTENTIAL FOR RECOVERY

14:15 **Barrett, L.**; Shamberger, K.; Hicks, T.; Hooper, M.; Correa, A.; Davies, S.; Santiago-Vazquez, L.; Sylvan, J.: POST-HARVEY CARBONATE CHEMISTRY AT THE FLOWER GARDEN BANKS CORAL REEFS

14:30 **McAskill, S.**; Zapfe, G.; Geist, S.: EFFECTS OF HURRICANE HARVEY ON LARVAL FISH DISTRIBUTION IN THE GULF OF MEXICO

14:45 **Simmons, K.**; Eggleston, D.; Bohnenstiehl, D.: ECOLOGICAL IMPACTS OF HURRICANE IRMA ON A CORAL REEF SOUNSCAPE

15:00 **Dunlap-Smith, A.**: PHOTOGRAPHIC SURVEYS AND COMPUTER VISION TRACK FRESHWATER DECAPOD CRUSTACEAN COMMUNITIES ON ST. JOHN, US VIRGIN ISLANDS.

15:45 **Wilson, S.**; Furman, B.; Hall, M.; Fourqurean, J.: ASSESSMENT OF HURRICANE IRMA IMPACTS ON SOUTH FLORIDA SEAGRASS COMMUNITIES USING LONG-TERM MONITORING PROGRAMS

16:00 **Crespo-Medina, M.**; Minnigh, H.; Rodríguez Curet, T.; Hertler, H.; Christian, R.; Frontera, T.; Ramírez Toro, G.: IMPACT OF HURRICANE MARIA ON A CARIBBEAN MANGROVE FOREST

16:15 **Patrick, C.**; Yeager, L.; Armitage, A.; Carvallo, F.; Congdon, V.; Dunton, K.; Fisher, M.; Hardison, A.; Hogan, J.; Hosen, J.; Hu, X.; Reese, B.; Kinard, S.; Kominoski, J.; Lin, X.; Liu, Z.; Montagna, P.; Pennings, S.; Walker, L.; Weaver, C.; Wetz, M.: DRIVING WIND AND TORRENTIAL RAIN: IMPACTS OF HURRICANE HARVEY ON COASTAL ECOSYSTEMS

16:30 **Carvallo, F.**; Kinard, S.; Gonzalez, D.; Hogan, J.; Reese, B.; Patrick, C.: RESPONSES OF COASTAL PLAIN STREAMS ALONG AN ARIDITY GRADIENT TO HURRICANE HARVEY

16:45 **Peri, F.**: HURRICANE MARIA EFFECTS ON BIODIVERSITY IN PUERTO RICAN FORESTS AS MEASURED BY TERRESTRIAL LIDAR SCANNER

17:00 **Kominoski, J.**; Kuhn, A.; Charles, S.; Pennings, S.; Weaver, C.; Maddox, T.; Armitage, A.: PLANT COMPOSITION AFFECTS ECOSYSTEM CONNECTIVITY DURING A CATASTROPHIC HURRICANE

17:15 **Rothenberger, M.**; Armstrong, A.; Spitz, M.: SOCIAL-ECOLOGICAL SYSTEM RESPONSES TO HURRICANE SANDY IN THE HUDSON-RARITAN ESTUARY

SS043 BEYOND THE NUMBERS: STRATEGIES FOR INCLUSIVE PRACTICES ACROSS THE AQUATIC SCIENCES

Chair(s): Brandon Jones, National Science Foundation (mbjones@nsf.gov)
Lisa Rom, National Science Foundation (elrom@nsf.gov)

Location: Room 208 C

09:00 **Jones, B.**: IT'S ALL GOOD: THE RECIPROCITY OF DIVERSITY IN STEM DISCIPLINES

09:15 **Batchelor, R.**; Kirsch, R.; King, B.; Habtes, S.; Crockett, J.: SPARKS FOR CHANGE: DEVELOPING DEI CHANGE AGENTS IN THE GEOSCIENCES

09:30 **Hodder, J.**; Macdonald, H.; Baer, E.; Beane, R.; Bragg, D.; Emerson, N.; Iverson, E.; McDaris, J.; O'Connell, K.; Ormand, C.: ENGAGING TWO-YEAR COLLEGE GEOSCIENCE FACULTY TO PROMOTE DIVERSITY, EQUITY, AND INCLUSION: THE SAGE 2YC PROJECT

09:45 **Schiebel, H.**: COMBINING CITIZEN SCIENCE AND SERVICE LEARNING IN THE UNDERGRADUATE CLASSROOM

10:00 **Aguilar, C.**; Cuhel, R.: INTERDISCIPLINARY EXPEDITIONARY RESEARCH SCIENCE EDUCATION PROMOTES ETHNIC, GENDER, SOCIOECONOMIC, AND DISCIPLINARY DIVERSITY IN AQUATIC SCIENCES

10:15 **Garza, C.**; Parrish, J.; Harris, L.; Posselt, J.: ASPIRE: ACTIVE SOCIETAL PARTICIPATION IN RESEARCH AND EDUCATION

14:00 **Hernandez Delgado, E.**; Suleiman Ramos, S.: COMMUNITY-BASED CORAL REEF REHABILITATION RESEARCH IN PUERTO RICO: AN ALTERNATIVE PATHWAY INTO PARTICIPATORY SCIENCE FOR TRADITIONALLY UNDERREPRESENTED GROUPS

14:15 **Brandon, J.**; Peach, C.; Arnold, E.; Paulenich, C.; Sickler, S.: PRICE OUTREACH'S BEACH SCIENCE PROGRAM: BRIDGING THE GAP BETWEEN TRAINING SCIENCE COMMUNICATORS AND ENGAGING HISTORICALLY UNDERSERVED AUDIENCES

14:30 **Jones, M.**; Jabanoski, K.: LESSONS LEARNED IN ENGAGING UNDERSERVED AUDIENCES IN NOAA EDUCATION PROGRAMS

14:45 Moser, F.; **Barberena Arias, M.**; Harris, L.; Maldonado, P.; Olivo Delgado, C.; Pierson, J.: BUILDING MARINE SCIENCE RESEARCH CAPACITY IN EARLY STAGE HISPANIC UNDERGRADUATE STUDENTS

15:00 Abernethy, E.; Arismendi, I.; **Boegehold, A.**; Colon-Gaud, C.; Cover, M.; Larson, E.; Moody, E.; Penaluna, B.; Shogren, A.; Webster, A.: STRIVING TOWARDS A MORE DIVERSE AND INCLUSIVE SCIENTIFIC SOCIETY: PROGRESS MADE IN THE SOCIETY FOR FRESHWATER SCIENCE

15:15 **Webster, A.**; Zemenick, A.; Jones, S.: PROJECT BIODIVERSIFY: A REPOSITORY OF TEACHING MATERIALS AND METHODS TO MAKE BIOLOGICAL AND NATURAL SCIENCE CLASSROOMS MORE INCLUSIVE

SS059 SOCIO-ECOLOGICAL RESEARCH FOR ACTIONABLE SUSTAINABLE SOLUTIONS: EXAMPLES, PERSPECTIVES AND CHALLENGES

Chair(s): Nicolas F. St-Gelais, Université de Montréal (nicolas.fstgelais@gmail.com)
Jean-Olivier Goyette, Université de Montréal (jean-olivier.goyette@umontreal.ca)
Julie Talbot, Université de Montréal (julie.talbot@gmail.com)
Michelle McCrackin, Stockholm University (michelle.mccrackin@su.se)
Roxane Maranger, Université de Montréal (r.maranger@umontreal.ca)

Location: Room 209 C

14:00 **McCrackin, M.**: HOW CAN NATURAL SCIENTISTS INFORM POLICY? LESSONS FROM A UNIVERSITY-BASED BOUNDARY ORGANIZATION[T]

14:30 **Messager, M.**; Davies, I.; Levin, P.: SUFFICIENT AND EFFICIENT SOLUTIONS FOR TOXIC URBAN STORMWATER

14:45 **Plante, S.**: ADAPTATIONS BASED ON SOCIAL-ECOLOGICAL SYSTEMS: THE ROLE OF GOVERNANCE FOR COASTAL AND RIPARIAN COMMUNITIES.

15:00 **Chen, R.**; Bosma, K.; Borrelli, M.; Kirshen, P.; Lockwood, L.: EXPLORING THE USE OF DREDGE SPOILS TO ENHANCE THE ECOSYSTEM SERVICES PROVIDED BY THE BOSTON HARBOR ISLANDS

15:45 **Salk, K.**; Denny, R.; Greif, J.: SOCIAL-ECOLOGICAL INTERACTIONS OF NITROGEN MANAGEMENT IN THE MISSISSIPPI RIVER BASIN AND THE ROLE OF POLICY

16:00 **Maranger, R.**; Fortin St Gelais, N.; Clarke, E.; Franssen, J.; Lapierre, J.; Messer, P.; Poisot, T.; Slade, S.; Talbot, J.: MAKING ECOSYSTEM SCIENCE MATTER: STAKEHOLDER-ENGAGED RESEARCH THROUGH CO-DESIGN AND SOCIAL-ECOLOGICAL SYNTHESIS

16:15 **Fortin St-Gelais, N.**; Lapierre, J.; Siron, R.; Maranger, R.: IDENTIFYING THRESHOLDS OF THE EFFECT OF LAND USE AND CLIMATE CHANGE ON AQUATIC ECOSYSTEM SERVICES.

16:30 **Atakpa, D.**: BLUE ECONOMY MECHANISMS FOR ACTIONABLE SUSTAINABLE SOLUTIONS

SS060 THE ADVENT OF SAMPLING BIOLOGICAL PROCESSES IN AQUATIC SYSTEMS USING AUTONOMOUS PLATFORMS

Chair(s): Oscar Schofield, Rutgers University (oscar@marine.rutgers.edu)
Josh Kohut, Rutgers University New Brunswick (kohut@marine.rutgers.edu)
Craig Lee, Applied Physics Laboratory (craig@apl.washington.edu)
Kelly Benoit Bird, MBARI (kbb@mbari.org)
Mark Moline, University of Delaware (mmoline@udel.edu)

Location: Room 209 A/B

15:45 **Schofield, O.**; Glenn, S.; Kohut, J.; Miles, T.; Saba, G.: PLATFORMS ARE MATURE BUT WHAT SENSORS AND SAMPLING DO WE NEED TO SAMPLE MARINE FOOD WEBS?

16:00 **Breier, J.**; Jakuba, M.; Saito, M.; Chan, E.; Mcilvin, M.; Moran, D.; Alanis, B.; Johnson, R.: REVEALING OCEAN BIOCHEMICAL STRUCTURE WITH HIGH-RESOLUTON SAMPLING FROM AN AUTONOMOUS UNDERWATER PROFILING VEHICLE: CLIO

16:15 **Forget, G.**; Dutkiewicz, S.; Jahn, O.; Follows, M.: SIMULATED BIOGEOCHEMICAL PROFILING FLOAT ARRAYS IN CBIOMES

16:30 **Villareal, T.**; Wilson, C.: LONG-DURATION SAMPLING OF THE PACIFIC GYRE USING HOLOGRAPHY AND ACTIVE FLUORESCENCE ON A WAVE GLIDER SV2

16:45 **Penta, B.**; Rice, A.; Shulman, I.; Anderson, S.; Martens, I.; Goode, W.; Cohen, J.; Hudson, K.; Greer, A.; Lehrter, J.; Hagemeyer, A.; Binder, B.; McFarland, M.; Nayak, A.; Stockley, N.: AUTONOMOUS LAGRANGIAN OBSERVATIONS OF OCEAN BIOPHYSICAL PARAMETERS OFFSHORE OF DELAWARE BAY, USA

17:00 **Shulman, I.**; Penta, B.; Moline, M.; Oliver, M.; Anderson, S.; Hudson , K.; Cohen, J.: BIOLUMINESCENCE POTENTIAL DYNAMICS AT FRONTAL STRUCTURES IN THE DELAWARE BAY AREA.

17:15 **Birch, J.**: AUTONOMOUS TARGETED SAMPLING WITH MOBILE ECOGENOMIC SAMPLERS.

SS063 OCEAN, COASTAL, AND FRESHWATER ACIDIFICATION: RESEARCH AND EDUCATION

Chair(s): Robert Chen, University of Massachusetts Boston (bob.chen@umb.edu)
Shannon Davis, University of Massachusetts Boston (Shannon.Davis002@umb.edu)
Joseph Salisbury, University of New Hampshire, Ocean Processes Analyses Lab (joe.salisbury@unh.edu)

Location: Room 103 A

14:00 **Feely, R.**; Carter, B.; Bednarsek, N.; Cai, W.; Alin, S.; Okazaki, R.; Greeley, D.: THE COMBINED EFFECTS OF ACIDIFICATION AND HYPOXIA ALONG THE WEST COAST OF NORTH AMERICA[T]

14:30 **Falkenberg, L.**: USING ACADEMIC WRITING TO FACILITATE OUTREACH AND EDUCATION REGARDING OCEAN ACIDIFICATION RESEARCH ACROSS TRADITIONAL DISCIPLINARY LINES

14:45 **San Antonio, C.**; Tlusty, M.; Poynton, H.; Krick, K.; Hannigan, R.: EXAMINING THE EFFECTS OF OCEAN WARMING AND ACIDIFICATION ON JUVENILE AMERICAN LOBSTERS

15:00 **Gao, Y.**; Svec, R.: CLIMATE CHANGE AND OCEAN ACIDIFICATION REVEALED FROM ISOTOPIC SIGNATURES OF CARBONATE SHELLS

15:15 **Brandt, S.**; Cary, C.; Pilditch, C.; Vopel, K.; Hartland, A.; Lee, C.; Marshall, A.; Laverock, B.: THE EFFECTS OF OCEAN ACIDIFICATION ON MICROBIAL NUTRIENT CYCLING AND PRODUCTIVITY IN COASTAL MARINE SEDIMENTS

16:00 **Wyatt, S.**; Varela, D.; Grundle, D.: OCEAN ACIDIFICATION EFFECTS ON SILICEOUS PHYTOPLANKTON FROM THE SUBARCTIC NORTHEAST PACIFIC AND SUBTROPICAL NORTH ATLANTIC OCEANS

16:30 **Stevenson, A.**; Archer, S.; Dunham, A.; Marliave, J.; Schultz, J.; Martone, P.; Harley, C.: CLIMATE CHANGE PETRIFIES GLASS SPONGE REEFS: FUTURE PROJECTIONS WILL IMPAIR FILTRATION, INCREASE NECROSIS, AND WEAKEN INFRASTRUCTURE OF *APHROCALLISTES VASTUS*

16:45 **Melendez, M.**; Salisbury, J.; Federal, D.; Langdon, C.; Morell, J.; Manzello, D.; Musielewicz, S.; Sutton, A.: ARE CORAL REEF AREAS IN PUERTO RICO IN A NET DISSOLUTION STATE? OBSERVATIONS OF OCEAN ACIDIFICATION AND NET ECOSYSTEM METABOLISM FROM LA PARGUERA, PUERTO RICO

17:00 **Fitz, K.**; Habeck, E.; Boyce, D.; Chambers, R.: INTER FEMALE DIFFERENCES IN OFFSPRING QUALITY IN ATLANTIC SILVERSIDE (MENIDIA MENIDIA) AND THEIR SCOPE OF RESPONSE TO ELEVATED CO2

TUESDAY

[T] REPRESENTS INVITED PRESENTATIONS

TUESDAY

SS070 ENGAGING UNDERREPRESENTED STUDENTS IN OCEAN SCIENCE TO PROMOTE A DIVERSE INCLUSIVE WORKFORCE

Chair(s): Andrea Johnson, National Science Foundation (andjohns@nsf.gov)
Todd Christenson, NOAA Office of Education (todd.christenson@noaa.gov)
Corey Garza, California State University, Monterey Bay (cogarza@csumb.edu)

Location: Room 208 C

15:45 **Garza, C.**: PATHWAYS TO DIVERSITY AND INCLUSION IN 21ST CENTURY OCEAN SCIENCE

16:00 **Burgos Rodriguez, J.**: CREATING OUR OWN REVOLVING DOOR: BUILDING INFLUENCE TO AFFECT CHANGE (CREANDO NUESTRA PROPIA PUERTA GIRATORIA: CONSTRUYENDO INFLUENCIA PARA CREAR CAMBIO)

16:15 **Gibson, D.**.: UNDERREPRESENTED BUT NOT FORGOTTEN: THE MAKING OF A MARSHALL SCHOLAR

16:30 **George, A.**; Liles, G.; Jearld, A.; Gutierrez, B.; Harden, B.; Scott-Price, O.; Johnson, K.: TEN YEARS OF SUCCESS IN INCREASING THE PRESENCE OF UNDERREPRESENTED MINORITIES IN THE GEOSCIENCES: THE WOODS HOLE PARTNERSHIP EDUCATION PROGRAM MODEL AND OUTCOMES

16:45 **Forbes, H.**; Stout, J.; Wilson Grimes, K.; Guannel, M.; Jones, N.; Brandt, M.; Bucklin, C.; Medina, M.: YOUTH OCEAN EXPLORERS: A PATHWAY TO DIVERSIFYING THE FIELD OF AQUATIC SCIENCE IN THE U.S. VIRGIN ISLANDS

17:00 Guannel, M.; **Monrose, N.**; Plyley, C.: PROFESSIONAL LEARNING COMMUNITIES IN THE UNITED STATES VIRGIN ISLANDS PROVIDE SUPPORT THROUGH TEACHER TRAINING AND PROJECT-BASED LEARNING IN AQUATIC SCIENCES

196 papers

TUESDAY POSTERS

All poster sessions are held in Exhibit Hall B.

CS004 URBAN ECOSYSTEMS

141 **Maldonado-Benítez, N.**; Ramirez, A.: EFFECT OF URBANIZATION ON TROPICAL STREAM HABITATS AND THEIR ASSOCIATED DRAGONFLY FAUNA IN PUERTO RICO

142 **Dearden, E.**; Matiasek, S.: EVALUATION OF POLLUTANT REMOVAL AND RUNOFF REDUCTION OF A STORMWATER BIOFILTRATION SYSTEM

143 **Duval, T.**; TRACE METALS IN URBANIZING HEADWATER STREAMS: DIFFERENT RESPONSES FOR DIFFERENT METALS

144 **Walteros, J.**; Ramirez, A.: URBAN STREAM IN LATIN AMERICA: AN ASSESSMENT OF CURRENT CONDITIONS AND RESEARCH NEEDS

CS005 ACIDIFICATION

71 **Bockmon, E.**; Dickson, A.: A REPEATED INTER-LABORATORY COMPARISON OF CARBON DIOXIDE MEASUREMENTS: AN ASSESSMENT OF UNCERTAINTY

73 **Boisen, O.**; Goeltz, J.: IN SITU PH MEASUREMENTS USING SELF-CALIBRATING IROX ELECTRODES

75 **Davis, A.**; Thunell, R.; Tappa, E.; Benitez-Nelson, C.; Black, D.; Osborne, E.; Davis, C.: RECONSTRUCTING OCEAN ACIDIFICATION USING FOSSIL PLANKTONIC FORAMINIFERA IN THE CARIACO BASIN

76 Hideshima, K.; **Takeda, S.**: SEASONAL CHANGE IN THE RELATIONSHIP BETWEEN PH AND AOU IN COASTAL WATERS OFF WESTERN KYUSHU, JAPAN

77 **Schockman, K.**; Byrne, R.: SPECTROPHOTOMETRIC DETERMINATION OF CARBONATE DISSOCIATION CONSTANT, K_2, IN SEAWATER

78 **Tran, T.**: AMELIORATING OCEAN ACIDIFICATION: TOWARDS A MODEL RELATING PCO2, IRRADIANCE AND LEAF AREA INDEX OF ZOSTERA MARINA (EELGRASS) IN PADILLA BAY, WA

CS006 TRACE METALS

79 **Pessoa, I.**; Geraldes, M.: LEAD ISOTOPE COMPOSITION IN BIVALVE SHELLS FROM ILHA GRANDE BAY (RIO DE JANEIRO, BRAZIL)

81 **Wei, Y.**; Liu, G.; Wang, R.; Fu, B.: EMISSION CHARACTERISTICS AND ENVIRONMENT INFLUENCE OF SIZE-CLASSED PARTICULATE PB FROM A COAL-FIRED POWER PLANT EQUIPPED WITH CIRCULATING FLUIDIZED BED (CFB) BOILER

82 **Serrano, A.**; Richardson, C.: ASSESSING THE EFFECT OF A CARBON AMENDMENT ON TRACE METAL LEACHING FROM SEDIMENT CORES

CS007 HYPOXIA

57 Weinstock, J.; **Collin, R.**: EFFECT OF SEASONAL HYPOXIA ON MEROPLANKTON IN A CARIBBEAN ESTUARY

58 **Liu, C.**; Liu, X.; Wang, S.; Li, B.; Du, Y.: EFFECTS OF PERTURBATIONS BY DIFFERENT DISTURBANCE METHODS ON MITIGATION OF ANOXIC HYPOLIMNETIC IN THERMAL STRATIFIED RESERVOIRS

59 **Adelson, A.**; Pawlak, G.; Davis, K.; Collin, R.; Gaul, A.; Reed, V.; Altieri, A.; Giddings, S.: HYPOXIC EVENTS IN BAHÍA ALMIRANTE, A SEMI-ENCLOSED TROPICAL EMBAYMENT

60 **Okada, T.**; Imamura, M.; Nakamura, Y.: SULFUR PEAK MODELING AT THE MIDDLE LAYER OF DEAD ZONE IN TOKYO BAY

61 **Bouchard, C.**; Cambazoglu, K.; Dinniman, M.; Lau, Y.; Fitzpatrick, P.; Hoffman, E.; Wiggert, J.: TURBIDITY INFLUENCE ON BIOGEOCHEMICAL PROCESSES IN A RIVER-DOMINATED COASTAL ENVIRONMENT

CS009 GAS FLUXES

90 **Laubach, A.**; Weber, T.; Kessler, J.; Eisenstadt, A.: CONSTRAINING METHANE EMISSIONS FROM THE LAURENTIAN GREAT LAKES USING MACHINE LEARNING MODELS

91 **Dornblaser, M.**; Butman, D.; Bogard, M.; Striegl, R.: INTERIOR ALASKA LAKES – NET SOURCES OR SINKS OF CARBON GASES?

CS010 NITROGEN BIOGEOCHEMISTRY AND CYCLING

92 **Zhang, L.**; Lee, W.; Ji, C.; Liu, X.; Altabet, M.: COMPOUND SPECIFIC N ISOTOPE ANALYSIS OF AMINO ACIDS AND THE APPLICATIONS IN OCEANOGRAPHY AND ECOLOGY

93 **Chang, B.**; Sonnerup, R.; Bullister, J.: EQUATORIAL TRANSPORT OF NITROUS OXIDE FROM THE EASTERN TROPICAL PACIFIC OCEAN

94 **Presley, R.**; Giblin, A.; Algar, C.; O'Neill, S.; Rich, J.: FACTORS CONTROLLING NITROGEN REMOVAL AND RETENTION IN MARINE SEDIMENTS

95 **Thomas, R.**; Fawcett, S.; Forrer, H.; Knapp, A.: FIELD-BASED ESTIMATES OF THE 15N AND 18O ISOTOPE EFFECTS OF NITRATE ASSIMILATION IN THE SOUTHERN OCEAN

96 **Hansen, A.**; Kraus, T.; Downing, B.; Gelber, A.; Ahmed, Q.; Bergamaschi, B.: MAPPING NITROGEN CONCENTRATIONS IN THE SAN FRANCISCO-SAN JOAQUIN DELTA (CALIFORNIA, USA): RESULTS FROM A NEWLY DEVELOPED CONTINUOUS FLOW-THROUGH AMMONIUM ANALYZER

97 **Mathuri, M.**; Granger, J.; Boshers, D.: NITROGEN ISOTOPE FRACTIONATION DURING AMMONIUM ASSIMILATION BY MARINE PHYTOPLANKTON ISOLATES

98 **Mino, Y.**; Sukigara, C.; Honda, M.; Kawakami, H.; Fujiki, T.; Kitamura, M.; Wakita, M.: SEASONAL AND INTERANNUAL VARIATIONS IN THE UPPER LAYER NITROGEN AVAILABILITY ASSESSED BY NITROGEN ISOTOPIC COMPOSITIONS OF SETTLING PARTICLES IN THE WESTERN SUBTROPICAL NORTH PACIFIC

99 **Rollinson, V.**; Granger, J.; Vaudrey, J.; Koerting, C.; Tobias, C.; Jacksin, R.: SOURCES OF NUTRIENTS DISCHARGED SEASONALLY FROM A SOUTHERN NEW ENGLAND RIVER EVIDENCED BY NITRATE N AND O ISOTOPE RATIOS

CS012 CARBON FLUXES IN FW AND MARINE ENVIRONMENT

32 **Beraldo Bittar, T.**; Passow, U.; Harvey, E.: AN UPDATED METHOD FOR THE CALIBRATION OF TRANSPARENT EXOPOLYMER PARTICLE MEASUREMENTS

CS013 DISSOLVED ORGANIC MATTER - DOC, DON, DOP, FDOM, CDOM

105 **Govender, Y.**: "RESILIENCE OF LAGUNA GRANDE AT FAJARDO TO "BLACKOUTS AND HURRICANES": LONG-TERM MONITORING METHODS FOR BIOLUMINESCENCE IN PUERTO RICO"

106 **D'Andrilli, J.**: CHALLENGES OF FLUORESCENT DISSOLVED ORGANIC MATTER CHEMICAL INTERPRETATIONS: A CASE STUDY OF POLAR ICE CORES

TUESDAY

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

107 **Gao, L.**; Gao, Y.; Zong, H.; Guo, L.: ELUCIDATING THE HIDDEN NON-CONSERVATIVE BEHAVIOR OF DOM IN LARGE-RIVER ESTUARIES

108 **Retelletti Brogi, S.**; Kim, J.; Ryu, J.; Jin, Y.; Lee, Y.; Hur, J.: EXPLORING SEDIMENT PORE WATER DISSOLVED ORGANIC MATTER (DOM) IN A MUD VOLCANO

109 **Lopez, C.**; Bercovici, S.; Orellana, M.; Hansell, D.: SEASONAL DYNAMICS OF DISSOLVED ORGANIC CARBON IN THE DEEP EASTERN NORTH PACIFIC

110 **Liu, S.**; Huynh, N.; Comstock, J.; Opalk, K.; Halewood, E.; English, C.; Parsons, R.; Giovannoni, S.; Bolanos, L.; Carlson, C.: STABLE ISOTOPE PROBING IDENTIFIES BACTERIOPLANKTON LINEAGES CAPABLE OF UTILIZING DOM ACROSS A RANGE OF BIOAVAILABILITY

111 **Fiss, M.**; Osburn, C.; Paerl, R.: THE IMPACTS OF URBAN DOM ON COASTAL CHEMISTRY AND BACTERIOPLANKTON COMPOSITION

112 **Sanders, A.**; West, K.; Vermilyea, A.: THE ROLE OF LANDCOVER AND STORM EVENTS ON THE PHOTOCHEMICAL TRANSFORMATION OF DISSOLVED ORGANIC CARBON CONCENTRATION AND BIOAVAILABILITY

CS014 SEDIMENT DYNAMICS

396 **Ramirez-Irizarry, N.**; Sherman, C.: ASSESSMENT OF GEOLOGICAL AND GEOCHEMICAL PROXIES OF SOURCES OF TERRIGENOUS SEDIMENT REACHING REEFS ADJACENT TO THE GUANICA BAY WATERSHED, SOUTHWEST PUERTO RICO

397 **Kuhn, M.**; Stuhrenburg, J.; Broder, T.; Olefeldt, D.; Knorr, K.: GROUNDWATER CONNECTIVITY, SEDIMENT REDOX CONDITIONS, AND METHANE EMISSIONS FROM WESTERN CANADIAN PEATLAND LAKES

398 **Hansen, J.**; Kraus, T.; Downing, B.; Graham, N.; O'Donnell, K.; Bergamaschi, B.: IMPROVING BENTHIC NUTRIENT FLUX RATE DETERMINATIONS USING REAL-TIME, FIELD-BASED HIGH FREQUENCY MEASUREMENTS

399 **Boyko, V.**; Torfstein, A.: OXYGEN CONSUMPTION IN PERMEABLE AND COHESIVE SEDIMENTS OF THE OLIGOTROPHIC MARINE SYSTEM AFFECTED BY AEOLIAN DRY DEPOSITION

400 **Cortez, D.**; Ferner, M.: RELATIONSHIP BETWEEN TURBIDITY AND TOTAL SUSPENDED SOLIDS ACROSS A SALINITY GRADIENT IN THE SAN FRANCISCO ESTUARY

401 **Martinez, L.**; Santos Mercado, H.: USING PETROGRAPHIC ANALYSIS OF SANSTONES FROM OLIGOCENE AND MIOCENE AGE AND SANDS OF THE HOLOCENE FOR ISLAND ARC SEDIMENTATION EVOLUTION IN THE NORTH COAST TERTIARY BASIN OF PUERTO RICO

CS020 FISH AND FISHERIES

9 **Oramadike, C.**: ASSESSING THE AWARENESS OF ARTISAN FISHER'S FOLKS ON ANTIBIOTIC RESISTANCE IN MAKOKO FISH MARKET LAGOS STATE NIGERIA

10 **Martinez, E.**; Lamptey, D.; Colombo, R.; Menze, M.: ARE FISHES IN POWER PLANT COOLING RESERVOIRS GIVING US A SNEAK PEEK TO ENVIRONMENTAL WARMING?

11 **Molina, J.**; Waterhouse, L.; Mason, E.; Pattengill-Semmens, C.; McCoy, C.; Johnson, B.; Semmens, B.: ESTIMATING THE EFFORT REQUIRED FOR GENERATING A PHOTOGRAPHIC DATABASE ON NASSAU GROUPER: RECOMMENDATIONS FOR POTENTIAL MARK AND RECAPTURE STUDIES

12 **Hazlett, M.**; Walker, C.; Guo, C.: EXAMINING FISH SPECIES DIVERSITY IN NEARSHORE RIVER MOUTH HABITATS OF KACHEMAK BAY, ALASKA

13 **Gonzalez Colmenares, G.**; Harms-Tuohy, C.; Schizas, N.: INVESTIGATING THE USE OF EDNA SAMPLING TO LOCATE FISH SPAWNING AGGREGATIONS

14 **Sabal, M.**; Merz, J.; Workman, M.; Palkovacs, E.: PREY TRAITS AND HABITAT MODIFY ESCAPE DECISIONS IN MIGRATING JUVENILE SALMON

15 **Usman, A.**; Adeola, U.: STUDIES OF GENETIC VARIATIONS ON WILD AND CULTURED CICHLID POPULATIONS (SAROTHERODON MELANOTHERON) IN LAGOS STATE.

16 **Franqui Rivera, G.**; Schizas, N.: USING DNA BARCODING TO CHARACTERIZE THE DIVERSITY OF SHARKS IN PUERTO RICO

CS022 BIODIVERSITY

149 **Hashem, S.**; Kawai, K.: GENETIC RELATIONSHIPS OF CICHLID FISHES FROM LAKE MALAWI BASED ON MITOCHONDRIAL DNA SEQUENCES

150 **Mariani-Rios, A.**; Ramírez, A.: ENVIRONMENTAL FACTORS ASSOCIATED WITH DRAGONFLY ASSEMBLAGES AT EL YUNQUE NATIONAL FOREST

151 **Contreras, P.**; Rijos, C.; Rivera, M.; Rodríguez, C.: ABUNDANCE OF DIFFERENT SPECIES OF LAND CRAB LARVAE FOUND IN PUERTO RICO: MEGALOP STAGE

152 **Fidalgo De Souza, L.**: IDENTIFYING CATTLE EGRET (BUBULCUS IBIS) COLONIES IN MANGROVE ECOSYSTEMS THROUGH GUANO SPECTRAL SIGNATURE

153 **Leon Zayas, R.**; Youtsey, B.; Dixson, D.; Biddle, J.: UNCOVERING GENES ASSOCIATED WITH SECONDARY METABOLITE FROM MARINE SPONGES AND SEDIMENTS USING METAGENOMICS

154 **Dieunel, D.**; Rahman, Z.; Massey, E.: THE BLACK QUEEN, THE PROTEOMIC CONSTRAINT, AND THE DYNAMICS OF GENOME REDUCTION OF MARINE PHOTOSYNTHETIC PICOEUKARYOTES

155 Gasca-Pineda, J.; **Galindo-Sánchez, C.**; Arteaga, M.; Martínez, M.; Jiménez Rosenberg, S.; Herzka, S.; Nakamura, Y.: COMMUNITY STRUCTURE AND DIVERSITY OF ZOOPLANKTON IN THE PERDIDO REGION OF THE GULF OF MEXICO AN APPROACH WITH DNA METABARCODING

CS023 COMMUNITY ECOLOGY

156 **Esparra-Escalera, H.**; Santos-Flores, C.; Sotomayor-Ramírez, D.: TAXONOMIC ASSESSMENT AND PERIPHYTON BIOMASS IN PRESENCE OF TWO INVASIVE THIARIDAE SNAILS, CONTRASTING WITH EXTERNAL MORPHOLOGICAL TRAITS, ON FRESHWATER ARTIFICIAL SYSTEMS

157 **Gonzalez Hernandez, N.**; Padilla Maldonado, B.; Colon Molina, D.: MATING BEHAVIOR OF PHYSA ACUTA (GASTROPODA: PHYSIDAE) ACCORDING TO BODY-SIZE

158 **Lisboa, J.**; Padilla Maldonado, B.; Santos, C.: ASSOCIATION BETWEEN AUTOTROPH PERIPHYTON BIOMASS AND FRESHWATER SNAIL COMMUNITY IN STREAMS OF THE RIO ABAJO STATE FOREST, PUERTO RICO

159 **Santiago-Vera, J.**; Wilson, J.; Ramirez, A.: HURRICANE EFFECTS ON TROPICAL STREAM MEIOFAUNA: AN EXPERIMENTAL SIMULATION OF MAJOR ABIOTIC IMPACTS

160 **Benes, K.**; Karlen, J.: THE RE-STRUCTURING OF A COMMUNITY? THE DECLINE OF *CHONDRUS CRISPUS* ON GULF OF MAINE ROCKY SHORES

T REPRESENTS TUTORIAL PRESENTATIONS

CS027 VIRUSES

20 **Aleman, M.**; Ignacio Espinoza, J.; Long, A.; Fuhrman, J.: DYNAMICS OF VIRUSES ANALYZED METAGENOMICALLY IN RELATION TO OTHER PLANKTON DURING AN ALGAL BLOOM

CS028 ZOOPLANKTON ECOLOGY AND PHYSIOLOGY

21 **Dutz, J.**; Wasmund, N.; Huwer, B.; Dierking, J.: LONG-TERM INCREASE IN ZOOPLANKTON STOCKS AND INTERANNUAL VARIABILITY IN THE BALTIC SEA: DOES PHENOLOGY MATTER?

22 **Risley, S.**; Niemisto, M.; Fields, D.: FEEDING RATES OF HOMARUS AMERICANUS (AMERICAN LOBSTER) LARVAE AT DIFFERENT FOOD CONCENTRATIONS.

23 **Blanco Bercial, L.**: ANNUAL CYCLE OF THE MESOZOOPLANKTON COMMUNITY IN THE SARGASSO SEA

24 **Gray, D.**; Vucic, J.: WHICH PHYSICOCHEMICAL VARIABLES SHOULD ZOOPLANKTON ECOLOGISTS MEASURE WHEN THEY VISIT A LAKE?

25 Makovec, T.; Kogovšek, T.; Lučić, D.; Hure, M.; Onofri, V.; **Malej, A.**: JELLYCO - A SIMPLE IMAGING SYSTEM FOR IN SITU QUANTIFICATION OF JELLYFISH

26 **Fies, J.**; Wagner, Z.; Colin, S.; Costello, J.: DETERMINING PREY SELECTIVITY IN CASSIOPEA

27 **Wagner, Z.**; Costello, J.; Colin, S.: SCYPHOMEDUSAE FEEDING MECHANICS ON COPEPOD PREY: HIGH RESOLUTION PREY TRACKING THROUGH THE FEEDING PROCESS

28 **Maas, A.**; Cope, J.; Miccoli, A.; Stamieszkin, K.; Steinberg, D.: CHARACTERIZATION OF ZOOPLANKTON ACTIVE FLUX IN THE N.E. PACIFIC OCEAN

29 **Brown, P.**; Walsh, E.: LIFESTYLE, GENOME SIZE, AND BERGMANNS'S RULE IN AN UNDERSTUDIED INVERTEBRATE CLADE

30 **Brownlee, E.**; Gast, R.; Johnson, M.; Beaudoin, D.; Olson, R.; Sosik, H.: MOLECULES AND MORPHOLOGY: UNDERSTANDING SEASONAL PATTERNS OF PROTIST COMMUNITIES WITH COMPLEMENTARY TAXONOMIC ANALYSES

31 **Sommer, S.**; Ohman, M.: A TRAIT-BASED APPROACH TO TEST TOP-DOWN CONTROL OF ZOOPLANKTON COMMUNITIES IN THE SOUTHERN CALIFORNIA CURRENT ECOSYSTEM

SS004 UNDERGRADUATE RESEARCH IN AQUATIC SCIENCES POSTERS

235 **Luo, Z.**: EXAMINING CHALLENGES OF EFFECTIVE ACCOUNTABILITY MEASURES IN REGIONAL FISHERY MANAGEMENT

236 **Althoen, C.**: IMPACTS OF PHYTOPLANKTON BLOOMS ON SEA SPRAY AEROSOL

237 **Ma, S.**; Gonsior, M.; Powers, L.; Robey, N.; Townsend, T.; Cooper, W.: PHOTO-DEGRADATION AND AMMONIFICATION OF DISSOLVED ORGANIC MATTER IN LANDFILL LEACHATES COMPARED TO FRESHWATER

238 **Pardis, W.**; Colson, B.; Michel, A.; Nicholson, D.: A MINIATURE DISSOLVED MULTI-GAS SENSOR FOR AQUATIC IN-SITU MEASUREMENT

239 **Dawydiak, W.**; Lubelczyk, L.; Archer, S.; Posman, K.; Poulton, N.: A PREY SATURATION APPROACH AS AN IMPROVED METHOD FOR ESTIMATING MICROZOOPLANKTON GRAZING RATES

240 **Cosca-Baresh, A.**; Wall, C.: ACOUSTICAL ANALYSIS OF TWO SHALLOW WATER OCEAN NOISE REFERENCE STATION NETWORK SITES: AMERICAN SAMOA AND BUCK ISLAND

241 **Tossas Deida, T.**: ALTERED HAIR CELL DEVELOPMENT AND REGENERATION IN A ZEBRAFISH MUTANT LINE LACKING THE CANNABINOID RECEPTOR (CB2)

242 **Hauke, A.**; Wu, J.; Rogers, S.: ANAEROBIC CO-DIGESTION OF SEAWEEDS WITH ORGANIC WASTE STREAMS

243 **Sabate, K.**; Yoza, B.: ANALYSIS OF THE MICROBIAL COMMUNITIES ASSOCIATED WITH THE DEGRADATION OF INVASIVE ALGAE

244 **Parry, D.**; Davis, G.; Baumgartner, M.; Van Parijs, S.: ANALYZING SEI WHALE DISTRIBUTION IN THE WESTERN NORTH ATLANTIC USING PASSIVE ACOUSTICS

245 **Wong, M.**; Chapman, J.; Kaup, T.; Bobo-Shisler, C.; Dumbauld, B.: ARE THE LARGE MUD SHRIMP, UPOGEBIA PUGETTENSIS, LUCKY OR SMART?

246 **Conner, K.**; Rheuban, J.; McCorkle, D.; Long, M.: ASSESSING BENTHIC ECOSYSTEM CALCIFICATION AND PRODUCTION IN KEY LARGO REEF ENVIRONMENTS USING GRADIENT EXCHANGE SYSTEMS

247 **Dellaert, Z.**; Roberson, L.: ASSESSING CALCIFICATION OF ASTRANGIA UNDER DIFFERENT TEMPERATURE AND PHYSIOLOGICAL CONDITIONS USING SCANNING ELECTRON MICROSCOPY

248 **Keyes, P.**; Beirne, E.; Aeppli, C.: CAN PHOTO-OXIDIZED OIL BE BIODEGRADED BY MARINE MICROBES?

249 **Sebourn, W.**; Parker, A.; Sakuma, K.: CHARACTERIZATION OF OCEAN CONDITIONS IN MONTEREY BAY, CA TO SUPPORT FISHERIES ECOSYSTEM RESEARCH

250 **Perron, S.**; White, W.: COMPARING ALTERNATIVE METRICS OF METAPOPULATION 'PATCH VALUE' TO IDENTIFY EFFECTIVE MARINE PROTECTED AREA DESIGN STRATEGIES

251 **MacFarland, A.**; Schiebel, H.: COMPARING CHROMOPHORIC DISSOLVED ORGANIC MATTER METHODOLOGIES: IS BIGGER ALWAYS BETTER?

252 **Dorsey, K.**; Turner, C.; Brill, R.; Stojilovic, O.; Williams-McLeod, S.; Schwarz, M.; Hurst, T.; Horodysky, A.: COMPARISON OF THE VISUAL NEUROBIOLOGY OF TROPICAL AND COASTAL MARINE FISHES UNDER OCEAN ACIDIFICATION

253 **Hedberg, S.**; Richardson, K.; Sasaki, M.; Dam, H.: COMPLEX INTERACTIONS BETWEEN LOCAL ADAPTATION, PLASTICITY, AND SEX AFFECT VULNERABILITY TO WARMING IN A WIDESPREAD MARINE COPEPOD

254 **Simonetti, S.**; Neal, B.; Honisch, B.; Price, N.: COMPLEX INTERACTIONS IN CHANGING SEAS: AN EMERGING RELATIONSHIP BETWEEN A HOST AND A MARINE FUNGAL PATHOGEN

255 **Greenlee, S.**; Matrai, P.; Countway, P.: DMSP INFLUENCES FUNCTIONAL GENE ABUNDANCE IN ANTARCTIC BACTERIAL COMMUNITIES

256 **Kaup, T.**; Chapman, J.; Wong, M.; Bobo-Shisler, C.; Dumbauld, B.: DO UPOGEBIA PUGETTENSIS FARM THE LINING OF THEIR BURROWS?

257 **Batterbee, C.**; Batterbee, C.; Null, K.; Balmagia, J.; Clark, R.: EFFECTS OF A WETLAND-BIOREACTOR TREATMENT SYSTEM ON NUTRIENT LOADING IN A CENTRAL CALIFORNIA COAST WATERSHED

258 **Johnson, G.**; McManus, G.: EFFECTS OF HYPOXIA ON GROWTH AND MORTALITY OF LONG ISLAND SOUND CILIATES

259 **Ward, A.**; Franzè, G.; Barton, A.; Menden-Deuer, S.: EFFECTS OF TURBULENCE ON MICROZOOPLANKTON GROWTH AND GRAZING

261 **Kaplan, E.**; Sanford, L.: EXAMINING THE EFFECTS OF 3-D REEF STRUCTURES ON THE MIXING AND REAERATION OF HYPOXIC WATERS

TUESDAY

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

262 **Hockenberry, Z.**; Choi, F.: EXAMINING THE EFFECTS OF FLOOD-MITIGATING INFRASTRUCTURE ON THERMAL PERFORMANCE OF P. VULGARISM IN THE RUMNEY MARSH (REVERE, MASSACHUSETTS).

263 Borel, A.; Henslee, H.; McNab, M.; **Hoch, M.**: FATE OF A PLASTIC DEBRIS MICROBIOME DURING SIMULATED TRANSPORT FROM A MAYA MOUNTAIN RIVER TO THE MESOAMERICAN BARRIER REEF, BELIZE.

264 **Hamman, C.**; Carlson, J.; Mathers, A.; Moncrief-Cox, H.; Deacy, B.: FEEDING ECOLOGY OF SILKY SHARK, CARCHARHINUS FALCIFORMIS, IN THE NORTHWEST ATLANTIC OCEAN

265 **Overton, E.**; Franks, D.; Shaul, N.; Pedler, B.; Borgen, M.; Rawat, A.; Fenical, W.; Aluwihare, L.; Reddy, C.; Hahn, M.: HALOGENATED MARINE NATURAL PRODUCTS: A THREAT TO HUMAN HEALTH?

266 Archer, S.; **Brycki, J.**; Posman, K.: HALOPEROXIDASE-MEDIATED GAS EXCHANGE IN TWO DIATOM SPECIES

267 **Olive, D.**; Schwieterman, G.; Brill, R.; Bushnell, P.: HEMATOLOGICAL IMPACTS OF HYPOXIA ON THE CLEARNOSE SKATE (RAJA EGLANTERIA)

269 **Howard, H.**; Fenwick, I.; Betancourt, A.; Long, A.: HERBIVORY INCREASES ENERGY ALLOCATION TOWARDS REPRODUCTION IN SMALL TURBINARIA ORNATA

270 **Hurtado, G.**; Walsh, E.: HOW COMPETITIVE INTERACTIONS ARE ALTERED BY ENVIRONMENTAL TOXICANTS

272 **Jenkins, A.**; Trask, C.; Lawrence, J.; Dzieweczynski, T.: I HAVE MIXED FEELINGS ABOUT THIS: THE COMBINED EFFECTS OF VINCLOZOLIN AND ETHINYLESTRADIOL ON BOLDNESS IN MALE SIAMESE FIGHTING FISH

273 **Ríos-Guzmán, H.**; Imhof, K.; Banister, R.; Muller, E.: IDENTIFYING THE PUTATIVE PATHOGEN OF THE FLORIDA TISSUE LOSS DISEASE OUTBREAK

274 **Szewczyk, C.**; Testa, J.: IMPACTS OF OXYGEN DEPLETION ON PHOSPHORUS CYCLING IN ROCK CREEK, AN AERATED TIDAL TRIBUTARY OF THE PATAPSCO RIVER

275 **Bideau, S.**; Habtes, S.; Durgadeen, N.; Pott, C.; Stovall, A.; Terry, L.: IMPACTS OF WATER QUALITY ON CORAL BLEACHING INDICES IN ST. CROIX, USVI

276 **Tran, C.**; Bucheit, J.; Ibanez, A.: INCREASING NATURE-BASED TECHNIQUES FOR FUTURE SHORELINE MANAGEMENT

277 **Stuart, C.**; Suskiewicz, T.; Rasher, D.: KELP—A COMEBACK STORY: A GEOSPATIAL ANALYSIS OF MAINE'S KELP FORESTS OVER THE PAST QUARTER CENTURY

278 **Gonzales, D.**; Yvon-Lewis, S.; Dedrick, J.: METHANE EMISSIONS FROM GALVESTON BAY AND THE NORTHWESTERN GULF OF MEXICO

279 **Pavia, A.**; Liu, Y.: METHOD DEVELOPMENT FOR THE SIMULTANEOUS EXTRACTION OF EMERGING CONTAMINANTS AND DOM IN SEAWATER

280 **Montalvo-Rodriguez, P.**; Jones, R.; Orcutt, B.: MICROBES EATING ROCKS AT THE BOTTOM OF THE OCEAN: DETERMINING CHEMICAL PREFERENCES OF BASALT-ATTACHED MICROBIAL COMMUNITIES

281 **Agrawal, A.**; Laundry, J.; Lasdin, K.; Brander, S.: MICROPLASTIC INGESTION BY TWO DIFFERENT POPULATIONS OF ROCKFISH (SEBASTES MELANOPS) OFF OF THE COAST OF OREGON

282 **Anderson, K.**; Vohsen, S.; Fisher, C.; Baums, I.: OCCURRENCE AND DISTRIBUTION OF APICOMPLEXAN-RELATED SYMBIONTS IN CORALS ACROSS A WIDE DEPTH RANGE

283 **Coles, D.**; Antczak, O.: PSEUDO-NITZCHIA MONITORING AND DETECTION OF A NOVEL SPECIES IN THE GULF OF MAINE

284 **Antoine, J.**; Gutting, A.; Smith, T.; Brandt, M.: RECRUITMENT AND SUCCESSION IN SPACE CREATED BY CORAL MORTALITY

285 **Brady, C.**; Bitterwolf, K.; Paytan, A.: REFINING SEAWATER SR ISOTOPE RATIOS DURING THE EOCENE USING MARINE BARITE

286 **Vélez Negrón, V.**: SENSITIZED SENTINELS FOR ENVIRONMENTAL TOXICOLOGY SCREENS: ZEBRAFISH MUTANT LINES AFFECTED IN MAINTENANCE AND FUNCTION OF CNS AND PNS

287 **Pinard, Z.**; Weiss, S.; Davis, G.; Cholewiak, D.: SINGING GIANTS OF THE SEA: USING PASSIVE ACOUSTICS TO TRACK BLUE WHALES IN THE WESTERN NORTH ATLANTIC

288 **Villalon, B.**; Crowder, L.; Frawley, T.: SOCIO-ECOLOGICAL INTERACTIONS WITHIN SMALL-SCALE FISHERIES: ULTRALIGHT VMS TRACKING IN THE GULF OF CALIFORNIA

289 **Searles, A.**; Zink, I.; Browder, J.; Serafy, J.: TESTING BEHAVIORAL PREFERENCE OF THE BRACKISH GRASS SHRIMP, PALAEMONETES INTERMEDIUS, FOR DIFFERENT SEAGRASS SPECIES

290 **Jaffery, R.**; Leng, B.; Alderman, D.; Canava, K.; Costanza, C.; Davis, H.; Goehring, L.; Spore, P.; Duke, J.; Scott, T.: THE COMBINED EFFECT OF SUSPENDED SEDIMENT AND NITRATE CONCENTRATIONS ON WATER COLUMN DENITRIFICATION RATES OF FACULTATIVE ANAEROBES

291 **Rangel-Parra, L.**; Sánchez-Garcia, M.; Roberson, L.: COLD SHOCKS INDUCE ASYMBIOTIC COLONIES IN *ASTRANGIA POCULATA*, AN EMERGING CORAL MODEL

292 **Rodkey, T.**; Hillhouse, J.; Quigg, A.: THE EFFECTS OF CRUDE OIL AND COREXIT ON SILICA-REPLETE AND SILICA-LIMITED *PHAEODACTYLUM TRICORNUTUM*

293 **Glickman, O.**; Cruz Reyes, R.; Vohsen, S.; Fisher, C.: THE EFFECTS OF PROXIMITY TO ACTIVE HYDROCARBON SEEPS ON THE ABUNDANCE AND DIVERSITY OF FAUNA ASSOCIATED WITH DEEP WATER OCTOCORALS

294 **Figueredo, N.**; Myers, E.; Juhl, A.: THE IMPACT OF PARTICLE ASSOCIATION ON TEMPERATURE-DEPENDENT GROWTH, LIGHT-INDUCED MORTALITY AND SPECIATION OF VIBRIO SP. IN THE HUDSON RIVER ESTUARY

295 Portrais, K.; **Stevens, M.**; Trask, C.; Mundy, S.; Szetela, J.; Dzieweczynski, T.: THE OUTLOOK IS NOT SO BRIGHT: THE EFFECT OF BENZOPHENONE-3 ON THE BEHAVIOR OF MALE SIAMESE FIGHTING FISH

296 **Arjona, A.**; Mayot, N.; Matrai, P.: THE ROLE OF SEA ICE EXTENT ON PHYTOPLANKTON PHENOLOGY IN THE GREENLAND SEA

297 **Alvarado, E.**; Rauschenberg, S.; Twining, B.: TOTAL ARSENIC, LEAD, AND CADMIUM IN MAINE KELP MARKETED FOR HUMAN CONSUMPTION

298 **King, A.**: TOXICITY OF STORMWATER RUNOFF AND NATURAL SUNLIGHT TO EARLY LIFE STAGE FISH EMBRYOS

299 **Delpin Sosa, M.**; Burns, W.; Sherenco, F.; Schroth, A.: UNDERSTANDING THE TIMING AND INTENSITY OF CYANOBACTERIA BLOOMS IN ST. ALBANS BAY AND MISSISQUOI BAY

300 **Steinmetz, H.**; Mitchell, C.; Balch, W.: USING MULTISPECTRAL VERSUS HYPERSPECTRAL RADIOMETRY TO PREDICT FOREL-ULE OCEAN COLOR

T REPRESENTS TUTORIAL PRESENTATIONS

301 **Bowman, J.**; Kiffney, P.: VARIABILITY IN INVERTEBRATE PREY: CONSEQUENCES FOR STREAM-REARING SALMON IN THE CEDAR RIVER, WA

302 **Morag, E.**; Montana, C.; Schalk, C.: ASSESSING HISTORICAL FISH COMMUNITY STRUCTURE USING TAXONOMIC, FUNCTIONAL, AND HISTORICAL-CONTEMPORARY DATA.

303 **Gangrade, S.**; McCormick, L.; Carrière-Garwood, J.; Levin, L.: VISUAL LUMINOXYSCAPES OF FOUR INVERTEBRATE LARVAL SPECIES IN THE NEARSHORE SOUTHERN CALIFORNIA BIGHT

304 **Seltzer, P.**; Gutner-Hoch, E.; Roberson, L.: VITAMIN C TRANSPORTER GENE REGULATION IN EARLY DEVELOPMENTAL STAGES OF *PORITES ASTREOIDES*

305 **Alderman, D.**; Goehring, L.; Canava, K.; Costanza, C.; Davis, H.; Jaffery, R.; Leng, B.; Spore, P.; Duke, J.; Scott, T.: WATER COLUMN DENITRIFICATION BY FACULTATIVE ANAEROBES INCREASES WITH ORGANIC MATTER AND NITRATE CONCENTRATIONS

SS005 ASLOMP STUDENT SYMPOSIUM

306 **Rios, O.**; Regier, P.: BIO-REACTIVITY OF DISSOLVED ORGANIC MATTER FROM NUTRIENTS ALONG THE RIO GRANDE RIVER

307 **Roldan Ayala, Z.**; Velez Villamil, S.; Sastre Wirshing, M.: FIRST REPORT OF LEPIDODINIUM SP. IN PUERTO RICO

308 **Askin, S.**; Fisher, R.; McDowell, J.: A GENETIC ASSESSMENT OF CHANNELED WHELK BUSYCOTYPUS CANALICULATUS POPULATIONS IN THE MID-ATLANTIC

309 **Lugo, K.**: ADSORPTION OF MULTIPLE METAL CATIONS BY MN OXIDES

310 **Capone, D.**; Ashford, O.; Levin, L.; Mendoza, G.; Orphan, V.: ARE METHANE SEEP SEDIMENTS A HOT SPOT OF ECOSYSTEM-FUNCTIONING IN THE DEEP-SEA?

311 **Cannady, A.**; Schopflin, L.; Leon Zayas, R.; Biddle, J.: ASGARD ARCHAEA IN ESTUARINE SEDIMENTS OF DELAWARE AND OREGON

312 **Propson, B.**; Dimino, T.; McCanty, S.; Christian, A.: ASSESSING BIODIVERSITY RESPONSES TO A HEADWATER STREAM SYSTEM RESTORATION PROJECT

313 **Cozy, M.**; McCoy, C.; Pattengill-Semmens, C.; Johnson, B.; Heppell, S.; Semmens, B.: ASSESSING THE EFFECTIVENESS OF THE FACIAL RECOGNITION SOFTWARE, I3S, ON THE NASSAU GROUPER (EPINEPHELUS STRIATUS)

314 **Beraki, S.**; Barco, R.; Albino, D.; Broomell, S.; Canela, R.; Dingmon, T.; Estrada, S.; Fernandez, M.; Lopez, A.; Savalia, P.; Abuyen, K.; Tully, B.: CHARACTERIZATION OF STRAIN EBB-1: A NOVEL, MARINE, IRON-OXIDIZING MICROBE

315 **Freeman, D.**; Deshpande, A.; Lynch, J.; Lascelles, N.; Drayton, D.; Brignac, K.; Jung, M.; Hoskins-Brown, D.: COMPARING 2 METHODS TO CHARACTERIZE THE CHEMICAL COMPONENTS OF MARINE PLASTIC DEBRIS

316 **Santano, J.**; Milton, I.; Navarro, B.; Warren, R.: CORAL REEF STRUCTURAL COMPLEXITY IMPACTS ECOSYSTEM FUNCTIONS

317 **Estrada, C.**; DeLeon Sanchez, E.; Leach, T.; Strader, M.: EFFECTS OF REARING TEMPERATURES ON THE MORPHOLOGY AND PHYSIOLOGY OF THE PAINTED SEA URCHIN LYTECHINUS PICTUS

318 **Warren, R.**; Milton, I.; Navarro, B.; Santano, J.; Barber, P.; Fong, P.; Fong, C.: EXPLORING HOW CORAL REEF STRUCTURAL COMPLEXITY IMPACTS ECOSYSTEM FUNCTIONS

319 **Pena, S.**; Matassa, C.; Ewanchuk, P.: IDENTIFYING THE MECHANISTIC DRIVERS OF RAPID SALT MARSH LOSS IN A SOUTHERN NEW ENGLAND ESTUARY

320 **Ousley, A.**; Rom, E.: INCREASING STUDENT DIVERSITY: DATA TRENDS FROM THE NATIONAL SCIENCE FOUNDATION'S DIVISION OF OCEAN SCIENCES RESEARCH EXPERIENCES FOR UNDERGRADUATES PROGRAM

321 **Bullock, L.**; Cruz Reyes, R.; Fisher, C.: RECOVERY AND REGROWTH AFTER SAMPLING BY DEEP WATER OCTOCORALS IN THE GULF OF MEXICO

322 **Keep, M.**; Chant, R.: SALINITY AND VELOCITY STRUCTURE OF RARITAN BAY: IMPLICATIONS FOR MICROPLASTIC DISTRIBUTION

323 **Few, C.**; Lonhart, S.: SANCTUARY BIODIVERSITY ASSESSMENT FOR WEST COAST REGION

324 **Catalan, E.**; Alvarez, B.; Smith, N.; Barber, P.; Fong, P.; Fong, C.: SEA URCHIN IMPORTANCE IN CROPPING ALGAL TURF AND REMOVING SEDIMENT ON CORAL REEFS

325 **Alvarez Caraveo, B.**; Catalan, E.; Smith, N.; Barber, P.; Fong, C.; Fong, P.: SEA URCHIN IMPORTANCE IN CROPPING ALGAL TURF AND REMOVING SEDIMENT ON CORAL REEFS

326 **Smith, N.**; Alvarez, B.; Catalan, E.; Barber, P.; Fong, P.; Fong, C.: SEA URCHIN IMPORTANCE IN CROPPING ALGAL TURF AND REMOVING SEDIMENT ON CORAL REEFS

327 **Cipolla, K.**; Gilg, M.: SPATIAL ANALYSIS AND CHARACTERIZATION OF A FUNDULUS HYBRID ZONE IN NORTHEAST FLORIDA

328 **Sunstone, C.**; Olive, D.; Coleman, H.; Hourigan, T.; McGuinn, R.; Dornback, M.; King, R.: STORY MAPPING TO ILLUSTRATE DEEP-SEA CORAL AND SPONGE PROTECTIONS

329 **Mayorga, C.**; Gonneea, M.; Spivak, A.: THE EFFECT OF DITCHING ON CARBON CYCLING IN SALT MARSHES

330 **Rabsatt, T.**; Webb, S.; Erazo, N.; Bowman, J.: THE IMPACT OF SEAGRASS BEDS ON MICROBIAL COMMUNITY STRUCTURE AND PATHOGEN LOAD IN AND ADJACENT TO SAN DIEGO BAY

331 **Lee, J.**; Chiang, K.; Tsai, S.: THE OPTIMAL CULTURE TIME FOR MEASURING THE GROWTH RATE OF NOCTILUCA SCINTILLANS

332 **Zerecero Marin, E.**; Barber, P.: TURBINARIA ORNATA, A MACROALGA THAT DOMINATES MANY HUMAN-IMPACTED REEFS IN FRENCH POLYNESIA, HOSTS HIGHLY ABUNDANCES OF DINOFLAGELLATES THAT CAUSE CIGUTERA

333 **Giandonato, T.**; Suca, J.; Wiley, D.; Llopiz, J.: VARIABILITY IN CONDITION FOR SAND LANCE (AMMODYTES SPP.) ON STELLAWAGEN BANK

334 **Truong Phan, P.**; Shearman, R.; Sanchez-Rios, A.: WINTERTIME CIRCULATION IN THE SOUTH CHINA SEA: COMPARISON BETWEEN TAIWAN AND VIETNAM

SS010 EXPLORING WHAT MAKES UNDERGRADUATE RESEARCH EXPERIENCES WORK: EVIDENCE FROM STUDENTS AND MENTORS

337 **Asch, R.**: LESSONS LEARNED FROM A COURSE-BASED UNDERGRADUATE RESEARCH EXPERIENCE ON THE ECOLOGICAL IMPACTS OF MICROPLASTIC POLLUTION IN URBANIZED AND RURAL ESTUARIES

338 **Cheung, I.**; Yalcin, K.: HOW INTEGRATED IS YOUR UNDERGRADUATE RESEARCH EXPERIENCE (URE)? A REFLECTION ON A MARINE SCIENCES RESEARCH EXPERIENCE FOR UNDERGRADUATES PROGRAM

[1] REPRESENTS INVITED PRESENTATIONS

TUESDAY

339 **Matlaga, D.**; Tobin, T.; Holt, J.; Peeler, M.; Persons, M.: ASSESSING THE RELATIVE MERITS OF SEMESTER-LONG VS. YEAR-LONG RESEARCH EXPERIENCES AT A LIBERAL ARTS INSTITUTION

340 **Watson, E.**; Courtney, S.: CO-OPERATIVE EDUCATION AS A VEHICLE FOR IMMERSIVE RESEARCH EXPERIENCES FOR UNDERGRADUATE STUDENTS

SS011 ANATOMY OF A BLOOM: UNRAVELING DRIVERS OF BIOMASS CHANGE AND CARBON DYNAMICS OVER THE ANNUAL CYCLE

33 **Eberhard, E.**; D'souza, N.; Passow, U.: THERMAL TOLERANCE AND PHOTOACCLIMATION OF THE DIATOM THALASSIOSIRA PSEUDONANA AT HIGH PCO2 LEVELS

SS013 ADAPTATION OF AQUATIC BIODIVERSITY TO GLOBAL CHANGE

67 **Elmarsafy, M.**; Gray, D.: WILL ZOOPLANKTON ON THE GREAT PLAINS BE ABLE TO KEEP UP WITH CLIMATE CHANGE?

SS024 RECONSTRUCTING ADAPTIVE RESPONSES IN AQUATIC ECOSYSTEMS USING ANCIENT DNA AND RESURRECTION ECOLOGY

202 **Ellegaard, M.**: LONG-TERM PHYTOPLANKTON SEEDBANKS AND THEIR ROLE IN ECOSYSTEM FUNCTION AND RESILIENCE

203 Legrand, B.; Miras, Y.; **Latour, D.**: POTENTIAL OF CYANOBACTERIAL AKINETES REVIVISCENCE THROUGH A 6700 YEARS CORE IN A EUTROPHIC LAKE

SS036 ECOLOGICAL APPLICATIONS OF EARTH SYSTEM MODELS AND REGIONAL CLIMATE MODELS

68 **Cordero Quiros, N.**; Miller, A.; Subramanian, A.; Luo, J.: CALIFORNIA CURRENT SYSTEM RESPONSE TO EL NIÑO: A PERSPECTIVE FROM OCEAN MODELS.

69 **Bartlett, B.**; Asch, R.; Erisman, B.: PROJECTIONS OF CHANGES IN THE DISTRIBUTION OF NASSAU GROUPER SPAWNING HABITAT USING AN ENSEMBLE OF EARTH SYSTEM MODELS

SS037 TERRESTRIAL MATERIALS IN PLANET WATER: TRACKING INFLUENCES ALONG THE LAND-OCEAN CONTINUUM

42 **Riise, G.**; Xiao, Y.: COUPLING BETWEEN INCREASED LAKE COLOR AND IRON

43 **Salinas, C.**; Adame, M.; Lopez-Merino, L.; Masque, P.; Masque, P.; Masque, P.; Gorham, C.; Lavery, P.; Serrano, O.: RECONSTRUCTION OF HUMAN- AND NATURAL-INDUCED ENVIRONMENTAL CHANGE IN AUSTRALIA USING SEAGRASS SEDIMENTARY ARCHIVES

44 **Ahrens, J.**; Beck, M.; Marchant, H.; Ahmerkamp, S.; Waska, H.; Simon, H.; Schnetger, B.; Brumsack, H.: SEASONALITY OF PORE WATER GEOCHEMISTRY IN A HIGH-ENERGY SANDY BEACH: IMPLICATIONS FOR THE FATE OF INORGANIC NITROGEN AND METALS

45 **Xiao, Y.**; Rohrlack, T.; Riise, G.: UNRAVELING THE OPTICAL PROPERTIES OF SEDIMENT ORGANIC MATTER IN SHAPING HISTORICAL CHANGE OF LAKE WATER COLOR

46 **Burns, W.**; Stockwell, J.; Wilkes, A.; Marti de Ocampo, C.; Herdman, L.; Schroth, A.: USING BOTH LONG-TERM MONITORING DATA AND HIGH-FREQUENCY BUOY DATA TO BETTER UNDERSTAND HOW TWO SHALLOW, EUTROPHIC BAYS IN LAKE CHAMPLAIN RESPOND TO PHENOLOGICAL AND EPISODIC EVENTS

SS038 REAL WORLD ECOLOGY LANDSCAPES – UP-SCALING COMMUNITY ECOLOGY EXPERIMENTS IN AQUATIC SYSTEMS

204 **Scharnweber, K.**; Chacagueda, F.; Dalman, E.; Tranvik, L.; Eklöv, P.: THE EMERGENCE OF FATTY ACIDS - CHIRONOMIDAE AS VECTORS

SS039 EFFECTS OF STORM EVENTS ON AQUATIC, COASTAL, AND OCEANIC ENVIRONMENTS AND ECOSYSTEMS: FROM PHYSICAL PROCESSES TO FOOD WEBS

423 **Li, Y.**; Kelly, D.; Zhang, K.; Robertson, C.: A NEW INTEGRATED STORM SURGE AND FRESHWATER OVERLAND FLOODING MODEL WITH VERIFICATIONS AT SOUTH FLORIDA

424 **Endo, Y.**: CHANGE IN MESOZOOPLANKTON COMPOSITION IN A BAY BEFORE AND AFTER THE 2011 TOHOKU EARTHQUAKE TSUNAMI

425 **López-Lloreda, C.**; McDowell, W.; Potter, J.: DISSOLVED GREENHOUSE GASES IN STREAMS AND THEIR RESPONSE TO HURRICANES IRMA AND MARIA IN A TROPICAL FOREST IN PUERTO RICO

426 **Geist, S.**; McAskill, S.: DOES VERTICAL DISTRIBUTION OF FISH LARVAE SUGGEST AVOIDANCE OF A FRESHWATER PLUME IN THE GULF OF MEXICO CAUSED BY HURRICANE HARVEY?

427 **Walker, L.**; Montagna, P.: IMPACT OF HURRICANE HARVEY ON THE WATER QUALITY OF TEXAS ESTUARIES

428 Cobleigh, K.; Townsend, J.; Hollister, K.; Nagel, J.; Jerris, K.; Clower, P.; **Raymore, M.**; Afshar, M.; Funk, K.; Roper, Z.; Farchette, A.; Packard, A.: IMPACTS OF HURRICANES IRMA AND MARIA ON CORAL RESTORATION AND ASSOCIATED FISH DIVERSITY IN ST. THOMAS, U.S. VIRGIN ISLANDS

429 **Viggiano, M.**; Jensen, B.; Harrison, N.; Parker, S.; Goodman, K.: MEASURING ECOLOGICAL RESPONSES TO HURRICANE MARIA AT NEON SUBTROPICAL STREAM SITES

430 **Garcia, E.**; Rivera Casillas, P.; Chardon Maldonado, P.: MEASURING THE TIME-VARYING FREE SURFACE WITHIN THE INNER SURF AND SWASH ZONES USING A LIDAR

431 **Rodriguez-Abudo, S.**; Rodriguez-Abudo, S.; Chardon-Maldonado, P.; Canals-Silander, M.; Morell, J.: OBSERVATIONS OF NEARSHORE WAVES AND CURRENTS IN RINCÓN, PR DURING HURRICANE MARÍA

432 **Ogurcak, D.**; Price, R.: SALINIZATION OF A FRESHWATER LAKE FOLLOWING HURRICANE IRMA STORM SURGE IN THE FLORIDA KEYS

433 **Robinson, K.**; Topor, Z.; Tucru, A.: SPATIAL VARIABILITY IN GELATINOUS PLANKTON IN THE WAKE OF HURRICANE HARVEY

434 **Kurtay, G.**; Pathare, M.; Stauffer, B.: THE IMPACTS OF HURRICANE HARVEY ON PHYTOPLANKTON FUNCTIONAL GROUPS IN NORTHERN GULF OF MEXICO

435 **Harrison, N.**; Slater, M.; Krauss, R.; Monahan, D.; Viggiano Beltrocco, M.: THE NATIONAL ECOLOGICAL OBSERVATORY NETWORK: MONITORING CHANGES IN STREAM MORPHOLOGY AND BIOLOGICAL HABITAT IN TWO SUB-TROPICAL WATERSHEDS OF PUERTO RICO

T REPRESENTS TUTORIAL PRESENTATIONS

SS040 RADIONUCLIDES IN AQUEOUS SYSTEMS

117 **Brooks, G.**; Larson, R.; Schwing, P.; Armenteros-Almanza, M.; Martinez-Suarez, A.; Diaz-Asencio, M.; Jimenez-Reyes, D.: HIGH RESOLUTION SEDIMENT DISTRIBUTION AND ACCUMULATION PATTERNS OVER THE PAST ~100 YEARS: NW CUBA

115 **Ruben, Z.**; Murgulet, D.; Douglas, A.: THE PRESENCE AND SIGNIFICANCE OF RADIUM AND RADON ISOTOPES IN SOUTH TEXAS ESTUARIES

118 **Smith, C.**; Everhart, C.; Zaremba, N.; Forde, A.; McBride, E.; Jacoby, C.: TRANSIENT VARIABILITY OF GROUNDWATER-SURFACE WATER EXCHANGE OBSERVED THROUGH REPEATED SURVEYS OF GROUNDWATER-BORNE TRACERS.

119 **Lapid, G.**; Torfstein, A.: DISSOLVED 230TH AND 232TH AS TRACERS OF THE IMPACT OF DUST STORMS IN THE GULF OF AQABA, RED SEA

120 **Buesseler, K.**; Benitez-Nelson, C.; Burt, W.; Clevenger, S.; Drysdale, J.; Francois, R.; Roca Marti, M.; Pike, S.; Resplandy, L.; Umhau, B.; Wyatt, A.: EXPORTS: USING HIGH RESOLUTION STUDIES OF THORIUM-234 AT OCEAN STATION PAPA TO ELUCIDATE SPATIAL AND TEMPORAL VARIABILITY IN PARTICLE EXPORT AND ATTENTION

121 **Waples, J.**: MEASURING BISMUTH-210 ACTIVITY IN AQUEOUS SYSTEMS

SS043 BEYOND THE NUMBERS: STRATEGIES FOR INCLUSIVE PRACTICES ACROSS THE AQUATIC SCIENCES

346 **Ortiz, G.**; Ramírez, A.: INVOLVING COMMUNITIES LIVING ON SAN JUAN BAY ESTUARINE BASIN IN MACROINVERTEBRATE BIOMONITORING FOR AQUATIC ECOSYSTEMS CONSERVATION AND SUSTAINABILITY

SS050 IMPROVING CORAL REEF RESILIENCE WITH TRANSFORMATIONAL SCIENCE

138 **Combs, I.**; Voss, J.; Beal, J.: CHARACTERIZING THE IMPACTS OF SCLERACTINIAN TISSUE LOSS DISEASE ON CORALS IN SOUTHEAST FLORIDA

139 **Mizrahi, D.**; Lopes, R.: DESICCATION TOLERANCE OF SUN CORAL, TUBASTRAEA SPP.

140 **Shilling, E.**; Voss, J.; Beal, J.: IMPLEMENTING DISEASE INTERVENTION STRATEGIES ON CORALS IN THE SOUTHEAST FLORIDA REEF TRACT AND ASSESSING THEIR POTENTIAL IMPACT ON MUCUS MICROBIAL COMMUNITIES

SS051 NEW VIEWS ON THE BIOLOGICAL TRANSFORMATION OF METALS IN THE MARINE ENVIRONMENT

123 **Ruacho, A.**; Barbeau, K.; Bundy, R.: A NOVEL METHOD FOR THE PROCESSING OF MULTIPLE ANALYTICAL WINDOW ELECTROCHEMICAL TITRATION METAL SPECIATION DATA

124 **Barbeau, K.**; Manck, L.; Dupont, C.: CHARACTERIZATION OF ORGANIC IRON TRANSPORTERS IN A COPIOTROPHIC MARINE BACTERIUM

125 **Chmiel, R.**; Oldham, V.; Rao, D.; DiTullio, G.; Saito, M.: COBALT SPECIATION IN THE AMUNDSEN AND ROSS SEAS DURING BLOOM CONDITIONS

126 **Wallace, Z.**; Spitz, Y.; Combes, V.: INSIGHTS INTO THE ROLE OF THE MICROBIAL LOOP ON THE IRON CYCLE IN PATAGONIAN SHELF AND ADJACENT SOUTHERN OCEAN WATERS

127 **Mori, C.**; Striebel, M.; Schnetger, B.; Pahnke, K.; Brumsack, H.: NON-CONSERVATIVE BEHAVIOR OF MOLYBDENUM AND THALLIUM INDUCED BY A PHYTOPLANKTON SUMMER BLOOM?

128 **Babcock-Adams, L.**; Follett, C.; Bundy, R.; Repeta, D.: SIDEROPHORE PRODUCTION IN ASSOCIATION WITH LARGE CELL FRACTION DIAZOTROPHS

129 **Kustka, A.**; Bornhorst, R.; Shrestha, R.; Hildebrand, M.; Stelmark, K.: THE ROLE OF THE DIVALENT METAL TRANSPORTER, NRAMP, IN IRON AND MANGANESE HOMEOSTASIS IN THE DIATOM THALASSIOSIRA PSEUDONANA.

130 **Li, J.**; Babcock-Adams, L.; Boiteau, R.; McIlvin, M.; Repeta, D.: TOWARDS A MORE QUANTITATIVE DETERMINATION OF FE BINDING LIGANDS IN SEAWATER

SS057 EASTERN BOUNDARY UPWELLING SYSTEMS IN A CHANGING OCEAN: RECENT INSIGHTS AND FUTURE PERSPECTIVES

441 **Closset, I.**; Brzezinski, M.; Krause, J.; Thamatrakoln, K.; Bruland, K.; Jones, J.: A MOSAIC OF SILICA PRODUCTION REGIMES ALONG THE CALIFORNIA COAST: WHEN SPATIAL AND TEMPORAL HETEROGENEITY ENHANCE DIATOMS RESILIENCE TO CLIMATE FORCING

442 **Fulton, K.**; Barbeau, K.; Forsch, K.; Manck, L.; Krause, J.; Stukel, M.: ATTENUATION OF DIATOM-MEDIATED EXPORT IN A COASTAL UPWELLING FILAMENT IN THE CALIFORNIA CURRENT ECOSYSTEM

443 **Fumo, J.**; Carter, M.; Flick, R.: DESCRIPTION AND CONTEXT OF THE SUMMER 2018 SOUTHERN CALIFORNIA MARINE HEAT WAVE

444 **Muñoz, T.**; Berger, T.; Auger, P.; Bento, J.; Gorgues, T.; Di Lorenzo, E.; Hormazábal, S.: MESOSCALE EDDY EFFECTS ON THE SPATIAL DISTRIBUTION OF PRIMARY PRODUCTIVITY AND DISSOLVED OXYGEN IN THE EASTERN SOUTH PACIFIC

SS060 THE ADVENT OF SAMPLING BIOLOGICAL PROCESSES IN AQUATIC SYSTEMS USING AUTONOMOUS PLATFORMS

445 **Fujiki, T.**; Inoue, R.; Honda, M.; Wakita, M.; Sukigara, C.; Mino, Y.; Abe, O.: RELATIONSHIP BETWEEN PHYTOPLANKTON PRODUCTIVITY AND OXYGEN CYCLE IN THE SUBTROPICAL WESTERN NORTH PACIFIC

SS063 OCEAN, COASTAL, AND FRESHWATER ACIDIFICATION: RESEARCH AND EDUCATION

131 **Higgins, J.**; Bartoloni, S.; Wewerka, S.; Walter, R.; Bockmon, E.: CARBONATE CHEMISTRY SPATIAL VARIABILITY AND SEAGRASS LOSS IN A LOW-INFLOW ESTUARY

132 **Wolfe, W.**; Martz, T.; Goericke, R.; Dickson, A.; Send, U.: ANALYSIS OF FOUR YEARS OF CARON SYSTEM MEASUREMENTS MADE OFF OF THE COAST OF SOUTHERN CALIFORNIA

133 **Pardis, W.**; Long, D.; DeGrandpre, M.; Sensors, S.: MEASURING PROTONS WITH PHOTONS: A SPECTROPHOTOMETRIC PH ANALYZER FOR NATURAL AQUATIC SYSTEMS

134 **Xu, Y.**; Cai, W.: NORTHWEST ATLANTIC COASTAL ACIDIFICATION UNDER CLIMATE CHANGE

135 **Hudson-Heck, E.**; Byrne, R.: PURIFICATION AND CHARACTERIZATION OF THYMOL BLUE FOR SPECTROPHOTOMETRIC PH MEASUREMENTS IN RIVERS, ESTUARIES, AND SEAWATER

TUESDAY

[1] REPRESENTS INVITED PRESENTATIONS

136 **García-Troche, E.**; Morell, J.; Meléndez, M.; Salisbury, J.: SEASONAL INORGANIC CARBON DYNAMICS AT A SEMI-ENCLOSED TROPICAL MANGROVE LAGOON: A COASTAL OCEAN ACIDIFICATION HOTSPOT

137 **Awad, M.**; Shaltout, N.; Madkour, F.; Abuel-Regal, M.; El-Wazzan, E.: SENSITIVITY OF THE GROOVED CARPET SHELL CLAM (RUDITAPES DECUSSATUS) TO OCEAN ACIDIFICATION

231 papers

WEDNESDAY ORALS

CS005 ACIDIFICATION

Chair(s): Chris Langdon, University of Miami (clangdon@rsmas.miami.edu)
George Waldbusser, Oregon State University (waldbuss@coas.oregonstate.edu)

Location: Room 102 A/B/C

09:00 **Waldbusser, G.**; Smith, S.; Hales, B.: EFFECTS OF SEAGRASS AND TIDES ON CARBONATE CHEMISTRY EXPOSURE IN JUVENILE OYSTERS

09:15 **Bednarsek, N.**; Feely, R.; Saenger, C.; Alin, S.; Nezlin, N.; Roethler, M.: CARAPACE DISSOLUTION OBSERVED UNDER CURRENT LOW PH IS A HIGH- RISK STRATEGY FOR DUNGENESS CRAB LARVAL STAGES

09:30 **Lowder, K.**; Taylor, J.: AS LONG AS THE WATER'S WARM: CALIFORNIA SPINY LOBSTER LARVAE GET BIGGER FASTER IN WARMER WATER DESPITE DECREASES IN PH

09:45 Ragg, N.; **Delorme, N.**; Gale, S.; Camara, M.; King, N.; Frost, E.; Jury, J.; Hilton, Z.; Finnie, B.; Fielder, C.; Peychers, C.: ACCLIMATION AND ADAPTATION POTENTIAL TO OCEAN ACIDIFICATION IN THE GREENSHELL MUSSEL PERNA CANALICULUS

10:00 **Liberti, K.**; Gray, M.; Mayer, L.; Testa, J.; Brady, D.: THE CARBONATE CARRYING CAPACITY OF AN INTENSIVE OYSTER AQUACULTURE AREA

10:15 **Zayas-Santiago, C.**; Rivas-Ubach, A.; Ward, N.; Kuo, L.; Zimmerman, R.: METABOLOMICS REVEAL BIOCHEMICAL PATHWAYS RESPONSIBLE FOR EELGRASS RESPONSE TO CLIMATE CHANGE

14:00 **Langdon, C.**: FACTORIAL EXPERIMENT LOOKING AT THE EFFECTS OF TEMPERATURE AND CO2 ON THE CORAL-ALGAL SYMBIOSIS IN ACROPORA CERVICORNIS [1]

14:15 McNicholl, C.; **Koch, M.**; Hofmann, L.: THE ROLE OF PHOTOSYNTHESIS, LIGHT, AND OA CONDITIONS ON THALLI SURFACE O2 AND PH DYNAMICS: MICROSENSOR STUDIES ON TROPICAL REEF MACROALGAE

14:30 **Schroeder, H.**; Hintz, C.: NOVEL LONG-TERM ACCLIMATION APPROACH TO INVESTIGATING OCEAN ACIDIFICATION IN TROPICAL CORAL CULTURE

14:45 **Monroe, A.**; Jarrold, M.; Munday, P.; Ravasi, T.: COMBINED EFFECTS OF ELEVATED TEMPERATURES AND PCO2 ON THE TRANSCRIPTOMIC RESPONSE OF A CORAL REEF FISH

15:00 **Griffin, A.**; Andersson, A.; Cyronak, T.; Eyre, B.; Stoltenberg, L.: FROM THE BEAKER TO THE REEF: A COMPARISON OF BIOGENIC CARBONATE SEDIMENT DISSOLUTION RATES IN THE LABORATORY AND THE FIELD

15:15 **Doo, S.**; Edmunds, P.; Carpenter, R.: RAPID NIGHT-TIME COMMUNITY READJUSTMENT ATTENUATES EFFECTS OF OCEAN ACIDIFICATION ON A CORAL REEF

15:45 **Gobler, C.**; Young, C.: THE ABILITY OF COASTAL ACIDIFICATION TO ESTABLISH TOP-DOWN AND BOTTOM-UP CONTROLS ON MACROALGAE POPULATIONS

16:00 **Sharp, J.**; Byrne, R.: CARBONATE ION DETERMINATIONS IN SEAWATER: A DECADE OF METHODOLOGICAL DEVELOPMENT

16:15 **Bushnell, M.**: QARTOD STATUS AND PLANS TO DEVELOP A MANUAL FOR THE REAL-TIME QUALITY CONTROL OF PH DATA

16:30 **Pacella, S.**; Brown, C.; Kaldy, J.; Labiosa, R.; Hales, B.; Mochon Collura, T.; Stecher, H.; Waldbusser, G.: OCEANIC AND WATERSHED CONTROLS OF COASTAL ACIDIFICATION IN A SMALL, AGRICULTURALLY DEVELOPED CALIFORNIA CURRENT ESTUARY

16:45 **McNicholl, C.**; Koch, M.; Swarzenski, P.; Metian, M.; Oberhaensli, F.; Taylor, A.; Gomez, M.: DRIVERS OF TROPICAL REEF MACROALGAL CALCIFICATION UNDER OCEAN ACIDIFICATION

17:00 **Diamond, S.**; Harvey, R.; Heathcote, A.; Morales-Williams, A.: BROWNING OR GREENING? LONG-TERM CHANGES IN PHYTOPLANKTON PRODUCTIVITY AND COMMUNITY COMPOSITION IN ACID-IMPARED LAKES

CS009 GAS FLUXES

Chair(s): Marco Aurelio Santos, Energy Planning Program/COPPE/UFRJ (aurelio@ppe.ufrj.br)
Jorge Machado Damazio, CEPEL (damazio@cepel.br)
Tonya Delsontro, University of Geneva, Switzerland (tdelsontro@gmail.com)
Jake Beaulieu, US EPA (Beaulieu.Jake@epa.gov)

Location: Room 201

09:00 **Beaulieu, J.**; Barnett, W.; Waldo, S.; Balz, D.; Platz, M.; White, K.: PREDICTING CARBON DIOXIDE AND METHANE EMISSION RATES FROM U.S. RESERVOIRS USING BOOSTED REGRESSION TREES

09:15 **Battin, T.**; Horgby, A.; Ulseth, A.: MON DIEU, HAVE WE FORGOTTEN THE MOUNTAIN CHIMNEYS?

09:30 **Weber, T.**; Wiseman, N.: GLOBAL OCEANIC METHANE EMISSIONS ESTIMATED FROM STATISTICAL MAPPING

09:45 **Grinham, A.**; Dunbabin, M.; Deering, N.; Sturm, K.; Albert, S.: RAPID, LARGE-SCALE ASSESSMENT OF METHANE EBULLITION USING HIGH RESOLUTION AERIAL DRONE IMAGERY

10:00 **Scholz, K.**; Gioli, B.; Miglietta, F.; Sommaruga, R.; Tomelleri, E.; Tonon, G.; Zaldei, A.; Wohlfahrt, G.: SETTING SAIL TO MEASURE LAKE-ATMOSPHERE GREENHOUSE GAS EXCHANGE

10:15 **Keller, P.**: DRYFLUX: A GLOBAL SURVEY OF CO2 EMISSIONS FROM DRY SEDIMENTS

14:00 **Peeters, F.**; Encinas Fernandez, J.: ON THE SOURCE OF THE CH4 EMITTED FROM SURFACE WATERS OF LAKES

14:15 **Sieczko, A.**; Schenk, J.; Nguyen, T.; Rudberg, D.; Pajala, G.; Sawakuchi, H.; Sundgren, I.; Bastviken, D.: EFFECT OF PRECIPITATION EVENTS ON CH4 EMISSIONS FROM BOREAL AND SUBARCTIC LAKES

14:30 **Rust, F.**; Tremblay, A.; del Giorgio, P.: SPATIOTEMPORAL VARIABILITY OF pCO2 AND pCH4 AND EMISSIONS OF THE RECENTLY BUILT LA ROMAINE HYDROPOWER COMPLEX (QUÉBEC, CANADA)

14:45 Webb, J.; **Finlay, K.**; Leavitt, P.; Baulch, H.; Simpson, G.; Haig, H.; Hodder, K.: GREENHOUSE GAS DYNAMICS IN SMALL AGRICULTURAL WATER BODIES IN THE NORTHERN GREAT PLAINS [1]

15:00 **Whitfield, C.**: THE IMPORTANCE OF EBULLITION IN SMALL, SHALLOW AGRICULTURAL RESERVOIRS

15:15 **Koschorreck, M.**; Hentschel, I.; Keller, P.; Boehrer, B.; Obrador, B.: OXYGEN EBULLITION FROM LAKES

CS010 NITROGEN BIOGEOCHEMISTRY AND CYCLING

Chair(s): Julie Granger, University of Connecticut (julie.granger@uconn.edu)

Location: Room 208 A/B

09:00 **Eberhard, E.**; Marcarelli, A.; Baxter, C.; Techtmann, S.: PATCH DYNAMICS OF NITROGEN FIXATION AND DENITRIFICATION IN STREAMS

[1] REPRESENTS INVITED PRESENTATIONS

09:15 **Nevorski, K.**; Marcarell, A.: VVARIATION IN DENITRIFICATION, NITROGEN FIXATION, AND NITRIFICATION OVER DAILY TO SEASONAL TIME SCALES IN A FORESTED STREAM

09:30 **Haas, S.**; LaRoche, J.; Desai, D.; Pawlowicz, R.; Wallace, D.: EVIDENCE FOR UNUSUAL NITROGEN CYCLING IN A LONG-TERM STRATIFIED LAKE FROM COMBINED MOLECULAR AND STABLE ISOTOPE APPROACHES

09:45 **Davis, H.**; Costanza, C.; Alderman, D.; Canava, K.; Goehring, L.; Jaffery, R.; Leng, B.; Spore, P.; Duke, J.; Scott, T.: DENITRIFICATION BY FACULTATIVE ANAEROBES IN OXIC WATER COLUMNS

10:00 Boedecker, A.; McCarthy, M.; Niewinski, D.; Chaffin, J.; **Newell, S.**: EVALUATING SEDIMENTS AS AN ECOSYSTEM SERVICE IN WESTERN LAKE ERIE THROUGH QUANTIFICATION OF NITROGEN CYCLING PATHWAYS

10:15 **Boedecker, A.**; Scott, T.: SATURATION LEVELS OF DINITROGEN GAS IN THE WATER COLUMNS OF MULTIPLE THERMALLY STRATIFIED RESERVOIRS

14:00 **Wilson, S.**; Song, B.; Anderson, I.: COMPARISON OF NITRIFYING COMMUNITIES ALONG THE VERTICAL GRADIENTS OF AMMONIUM, OXYGEN, AND SALINITY IN A SUBTERRANEAN ESTUARY

14:15 **Qin, W.**; Zheng, Y.; Zhao, F.; Stahl, D.; Ingalls, A.: GENOMIC CENSUS OF THE BIOGEOGRAPHY OF GLOBALLY ABUNDANT MARINE AMMONIA-OXIDIZING ARCHAEA

14:30 **Rich, J.**; O'Neill, S.; Mackin-McLaughlin, J.; Presley, R.: INTRACELLULAR NITRATE CONTROLS THE OVERALL FATE OF NITRATE IN PHOTOSYNTHETIC SUBTIDAL MARINE SEDIMENTS

14:45 **Wu, J.**; Rogers, S.: SEAWEED CULTIVATION AS AN ALTERNATIVE NUTRIENT MANAGEMENT STRATEGY FOR WASTEWATER TREATMENT AND RECOVERY FACILITIES: A CASE STUDY IN BOOTHBAY HARBOR, MAINE

15:00 **Cornwell, J.**; Owens, M.; Jackson, M.; Kellogg, L.: BIVALVES AND NUTRIENT ECOSYSTEM SERVICES: THE CENTRAL ROLE OF BIODEPOSITS

15:15 **Ji, Q.**; Grundle, D.: LASER-BASED NITROGEN ISOTOPIC ANALYSIS AT NANO-MOLAR LEVEL

CS020 FISH AND FISHERIES

Chair(s): Brent A. Murry, U.S. Fish and Wildlife Service (brent.murry@gmail.com)

Location: Room 202

09:00 **Herbig, J.**; Keller, J.; Morley, D.; Walter, K.; Barbera, P.; Acosta, A.: MOVEMENT PATTERNS AND HABITAT USE OF YELLOWTAIL SNAPPER IN THE DRY TORTUGAS, FLORIDA

09:15 **Keller, J.**; Herbig, J.; Morley, D.; Wile, A.; Barbera, P.; Acosta, A.: BLACK GROUPER MOVEMENT PATTERNS IN THE LOWER FLORIDA KEYS

09:30 **Sancho, G.**; Frazier, B.; Edman, R.; Shaw, A.; Bubley, W.: TIGER SHARK HABITAT UTILIZATION IN COASTAL ENVIRONMENTS OFF SOUTH CAROLINA, USA

09:45 **Perle, C.**; Watkins, J.; Lepak, J.; Miller, R.; Weidel, B.; Lantry, B.; Rudstam, L.: LAKE ONTARIO- UNWINDING THE EFFECTS OF HYDROLOGY AND PREY AVAILABILITY ON AN INTRODUCED SPECIES TO MAKE PREDICTIONS ABOUT ITS LONG TERM ECOLOGICAL ROLE AS A TOP PREDATOR.

10:00 **Galvez, B.**; Smith, S.; Townsend, H.: TROPHIC ECOLOGY OF WEAKFISH (CYNOSCION REGALIS) FROM THE DELAWARE BAY USING STABLE ISOTOPE AND STOMACH CONTENT ANALYSES

10:15 **Duran, A.**; Trexler, J.: SYNCHRONY AND SPATIOTEMPORAL DYNAMICS OF WETLAND FISHES OF SOUTH FLORIDA

CS022 BIODIVERSITY

Chair(s): Robert Ptacnik, WasserCluster Lunz (robert.ptacnik@wcl.ac.at)

Location: Room 103 B

09:00 **Arocho Hernández, N.**; Santos, C.: AQUATIC INVERTEBRATE COMMUNITIES IN A PUERTO RICAN MANGROVE FOREST: EFFECTS OF PRESENCE OF FLOATING FERNS (SALVINIA SPP.) AND SAMPLING GEAR-TYPE

09:15 **Ruiz-Ramos, D.**; Curd, E.; Schweizer, T.; Lin, M.; Gold, Z.; Meyer, R.; Dawson, M.: PATTERNS OF BIODIVERSITY ALONG THE CALIFORNIA COAST FROM ENVIRONMENTAL DNA

09:30 **Bouissou, A.**; Brodie, S.; Carroll, G.; Dodson, S.; Field, J.; Hazen, E.; Jacox, M.; Santora, J.; Satterthwaite, E.; Savoca, M.; Schroeder, I.; Welch, H.; Bograd, S.: RELATIONSHIPS BETWEEN THE ENVIRONMENT, PREY, AND TOP MARINE PREDATORS IN THE CENTRAL CALIFORNIA CURRENT ECOSYSTEM

09:45 **Carvalho, S.**; Pearman, J.; Anlauf, H.; Coker, D.; Villalobos, R.; Aylagas, E.; Curdia, J.; Lindo, R.; Kattan, Y.; Qurban, M.; Jones, B.; Berumen, M.: SIZE MATTERS: RED SEA CORAL REEF BIODIVERSITY IN RESPONSE TO ENVIRONMENTAL GRADIENTS SHOW INCONSISTENT PATTERNS FROM MICROBES TO FISH

10:00 **Shapiro, S.**: IDENTIFICATION OF MORPHOLOGICAL CHARACTERISTICS OF DEEP-SEA CORALS

10:15 **Rodriguez-Matos, P.**; Amaro, C.; Gonzalez, G.; Schizas, N.; Nadathur, G.: BIODIVERSITY OF DEEP-SEA ZOOPLANKTON OFF THE SOUTHWEST COAST OF PUERTO RICO

14:00 **Ptacnik, R.**; Vad, C.; Bengtsson, M.; Abonyi, A.; Eggers, L.; Preiler, C.: ON THE RELEVANCE OF DISPERSAL LIMITATION IN PHYTOPLANKTON – LESSONS FROM MESOCOSM EXPERIMENTS

14:15 **Bengtsson, M.**; Eggers, L.; Lumpi, T.; Horvath, Z.; Preiler, C.; Ptacnik, R.: CONNECTIVITY BOOSTS BIODIVERSITY AND ALTERS ECOSYSTEM FUNCTION THROUGH MICROBIAL INTERACTIONS IN FRESHWATER METACOMMUNITIES

14:30 **Morales-Williams, A.**; Sheik, C.; Kellerman, A.; Cotner, J.: SEASONAL SYNCHRONY OF CHEMICAL AND MICROBIAL DIVERSITY ACROSS LAKE TROPHIC GRADIENTS

15:00 Marroquim, A.; **Perry, R.**; Mathes, C.; Valente, R.: ENVIRONMENTAL BASELINE SURVEY DATA IN PERDIDO AND BAY OF CAMPECHE (SOUTHERN GULF OF MEXICO)

15:15 **Valente, R.**; Marroquim, A.; Perry, R.; Mathes, C.: BASELINE CHARACTERIZATION OF SOFT-BOTTOM BENTHIC COMMUNITIES IN SUPPORT OF REGULATORY PERMITTING FOR DRILLING IN THE DEEPWATER MEXICAN GULF OF MEXICO

CS024 MICROBIAL ECOLOGY AND PHYSIOLOGY

Chair(s): Barbara Campbell, Clemson University (bcampb7@clemson.edu)

Location: Room 101 A/B

09:00 Sanfilippo, J.; Nguyen, A.; Grébert, T.; Garczarek, L.; Karty, J.; Pokhrel, S.; Strnat, J.; Partensky, F.; Schluchter, W.; **Kehoe, D.**: THE MOLECULAR BASIS FOR THE ADAPTATION AND ACCLIMATION OF *SYNECHOCOCCUS* TO CHANGING LIGHT COLOR ENVIRONMENTS[I]

09:15 **Maresca, J.**; Hempel, P.; Keffer, J.; Singh, A.; Polson, S.: LIGHT-ENHANCED GROWTH IN NON-PHOTOTROPHIC ACTINOBACTERIA [I]

[I] REPRESENTS TUTORIAL PRESENTATIONS

09:30 **Pontiller, B.**; Lundin, D.; Martinez Garcia, S.; Pinhassi, J.: DISTINCT RESPONSES IN BACTERIOPLANKTON COMMUNITY STRUCTURE, ACTIVITY, AND GENE EXPRESSION TO LABILE DISSOLVED ORGANIC MATTER COMPOUNDS OF DIFFERENT COMPLEXITY

09:45 **Campbell, B.**; Lim, S.; Kirchman, D.: SAR11 ABUNDANCE, POTENTIAL FUNCTION AND ACTIVITY IN TWO CONTRASTING ESTUARIES[I]

10:00 **Noell, S.**; Giovannoni, S.: SAR11 CELLS HAVE A MULTIFUNCTIONAL AND EXTRAORDINARILY HIGH AFFINITY GLYCINE BETAINE TRANSPORTER

10:15 **Long, A.**; Ignacio Espinoza, J.; Fuhrman, J.: ESTIMATING MARINE BACTERIAL GROWTH RATES USING METAGENOMICS AND METATRANSCRIPTOMICS

14:00 **Tully, B.**: METABOLIC DIVERSITY WITHIN THE GLOBALLY ABUNDANT MARINE GROUP II EURYARCHAEA OFFERS INSIGHT INTO ECOLOGICAL PATTERNS

14:15 **Vollmer, S.**: USING SPATIAL AND TEMPORAL VARIATION IN STAGHORN CORAL MICROBIOMES TO IDENTIFY CORAL PATHOGENS

14:30 **Meyer, J.**; Paul, V.; Ushijima, B.; Aeby, G.; Häse, C.: MICROBIAL COMMUNITY SHIFTS ASSOCIATED WITH THE ON-GOING STONY CORAL TISSUE LOSS DISEASE OUTBREAK ON THE FLORIDA REEF TRACT

14:45 **Cobleigh, K.**; Macknight, N.; Medina, M.; Mydlarz, L.; Brandt, M.: EVALUATING DIFFERENCES IN MICROBIAL COMMUNITIES OF WHITE PLAGUE DISEASED AND HEALTHY COLONIES OF MAJOR REEF BUILDING CARIBBEAN CORALS

15:00 **Miller, N.**; Meyer, J.; Frazer, T.; Maneval, P.: MICROBIOME STRUCTURE OF NURSERY-REARED ACROPORA CERVICORNIS

15:15 **Rivera-Vicéns, R.**; Juravel, K.; Wöerheide, G.: BACTERIAL COMMUNITIES FROM THREE DIFFERENT SPONGES SPECIES FROM THE GENUS TETHYA.

15:45 **Da Silva, L.**; Ossai, S.; Chigbu, P.; Parveen, S.: CHARACTERIZATION OF VIBRIO VULNIFICUS ISOLATED FROM ENVIRONMENTAL AND BLUE CRAB SAMPLES COLLECTED FROM MARYLAND COASTAL BAYS

16:00 **Varg, J.**; Svanback, R.; Boughman, J.: PARASITES AND THE MICROBIOME OF THREESPINE FISH GASTEROSTEUS ACULEATUS

16:15 **Sehein, T.**; Gast, R.; Pachiadaki, M.; Guillou, L.; Follows, M.; Edgcomb, V.: TRACKING TROJAN HORSES IN THE MARINE REALM: A HIGH-RESOLUTION DATASET TO INVESTIGATE THE IMPACTS OF PROTIST PARASITISM ON COASTAL MARINE MICROBIAL COMMUNITIES

16:30 **Erazo, N.**; Giddings, S.; Bowman, J.: MANGROVES AND MICROBES, COORDINATED ECOSYSTEM ENGINEERS IN COASTAL ECUADOR

16:45 **Comstock, J.**; Carlson, C.; James, A.; Nelson, C.: LINKAGES BETWEEN BACTERIAL AND ARCHAEAL COMMUNITY STRUCTURE AND BIOGEOCHEMICAL VARIABILITY AROUND THE ISLANDS OF MO'OREA AND TAHITI

17:00 **Hall, E.**: THE EFFECT OF HEAVY METALS ON THE BACTERIAL BIOGEOGRAPHY OF A LARGE, TROPICAL WATERSHED

17:15 **Turner, S.**; Lopez-Fernandez, M.; Lundin, D.; Brambilla, D.; Peura, S.; Bertilsson, S.; Dopson, M.: METAGENOMIC ANALYSIS REVEALS NOVEL BACTERIAL CANDIDATE TAXA FROM DEEP AQUIFERS

CS026 PRIMARY PRODUCTION

Chair(s): Euan Reavie, University of Minnesota (ereavie@d.umn.edu)

Location: Room 208 A/B

15:45 **Lopez Sandoval, D.**; Delgado-Huertas, A.; Duarte, C.; Agustí, S.: THE USE OF 13C TO MEASURE PRIMARY PRODUCTION IN OLIGOTROPHIC MARINE SYSTEMS

16:00 **Morelle, J.**; Richard, A.; Maire, O.; Slimani, A.; Orvain, F.: BIOTURBATION AND DENSITY INFLUENCE FROM BENTHIC MACROFAUNA ON MICROPHYTOBENTHIC PRIMARY PRODUCTION IN THE SEINE ESTUARY: STUDY OF MICROSCALE PROCESSES

16:15 **McCormick, A.**; Phillips, J.; Botsch, J.; Ives, A.: SPATIOTEMPORAL VARIATION IN THE RELATIVE MAGNITUDES OF BENTHIC AND PELAGIC PRODUCTION IN A SHALLOW EUTROPHIC LAKE

16:30 **Phillips, J.**: TIME-VARYING RESPONSES OF ECOSYSTEM METABOLISM TO LIGHT AND TEMPERATURE

16:45 **Krishna, S.**: COUPLED 1D PHYSICAL-BIOGEOCHEMICAL MODEL SYSTEM TO SIMULATE PRIMARY PRODUCTION IN LAKE GENEVA

17:00 **Reavie, E.**; Cai, M.; Meyer-Jacob, C.; Smol, J.; Werne, J.: PALEOLIMNOLOGY OF PRODUCTIVITY IN THE LAURENTIAN GREAT LAKES: A COMPARISON OF GEOCHEMICAL METHODS[I]

CS037 COASTAL ECOSYSTEMS

Chair(s): Helene Frigstad, Norwegian Institute for Water Research (helene.frigstad@niva.no)
Andrew King, Norwegian Institute for Water Research (andrew.king@niva.no)

Location: Room 104

09:00 **Oczkowski, A.**; Santos, E.; Martin, R.; Hanson, A.; Huertas, E.; Watson, E.; Wigand, C.: NUTRIENT DYNAMICS IN A TROPICAL URBAN ESTUARY INDICT UNANTICIPATED NITROGEN SOURCES AS CONTRIBUTING TO ENVIRONMENTAL HEALTH PROBLEMS—A CASE STUDY FROM SAN JUAN, PUERTO RICO[IT]

09:30 **Karlsson, C.**; Pontiller, B.; Teikari, J.; Traving, S.; Happel, E.; Henke, B.; Huchaiah, V.; Nilsson, E.; Alneberg, J.; Lundin , D.; Sivonen , K.; Andersson, A.; Riemann, L.; Middelboe, M.; Kisand , V.; Pinhassi, J.: METATRANSCRIPTOMIC ANALYSIS UNCOVERS DIVERGENT RESPONSES OF BALTIC SEA BACTERIA TO FOREST AND AGRICULTURE RIVER LOADINGS.

09:45 **Frigstad, H.**; Naustvoll, L.; Andersen, G.; Trannum, H.; Kaste, Ø.; Deininger, A.; Hjermann, D.: COASTAL ECOSYSTEM RESPONSE TO OLIGOTROPHICATION AND CHANGING RIVERINE LOADS – A NORWEGIAN CASE-STUDY

10:00 **Moore, T.**; Fong, P.: ANTHROPOGENIC DISTURBANCES ENHANCE NUTRIENT-DRIVEN ACIDIFICATION IN TWO CALIFORNIA ESTUARIES

10:15 **Szymczycha, B.**; Borecka, M.; Białk-Bielińska, A.: CAN SUBMARINE GROUNDWATER DISCHARGE BE A SOURCE OF PHARMACEUTICALS TO THE MARINE ENVIRONMENT?

14:00 **Arula, T.**; Woodland, R.; Houde, E.: DEFINING THE TROPHIC ROLE OF A KEY FORAGE SPECIES IN A LARGE COASTAL ECOSYSTEM

14:15 **Volaric, M.**: OYSTER REEF ACTIVITY INVESTIGATED USING A NOVEL COMBINATION OF BIOACOUSTICS AND AQUATIC EDDY COVARIANCE FLUX MEASUREMENTS

14:30 **Wenker, R.**; Stevens, B.: SEA WHIP CORAL (LEPTOGORGIA VIRGULATA) IN THE MID-ATLANTIC BIGHT: AGE, COLONY COMPLEXITY, AND DISTRIBUTION

[I] REPRESENTS INVITED PRESENTATIONS

14:45 **Clower, P.**; Grimes, K.; Hartwell, I.; Brandt, M.; Nick, S.: ASSESSING THE LEVELS OF POLLUTANTS IN MANGROVE LAGOON, ST. THOMAS EAST END RESERVES, U.S. VIRGIN ISLANDS, BETWEEN 2011 AND 2018 AND THE ABILITY OF MANGROVES TO INTERCEPT POLLUTANTS

CS045 PHYSICAL DYNAMICS

Chair(s): Katherine Fitzenreiter, University of Maryland Eastern Shore (kgfitzenreiter@umes.edu)

Location: Room 104

15:45 **Walter, R.**; Rainville, E.; O'Leary, J.: HYDRODYNAMICS IN A SHALLOW SEASONALLY LOW-INFLOW ESTUARY FOLLOWING EELGRASS COLLAPSE

16:00 **Amador, A.**; Pawlak, G.; Giddings, S.; Feddersen, F.; Merrifield, M.: TURBULENT REYNOLDS STRESSES IN A TIDALLY DRIVEN ALONGSHORE FLOW

16:15 **Fitzenreiter, K.**: SURFACE CIRCULATION PATTERNS AND CURRENT VELOCITIES DERIVED FROM DRIFTERS IN COASTAL WATERS OF MARYLAND AND VIRGINIA

16:30 **Haak, C.**; Cowles, G.; Danylchuk, A.: WAVE AND TIDE-DRIVEN FLOW SHAPE THE DISTRIBUTION OF A JUVENILE FISH

16:45 **Ladah, L**: HOW DO INTERNAL WAVES CONTROL COASTAL PRODUCTIVITY?

17:00 **Fujimura, A.**; Reniers, A.; Paris, C.; Shanks, A.; MacMahan, J.; Morgan, S.: BIOLOGICAL-PHYSICAL INTERACTIONS OF COASTAL PLANKTON DYNAMICS

17:15 **Savelli, R.**; Bertin, X.; Orvain, F.; Gernez, P.; Dale, A.; Coulombier, T.; Pineau, P.; Lachaussee, N.; Polsenaere, P.; Dupuy, C.; Le Fouest, V.: FROM CHRONIC TO CATASTROPHIC RESUSPENSION MECHANISMS OF MICROPHYTOBENTHOS ON A TEMPERATE INTERTIDAL MUDFLAT: A MODELLING STUDY

SS007 HALOPHILA STIPULACEA: ECOLOGY AND MANAGEMENT OF THE GLOBALLY INVASIVE SEAGRASS

Chair(s): Demian Willette, Loyola Marymount University (demian.willette@lmu.edu)
Edwin Cruz-Rivera, University of the Virgin Islands (edwin.cruzrivera@uvi.edu)

Location: Room 209 C

09:00 **Willette, D.**; Muthukrishnan, R.; Chiquillo, K.; Cross, C.; Fong, P.; Kelly, T.; Toline, A.: RESILIENCE TO DISTURBANCE AND RAPID GROWTH OF FRAGMENTS DEMONSTRATE INDO-PACIFIC SEAGRASS'S CAPACITY TO SPREAD AS AN INVASIVE SPECIES

09:15 **Muthukrishnan, R.**; Willette, D.; Chiquillo, K.; Cross, C.; Fong, P.; Toline, C.: SHIFT IN FOUNDATION SPECIES AS A RESULT OF INVASION ALTERS ECOSYSTEM FUNCTIONING IN A TROPICAL SEAGRASS COMMUNITY

09:30 **Chiquillo, K.**; Fong, P.; Willette, D.; Cruz-Rivera, E.; Barber, P.: LEAGUE OF SEAGRASSES: THE OUTCOME OF INTERSPECIFIC INTERACTIONS BETWEEN NATIVE VERSUS INVASIVE SEAGRASSES SHOWS A POSITIVE EFFECT ON THE INVASIVE, HALOPHILA STIPULACEA

09:45 **Cassell, J.**; Cruz-Rivera, E.; Jobsis, P.: SEASONAL NUTRIENT PATTERNS OF THREE SEAGRASS SPECIES IN A SMALL BAY ON ST. THOMAS, USVI

10:00 **Whitman, E.**; Heithaus, M.; Quiquempois, V.; Niviere, M.; Bernus, J.; Kiszka, J.: LARGE GRAZER FORAGING HABITAT USE AND RESOURCE SELECTION IS AFFECTED BY THE INVASIVE SEAGRASS, HALOPHILA STIPULACEA, IN THE FRENCH WEST INDIES

10:15 **Cross, C.**; Willette, D.: EVALUATING THE CARBON SEQUESTERING CAPACITY OF SEAGRASS HALOPHILA STIPULACEA IN ITS INVASIVE CARIBBEAN RANGE

14:00 **Cruz-Rivera, E.**; Abdul-Samad, H.; Madera, A.: DOES *HALOPHILA STIPULACEA* SIMPLY TASTE BAD? LETTING SOME INVERTEBRATES DECIDE

14:15 Jensen, A.; **Grimes, K.**; Smith, E.; Brandt, M.: *HALOPHILA STIPULACEA* PROVIDES UNEXPECTED BLUE CARBON BENEFIT IN THE U.S. VIRGIN ISLANDS

14:30 **Stanford, A.**; Lewis, S.; Forbes, K.; Romney, C.; Wyllie-Echeverria, S.; Turner, T.: MULTIPLE INVASIONS OF *HALOPHILA STIPULACEA* IN THE US VIRGIN ISLANDS

15:00 **Wesselmann, M.**; Anton, A.; Hendriks, I.; Duarte, C.; Agustí, S.; Apostolaki, E.; Savva, I.; Marbà, N.: SHIFTING THERMAL TOLERANCE FROM NATIVE TO EXOTIC SEAGRASS (HALOPHILA STIPULACEA) POPULATIONS

15:15 **Winters, G.**; Manh Nguyen, H.; Singh Yadav, N.; Barak, S.; Lima, F.; Sapir, Y.: RESPONSES OF INVASIVE AND NATIVE HALOPHILA STIPULACEA POPULATIONS TO CLIMATE CHANGE - A MULTIDISCIPLINARY APPROACH INTEGRATING PHYSIOLOGY AND GENE EXPRESSION

SS012 TRACING THE NATURAL AND ANTHROPOGENIC CARBON CYCLE ACROSS AQUATIC ENVIRONMENTS

Chair(s): Andrew Margolin, Virginia Institute of Marine Science (armargolin@vims.edu)
Ryan Woosley, Massachusetts Institute of Technology (rwoosley@mit.edu)

Location: Room 209 A/B

14:00 **Margolin, A.**: ENVIRONMENTAL IMPACTS ON CARBON BIOGEOCHEMISTRY IN MARGINAL SEAS

14:15 **Savoie, A.**; Rosen, J.; Howden, S.; Hayes, C.: CARBON DYNAMICS OFF THE CONTINENTAL SHELF OF THE GULF OF MEXICO

14:30 **Chu, S.**; Sutton, A.; Warren, T.; Parker, V.: SURFACE AND SUBSURFACE COMPARISON OF CARBONATE SYSTEM OBSERVATIONS IN U.S. COASTAL AND CORAL ECOSYSTEMS

14:45 **Brooks, M.**; Lueker, T.; Dickson, A.; Emanuele, G.; Long, M.; Keeling, R.: CHANGES IN THE 13C/12C RATIO OF DISSOLVED INORGANIC CARBON NEAR BERMUDA AND HAWAII OVER THE PAST THREE DECADES

15:00 **Macovei, V.**; Hartman, S.; Schuster, U.; Torres-Valdes, S.; Moore, C.; Sanders, R.: TEMPORAL VARIATIONS IN THE CONTRIBUTION OF PHYSICAL AND BIOLOGICAL FACTORS ON THE CARBONATE SYSTEM OF THE MID-LATITUDE NORTH ATLANTIC

16:00 **Neary, L.**; Remmer, C.; Klemt, W.; Kay, M.; Wolfe, B.; Hall, R.: EFFECTS OF SPRING FLOODING AND PRIMARY PRODUCTIVITY ON LAKE CARBON BALANCE IN THE PEACE-ATHABASCA DELTA, ALBERTA (CANADA)

16:15 **Krause, J.**; Watson, E.; Gray, A.; McDonnell, J.; Hinojosa, A.: BLUE CARBON IN AN EASTERN PACIFIC REVERSE ESTUARY: ORGANIC MATTER CYCLING LINKS SALT MARSHES AND SEAGRASS BEDS

16:30 **Kaldy, J.**; Brown, C.; Pacella, S.; Young, M.; Mochon Collura, T.; Stecher, J.: SPATIAL AND TEMPORAL PATTERNS OF DISSOLVED ORGANIC AND INORGANIC CARBON DELIVERY TO TILLAMOOK ESTUARY, OREGON (USA)

WEDNESDAY

T REPRESENTS TUTORIAL PRESENTATIONS

16:45 **King, A.**; Frigstad, H.; Norli, M.; Bellerby, R.; Kaste, Ø.; Sørensen, K.: DOM AND NUTRIENTS FROM RIVERS TO COAST: OBSERVATIONS AND EXPERIMENTAL RESULTS

SS014 ECOSYSTEM BASED MANAGEMENT: HOLISTIC APPROACHES TO EFFECTIVE MANAGEMENT OF REGIONAL ECOSYSTEM

Chair(s): Michael Roman, Univ. of Maryland Center for Environmental Science (roman@umces.edu)
Felix Martinez, NOS/NCCOS/NOAA (Felix.Martinez@noaa.gov)
Amie West, Univ. of Maryland Center for Environmental Science (awest@umces.edu)

Location: Room 209 C

15:45 **Schwing, F.**; Link, J.; Brady, P.: PLANNING AND IMPLEMENTING A REGIONAL ECOSYSTEM APPROACH TO FISHERIES MANAGEMENT: A CASE STUDY FOR FEDERAL ECOSYSTEM-BASED MANAGEMENT

16:00 **Marshak, T.**; Link, J.: CHARACTERIZING AND COMPARING MARINE FISHERIES ECOSYSTEMS IN THE UNITED STATES – SUCCESSFUL FACTORS IN MOVING TOWARD ECOSYSTEM-BASED FISHERIES MANAGEMENT

16:15 **Strand, E.**; Goodbody-Gringley, G.: CHARACTERIZING BERMUDA'S BAITFISH POPULATIONS TO IMPROVE MANAGEMENT AND FISHERY SUSTAINABILITY

16:30 **Williams, N.**; Rose, K.: TOWARDS ECOSYSTEM-BASED MANAGEMENT FOR COASTAL LOUISIANA: EVALUATION OF DATA AVAILABILITY AND CONSTRAINTS

16:45 **Martin Bras, M.**; Cabral Guadalupe, A.; Fernandez Porto, J.; Latz, M.; Carter, M.: SUPPORTING LOCAL EDUCATION AND CONSERVATION BY INTEGRATING RESEARCHERS ON SCIENCE EDUCATION AND TRAINING LOCAL RESEARCHERS TO BUILD TRANSFERABLE CAPACITY

17:00 **Dennison, W.**; Orth, R.; Lefcheck, J.; Landry, B.: CREATING AND COMMUNICATING ENVIRONMENTAL INTELLIGENCE FOR CHESAPEAKE BAY RESEARCH AND MANAGEMENT

17:15 **West, A.**; Roman, M.; Martinez, F.; Dennison, W.; Miller, T.; Moser, F.; Rose, K.; Wainger, L.: PERCEPTIONS OF ECOSYSTEM-BASED MANAGEMENT: TALES FROM THE FIELD

SS022 PHYTOPLANKTON-BACTERIA INTERACTIONS: MOLECULAR INSIGHTS, CHEMICAL DRIVERS, AND BEHAVIORAL DYNAMICS

Chair(s): Kristen Whalen, Haverford College (kwhalen1@haverford.edu)
Elizabeth Harvey, Skidaway Institute of Oceanography (elizabeth.harvey@skio.uga.edu)
David Rowley, University of Rhode Island (drowley@mail.uri.edu)

Location: Room 209 A/B

09:00 **Loftus, S.**; Johnson, Z.: DYNAMICS OF DISSOLVED ORGANIC CARBON AND PHYTOPLANKTON GROWTH IN SPENT MEDIA

09:30 **Sison-Mangus, M.**; Kempnich, M.: THE INFLUENCES OF ASSOCIATED BACTERIA ON THE DOMOIC ACID SYNTHESIS OF THE TOXIC DIATOM PSEUDO-NITZSCHIA

09:45 **Whalen, K.**; Becker, J.; Pollara, S.; Chase, D.; Downing, G.; Overton, E.; Harvey, E.: DECIPHERING THE OCEAN'S CHEMOINFORMATIC LANGUAGE: A TALE OF TWO MOLECULES

10:00 **Erdner, D.**; Severin, T.; Jauzein, C.: STRUCTURE AND FUNCTION OF THE MICROBIOME OF THE TOXIC DINOFLAGELLATE *ALEXANDRIUM TAMARENSE*

10:15 Shibl, A.; Ochsenkühn, M.; Cardenas, A.; Fei, C.; Voolstra, C.; **Amin, S.**: ROSEOBACTER GROUP BACTERIA DOMINATE INTERACTIONS BETWEEN A COSMOPOLITAN DIATOM AND ITS MICROBIOME

SS023 CARBON CYCLING ACROSS GRADIENTS IN THE LAND-OCEAN-CONTINUUM

Chair(s): Michael Seidel, University of Oldenburg, Germany (m.seidel@uni-oldenburg.de)
Nicholas D. Ward, Pacific Northwest National Laboratory, USA (nickdward@gmail.com)
Sairah Y. Malkin, University of Maryland, USA (smalkin@umces.edu)
Patricia M. Medeiros, University of Georgia, USA (medeiros@uga.edu)

Location: Room 208 C

09:00 **Kujawinski, E..**: DISSOLVED AND PARTICULATE MICROBIAL METABOLITES REFLECT DISTINCT METABOLIC PROCESSES ACROSS SPATIAL AND TEMPORAL OCEAN GRADIENTS[1]

09:30 **Hawkes, J.**; D'Andrilli, J.: QUALITY CONTROL IN DISSOLVED ORGANIC MATTER COMPOSITION ASSESSMENT BY HIGH RESOLTUION MASS SPECTROMETRY, AN INTERNATIONAL LABORATORY COMPARISON

09:45 **Wissel, B.**; Quinones-Rivera, Z.; Nanayakkara, L.; Leavitt, P.; Bateson, D.: SEASONAL AND INTERANNUAL PATTERNS ACROSS CANADIAN HARDWATER LAKES: AN ISOTOPE PERSPECTIV

10:00 **Marcé, R.**; Liight, T.; Catalán, N.; Obrador, B.; Koschorreck, M.; Montes-Pérez, J.; Barbosa, L.; Moreno-Ostos, E.: CARBON EMISSIONS FROM EXPOSED SEDIMENTS IN DRYING HARDWATER LAKES ARE DECOUPLED FROM ORGANIC MATTER LOSS

10:15 **Nydahl, A.**; Wallin, M.; Weyhenmeyer, G.: REASONS BEHIND A LONG-TERM PCO2 INCREASE IN BOREAL LAKES AND STREAM

14:00 **Hernes, P.**; Eckard, R.: OVERPRINTING RIVERINE DOM WITH NON-POINT SOURCES

14:15 **Regier, P.**; Gonzalez-Pinzón, R.; Van Horn, D.; Khandelwal, A.; Nichols, J.; Reale, J.: IMPACTS OF URBAN STORMWATER RUNOFF ON DOWNSTREAM WATER QUALITY IN THE RIO GRANDE

14:30 **Tan, Z.**; Leung, L.; Li, H.; Tesfa, T.; Zhu, Q.; Huang, M.: SOIL EROSION CAUSES SUBSTANTIAL LOSS OF TERRESTRIAL ORGANIC CARBON AND NUTRIENTS FROM THE CONTERMINOUS UNITED STATES TO THE COASTAL ZONE

14:45 Bianucci, L.; **Balaguru, K.**; Smith, R.; Leung, L.; Moriarty, J.: CONTRIBUTION OF HURRICANE-INDUCED SEDIMENT RESUSPENSION TO COASTAL OXYGEN DYNAMICS

15:00 **Kadjeski, M.**; Fasching, C.; Xenopoulos, M.: SEASONALITY AND LANDSCAPE FACTORS DRIVE ORGANIC MATTER EXPORT IN STREAMS OF VARYING LAND USE

15:15 **Benk, S.**; Li, Y.; Roth, V.; Gleixner, G.: CHASING SURFACE INPUTS OF YOUNG DISSOLVED ORGANIC MATTER THROUGH A KARSTIC AQUIFER SYSTEM

WEDNESDAY

[1] REPRESENTS INVITED PRESENTATIONS

WEDNESDAY

SS041 THE NEXT GENERATION: UNDERGRADUATE RESEARCH IN PUERTO RICO AND THE US VIRGIN ISLANDS

Chair(s): Maria Barberena-Arias, Universidad del Turabo (mbarberena1@suagm.edu)
Pedro Maldonado, Universidad Metropolitana (um_pmaldonado@suagm.edu)
Lora Harris, U of MD Center for Environmental Science (harris@umces.edu)
James Pierson, U of MD Center for Environmental Science (jpierson@umces.edu)

Location: Room 204

14:00 **Luyando-Flusa, S.**; Ocasio Torres, M.; Crowl, T.: EFFECTS OF ENVIRONMENTAL VARIABLES ON SHRIMP REPRODUCTION IN A FRESHWATER SYSTEM IN PUERTO RICO

14:15 **Barberena, K.**;Rivera, R.; Diaz, K.; Barberena-Arias, M.: ABUNDANCE OF DIFFERENT SIZE CLASSES OF PLASTICS IN HIGH AND LOW ENERGY BEACHES

14:30 **Sanchez, J.**; Ruiz, C.; Toledo, C.: COULD RISES IN WATER TEMPERATURE PREDICTED BY CLIMATE CHANGE DECREASE CORAL REEF COMPLEXITY BY REDUCING BRANCHING CAPACITY OF ACROPORA CERVICORNIS?

14:45 **Valdes, D.**; Maldonado, P.: SMALL-SCALED LOW-COST PORTABLE SENSOR PLATFORM GEO-LOCATED

15:00 **Torres Rivera, A.**; Barberena-Arias, M.: ARTHROPOD DIVERSITY VARIATION DURING RED MANGROVE (RHIZOPHORA MANGLE) LEAVES DECOMPOSITION IN DRY AND WET SEASONS

15:15 **Del Valle, T.**; Barberena-Arias, M.: ABUNDANCE OF PLASTIC DEBRIS ON SEA TURTLE NESTING BEACHES ON THE NORTHERN REGION OF PUERTO RICO

15:45 **Medero, L.**; Schott, E.; Zhao, M.: GENETIC VARIATION OF A BLUE CRAB VIRUS AS A TOOL TO UNDERSTAND CRAB MOVEMENT

16:00 **Vidal Geraldino, P.**; Gomez Arias, S.; López, G.; Allen, M.: HYDROGRAPHY AT LAGUNA GRANDE COMPARED TO THE RED MANGROVE LAGOON

16:15 **Betancourt Gomez, N.**; Aquino-Pereira, A.; Torres Rivera, A.; Perez-Cruz, I.: ARTHROPODS AND CDOM ASSOCIATED TO MANGROVES IN LAGUNA GRANDE, FAJARDO, P.R

16:30 **Mercado, A.**; Vicente Ramos, K.; Alvarez-Rosario, J.; Rivera-Carrasquillo, Y.: PRIMARY PRODUCTIVITY AND COLOR DISSOLVED ORGANIC MATTER IN, LAGUNA GRANDE, A TROPICAL COASTAL LAGOON.

16:45 **Rivas Rivera, Y.**; Colón, F.; Rios Morales, A.; Torres-Figueroa, A.; Pierson, J.: ZOOPLANKTON GRAZING ON PHYTOPLANKTON IN THE BIOLUMINESCENT LAGOON AT LAGUNA GRANDE, FAJARDO, P.R.

17:00 **Irizarry, Y.**; del Valle, C.; Harris, L.: INVESTIGATING THE RELATIONSHIP BETWEEN COLORED DISSOLVED ORGANIC MATTER AND PYRODINIUM BAHAMENSE USING A BIOASSAY APPROACH

17:15 **Torres-Perez, J.**: US VIRGIN ISLANDS ECOLOGICAL FORECASTING AND WATER RESOURCES: A DEMONSTRATION OF COLLABORATIVE OPPORTUNITIES FOR CAPACITY–BUILDING OF CARIBBEAN-BASED STUDENTS

SS055 THE 2017 HURRICANE SEASON: CHALLENGES, INNOVATIONS, AND RESILIENCY IN FORMAL AND INFORMAL EDU

Chair(s): Michele Guannel, University of the Virgin Islands (michele.guannel@uvi.edu)
Howard Forbes, University of the Virgin Islands (howard.forbes@uvi.edu)
Jarvon Stout, University of the Virgin Islands (jarvon.stout@uvi.edu)

Location: Room 201

15:45 **Edwards, K.**; Forbes, Jr, H.: THROUGH THE MAGNIFYING GLASS: USING NATURAL DISASTERS TO EDUCATE ON WASTE IN THE UNITED STATES VIRGIN ISLANDS (USVI)

16:00 **Skerrit, C.**; White, T.; Monrose Mills, N.: RECOVERY WITH RENEWABLE ENERGY: PROJECT BASED LEARNING IN A MATHEMATICS CLASSROOM

16:15 **Salem Jubran, Y.**; Callwood, V.: CLASSROOM TEACHERS' PERCEPTION OF THE IMPACT OF HURRICANES IRMA AND MARIA ON STUDENT BEHAVIOR IN HIGH SCHOOL CLASSROOMS IN ST. THOMAS, UNITED STATES VIRGIN ISLANDS

16:30 **Charles, M.**: NORMALCY IN EDUCATION: HURRICANE RECOVERY AT A SCHOOL IN THE UNITED STATES VIRGIN ISLANDS

16:45 **Guannel, M.**; Daniel, I.: TRANSFORMATION OF A FRESHMAN-LEVEL UNIVERSITY SCIENCE COURSE TO INCORPORATE REAL-TIME HURRICANE RESPONSE, RECOVERY, AND RESILIENCY

17:00 Brown, L.; **Francis, G.**: THE IMPACTS OF HURRICANES IRMA AND MARIA IN THE VIRGIN ISLANDS AND THE RECOVERY PROCESS THROUGH ACTION IN SCIENCE

17:15 **Jobsis, P.**: VIERS BEFORE AND AFTER IRMA AND MARIA: PAST, PRESENT & FUTURE OF RESEARCH, EDUCATION AND ENGAGEMENT AT THE VIRGIN ISLANDS ENVIRONMENTAL RESOURCE STATION

SS062 MIXOTROPHIC PROTISTS: AN UNDERRATED MAJORITY IN MARINE AND FRESHWATER ECOSYSTEMS?

Chair(s): Mia Bengtsson, University of Greifswald (mia.bengtsson@uni-greifswald.de)
Robert Fischer, WasserCluster Lunz (robert.fischer@wcl.ac.at)
Robert Ptacnik, WasserCluster Lunz (robert.ptacnik@wcl.ac.at)
Stella Berger, IGB Berlin (berger@igb-berlin.de)
Jens Nejstgaard, IGB Berlin (nejstgaard@igb-berlin.de)
Susanne Wilken, University of Amsterdam (s.wilken@uva.nl)

Location: Room 103 B

15:45 **Leles, S.**; Polimene, L.; Bruggeman, J.; Blackford, J.; Mitra, A.; Flynn, K.: MIXOTROPHIC FUNCTIONAL DIVERSITY AND IMPLICATIONS FOR ECOSYSTEM FUNCTIONING

16:00 **Princiotta, S.**; Hendricks, S.; White, D.: CAN MIXOTROPHIC PROTISTS GRAZE THE BLOOM? TROPHIC INTERACTIONS BETWEEN A PHAGOTROPHIC PHYTOFLAGELLATE AND A TOXIN-PRODUCING CYANOBACTERIA

16:15 **Cohen, N.**; McIlvin, M.; Moran, D.; Held, N.; Saunders, J.; Hawco, N.; DiTullio, G.; Lamborg, C.; McCrow, J.; Dupont, C.; Allen, A.; Saito, M.: CONTRASTING METABOLIC SIGNATURES IN MIXOTROPHIC DINOFLAGELLATES ACROSS GEOCHEMICAL GRADIENTS IN THE CENTRAL PACIFIC OCEAN

16:30 **Jakobsen, H.**; Haraguchi, L.; Armengol, L.: SUCCESSION OF PLANKTONIC CILIATES IN A SHALLOW TEMPERATE ESTUARY: SPECIES DYNAMICS, TROPHY AND GRAZING

16:45 **Ross, B.**; Hallock, P.: DORMANCY, SURVIVAL, AND RECOVERY OF THE PHOTOSYMBIOTIC FORAMINIFERA AMPHISTEGINA GIBBOSA FOLLOWING LONG-TERM APHOTIC INCUBATION

17:00 **Sommaruga, R.**: MORE MIXOTROPHS IN OLIGOTROPHIC CLEAR MOUNTAIN LAKES, DURING THE ICE-COVERED PERIOD BUT ALSO IN GLACIER-FED TURBID LAKES

17:15 **Fischer, R.**; Ivankovic, M.; Nejstgaard, J.; Berger, S.; Grossart, H.; Bengtsson, M.; Vad, C.; Ptacnik, R.: MIXOTROPHS & CLIMATE CHANGE: WHAT THE FUTURE MIGHT HOLD

SS065 TURNING THE LIGHTS ON FOR DEEP-SEA ECOSYSTEMS IN THE CARIBBEAN, GULF OF MEXICO, AND US SE ATLANTIC

Chair(s): James Murphy, NOAA Office of Ocean Exploration and Research (jwmurphy@hawaii.edu)
Amanda Netburn, NOAA Office of Ocean Exploration and Research (amanda.netburn@noaa.gov)
Peter Etnoyer, NOAA Center for Coastal Monitoring and Assessment (peter.etnoyer@noaa.gov)
Shirley Pomponi, FAU - Harbor Branch Oceanographic Institute (SPomponi@fau.edu)
Caitlin Adams, NOAA Office of Ocean Exploration and Research (caitlin.adams@noaa.gov)
Kasey Cantwell, NOAA Office of Ocean Exploration and Research (kasey.cantwell@noaa.gov)

Location: Room 103 A

09:00 **Schwing, P.**; O'Malley, B.; Machain-Castillo, M.; Armenteros, M.; Martínez-Colón, M.; Hollander, D.: BENTHIC FORAMINIFERA BASELINE MEASUREMENTS AND INTER-REGIONAL COMPARISONS FOR THE DEEP GULF OF MEXICO

09:15 **Vohsen, S.**; Fisher, C.; Baums, I.: A SURVEY OF DEEP_SEA CORAL MICROBIOMES IN THE GULF OF MEXICO REVEALS POTENTIAL SYMBIOSES AND PROVIDES INSIGHT FOR MICROBES FOUND IN SHALLOW-WATER CORALS

09:30 Demopoulos, A.; **Bourque, J.**; Durkin, A.; Cordes, E.: THE INFLUENCE OF SEEP HABITATS ON SEDIMENT MACROFAUNAL BIODIVERSITY AND FUNCTIONAL TRAITS

09:45 **Amon, R.**; Herguera, J.; Herzka, S.; Sheinbaum, J.; Ochoa, J.; Candela, J.; Perez Brunius, P.; Hernandez Ayon, M.; Camacho Ibar, V.: INSIGHTS INTO THE DEEP CIRCULATION OF THE GULF OF MEXICO AND NORTHERN CARIBBEAN BASED ON DIC RADIOCARBON DEPTH PROFILES, HYDROGRAPHY AND VOLUME TRANSPORT ESTIMATES

10:00 **Jiménez Rosenberg, S.**; Arteaga, M.; Galindo-Sánchez, C.; Saavedra-Flores, A.; Robles-Flores, J.; Gasca, J.; Martínez, M.: FISH LARVAE DIVERSITY IN THE DEEP WATER REGION OF THE GULF OF MEXICO

10:15 **McKenzie, T.**; Russell, B.; Dalgleish, F.; Mike Twardoski, M.; Boswell, K.; Nayak, A.: MULTI-SCALE CHARACTERIZATION OF MESOPELAGIC ORGANISM DISTRIBUTIONS AND PATCHINESS IN THE NORTH CENTRAL GULF OF MEXICO

14:00 **Auscavitch, S.**; Lunden, J.; Demopoulos, A.; Quattrini, A.; Cordes, E.: DISTRIBUTION PATTERNS OF DEEP-WATER CORALS ON SEAMOUNTS IN THE ANEGADA PASSAGE: TOWARD A BETTER UNDERSTANDING OF DEEP-SEA CORAL BIOGEOGRAPHY IN THE CARIBBEAN

14:15 **Mienis, F.**; Hanz, U.; Lavaleye, M.; Van Haren, H.; Duineveld, G.: HYDRODYNAMIC AND FAUNAL CHARACTERISTICS OF THE DEEP FLANKS OF THE SABA BANK

14:30 **Wagner, D.**; Williams, S.; Auscavitch, S.; Sowers, D.; France, S.: OCEANO PROFUNDO 2018: EXPLORATION OF DEEP-SEA HABITATS OF PUERTO RICO AND THE U.S. VIRGIN ISLANDS ABOARD NOAA SHIP OKEANOS EXPLORER

14:45 **Messing, C.**; Syverson, V.; Veitch, M.; Baumiller, T.: ADVANCES IN UNDERSTANDING THE DEEP TROPICAL WESTERN ATLANTIC CRINOID FAUNA (ECHINODERMATA)

15:00 **Ford, M.**: OBSERVATIONS OF AN UNDESCRIBED CTENOPHORE FROM A 3,910 M DEPTH OFF PUERTO RICO

15:15 **Flood, R.**; Sautter, L.; Morrison, C.: ROV STUDIES OF ABYSSAL BED FORMS (FURROWS) ON THE BLAKE BAHAMA OUTER RIDGE

15:45 **Cordes, E.**; Gasbarro, R.; Davies, A.: DISCOVERY OF AN EXTENSIVE CORAL REEF ECOSYSTEM OFF OF THE US SOUTHEAST ATLANTIC COAST

16:00 **Morrison, C.**: PUTTING AN EXTENSIVE AND PREVIOUSLY UNDETECTED LOPHELIA REEF INTO CONTEXT: GENETIC CONNECTIVITY AMONG NORTHWESTERN ATLANTIC LOPHELIA POPULATIONS

16:15 **Lunden, J.**; Keller, A.; Demopoulos, A.; Joye, S.; Cordes, E.: REGIONAL OCEANOGRAPHIC INFLUENCES ON SPECIALIZED DEEP-SEA ECOSYSTEMS ALONG THE U.S. ATLANTIC MARGIN

16:30 **Demopoulos, A.**; McClain-Counts, J.; Bourque, J.; Prouty, N.; Smith, B.; Brooke, S.; Ross, S.; Ruppel, C.: STABLE ISOTOPIC INSIGHTS INTO BATHYMODIOLUS CHILDRESSI PHYSIOLOGY, UNDERLYING GEOCHEMISTRY AND FOOD-WEB LINKAGES AT TWO SEEPS IN THE U.S. ATLANTIC MARGIN

16:45 **Vokhshoori, N.**; McCarthy, M.; Close, H.; Demopoulos, A.; Prouty, N.: DETERMINING VARIABILITY OF FOOD RESOURCES TO CHEMOSYMBIOTIC MUSSELS ALONG ENVIRONMENTAL GRADIENTS AT METHANE SEEPS USING AMINO ACID NITROGEN ISOTOPES

17:00 **Close, H.**; Vokhshoori, N.; Doherty, S.; Demopoulos, A.; McCarthy, M.; Prouty, N.: NOVEL COMPOUND-SPECIFIC ISOTOPIC FINGERPRINTS OF METHANE METABOLISM AND DIETARY RELATIONSHIPS IN *BATHYMODIOLUS* AT SEAFLOOR COLD SEEPS

17:15 **Jiang, M.**; Pan, C.; Barbero, L.; Reed, J.; Salisbury, J.; VanZwieten, J.; Wanninkhof, R.: MODELING AND OBSERVATIONS OF ENVIRONMENTAL CONDITIONS OVER DEEP CORAL REEFS IN THE FLORIDA STRAITS

SS066 LARGE RIVERS OF THE WORLD AS PIPES, CHIMNEYS AND REACTORS

Chair(s): Francois Guillemette, Université du Québec à Trois-Rivières (francois.guillemette3@uqtr.ca)
Jean-François Lapierre, University of Quebec at Montreal (jean-francois.lapierre.1@umontreal.ca)
Thomas Bianchi, University of Florida (tbianchi@ufl.edu)
Suzanne Tank, University of Alberta (suzanne.tank@ualberta.ca)

Location: Room 202

14:00 **Striegl, R.**; Foks, S.; Allen, G.: CARBON DIOXIDE AND METHANE EXCHANGE WITH RIVERS AND STREAMS IN THE UPPER MISSISSIPPI RIVER NETWORK[T]

14:30 **Duvert, C.**; Bossa, M.; Tyler, K.; Munksgaard, N.; Wynn, J.; Bird, M.; Setterfield, S.; Hutley, L.: GROUNDWATER-DERIVED CARBON INPUTS ENHANCE FLUVIAL CO2 EVASION IN A LARGE TROPICAL RIVER

14:45 **Fink-Mercier, C.**; Lapierre, J.; Amyot, M.; del Giorgio, P.: THE COUPLING BETWEEN DOM AND HG EXPORT IN LARGE BOREAL RIVERS ALONG A LATITUDINAL GRADIENT IN THE JAMES BAY REGION OF QUEBEC

[T] REPRESENTS INVITED PRESENTATIONS

15:00 **Guillemette, F.**; Grater, E.; Lapierre, J.; Maisonneuve, P.; del Giorgio, P.; Cabana, G.: SQUEEZING THE PIPE: ARCHIPELAGOS AS ACTIVE SITES OF CARBON PROCESSING ALONG A LARGE RIVER CONTINUUM

15:15 **Gerardin, M.**; del Giorgio, P.: FACTORS INFLUENCING THE BALANCE BETWEEN C EMISSION AND C TRANSPORT TO THE OCEAN BY LARGE BOREAL RIVERS IN NORTHERN QUEBEC

15:45 **Stubbins, A.**; Wagner, S.; Codden, C.; Martinez, J.; Raymond, P.; Payet, J.; Crump, B.: MOLECULAR INSIGHTS IN TO THE PLASTICITY OF MICROBIAL ORGANIC MATTER UTILIZATION[T]

16:00 Guillemette, F.; **Grater, E.**; del Giorgio, P.: A BACTERIAL METABOLIC JOURNEY THROUGH THE ST. LAWRENCE RIVERSCAPE

16:15 **Twiss, M.**; Brahmstedt, E.; Guillemette, F.; Cabana, G.: PHYTOPLANKTON DEVELOPMENT ALONG A 600 KM THALWEG OF THE ST. LAWRENCE RIVER

16:30 **Rose, V.**; Rollwagen-Bollens, G.; Bollens, S.; Zimmerman, J.: DECADAL-SCALE CHANGES IN LOWER COLUMBIA RIVER PHYTOPLANKTON

16:45 **Bergbusch, N.**; Quiñones-Rivera, Z.; Haig, H.; Swarbrick, V.; Hayes, N.; Leavitt, P.: UNINTENDED EFFECTS OF NITROGEN-RICH DISCHARGE FROM A TERTIARY WASTEWATER TREATMENT PLANT ON PRIMARY PRODUCERS IN PHOSPHORUS-RICH STREAMS OF THE NORTHERN GREAT PLAINS

17:00 **Iannino, A.**; Vosshage, A.; Weitere, M.; Fink, P.: HIGH NUTRIENT AVAILABILITY LEADS TO WEAKER TOP-DOWN CONTROL OF STREAM PERIPHYTON: COMPENSATORY FEEDING IN *ANCYLUS FLUVIATILIS*

17:15 **Yoshimura, M.**: CHANGE OF RADIOACTIVE CS CONCENTRATION IN FRESHWATER BIOTA IMPACTED BY THE FUKUSHIMA NUCLEAR POWER PLANT ACCIDENT

SS070 ENGAGING UNDERREPRESENTED STUDENTS IN OCEAN SCIENCE TO PROMOTE A DIVERSE INCLUSIVE WORKFORCE

Chair(s): Andrea Johnson, National Science Foundation (andjohns@nsf.gov)
Todd Christenson, NOAA Office of Education (todd.christenson@noaa.gov)
Corey Garza, California State University, Monterey Bay (cogarza@csumb.edu)

Location: Room 204

09:00 Barral, A.; Simmons, R.; **Bowman, J.**; DeForce, E.: ENGAGING NONTRADITIONAL STUDENTS BY CURE-ING MICROBES ON OCEAN PLASTICS

09:15 **Grimes, K.**; Brandt, M.; Bucklin, C.; Jones, N.; Medina, M.; Forbes, Jr., H.; Guannel, M.: SUPPORTING EMERGING AQUATIC SCIENTISTS (SEAS) YOUR TOMORROW: INCREASING DIVERISTY & INCLUSION IN THE MARINE SCIENCES IN THE U.S. VIRGIN ISLANDS

09:30 **Arnold, R.**; Peacock, M.: TACKLING CULTURALLY-RELEVANT RESEARCH AT NORTHWEST INDIAN COLLEGE: PREPARING THE NEXT GENERATION OF INDIGENOUS RESEARCHERS IN GENOMICS AND THE OCEAN SCIENCES

09:45 **Schmidt, W.**; Jimenez, M.: BUILD IT AND THEY WILL COME: INTRODUCING ENGINEERING STUDENTS TO OCEANOGRAPHY, PART 2

10:00 **Muller-Parker, G.**: FEDERAL FELLOWSHIPS AND GRADUATE EDUCATION: A FOCUS ON NSF PROGRAMS [T]

SS073 THE CHALLENGE OF CORAL REEF REHABILITATION IN THE CONTEXT OF CLIMATE CHANGE

Chair(s): Edwin Hernandez-Delgado, Sociedad Ambiente Marino (edwin.hernandezdelgado@gmail.com)
Samuel E. Suleiman-Ramos, Sociedad Ambiente Marino (samuelsuleiman@gmail.com)
Michael Nemeth, NOAA Restoration Center (michael.nemeth@noaa.gov)
Nilda Jimenez, Department of Natural Environmental Resources (njimenez@drna.pr.gov)

Location: Room 208 C

15:45 **Hernandez Delgado, E.**; Canals-Silander, M.; Suleiman Ramos, S.; Cuevas, E.: DEMOGRAPHIC AND OCEANOGRAPHIC MODELING TOOLS IN SUPPORT CORAL REEF REHABILITATION IN THE ANTHROPOCENE

16:00 **Canals Silander, M.**; Hernandez Delgado, E.: GEOMETRIC AND HYDRODYNAMIC OPTIMIZATION OF CORAL REEF RESTORATION PROJECTS TO ENHANCE WAVE POWER REDUCTION IN FRINGING REEFS

16:15 **Hsieh, Y.**; Sturaro, N.; Wang, P.; Denis, V.: CORAL TROPHIC PLASTICITY ACROSS CONTRASTED HABITATS

16:30 **Weinnig, A.**; Hallaj, A.; Cordes, E.: INFLUENCE OF TEMPERATURE AND PH ON THE PHENOTYPIC AND TRANSCRIPTOMIC RESPONSE OF A COLD-WATER CORAL (LOPHELIA PERTUSA) TO OIL AND DISPERSANT EXPOSURE

17:00 **Gomez-Andujar, N.**; Hernandez Delgado, E.: PARTICIPATORY MAPPING OF CORAL REEF VULNERABILITY: A TOOL TO SUPPORT ECOSYSTEM-BASED MANAGEMENT IN A TOURISM-IMPACTED CARIBBEAN ISLAND

17:15 **Suleiman Ramos, S.**; Hernandez Delgado, E.: CORAL REEF REHABILITATION IN THE ANTHROPOCENE: MOVING FROM CORAL TO LANDSCAPE AND ECOLOGICAL PROCESSES

212 papers

[T] REPRESENTS TUTORIAL PRESENTATIONS

THURSDAY ORALS

AS001 SUCCESS THROUGH SCIENCE: USING LIMNOLOGY AND OCEANOGRAPHY TO TACKLE DIFFICULT MANAGEMENT QUESTIONS

Chair(s): Lisa Campbell, Texas A&M University (lisacampbell@tamu.edu)
Bill McDowell, University of New Hampshire (bill.mcdowell@unh.edu)

Location: Room 204

09:15 **Barouillet, C.**; Meyer-Jacob, C.; Mushet, G.; Hennessy, S.; Bertin, A.; Cumming, B.: DISSOLVED ORGANIC CARBON CONCENTRATIONS EXERT A STRONGER CONTROL ON THE CLADOCERAN COMMUNITY COMPOSITION OF BOREAL LAKES THAN WARMING

09:30 **McCarthy, M.**; Boedecker, A.; Chaffin, J.; Gardner, W.; Hampel, J.; Hoffman, D.; Xu, H.; Zhang, L.; Zhu, G.; Newell, S.: CHEMICALLY REDUCED NITROGEN FORMS AS DRIVERS OF NON-N-FIXING CYANOBACTERIA BLOOMS IN LARGE LAKES ERIE AND TAIHU: MANAGEMENT IMPLICATIONS

09:45 **Owca, T.**; Kay, M.; Faber, J.; Wiklund, J.; Wolfe, B.; Hall, R.: USE OF PRE-INDUSTRIAL BASELINES TO MONITOR FOR SOURCES AND PATHWAYS OF METALS IN SURFACE SEDIMENT OF FLOODPLAIN LAKES IN THE PEACE-ATHABASCA DELTA (ALBERTA, CANADA)

10:00 Stow, C.; **Evans, M.**; Gibbons, K.: SCIENCE - MANAGEMENT ENGAGEMENT FOR LAKE ERIE NUTRIENT TARGETS

10:15 **Aguiar, D.**; Wiegner, T.; Abaya, L.; Stewart, J.; Burns, J.; Beets, J.; Couch, C.; Colbert, S.; Nelson, C.: SEWAGE POLLUTION EFFECTS ON A HAWAIIAN CORAL REEF

14:00 **Bednarsek, N.**; Feely, R.; Hunt, B.; Kessouri, F.; McLaughlin, K.: SYNTHESIS OF THRESHOLDS OF OCEAN ACIDIFICATION EFFECTS ON PELAGIC MOLLUSCKS

14:15 **Meunier, C.**; Groß, E.; Köring, M.; Duarte Moreno, H.; Tremblay, N.; Boersma, M.: GLOBAL CHANGE VULNERABILITY OF NORTH SEA PLANKTON AND ASSOCIATED ECOSYSTEM SERVICES: TOWARDS OPTIMIZED MANAGEMENT

14:30 **Gordon, R.**; Santana, N.: ASSESSING THE IMPACT OF THE LIONFISH THROUGH EDUCATION AND OUTREACH ACTIVITIES IN THE UNITED STATES VIRGIN ISLANDS

14:45 Kessouri, F.; **Sutula, M.**; McWilliams, J.; Bianchi, D.; Bednarsek, N.; McLaughlin, K.; Howard, E.; Deutsch, C.; Feely, R.; Weisberg, S.: OCEAN MODELING SUPPORTS MANAGEMENT CONVERSATIONS ON THE IMPACT OF LOCAL POLLUTION SOURCES ON ACIDIFICATION AND DEOXYGENATION IN SOUTHERN CALIFORNIA BIGHT

15:00 **Balser, A.**; Gaja, S.; Sanzone, D.; Murdoch, W.: SYSTEM-ORIENTED OCEAN SCIENCES IN THE GULF OF MEXICO: ADAPTING TO ENERGY POLICY REFORM

15:15 **Czub, M.**; Kotwicki, L.; Popiel, S.; Nawala, J.; Brzeziński, T.; Maszczyk, P.; Beldowski, J.: ECOLOGICAL IMPACTS OF SEA DUMPED CHEMICAL WARFARE AGENTS ON AQUATIC ECOSYSTEMS

AS005 EXTREME EVENTS

Chair(s): Elizabeth Minor, University of Minnesota Duluth (eminor@d.umn.edu)
Carmen Aguilar, University of Wisconsin Madison (aguilar@uwm.edu)
Bill McDowell, University of New Hampshire (bill.mcdowell@unh.edu)

Location: Room 209 A/B

14:00 **McDowell, W.**: DISTURBANCE AND RESPONSE: UNDERSTANDING THE EFFECTS OF EXTREME EVENTS ON AQUATIC BIOGEOCHEMISTRY [T]

14:30 **Corman, J.**; Ramos, J.; Elser, J.: SPATIAL AND TEMPORAL VARIABILITY OF PHYSICOCHEMICAL CHARACTERISTICS IN THE HARDWATER AQUATIC ECOSYSTEMS OF CUATRO CIENEGAS, MEXICO

14:45 **Cuhel, R.**; Scheib, A.; Aguilar, C.; Evans, C.: MATCH-MISMATCH: EL NIÑO AND A COINCIDENT DERECHO STIMULATE YELLOW PERCH RECRUITMENT IN A PREVIOUSLY DECIMATED LAKE MICHIGAN FISHERY

15:00 Cooney, E.; **Minor, E.**: RAIN EVENT EFFECTS ON THE BIOGEOCHEMISTRY OF WESTERN LAKE SUPERIOR

15:15 **Fulweiler, R.**; Maguire, T.; Heiss, E.; Foster, S.; Fields, L.: EXTREME WARMING DECREASES CONTINENTAL SHELF NITROGEN REMOVAL CAPACITY

15:45 **Aijaz, S.**; Jeffrey, K.; Zieger, S.; Greenslade, D.: TROPICAL CYCLONE WIND MODELLING FOR NORTHWEST AUSTRALIA

16:00 **Pearson, R.**; Jinks, K.; Brown, C.; Schlacher, T.; Olds, A.; Connolly, R.: FUNCTIONAL MARINE RESPONSES TO CLIMATE STRESSORS ALONG A CONNECTIVITY GRADIENT

16:15 **Toledo-Hernandez, C.**; Hernandez-Delgado, E.; Ruiz, C.; Gomez-Andujar, N.; Medina-Muñiz, J.; Canals-Silander, M.; Suleiman-Ramos, S.: HURRICANE IMPACTS AND THE RESILIENCE OF INVASIVE AND NUISANCE SPECIES ON SEAGRASSES: A CASE STUDY FROM PUERTO RICO

16:30 **Jerris, K.**; Turner, T.: THE EFFECT OF HURRICANES IRMA AND MARIA ON THE INVASION OF HALOPHILA STIPULACEA

16:45 **Ortiz-Zayas, J.**; Santiago, J.; Figueroa, Y.; Torres, D.; Jordan, T.: TEACHING HYDROLOGY AFTER THE SHOCK: EXPERIENTIAL LEARNING OF HURRICANE MARIA IN PUERTO RICO

17:00 **Richards, R.**: IS THE SPECIFIC CONDUCTANCE OF GROUNDWATER IN PUERTO RICO RISING?

CS008 STABLE ISOTOPES

Chair(s): Patrick Fink, University of Cologne (patrick.fink@uni-koeln.de)
Björn Wissel, University of Regina (bjoern.wissel@uregina.ca)

Location: Room 209 C

15:45 **Weber, F.**; Zaliznyak, T.; Edgcomb, V.; Taylor, G.: STABLE ISOTOPE PROBING AND CONFOCAL RAMAN MICROSPECTROMETRY: A PROMISING TOOLKIT TO STUDY PREDATOR-PREY RELATIONSHIPS AMONG PROTISTS AND PROKARYOTES

16:00 **Yohannes, E.**; Kleiber, R.; Rothhaupt, K.: TEMPORAL CHANGES IN THE CONTRIBUTION OF METHANE-OXIDIZING BACTERIA (MOB) TO CONSUMER BIOMASS DETERMINED USING 13C-LABELLED METHANE, STABLE ISOTOPES AND DNA

16:15 **Bramburger, A.**; Reavie, E.; Sheik, C.: METAGENOMIC AND STABLE ISOTOPE CHARACTERIZATION OF GREAT LAKES PHYTOPLANKTON COMMUNITIES

16:30 **Smith, S.**; Sasai, Y.; Yoshikawa, C.: MODELING SEASONAL AND INTER-ANNUAL VARIABILITY OF TROPHIC TRANSFER AND 15N STABLE ISOTOPE ENRICHMENT WITHIN THE PLANKTONIC FOOD CHAIN

16:45 **Gugele, S.**; Yohannes, E.; Brinker, A.: TROPHIC CLASSIFICATION OF INVASIVE STICKLEBACKS IN LAKE CONSTANCE USING STABLE ISOTOPES

17:00 **Remmer, C.**; Owca, T.; Neary, L.; Wiklund, J.; Kay, M.; Wolfe, B.; Hall, R.: DELINEATING EXTENT AND MAGNITUDE OF RIVER FLOODING IN A NORTHERN FLOODPLAIN LANDSCAPE USING WATER ISOTOPE TRACERS

17:15 **Fackrell, J.**; Fackrell, J.; Richardson, C.; Paytan, A.; Kendall, C.; Kraus, T.: STABLE ISOTOPE VALUES OF C, N, P, AND S COMPOUNDS IN TREATED WASTEWATER EFFLUENT FROM FACILITIES OF VARYING CAPACITIES AND TREATMENT PRACTICES

[T] REPRESENTS INVITED PRESENTATIONS

CS011 PHOSPHORUS BIOGEOCHEMISTRY AND CYCLING

Chair(s): Dennis Swaney, Cornell University (dps1@cornell.edu)
Michelle McCrackin, Stockholm University (michelle.mccrackin@su.se)

Location: Room 103 A

15:45 **Sharpley, A.**: AGRICULTURE, PHOSPHORUS, & FOOD-ENERGY-WATER SECURITY NEXUS: A CONUNDRUM OF DEFICIENCY & EXCESS[TT]

16:15 **Basu, N.**; Van Meter, K.: BEYOND THE MASS BALANCE: A PROCESS BASED APPROACH TO MODELLING LEGACY PHOSPHORUS DYNAMICS[I]

16:30 **Van Meter, K.**; Tenkouano, G.; Parsons, C.; Hall, R.; Basu, N.: RESERVOIR PHOSPHORUS LEGACIES: RECONSTRUCTING LONG-TERM NUTRIENT DYNAMICS USING MULTI-PROXY PALEOLIMNOLOGICAL ANALYSES

16:45 **Jilbert, T.**; Slomp, C.; Egger, M.; Behrends, T.; Wolthers, M.; van Helmond, N.; Gustafsson, B.; Dijkstra, N.; Kraal, P.: IMPACTS OF A CHANGING GLOBAL PHOSPHORUS CYCLE ON COASTAL OCEAN DEOXYGENATION[I]

17:00 **Albright, E.**; Fleck, R.; Shingai, Q.; Wilkinson, G.: IDENTIFYING POTENTIAL HOTSPOTS OF INTERNAL PHOSPHORUS LOADING IN SHALLOW LAKES

17:15 **Muller-Karulis, B.**; McCrackin, M.; Swaney, D.; Howarth, R.: LEGACY PHOSPHORUS IN THE BALTIC SEA DRAINAGE BASIN – HOW LONG WILL PAST INPUTS AFFECT NUTRIENT LOADS?

CS017 HARMFUL BLOOMS

Chair(s): Ted Harris, Kansas Biological Survey (ted.daniel.harris@gmail.com)

Location: Room 102 A/B/C

09:00 **Harris, T.**: GLOBAL MICROCYSTIN AGGREGATION (GMA) GLEON PROJECT

09:15 **Crawford, J.**; FROM HIGHLY EUTROPHIC TO MESOTROPHIC. THE SUCCESS STORY OF DEER CREEK RESERVOIR

09:30 **Burnet, S.**; Wilhelm, F.: VERTICAL DISTRIBUTION OF CYANOBACTERIA TOXINS IN WILLOW CREEK RESERVOIR, OR.

09:45 **Lange, J.**; Demir, F.; Huesgen, P.; Baumann, U.; von Elert, E.; Pichlo, C.: CHARACTERIZATION OF A DIGESTIVE PROTEASE FROM DAPHNIA: A POSSIBLE ROLE IN SUSCEPTIBILITY TO TOXIC CYANOBACTERIA

10:00 **Kashian, D.**; Boegehold, A.; Gopalakrishnan, K.; Johnson, N.: CAN HARMFUL ALGAL BLOOMS INHIBIT QUAGGA MUSSEL REPRODUCTION?

10:15 **Younan, L.**: CHARACTERIZING ALGAL BLOOMS USING REAL-TIME FLUORESCENCE MONITORING

14:00 **Sterner, R.**; Lafrancois, B.; Reinl, K.; Brovold, S.: CYANOBACTERIA BLOOMS TURN LAKE SUPERIOR GREEN

14:15 **Heathcote, A.**; Edlund, M.; Engstrom, D.: LAKE PHYSICS AND HABS: INTERNAL NUTRIENT LOADING DELIVERS TOXIC BLOOMS

14:30 **Reinl, K.**; Sterner, R.; Lafrancois, B.: POTENTIAL SOURCES OF CYANOBACTERIA BLOOMS TO THE APOSTLE ISLANDS

14:45 **Wagner, N.**; Osburn, F.; Wang, J.; Boedecker, A.; Herrera, M.; Scott, T.: ECOPHYSIOLOGY OF NITROGEN LIMITATION IN A DIAZOTROPHIC CYANOBACTERIA

15:00 **Leung, T.**; Swanner, E.: MULTI-WAVELENGTH FLUORESCENCE: A RAPID METHOD TO TRACK HARMFUL ALGAL BLOOMS IN IOWA'S AGRICULTURALLY IMPACTED LAKES

15:15 **Larson, J.**; Costello, D.; Evans, M.; Stoll, J.; Fitzgibbon, A.; Bailey, S.; Stelzer, E.; Loftin, K.; Givens, C.; Duris, J.; Lenaker, P.; Fogarty, L.: NUTRIENT AND TRACE METAL CO-LIMITATION OF CYANOBACTERIAL BLOOMS IN THE GREAT LAKES

15:45 **Greenfield, D.**; Coyne, K.: SYNTHESIZING A MULTI-YEAR COMPARISON OF SANDWICH HYBRIDIZATION ASSAY AND QUANTITATIVE PCR FOR HETEROSIGMA AKASHIWO: IMPLICATIONS FOR BLOOM MANAGEMENT

16:00 **Wu, Z.**; Luo, H.; Yu, L.; Lee, W.; Lin, S.: CHARACTERIZING TOXIN- AND NON-TOXIN-PRODUCING STRAINS OF GAMBIERDISCUS BALECHII USING COMPARATIVE TRANSCRIPTOMICS

16:15 **Harley, J.**; Lanphier, K.; Kennedy, E.; Whitehead, C.; Bidlack, A.: MACHINE LEARNING APPROACHES TO PREDICT PARALYTIC SHELLFISH TOXIN EVENTS IN SOUTHEAST ALASKA USING ENVIRONMENTAL VARIABLES

16:30 **Ruiz-Cooley, R.**; Anderson, C.; Kudela, R.: 15N FROM AMINO ACIDS REVEAL TWO ROUTES OF DOMOIC ACID TRANSFER IN MONTEREY BAY DURING THE EXTREME 2015 TOXIC EVENT

16:45 Gravinese, P.; **Saso, E.**; Lovko, V.; Pierce, R.: LETHAL AND SUBLETHAL EFFECTS OF RED TIDE (KARENIA BREVIS) EXPOSURE ON STAGE-1 LARVAE OF THE FLORIDA STONE CRAB MENIPPE MERCENARIA

17:00 **Tominack, S.**; Coffey, K.; Wetz, M.: TRENDS IN THE FREQUENCY AND DURATION OF RED TIDE BLOOMS IN SOUTH TEXAS ESTUARIES

17:15 **Cira, E.**; Wetz, M.: SPATIAL-TEMPORAL VARIABILITY IN AUREOMBRA LAGUNENSIS ("BROWN TIDE") BIOMASS IN BAFFIN BAY, TEXAS: RELATIONSHIPS WITH ENVIRONMENTAL DRIVERS

CS019 REGIME SHIFTS

Chair(s): Grace Wilkinson, Iowa State University (wilkinso@iastate.edu)

Location: Room 204

15:45 **Hansen, C.**; Wilkinson, G.; Burian, S.: DEVELOPING AND IMPLEMENTING AN EARLY WARNING SYSTEM FOR HABS IN UTAH LAKE

16:00 **Ortiz, D.**; Wilkinson, G.: DETECTING EARLY WARNINGS OF HARMFUL ALGAL BLOOMS IN SHALLOW LAKES

16:15 **Buelo, C.**; Pace, M.; Carpenter, S.: TIME VS. SPACE: COMPARING EARLY WARNING INDICATORS OF ALGAL BLOOMS

16:30 **Wachnicka, A.**; Browder, J.; Frankovich, T.; Wingard, L.; Louda, W.: RESILIENCE OF SOUTH FLORIDA ESTUARINE SYSTEMS TO CLIMATIC AND ANTHROPOGENIC DISTURBANCES

16:45 **Gao, Y.**; Svec, R.: STABLE ISOTOPIC RECORDS OF PACIFIC HALIBUT OTOLITHS IN DETECTION OF REGIME SHIFTS

17:00 **Silva, L.**; Calleja, M.; Ivetic, S.; Roth, F.; Carvalho, S.; G. Morán, X.: EFFECT OF THE SHIFT FROM CORAL- TO ALGAL-DOMINATED DISSOLVED ORGANIC MATTER ON TROPICAL HETEROTROPHIC BACTERIOPLANKTON IN THE CENTRAL RED SEA

17:15 **Schmitt, R.**; Holbrook, S.; Davis, S.; Adam, T.; Brooks, A.: EXPERIMENTAL SUPPORT FOR MACROALGAE AS AN ALTERNATIVE BASIN OF ATTRACTION ON CORAL REEFS

THURSDAY

[T] REPRESENTS TUTORIAL PRESENTATIONS

CS033 CORAL REEF ECOSYSTEMS

Chair(s): Justin Baumann, University of North Carolina at Chapel Hill (baumannj@live.unc.edu)
Travis Courtney, Scripps Institution of Oceanography (traviscourtney@gmail.com)

Location: Room 104

15:45 **Eakin, C.**; Skirving, W.; Heron, S.; Liu, G.; Hughes, T.; Donner, S.; Geiger, E.; De La Cour, J.; Gomez, A.; Marsh, B.; Devotta, D.: THE 2014 GLOBAL CORAL BLEACHING EVENT AND GLOBAL CORAL BLEACHING IN THE ANTHROPOCENE

16:15 **Tierney, C.**; Sudek, M.: ANTHROPOGENIC INFLUENCE ON CORAL REEF HEALTH IN AMERICAN SAMOA WATERSHEDS

16:30 **Begin, C.**; Dahl, A.; McLuckie, S.; Stutzman, N.; Larson, R.; Brooks, G.: VARIATION IN REEF COMMUNITIES IN SAINT LUCIA WITH PROTECTION STATUS AND SEDIMENTATION RATE

16:45 **Sawall, Y.**; Miller, A.; Smith, K.; Flesher, D.; Hochberg, E.: LIGHT-USE-EFFICIENCY OF REEF CORALS AND ALGAE AND ITS IMPLICATION FOR REMOTE SENSING

17:00 **Peltier, S.**; Hochberg, E.: CHARACTERIZING DIFFUSE ATTENUATION OF REEF AND ADJACENT WATERS IN BERMUDA AND HAWAII

17:15 **Hochberg, E.**: A NEW PERSPECTIVE ON CORAL REEFS: CORAL REEF AIRBORNE LABORATORY

SS018 METABOLIC DIVERSITY IN MARINE BIOGEOCHEMICAL CYCLES IN PRESENT AND FUTURE OCEAN

Chair(s): Gerhard J. Herndl, University of Vienna, Austria and NIOZ (gerhard.herndl@univie.ac.at)
Eva Sintes, Instituto Español de Oceanografia (eva.sintes@ieo.es)
Federico Baltar, University of Vienna (federico.baltar@univie.ac.at)
Thomas Reinthaler, University of Vienna (thomas.reinthaler@univie.ac.at)

Location: Room 201

09:00 Saw, J.; Nunoura, T.; Stepanauskas, R.; Longnecker, K.; Kujawinski, E.; Landry, Z.; Carlson, C.; **Giovannoni, S.**: PANGENOMICS REVEAL DIVERSIFICATION OF ENZYME FAMILIES AND NICHE SPECIALIZATION IN GLOBALLY ABUNDANT DEEP-OCEAN BACTERIA

09:15 **Mena, C.**; Reglero, P.; Balbín, R.; Martín, M.; Sintes, E.: SPATIAL AND TEMPORAL DYNAMICS OF TOTAL AND ACTIVELY GROWING PROKARYOTIC COMMUNITIES IN THE WESTERN MEDITERRANEAN SEA

09:30 **Bunse, C.**; Baltar, F.; Israelsson, S.; Legrand, C.; Lindh, M.; Lundin, D.; Muthusamy, S.; Sjöstedt, J.; Pinhassi, J.: SEASONAL BALTIC SEA BACTERIOPLANKTON STOCKS, ACTIVITIES AND DIVERSITY

09:45 **Mocaer, P.**; Urvoy, M.; Thomas, F.; Le Gall, S.; Vernet, M.; Hassler, C.; Baudoux, A.: IMPACT OF VIRUSES ON THE BIOAVAILABILITY OF MARINE BACTERIAL EXOPOLYSACCHARIDES

10:00 **Frank, A.**; Kido Soule, M.; Longnecker, K.; Kujawinski, E.: THE MODE OF MORTALITY DETERMINES THE COMPOSITION OF PHYTOPLANKTON-DERIVED DOM

10:15 **Ferrer-González, F.**; Moran, M.: DIATOM METABOLITES SUPPORTING HETEROTROPHIC BACTERIAL GROWTH IN THE SURFACE OCEAN

14:00 Longnecker, K.; Becker, J.; Braakman, R.; Dooley, K.; **Kido Soule, M.**; Chisholm, S.; Kujawinski, E.: *PROCHLOROCOCCUS*' IMPACT ON MARINE BIOGEOCHEMICAL CYCLES FROM METABOLOMICS AND GENOMIC PREDICTIONS

14:15 **McLean, C.**; Swarr, G.; Longnecker, K.; Kido-Soule, M.; Haley, S.; Dyhrman, S.; Kujawinski, E.: HARMFUL ALGAL BLOOM FORMING RAPHIDOPHYTE HETEROSIGMA AKASHIWO SHOWS DISTINCT PHENOTYPES UNDER NUTRIENT LIMITATION

14:30 **Van Oostende, N.**; Carroll, J.; Ward, B.: APPLICATION OF GENOME-BASED METHODS FOR GROWTH RATE ESTIMATION TO IN SITU CYANOBACTERIAL POPULATIONS

14:45 **Laurenceau, R.**; Bliem, C.; Chisholm, S.: MOBILIZATION OF GENOMIC ISLANDS IN THE MARINE CYANOBACTERIUM PROCHLOROCOCCUS

15:00 **Wang, J.**; Coles, V.; Kelly, T.; Mason, O.; Shropshire, T.; Stukel, M.: OIL-MICROBE INTERACTIONS IN THE GULF OF MEXICO (GOM) MODELED WITH THE GENOME-BASED EMERGENT OCEAN MICROBIAL ECOSYSTEM MODEL

15:15 **Widner, B.**; Kido Soule, M.; Ferrer-Gonzalez, F.; Moran, M.; Kujawinski, E.: A NOVEL METHOD FOR QUANTIFICATION OF POLAR DISSOLVED METABOLITES IN SEAWATER

15:45 **Granzow, B.**; Sosa, O.; Cahir, C.; Karl, D.; Repeta, D.: A FLUORESCENT CHEMICAL ASSAY FOR C-P LYASE ACTIVITY ANALYSIS

16:00 **Heal, K.**; Carlson, L.; Durham, B.; Groussman, R.; Armbrust, E.; Ingalls, A.: ENVIRONMENTAL METABOLOMICS ACROSS NATURAL OCEANOGRAPHIC GRADIENTS EXPOSE UNDERAPPRECIATED COMPOUNDS ENGAGED IN THE MULTIPLE ELEMENTAL CYCLES

16:15 **Lanpher, K.**; Popendorf, K.: DEPTH DEPENDENT TRENDS IN ENVIRONMENTAL METABOLIC ENERGY POTENTIAL

16:30 **Balmonte, J.**; Simon, M.; Giebel, H.; Arnosti, C.: LATITUDINAL AND DEPTH GRADIENTS OF MICROBIAL ENZYME ACTIVITIES FROM THE SOUTH PACIFIC TO THE BERING SEA

16:45 **Kessler, A.**; Cook, P.; Greening, C.: FERMENTATION DOMINATES MICROBIAL METABOLISM IN COASTAL PERMEABLE SANDS

17:00 **Yin, H.**; Zhu, Q.; Aller, J.; Aller, R.: CABLE BACTERIA ACTIVITY IN MANGANESE AND IRON-DEPLETED CARBONATE DEPOSITS

17:15 **Johnson, W.**; Kelly, E.; Ellis, G.; Leary, D.; Vora, G.: OSMOLYTE PRODUCTION IN THE SALT MARSH BACTERIUM VIBRIO NATRIEGENS

SS020 LARGE LAKES OF THE WORLD: DETECTING CHANGES USING LONG-TERM MONITORING

Chair(s): Lyubov Burlakova, SUNY College at Buffalo (burlakle@buffalostate.edu)
Alexander Karatayev, SUNY College at Buffalo (karataay@buffalostate.edu)
Lars Rudstam, Cornell University (lgr1@cornell.edu)
Elizabeth Hinchey Molloy, US EPA, Great Lakes National Program Office (hinchey.elizabeth@epa.gov)

Location: Room 208 C

09:00 **Barbiero, R.**; Lesht, B.; Burlakova, L.; Rudstam, L.; Watkins, J.; Karatayev, A.; Reavie, E.; Hinchey, E.: THIRTY FIVE YEARS OF EPA MONITORING OF THE LOWER FOOD WEB IN THE LAURENTIAN GREAT LAKES: AN OVERVIEW OF SYSTEM-WIDE CHANGES[T]

09:15 **Karatayev, A.**; Burlakova, L.; Mehler, K.; Rowe, M.; Nalepa, T.; Elgin, A.: LAKE MORPHOMETRY DETERMINES DREISSENA INVASION DYNAMICS

09:30 **Aguilar, C.**; Cuhel, R.: TIME SERIES OF BENTHIC NITROGEN EXCRETION BY THE INVASIVE QUAGGA MUSSEL (*DREISSENA BUGENSIS*) IN NEARSHORE-OFFSHORE GRADIENTS IN LAKE MICHIGAN 2006-2018

[T] REPRESENTS INVITED PRESENTATIONS

THURSDAY

09:45 **Burlakova, L.**; Karatayev, A.; Daniel, S.; Barbiero, R.; Mehler, K.; Hinchey, E.; Karatayev, V.: DECADAL TRENDS IN BENTHIC COMMUNITY OF THE LAURENTIAN GREAT LAKES REVEALED BY LONG-TERM MONITORING

10:00 **Xenopoulos, M.**; Young, J.; Lee, K.: DETECTING ECOLOGICAL CHANGES IN LAKE SIMCOE USING LONG-TERM MONITORING

10:15 **Collingsworth, P.**; Winter, C.; Osantowski, E.; Kocovsky, P.; Peterson, P.; Song, Q.: TRENDS IN TOTAL PHOSPHORUS AND CHLOROPHYLL IN LAKE ERIE: INSIGHTS FROM TWO MONITORING PROGRAMS

SS021 TRANSITIONING SCIENTIFIC RESEARCH INTO MEANINGFUL APPLICATIONS

Chair(s): Michael Allen, Maryland Sea Grant (mallen@mdsg.umd.edu)
Mona Behl, Georgia Sea Grant (mbehl@uga.edu)
Rebecca Briggs, NOAA National Sea Grant Office (rebecca.briggs@noaa.gov)

Location: Room 202

14:00 **Charlebois, P.**; Foley, C.; Nigrelli, C.; Salazar, K.; Hook, T.: FROM RESEARCH TO APPLICATION: CASE STUDIES FROM ILLINOIS-INDIANA SEA GRANT

14:15 **Harris, L.**; Dahlenburg, C.; Schott, E.; Woodland, R.: JUMP-STARTING SCIENTIFIC CO-PRODUCTION IN BALTIMORE: RESULTS OF THE "HARBOR SCIENCE" WORKSHOP

14:30 **Simmons, M.**; Bielmyer-Fraser, G.; Stalker, J.: MONITORING AND EVALUATION OF ANTHROPOGENIC IMPACTS ON THE LOWER ST. JOHNS RIVER

14:45 **Rosengard, S.**; Tortell, P.; Dowd, M.; Freshwater, C.: OPTICALLY-DERIVED TIME-SERIES OF PHYTOPLANKTON COMMUNITY COMPOSITION FOR MANAGEMENT OF WILD SALMON FISHERIES IN THE NORTHEAST PACIFIC OCEAN

15:00 **Anderson, C.**; Newton, J.; Ruhl, H.; MacCready, P.; Siedlecki, S.: CONNECTING STAKEHODLERS TO ECOSYSTEM CHANGE WITH ECOLOGICAL FORECAST MODELS IN THE CALIFORNIA CURRENT SYSTEM

15:15 **Peacock, M.**; Arnold, R.: COLLABORATIVE RESEARCH WITH OUR TRIBAL PARTNERS: STARTING AT THE PROPOSAL STAGE

16:00 **Lassiter, M.**; Felker-Quinn, E.; Greaver, T.: KEY ELEMENTS OF AQUATIC RESEARCH SUPPORTING THE REVIEW OF THE U.S. ENVIRONMENTAL PROTECTION AGENCY NATIONAL AMBIENT AIR QUALITY STANDARDS

16:15 **Turner, R.**; Swarzenski, C.; Bodker, E.: SOIL STRENGTH LOSSES IN FRESH MARSHES WITH INCREASED N AND P LOADING

16:45 **Fiori, E.**: CHALLENGES OF MICROBIAL WATER QUALITY MONITORING AT MONTEREY BAY AQUARIUM

17:00 **Chardon-Maldonado, P.**; Morell, J.; Rodriguez, S.; Canals, M.; Capella, J.: WHO NEEDS COASTAL OCEAN OBSERVATIONS IN THE US CARIBBEAN?

17:15 **Schmidt, W.**; Jimenez, M.; Rojas, D.; Smith, R.; Xu, H.: FREE VEHICLES: AN OCEANOGRAPHIC ANALOG TO WEATHER BALLOONS?

SS023 CARBON CYCLING ACROSS GRADIENTS IN THE LAND-OCEAN-CONTINUUM

Chair(s): Michael Seidel, University of Oldenburg, Germany (m.seidel@uni-oldenburg.de)
Nicholas D. Ward, Pacific Northwest National Laboratory, USA (nickdward@gmail.com)
Sairah Y. Malkin, University of Maryland, USA (smalkin@umces.edu)
Patricia M. Medeiros, University of Georgia, USA (medeiros@uga.edu)

Location: Room 104

09:00 **Wagner, S.**; Harvey, E.; Huynh, N.; McNair, H.; Arrington, E.; Stubbins, A.: SMOKE ON THE WATER: WILDFIRE AEROSOLS CONTRIBUTE TO MARINE DISSOLVED BLACK CARBON[T]

09:15 **Twilley, R.**; Edmonds, D.; Rovai, A.: CARBON SEQUESTRATION BY DELTAS WORLDWIDE

09:30 **Zimmerman, R.**; Hill, V.; Li, J.; Perez Ibanez, D.; Islam, K.; Schaeffer, B.; Megan, C.: TOWARD AN ESTIMATE OF SHALLOW WATER CARBON BURIAL FROM SPACE: QUANTITATIVE REMOTE SENSING OF SEAGRASS DISTRIBUTION AND DENSITY USING HIGH SPATIAL RESOLUTION MULTISPECTRAL IMAGERY

09:45 **Dai, M.**; Cao, Z.; Yang, W.; Guo, X.; Yin, Z.; Zhao, Y.: DIAGNOSIS OF CO2 FLUXES IN THE COASTAL OCEAN

10:00 **Canning, A.**; Körtzinger, A.; Fietzek, P.; Maier, M.; Wehrli, B.: BREAKING THE CARBON BOUNDARIES WITH THE USE OF MEMBRANE TECHNOLOGY – FROM OCEAN TO LAND

10:15 **Bridgham, S.**; Schulz, M.; Sadofsky, E.; Blount, K.; Fitch, A.: CARBON CYCLING AND TRACE GAS DYNAMICS IN DISTURBED, RESTORED, AND REFERENCE WETLANDS ALONG A SALINITY GRADIENT IN OREGON, USA

14:00 **Rutherford, K.**; Fennel, K.; Laurent, A.; Brickman, D.; John, J.: SHIFTING CIRCULATION UNDER A CHANGING CLIMATE: BIOGEOCHEMICAL IMPACTS IN THE NORTHWEST NORTH ATLANTIC

14:15 **Schlesinger, D.**; Myneni, S.: ORGANOCHLORINE DEGRADATION IN COASTAL WETLANDS IMPACTED BY SEA-LEVEL RISE

14:30 **Ward, N.**; Indivero, J.; Yabusaki, S.; McDowell, N.; Wang, W.; Stegen, J.; Bond-Lamberty, B.; Sengupta, A.; Pennington, S.; Zhang, P.; Gunn, C.; Bailey, V.: ABOVE AND BELOWGROUND BIOGEOCHEMICAL LINKAGES ALONG A SALT-IMPACTED COASTAL TERRESTRIAL-AQUATIC INTERFACE IN THE PACIFIC NORTHWEST, USA

14:45 **Morrissette, H.**; Pinsonneault, A.; Hood, R.: DISSOLVED ORGANIC CARBON SORPTION RATE KINETIC EXPERIMENTS TO VALIDATE A SEDIMENT FLUX MODEL

15:00 **Kazmiruk, Z.**; Kuzyk, Z.; Papakyriakou, T.; Rysgaard, S.; Guéguen, C.; Wang, F.: MICROBIAL DEGRADATION OF DISSOLVED ORGANIC CARBON IN RIVERINE AND COASTAL HUDSON BAY WATERS UNDER LANDFAST ICE AS INFERRED FROM INCUBATION EXPERIMENTS

15:15 **Yoshimura, C.**; Natsuike, M.; Fujii, M.; Endo, Y.; Allam, A.: ROLE OF RIVER BASINS IN IRON SUPPLY TO COASTAL ECOSYSTEM IN NORTHEAST JAPAN

SS027 SMALL SCALE SPATIAL AND TEMPORAL PATTERNS IN PARTICLES, PLANKTON, AND OTHER ORGANISMS

Chair(s): Aditya Nayak, Florida Atlantic University (anayak@fau.edu)
David Murphy, University of South Florida (davidmurphy@usf.edu)
Malcolm McFarland, Florida Atlantic University (mmcfarland@fau.edu)

Location: Room 103 B

09:00 **James, A.**; English, C.; Carlson, C.; Wilbanks, E.: THE KELP MICROBIOME – WHOSE THERE AND WHAT ARE THEY CAPABLE OF?

09:15 **Yang, Y.**: ENVIRONMENTAL IMPROVEMENT AND PLANKTON ECOLOGICAL SUCCESSION EFFECTS OF CULTIVATION OF SEAWEED GRACILARIA LEMANEIFORMIS – A CASE STUDY IN NANAO COASTAL WATERS, SOUTH CHINA

[T] REPRESENTS TUTORIAL PRESENTATIONS

09:30 **Fadeev, E.**; Iversen, M.; Wekerle, C.; Rogge, A.; Waite, A.; Bienhold, C.; Salter, I.; Hehemann, L.; Boetius, A.: ARCTIC OCEAN SEA ICE ENHANCES VERTICAL CONNECTIVITY OF MICROBIAL COMMUNITIES THROUGH SINKING MARINE AGGREGATES

09:45 **Pfreundt, U.**; Sengupta, A.; Ackermann, M.; Stocker, R.: RAPID ACTIVE AGGREGATE RESHAPING IN TRICHODESMIUM

10:00 **Nayak, A.**; McFarland, M.; Sullivan, J.; Moore, T.; Twardowski, M.; Stockley, N.: CHARACTERIZING VERTICAL DISTRIBUTIONS OF TWO CYANOBACTERIAL SPECIES DURING BLOOM CONDITIONS IN LAKE ERIE USING DIGITAL HOLOGRAPHIC IMAGING

10:15 **McFarland, M.**; Nayak, A.; Stockley, N.; Sullivan, J.: HIGH RESOLUTION VERTICAL DISTRIBUTIONS OF PHYTOPLANKTON AND MICROZOOPLANKTON IN COASTAL SHELF WATERS DETERMINED WITH AN IN SITU HOLOGRAPHIC MICROSCOPE

14:00 **True, A.**; Webster, D.; Weissburg, M.; Yen, J.: COPEPOD AVOIDANCE OF THIN LAYERS OF HARMFUL ALGAL COMPOUNDS

14:15 **Felipe, Y.**; Neves, F.; De-La-Cruz, L.; Costello, J.; Strickler, J.; Lopes, R.: THE SWIMMING BEHAVIOR OF THE CHAETOCEROS-VORTICELLA CONSORTIUM AND ITS ECOLOGICAL IMPLICATIONS

14:30 **Webster, D.**; Elmi, D.; Soumya, S.; Fields, D.: BEHAVIOR OF A CRUISE-SWIMMING COPEPOD IN MIMICS OF TURBULENT EDDIES

14:45 **Murphy, D.**; Olsen, D.; Kanagawa, M.; King, R.; Kawaguchi, S.; Osborn, J.; Webster, D.; Yen, J.: THE DENSITY AND STRUCTURE OF ANTARCTIC KRILL SCHOOLS IN THE LABORATORY

15:00 **Whitmore, B.**; Ohman, M.: USING THE ZOOGLIDER TO ADDRESS THE COVARIABILITY OF MESOZOOPLANKTON AND CHLOROPHYLL-A VERTICAL MICROSTRUCTURE WITH WATER COLUMN STABILITY

15:15 **Ellen, J.**: WHERE, WHEN, AND WATER CONDITIONS - PLANKTON IMAGE CLASSIFICATION WITH CONTEXT METADATA

15:45 **Besterman, A.**; Pace, M.: GEOMORPHOLOGY EXERTS BOTTOM-UP CONTROL ON INTERTIDAL FLAT BIOMASS

16:00 **Rakotomalala, C.**; Thouvenin, B.; Guizien, K.; Le Hir, P.; Orvain, F.: A ONE-DIMENSIONAL MODEL OF INTERTIDAL MICROPHYTOBENTHOS FUNCTIONING

16:15 **Lofton, M.**; Leach, T.; Beisner, B.; Carey, C.: SEASONAL VARIATION IN TOP-DOWN VERSUS BOTTOM-UP CONTROL OF PHYTOPLANKTON VERTICAL DISTRIBUTION IN NORTH TEMPERATE LAKES

16:30 **Strass, V.**; von Appen, W.: SMALL-SCALE VERTICAL AND HORIZONTAL VARIABILITY OF PHYSICO-BIOGEOCHEMICAL PROPERTIES ACROSS THE EQUATORIAL UNDERCURRENT IN THE ATLANTIC OCEAN

16:45 **Papiol Nieves, V.**; Enriquez, C.; Aragón-González, J.; Dávila, Y.; Cabrera, J.; Chiappa, X.: THE ROLE OF HYDRODYNAMICS AS DRIVER OF CHANGES IN PARTICULATE MATTER AND BIOTIC COMPOSITION IN A TROPICAL LAGOON

17:00 **Gould, R.**; Lewis, D.; Smith, G.; Smith, T.; Shulman, I.; Wijesekera, H.; Wang, D.; Anderson, S.: SMALL SCALE COASTAL PHYSICAL/BIO-OPTICAL VARIABILITY RELATED TO ATMOSPHERIC AND TIDAL FRONTS

17:15 **Dursun, F.**; Taş, S.; Ediger, D.: THE USE OF PIGMENT SIGNATURES TO ASSESS PHYTOPLANKTON ASSEMBLAGE STRUCTURE IN THE SEA OF MARMARA, TURKEY

SS035 EXPLORING MICROBIAL INTERACTIONS AND ORGANIC MATTER TRANSFORMATIONS WITHIN OXYGEN MINIMUM ZONES

Chair(s): Brett Walker, University of Ottawa (brett.walker@uottawa.ca)
Karl Kaiser, Texas A&M University Galveston Campus (kaiserk@tamug.edu)
Hussain Abdulla, Texas A&M University Corpus Christi (Hussain.Abdulla@tamucc.edu)
Andrew Babbin, Massachusetts Institute of Technology (babbin@mit.edu)
Frank Stewart, Georgia Institute of Technology (frank.stewart@biology.gatech.edu)
Clara Fuchsman, University of Maryland Horn Point Laboratory (cfuchsman@umces.edu)

Location: Room 208 A/B

09:00 **Algar, C.**; Rakshit, S.: A COUPLED BENTHIC-PELAGIC BIOGEOCHEMICAL MODEL FOR ASSESSING CARBON AND NUTRIENT DYNAMICS IN LOW OXYGEN WATERS

09:15 **Taylor, G.**; Medina, L.; Iabichella-Armas, M.; Lin, X.; Cernadas-Martin, S.; Lopez-Gasca, M.; Varela, R.; Astor, Y.; Muller-Karger, F.; Lorenzoni, L.; Rueda, D.; Thunell, R.; Scranton, M.; Suter, E.: ENVIRONMENTAL DRIVERS OF VARIABILITY IN MICROBIAL COMMUNITIES AND ACTIVITIES IN SURFACE AND DEEP WATERS DURING THE 21 YEAR CARIACO BASIN TIME-SERIES

09:30 **Suter, E.**; Pachiadaki, M.; Taylor, G.; Edgcomb, V.: KEY MICROBIAL TAXA LINK CHEMOAUTOTROPHIC CARBON FIXATION TO HIGHER TROPHIC LEVELS IN THE CARIACO BASIN FOOD WEB

09:45 **Saunders, J.**; McIlvin, M.; Moran, D.; Held, N.; Webb, E.; Santoro, A.; Dupont, C.; Saito, M.: FUNCTIONAL DIVERSITY OF BIOENERGETIC PATHWAYS ACROSS OXYGEN GRADIENTS IN THE CENTRAL PACIFIC

10:00 **Pavia, F.**; Anderson, R.; Lam, P.; Cael, B.; Vivancos, S.; Fleisher, M.; Lu, Y.; Zhang, P.; Cheng, H.; Edwards, R.: DISTINCT POC REGENERATION REGIMES IN THE PERU OMZ COMPARED TO OXIC WATERS OF THE EASTERN TROPICAL SOUTH PACIFIC

10:15 **Karthäuser, C.**; Ahmerkamp, S.; Bristow, L.; Hauss, H.; Iversen, M.; Kiko, R.; Lavik, G.; Kuypers, M.: THE INFLUENCE OF MARINE SNOW ON NITROGEN CYCLING IN OXYGEN MINIMUM ZONES

14:00 **Fuchsman, C.**; Carlson, M.; Lindell, D.; Devol, A.; Keil, R.; Rocap, G.: VIRAL INFECTION OF CYANOBACTERIA FUELS N2 PRODUCTION IN THE EASTERN TROPICAL NORTH PACIFIC OXYGEN DEFICIENT ZONE

14:15 **Beman, J.**: FINELY-TUNED AND TIGHTLY-COUPLED MICROBIAL CARBON AND NITROGEN CYCLING IN THE EASTERN TROPICAL NORTH PACIFIC OCEAN OXYGEN MINIMUM ZONE

14:30 **Suominen, S.**; Villanueva, L.; Gomez-Saez, G.; Sinninghe-Damste, J.: INTERPLAY BETWEEN MICROBIAL FUNCTIONAL POTENTIAL AND DISSOLVED ORGANIC MATTER COMPONENTS THROUGHOUT THE BLACK SEA WATER COLUMN

14:45 **Kaiser, K.**; Abdullah, H.; Walker, B.: DYNAMIC CYCLING OF DISSOLVED ORGANIC MATTER IN OXYGEN MINIMUM ZONES REVEALED BY SPECTROSCOPIC AND MOLECULAR-LEVEL ANALYSES

15:00 **Ruiz, P.**; Ramírez-Flandes, S.; Rodríguez-Léon, E.; Ulloa, O.: PARTITIONING OF AUTOTROPHIC CARBON FIXATION PATHWAYS ALONG THE REDOX GRADIENT IN ANOXIC MARINE ZONES

15:15 **Walker, B.**; De Pol-Holz, R.; Vargas, C.; Walker, J.; Xu, X.; Druffel, E.; Griffin, S.; Ulloa, O.: ISOTOPIC EVIDENCE FOR THE PRODUCTION OF METHANOGENIC DISSOLVED ORGANIC CARBON IN THE CHILEAN ANOXIC MARINE ZONE

[1] REPRESENTS INVITED PRESENTATIONS

SS042 LONG-TERM ECOLOGICAL STUDIES OF FRESHWATER ECOSYSTEMS IN THE GREATER CARIBBEAN BIOGEOGRAPHIC REGION

Chair(s): Omar Perez-Reyes, University of Puerto Rico (omar.perez15@upr.edu)
Tamara Heartsill-Scalley, International Institute of Tropical Forestry (theartsill@fs.fed.us)
Jesús Gómez-Carrasquillo, University of Puerto Rico (jesuslobo06@gmail.com)

Location: Room 103 A

14:00 **Marzolf, N.**; Ardón, M.; Ramírez, A.; Pringle, C.: LONG-TERM TRENDS IN WATER CHEMISTRY REVEAL EFFECTS OF EL NIÑO CLIMATE CYCLES IN GEOTHERMALLY-MODIFIED GROUNDWATER INFLUENCED LOWLAND TROPICAL STREAMS

14:15 **Murry, B.**; Olmeda, M.; Lilyestrom, C.; Garcia-Bermudez, M.: LONG-TERM ASSESSMENT (2007-2018) OF FISH ASSEMBLAGE DYNAMICS OF PUERTO RICO RESERVOIRS: APPLICATION OF COMMUNITY SIZE SPECTRA TO NOVEL SYSTEMS

14:30 **Gutierrez-Fonseca, P.**; Ramirez, A.; Pringle, C.; Torres, P.; Covich, A.; Crowl, T.; McDowell, W.; Ballantyne, F.; Perez-Reyes, O.: ECOLOGICAL RESPONSE OF STREAM ECOSYSTEMS TO EXTREME CLIMATE EVENTS IN PUERTO RICO

14:45 **Ramirez, A.**; Gutierrez-Fonseca, P.: EXTREME CLIMATIC EVENTS AND URBAN STREAMS IN PUERTO RICO: EFFECTS ON MACROINVERTEBRATES AND FISH ASSEMBLAGES

15:00 **Trexler, J.**: INTRA-COMMUNITY DIVERSITY OF INVASIVE SPECIES IMPACTS IN SPACE AND TIME: SCALING UP TO ECOSYSTEM FUNCTION

SS044 CONNECTING WATERSHED CHARACTERISTICS TO FLUVIAL EXPORTS AND IN-STREAM BIOGEOCHEMICAL PROCESSES

Chair(s): Adam Wymore, University of New Hampshire (adam.wymore@unh.edu)
William McDowell, University of New Hampshire (bill.mcdowell@unh.edu)

Location: Room 208 C

14:00 **Brookshire, J.**: EXAMINING THE ROLE OF TREE DIVERSITY ON HYDROLOGIC NUTRIENT EXPORTS FROM LOWLAND TROPICAL WATERSHEDS[T]

14:15 **Shattuck, M.**; McDowell, W.: HUMAN IMPACTS ON STREAM NITROGEN CHEMISTRY AND WATERSHED N RETENTION ACROSS A WIDE RANGE OF RURAL TO URBAN CATCHMENTS

14:30 **Skerlep, M.**; Kritzberg, E.: AFFORESTATION A MAJOR FACTOR BEHIND LONG TERM WATER COLOR TRENDS IN SOUTHERN SWEDEN

15:00 **Pugh, E.**; Devito, K.; Olefeldt, D.: LANDSCAPE CHARACTERISTICS INFLUENCING SPATIO-TEMPORAL VARIABILITY OF WATER AVAILABILITY AND QUALITY OF BOREAL PLAINS SHALLOW LAKES

15:15 **Hutley, N.**; Albert, S.; Grinham, A.; Gibbes, B.: COUPLED CATCHMENT-COASTAL HYDRONAMIC MODEL PROVIDES INSIGHT INTO PULSED SEDIMENT LOADING FROM UNMODIFIED TROPICAL ISLAND

15:45 **Marcarelli, A.**; Nevorski, K.; Eberhard, E.: WATERSHED TO REACH-SCALE CONTROLS OF N FIXATION AND DENITRIFICATION IN A CROSS-ECOREGION COMPARISON OF US STREAMS.[T]

16:00 **Tromboni, F.**; Chandra, S.; Dodds, W.; Schechner, A.; Poulson, S.: MEASURING RESPIRATION USING OXYGEN STABLE ISOTOPES IN DIFFERENT FUNCTIONAL PROCESS ZONES, ECOREGIONS, AND BIOMES OF THE WORLD

16:15 **Fazekas, H.**; Potter, J.; Snyder, L.; Shattuck, M.; McDowell, W.: HIGH-FREQUENCY IN SITU OPTICAL SENSORS REVEAL SPATIAL, SEASONAL, AND ANNUAL TRENDS IN NITRATE EXPORT TO THE GREAT BAY ESTUARY

16:30 **Webster, A.**; Krehlik, A.; Harms, T.: TIME SCALES OF VARIATION IN STREAM BIOGEOCHEMISTRY ACROSS CONTRASTS OF PERMAFROST EXTENT

16:45 **Shanley, J.**; Saraceno, J.; Aulenbach, B.; Clow, D.; Mast, M.; Hunt, R.; Hood, K.; Murphy, S.; Torres, A.; Ryan, K.; Wickland, K.; Scholl, M.: PROCESSES CONTROLLING STREAM DOC ACROSS FIVE DIVERSE HEADWATER LANDSCAPES

17:00 **Schroth, A.**; Vaughan, M.; Seybold, E.; Bowden, B.; Kincaid, D.; Wilkes, A.; Shanley, J.; Vermilyea, A.; Gold, A.; Inamdar, S.; Birgand, F.; Addy, K.: LAND USE CONTROLS ON SOLUTE EXPORT DYNAMICS: INSIGHT GLEANED FROM IN-SITU SPECTROPHOTOMETERS

SS045 DRONE REMOTE SENSING FOR AQUATIC SCIENCES: CHALLENGES AND SUCCESSES

Chair(s): Roy Armstrong, University of Puerto Rico (roy.armstrong@upr.edu)
William Hernandez, City University of New York - NOAA CESSRST Center (william.hernandez@upr.edu)

Location: Room 209 A/B

09:00 **Albert, S.**; Fluggen, M.; Deering, N.; Souter, J.; Amod, R.; Starr, B.; Grinham, A.: AUTONOMOUS AERIAL PLATFORMS TO MONITOR WATER QUALITY FROM INACCESSIBLE WATER BODIES

09:15 Ruiz, H.; **Sabater, J.**: ASSESSING THE VIABILITY OF USING UNMANNED AIRCRAFT SYSTEMS (UAS) AS A TOOL FOR SURVEYING THE ANTILLEAN MANATEE POPULATION IN PUERTO RICO

09:30 **Hernandez, W.**; Torres-Perez, J.; Viqueira, R.; Armstrong, R.; Lopez, O.: COMBINATION OF VERY HIGH RESOLUTION (VHR) SATELLITE AND DRONE IMAGERY FOR BENTHIC HABITAT MAPPING: A CASE STUDY FOR GUÁNICA, MANATÍ AND VEGA BAJA IN PUERTO RICO

09:45 **Ruiz, H.**; Scharer, M.; Llerandi-roman, I.; Prada, M.; Jackson, C.; Pacheco, C.: THE USE OF DRONES FOR HABITAT MAPPING AND VULNERABILITY ASSESSMENTS IN CAYS OF PUERTO RICO AND U.S. VIRGIN ISLANDS

10:00 **Chan, E.**; Breier, J.: TOWARD SYNOPTIC AND COORDINATED AUTONOMOUS AERIAL AND SURFACE DRONE OBSERVATIONS OF ESTUARINE BIOGEOCHEMISTRY

10:15 **Kislik, C.**; Dronova, I.; Kelly, M.: UAVS IN SUPPORT OF ALGAL BLOOM RESEARCH: A REVIEW OF CURRENT APPLICATIONS AND FUTURE OPPORTUNITIES

SS048 BASIC AND APPLIED RESEARCH FOR RESOURCE MANAGEMENT AND ENVIRONMENTAL COMPLIANCE

Chair(s): Ernesto Otero-Morales, Department of Marine Sciences/ UPRM (ernesto.otero3@upr.edu)
Jorge Bauzá-Ortega, San Juan Bay Estuary Program (oceanus.bauza@gmail.com)

Location: Room 202

09:00 **Straight, B.**; McKnight, D.: A COMPARATIVE STUDY OF DEEP-WATER SAMPLING WITH AN UNMANNED AERIAL WATER SAMPLING SYSTEM (UAWSS) TO TRADITIONAL SAMPLING METHODS: A CASE STUDY FROM DILLON RESERVOIR, SUMMIT COUNTY, COLORADO

09:15 **Otero Morales, E.**; Rodriguez, L.: DETAILED MAPPING OF SEAGRASS HABITAT DISTRIBUTION ASSOCIATED WITH COASTAL INDUSTRIAL FACILITIES.

[T] REPRESENTS TUTORIAL PRESENTATIONS

09:30 **Soler-Figueroa, B.**; Fontaine, D.; Carney, K.; Ruiz, G.; Tamburri, M.: CHARACTERISTICS OF GLOBAL PORT PHYTOPLANKTON AND IMPLICATIONS FOR BALLAST WATER REGULATIONS

09:45 **Myers, E.**; Juhl, A.: COMPARISON OF THE EFFECT OF PARTICLE ASSOCIATION ON THE PERSISTENCE OF EXOGENOUS FECAL INDICATOR BACTERIA AND SEWAGE-DERIVED PATHOGENS

10:00 **Rosa-Marin, A.**; Martinez-Colon, M.; Jagoe, C.; Woodley, C.: ENVIRONMENTAL ASSESMENT IN CORAL REEFS AT JOBOS BAY, PUERTO RICO

10:15 **Szafraniec, M.**; Waters, M.; Kenney, W.; Whitmore, T.; Lauterman, F.; Coveney, M.; Long, M.; Davenport, J.; Burnes, R.: RECONSTRUCTING OVER A CENTURY OF ECOLOGICAL AND WATER QUALITY CONDITIONS IN A SUB-TROPICAL LAKE IN CENTRAL FLORIDA USING AN ARRAY OF PALEOLIMNOLOGICAL VARIABLES

SS052 CARBON CYCLING WITHIN COASTAL WETLANDS AND WATER BODIES

Chair(s): Marshall Bowles, LUMCON (mbowles@lumcon.edu)
Brandi Kiel Reese, TAMUCC (brandi.reese@tamucc.edu)
Melitza Crespo-Medina, Inter-American University of Puerto Rico (melitzacm@inter.edu)

Location: Room 103 A

09:00 **Osborne, T.**; Simpson, L.; Schafer, T.: CARBON BIOGEOCHEMICAL PROCESSES ALONG A MANGROVE-SALTMARSH ECOTONE[T]

09:15 **Weisend, R.**; McGuire, S.; Denney, B.; Zhang, L.; Reese, B.: SEASONAL AND DIURNAL VARIATION OF METHANE FLUX IN MANGROVE ECOSYSTEMS

09:30 **Hargan, K.**; Williams, B.; Nuangsaeng, B.; Siriwong, S.; Tassawad, P.; Chaiharn, C.; Mcadoo, B.; Los Huertos, M.: ORGANIC CARBON AND NITROGEN SOURCES TO A THAI MANGROVE FOODWEB RECEIVING SHRIMP FARM EFFLUENT

09:45 **Raut, Y.**; Morando, M.; Capone, D.: SHIFTS IN MICROBIAL COMMUNITY COMPOSITION ASSOCIATED WITH DECOMPOSING MACROALGAE AND THE ROLE OF NITROGEN FIXATION IN THESE DETRITAL COASTAL SYSTEMS

10:00 **Schopflin, L.**; Cannady, A.; Leon-Zayas, R.; Biddle, J.: ENRICHMENTS OF ARCHAEA AND BACTERIA FROM DELAWARE'S INLAND BAYS

10:15 **van Grinsven, S.**; Villanueva, L.; Harrison, J.; Sinninghe-Damste, J.: METHANE OXIDATION IN ANOXIC LAKE WATERS MEDIATED BY NOVEL FACULTATIVE ANAEROBIC METHANOTROPHS

SS067 MICROBIAL PHOTOTROPHY: INTERACTIONS AND IMPACTS ON NUTRIENT CYCLING

Chair(s): Laura Gomez-Consarnau, CICESE, Mexico / U. Southern California, USA (gomezcon@usc.edu)
Xavier Mayali, Lawrence Livermore National Laboratory (mayali1@llnl.gov)

Location: Room 209 C

09:00 **Saito, M.**; McIlvin, M.; Moran, D.; Chan, E.; Alanis, B.; Lopez, P.; Hawco, N.; Lomas, M.; Bates, N.; Sedwick, P.; Johnson, R.; Jakuba, M.; Breier, C.: METAPROTEOMIC CHARACTERIZATION OF SEASONAL DYNAMICS AT THE BERMUDA ATLANTIC TIME SERIES USING THE BIOGEOCHEMICAL AUV CLIO[I]

09:15 **Groussman, R.**; Coesel, S.; Durham, B.; Morales, R.; Armbrust, E.: DIEL GENE EXPRESSION OF EUKARYOTIC PHYTOPLANKTON IN THE NORTH PACIFIC SUBTROPICAL GYRE: TEMPORAL AND FUNCTIONAL NICHE DIFFERENTIATION FROM METATRANSCRIPTOMES

09:30 **Freyria, N.**; Joli, N.; Potvin, M.; Lovejoy, C.: SEASONAL AND MULTI-ANNUAL CHANGES IN ARCTIC MICROBIAL EUKARYOTIC COMMUNITIES IN NORTHERN BAFFIN BAY

09:45 **Biddanda, B.**; Duff, R.; Weinke, T.: THE FIRST TANGO: DAY-NIGHT SHIFT BETWEEN PHOTOSYNTHESIS AND CHEMOSYNTHESIS DRIVEN BY DIEL VERTICAL MIGRATION OF MICROBIAL MAT COMMUNITIES IN LAKE HURON'S SUBMERGED SINKHOLES

10:00 **Bowman, J.**; Wilson, J.: MICROBIAL ECOSYSTEM INSIGHTS FROM CONTINUOUS MEMBRANE INLET MASS SPECTROMETRY (MIMS) IN THE COASTAL OCEAN

10:15 **Wilson, J.**; Mühle, J.; Carter, M.; Bowman, J.: THE RELATIONSHIP BETWEEN KEY HARMFUL ALGAL TAXA AND VOLATILE ORGANIC COMPOUNDS AS ASSESSED BY A 6-YEAR TIME SERIES OFF THE COAST OF SOUTHERN CALIFORNIA

14:00 Pushkarev, A.; Inoue, K.; Kandori, H.; **Beja, O.**: TYPE-1 AND 3 RHODOPSINS: MAJOR GROUPS OF MICROBIAL RHODOPSINS DISCOVERED VIA METAGENOMICS[I]

14:15 **Gomez-Consarnau, L.**: PHOTOHETEROTROPHIC SCAVENGERS AND LIFE ON THE EDGE OF STARVATION

14:30 **Hassanzadeh, B.**; Sañudo-Wilhelmy, S.; Gómez-Consarnau, L.: MICROBIAL RHODOPSIN IS THE NEW CHLOROPHYLL OF THE OCEAN

14:45 **Mayali, X.**; Gomez-Consarnau, L.; Needham, D.; Fuhrman, J.; Weber, P.: LIGHT INFLUENCES TAXON-SPECIFIC BACTERIAL DEGRADATION OF PHYTODETRITAL PARTICLES

15:00 **Graham, E.**; Heidelberg, J.; Tully, B.: EFFORTS IN ENRICHMENT AND ISOLATION OF A NOVEL PHOTOAUTOTROPH IDENTIFIED THROUGH METAGENOMICS

SS069 CLIMATE ADAPTATION IN AQUATIC AND HUMAN SYSTEMS

Chair(s): Andrew Pershing, Gulf of Maine Research Institute (apershing@gmri.org)
Nicholas Record, Bigelow Laboratory for Ocean Sciences (nrecord@bigelow.org)
Katherine Mills, Gulf of Maine Research Institute (kmills@gmri.org)

Location: Room 208 A/B

15:45 **O'Donnell, J.**: THE ROLE OF A SALT MARSH IN FLOOD PROTECTION DURING A HURRICANE

16:00 **Pinto Pacheco, S.**; Hernandez FIgueroa, E.; Cuevas, E.: DOES SPATIAL AND TEMPORAL VARIABILITY MATTERS ON SALINITY, CONDUCTIVITY, TDS AND PH IN A COSTAL URBAN WETLAND?

16:15 **Pershing, A.**; Franklin, B.; Kennedy, B.; Mills, K.; Thomas, A.; Alexander, M.; Scott, J.: INCREASES IN SURPRISING OCEAN TEMPERATURES WILL CHALLENGE THE LIMITS OF ECOSYSTEMS AND PEOPLE TO ADAPT

16:30 **Peura, S.**; Garcia, S.; Szekely, A.; Bergvall, C.; Schattenhofer, M.: DECREASED SNOW COVER STIMULATES UNDER ICE PRIMARY PRODUCERS, BUT IMPAIRS METHANOTROPHIC CAPACITY

17:00 **Lacharité, M.**; Brown, C.; Church, I.; Keith, D.; Sameoto, J.; Alleosfour, A.; Curtis, B.; Gazzola, V.; Hines, J.; Kirchhoff, S.: MAPPING THE CHANGING OCEAN TO CHARACTERIZE RISK FOR BENTHOS IN THE BAY OF FUNDY, CANADA

[I] REPRESENTS INVITED PRESENTATIONS

17:15 **Record, N.**; Runge, J.; Pendleton, D.; Balch, W.; Davies, K.; Pershing, A.; Johnson, C.; Stamieszkin, K.; Ji, R.; Feng, Z.; Kraus, S.; Kenney, R.; Thompson, C.; Hudak, C.; Mayo, C.; Chen, C.: CLIMATE-DRIVEN CIRCULATION CHANGES UNDERMINE CONSERVATION OF ENDANGERED NORTH ATLANTIC RIGHT WHALES

SS072 METHODS IN AQUATIC SCIENCE EDUCATION

Chair(s): Paul Kemp, ASLO (lomethods-editor@aslo.org)
Robert Chen, University of Massachusetts Boston (bob.chen@umb.edu)
Cayelan Carey, Virginia Polytechnical Institute and State University (cayelan@vt.edu)
Nicole Poulton, Bigelow Lab for Ocean Sciences (npoulton@bigelow.org)

Location: Room 101 A/B

09:00 **Rosengard, S.**; Albuquerque, F.; Moura, J.; Browne, K.: BLENDING SCIENTIFIC AND ARTISTIC TECHNIQUES TO ADVANCE ENVIRONMENTAL LITERACY AND STEWARDSHIP: CASE STUDIES IN BRAZIL AND HONG KONG

09:15 **Yazzie, T.**: STEAM OUTREACH AND EDUCATION: BRIDGING SCIENCE AND ART WITH INDIGENOUS KNOWLEDGE

09:30 **White, T.**; Skerrit, C.; Peets-Allamby, O.; Monrose Mills, N.; Guannel, M.: FROM TRASH TO TREASURE TO TURN THE TIDES: PROJECT BASED LEARNING IN A MATHEMATICS CLASSROOM

09:45 **Habtes, S.**; Grimes, K.; Bucklin, C.; Guannel, M.; Forbes, H.; Taylor, M.; Stout, J.; Roper, Z.: HOW PLACE-BASED CURRICULA AND COMMUNITY PROJECTS CAN ADDRESS REAL-WORLD PROBLEMS AFFECTING THE US VIRGIN ISLANDS

10:00 Henry, M.; **Pompey, A.**; Lawrence, S.; Gibbs, A.; King, C.; Gilbert, S.; Williams, J.: A COMPARISON OF THE PRESENCE OF HEAVY METALS IN SOIL AND WATER FROM FRESH AS WELL AS BRACKISH WATER IN ST. CROIX

10:15 **Chase, J.**; Laverdure, J.; Harris, N.: MAKE NO BONES ABOUT IT: OCEAN LITERACY THROUGH INQUIRY-BASED LEARNING

14:15 **Kujawa, E.**: THE STATE OF THE LAKE: COMMUNICATING LAKE AND WATERSHED SCIENCE AND ENGAGING STAKEHOLDERS IN THE LAKE CHAMPLAIN BASIN

14:30 **Harvey, E.**; Maas, A.: COMMUNICATING CLIMATE CHANGE TO THE PUBLIC: EFFECTIVE METAPHORS FROM THE NATIONAL NETWORK FOR OCEAN AND CLIMATE CHANGE INTERPRETATION (NNOCCI)

14:45 **Margolin, A.**: ARCTIC ANDY'S TALES FROM A FLOE: DEVELOPING A YOUTUBE MINISERIES TO MAKE EDUCATION GO VIRAL

15:00 **Hurzeler, I.**: COMMUNICATING OUR RELIANCE ON THE EARTH'S OCEANS: A WORKAROUND

15:45 **MacVittie, S.**; Ibanez, A.; Nuss, S.: DIFFERENTIATING CURRICULUMS: MAKING SCIENCE ACCESSIBLE FOR ELEMENTARY SCHOOL STUDENTS

16:00 **Spanbauer, T.**; Theriot, E.; Naski Keffer, L.; Owen, P.: SCIENTISTS WORKING WITH K-12 EDUCATORS: A METHOD TO EXPAND RESEARCH-BASED INQUIRY IN THE CLASSROOM

16:15 **Matsumoto, G.**; Magnusson, J.: EDUCATION AND RESEARCH: TESTING HYPOTHESES. GETTING REAL DATA INTO THE CLASSROOMS

16:30 **Poulton, N.**: EXPERIENCIAL LEARNING IN THE AQUATIC SCIENCES: BIGELOW LABORATORY'S BLOOM PROGRAMS FOR HIGH SCHOOL STUDENTS AND TEACHERS

16:45 **Cuhel, R.**; Aguilar, C.: WATER SYS-STEM TECHNICAL COLLEGE PROGRAM COMBINES EXPEDITIONARY ACADEMIC RESEARCH AND DIVERSE WATER INDUSTRY INTERNSHIPS TO PRODUCE A CAPABLE WORKFORCE

17:00 **Carey, C.**; Farrell, K.: INTEGRATING SIMULATION MODELING INTO UNDERGRADUATE AQUATIC ECOLOGY COURSES INCREASES STUDENTS' UNDERSTANDING OF GLOBAL CHANGE ON LAKES

17:15 **Sterner, R.**; Schreiner, K.; Williams, J.; Katsev, S.; Steinman, B.: AN ACTIVE LEARNING, GROUP-FOCUSSED, LIMNOLOGY CURRICLUM FOR GRADUATE STUDENTS

214 papers

THURSDAY POSTERS

AS001 SUCCESS THROUGH SCIENCE: USING LIMNOLOGY AND OCEANOGRAPHY TO TACKLE DIFFICULT MANAGEMENT QUESTIONS

373 **Portilla, K.**; Velarde, E.; Reascos, D.; Muylaert, K.: BIOLOGICAL CONTROL OF PHYTOPLANKTON BY SUBTROPICAL MACROPHYTE ELODEA DENSA: A MESOCOSM STUDY IN POLLUTED WATER

374 Hong, A.; **Hargan, K.**; Williams, B.; Nuangsaeng, B.; Siriwong, S.; Chaiharn, C.; Tassawad, P.; McAdoo, B.; Los Huertos, M.: EXAMINING IMPROVEMENTS IN THE MANAGEMENT OF SHRIMP AQUACULTURE EFFLUENT – USING MANGROVE MOLLUSCS AND SEDIMENTS AS BIOMONITORING TOOLS FOR TRACE METALS

375 **Walteros, J.**; Castaño, J.; Uribe, M.; Berrio, C.: LIMNOLOGICAL STUDIES FOR THE OTUN AND CONSOTA WATERSHEDS

AS005 EXTREME EVENTS

55 **Lucena, Z.**: LONG-TERM MONITORING OF THE LAKE HOUSTON WATERSHED — EVALUATING THE WATER-QUALITY RESPONSE TO EXTREME HYDROLOGIC EVENTS IN A MUNICIPAL SUPPLY RESERVOIR

56 **Jackson Mojica, A.**; Seybold, E.; Adair, E.; Schroth, A.: EFFECTS OF WINTER EVENTS ON NUTRIENT MOBILIZATION AND LOADING ACROSS DIFFERENT LAND USES

CS008 STABLE ISOTOPES

83 **Fan, R.**; Chikaraishi, Y.; Sugimoto, A.; Maximov, T.: DIVERSITY IN THE FOLIAR δ13C VALUE OF WILLOWS IN A TERRESTRIAL-AQUATIC TRANSECT, ARCTIC RIPARIAN AREA

84 **Sabel, M.**; Yohannes, E.; Straile, D.; Rothhaupt, K.: LONG-TERM CHANGES IN THE ISOTOPIC SIGNATURE OF LITTORAL MACROZOOBENTHOS SPECIES IN A LARGE, PRE-ALPINE EUROPEAN LAKE

85 Chikaraishi, Y.; **Takizawa, Y.**; Choi, B.: METABOLIC COMPRESSION OF TROPHIC DISCRIMINATION FACTOR IN 15N/14N WITHIN AMINO ACIDS FOR HERBIVOROUS GASTROPODS

86 **Choi, B.**; Won, H.; Chikaraishi, Y.; Kim, J.; Yang, E.; Kang, S.; Shin, K.: NITROGEN ISOTOPIC FRACTIONATION FACTOR OF NITRATE DURING ASSIMILATION BY PHYTOPLANKTON IN WESTERN NORTH PACIFIC OCEAN: CSIA APPROACH

87 **Helmer, D.**; Jarczak, J.; Martin-Creuzburg, D.; Rothhaupt, K.; Yohannes, E.: RECONSTRUCTING LAKE ECOSYSTEM HISTORY USING CARBON ISOTOPES IN RESTING DAPHNIA EGGS

88 **Tsuchiya, K.**; Komatsu, K.; Shinohara, R.; Imai, A.; Matsuzaki, S.; Kohzu, K.: SPATIAL VARIATIONS IN CONTRIBUTIONS OF METHANE-DERIVED CARBON TO CHIRONOMID LARVAE AND OLIGOCHAETA IN A SHALLOW, EUTROPHIC LAKE KASUMIGAURA, JAPAN

89 **Alharbi, A.**; Moore, B.: SPATIOTEMPORAL VARIABILITY OF STABLE ISOTOPE HYDROLOGY IN A POLYMICTIC LAKE.

CS011 PHOSPHORUS BIOGEOCHEMISTRY AND CYCLING

101 **Swaney, D.**; Howarth, R.: REGIONAL VARIATION OF NET ANTHROPOGENIC PHOSPHORUS INPUTS (NAPI), CROP PHOSPHORUS USE EFFICIENCY (PUE) AND RELATED VARIABLES ACROSS THE CONTINENTAL US: 1987-2012

102 **Wilkes, A.**; Schroth, A.; Bostick, B.; Leduc, M.; Kurek, M.; Shukle, J.; Cade-Menun, B.; Druschel, G.: PHOSPHORUS MOBILITY AND SPECIATION UNDER FLUCTUATING REDOX CONDITIONS IN SEDIMENT-WATER SYSTEMS

103 **Leduc, M.**; Shukle, J.; Druschel, G.; Wilkes, A.; Kurek, M.; Schroth, A.: SEASONAL WATER COLUMN DYNAMICS EXERT STRONG CONTROL ON THE CHEMICAL PARTITIONING OF BENTHIC PHOSPHORUS POOLS OF SHALLOW EUTROPHIC FRESHWATER SYSTEMS

104 **Sedlak, J.**; Acker, M.; Johnson, C.; Granzow, B.; Repeta, D.: SOLVING THE MARINE METHANE PARADOX: ISOLATION, QUANTIFICATION AND ISOTOPIC ANALYSES OF METHYLPHOSPHONIC ACID IN DISSOLVED ORGANIC MATTER

CS012 CARBON FLUXES IN FW AND MARINE ENVIRONMENT

376 **Carlson, A.**; St. Laurent, K.; Bundy, M.: DEEP BLUE CARBON: ASSESSING CARBON STOCK IN SEDIMENT CORES IN TWO DELAWARE TRIBUTARIES

377 **Taub, F.**: CARBON (RELATIVE TO N) DEMAND IS MUCH GREATER THAT THE REDFIELD RATIO IN CLOSED ECOLOGICAL SYSTEMS OF ALGAE, DAPHNIA, AND MICROBES.

CS015 ENVIRONMENTAL CHANGE

62 **Luyando-Flusa, S.**; Hein, C.; Reeder-Myers, L.; Rick, T.; Connell, J.: CHANGING DEPOSITIONAL ENVIRONMENTS OF THE UPPER CHESAPEAKE BAY IN RESPONSE TO EUROPEAN COLONIZATION AND IMPLICATIONS FOR OYSTER POPULATIONS

63 **Sáez-Uribe, R.**; Ortiz Zayas, J.; Yu, M.; Ramírez, A.: LAND USE LEGACY ON STREAM HEALTH RECOVERY: A LOOK INTO A TROPICAL MOUNTAINOUS WATERSHED IN PUERTO RICO

64 **Louime, C.**: NEXT GENERATION SEQUENCING TECHNOLOGIES REVEALED DIVERGENT MICROBIOTA DYNAMICS ALONG A TROPICAL COASTAL MARINE ECOSYSTEM

65 **Gargur-Leal, I.**; França, M.; Albergaria-Barbosa, A.: TEMPORAL CHANGES OF ORGANIC MATTER SOURCES IN THE SANTO ANTÔNIO LAGOON SYSTEM (BRAZIL)

CS017 HARMFUL BLOOMS

1 **Sterling, A.**; Tully, E.; Vaverka, S.; Strock, J.; Kirk, R.; Bertin, M.; Jenkins, B.: ANNUAL TIME-SERIES OF PSEUDO-NITZSCHIA SPP AND DOMOIC ACID PRODUCTION IN NARRAGANSETT BAY, RHODE ISLAND, USA

2 **Reed, M.**; Strope, E.; Myers, J.; McCarthy, M.; Newell, S.: EFFECTS OF FILTRATION TIMING AND PORE SIZE ON MEASURED NUTRIENT CONCENTRATIONS IN NATURAL WATERS

3 **Vaverka, S.**; Sterling, A.; Tully, E.; Strock, J.; Kirk, R.; Bertin, M.; Jenkins, B.: INVESTIGATING PSEUDO-NITZSCHIA SPECIES COMPOSITION AND TOXIN PRODUCTION IN NARRAGANSETT BAY, RI

4 **Dabney, B.**; Patiño, R.: LOW-DOSE STIMULATION OF GROWTH OF THE HARMFUL ALGA, PRYMNESIUM PARVUM, BY GLYPHOSATE AND GLYPHOSATE-BASED HERBICIDES

5 **Mapes, S.**; Kamalanathan, M.; Quigg, A.: MICROCOSM AND LABORATORY COMPARISON STUDY TO DETERMINE THE RESPONSE OF KARENIA BREVIS GROWTH AND BREVETOXIN PRODUCTION TO OIL AND DISPERSANT EXPOSURE

6 **Dietz, M.**; Peeters, F.; Dietrich, D.; Martin-Creuzburg, D.: OPTIMIZATION OF DIFFERENT FLUORESCENCE MEASURING PROBES FOR IN SITU DETECTION OF HARMFUL CYANOBACTERIA

[1] REPRESENTS INVITED PRESENTATIONS

7 **Hunter, R.**; Peacock, M.: RED TIDE, PASSED DOWN THROUGH THE GENERATIONS

8 **Scriven, V.**; Campbell, L.; Montressor, M.; Sarno, D.: THE POTENTIALLY TOXIC PSEUDONITZSCHIA MULTISTRIATA IN THE GULF OF MEXICO?

CS019 REGIME SHIFTS

66 **Wunsch, C.**; Martin-Creuzburg, D.; Rothhaupt, K.: THE ROLE OF MIXOTROPHS FOR CARBON FLOW DYNAMICS AND NUTRIENT REGENERATON IN A LARGE, EUROPEAN PRE-ALPINE LAKE

CS021 AQUATIC FOOD WEBS

145 **Patel, C.**; Ignoffo, T.; Slaughter, A.; Jungbluth, M.; Kimmerer, W.: COMPARING COPEPOD CONSUMPTION OF CYANOBACTERIA AND CRYPTOPHYTES IN SAN FRANCISCO ESTUARY USING QPCR

146 Craig, C.; Kimmerer, W.; **Cohen, S.**: DNA ANALYSIS REVEALS ONTOGENETIC DIET SHIFTS FROM LARVAE TO ADULTS OF A PREDATORY COPEPOD

147 **DeWeese, N.**; Branstrator, D.; Reavie, E.: HOW LONG HAS IT BEEN IN THE LAKE? SEDIMENT REVEALS INVASION HISTORY OF SPINY WATER FLEA

148 **Calbet, A.**; Grigoropoulou, A.; Duarte Ferreira, G.; Saiz, E.: THERMAL ACCLIMATION AND ADAPTATION IN MARINE MICROZOOPLANKTON

CS024 MICROBIAL ECOLOGY AND PHYSIOLOGY

161 **Agosto Shaw, K.**; Rossi, R.; Roberts, B.: FUNGAL PATHOGEN PRESENCE AND DIVERSITY OF FOUR SALT MARSH PLANTS IN LOUISIANA

162 **Segui, L.**; Varg, J.; Svanbäck, R.: DIET AND ONTOGENY EXPLAIN DIFFERENCES IN CRAYFISH GUT MICROBIAL COMMUNITY STRUCTURE

163 **Souflis, C.**; Palenik, B.; Nagarkar, M.: INVESTIGATIONS OF SEAFOAM FORMATION AND ASSOCIATED MICROBIAL COMMUNITIES IN SAN DIEGO, CA

164 **Ortiz-Villa, E.**; Kim, D.; Jones, A.: SPATIAL DIFFERENCES IN THE BACTERIAL MICROBIOME ASSOCIATED WITH MACROCYSTIS PYRIFERA ACROSS BLADE SIZE AND AGE

165 Guilherme de Oliveira, D.; Schneider, G.; Da Silva, A.; Rörig, L.; De Barros, M.; **Guimaraes, M.**; Barufi, J.: UV RADIATION EFFECT ON EXTREMOPHILE METAPHYTIC ALGAE FROM ALTITUDE OLIGOTROPHIC LAKES IN BRAZIL

166 **Labarre, A.**; Obiol, A.; Massana, R.: EXPRESSION OF GENES INVOLVED IN PHAGOCYTOSIS IN UNCULTURED HETEROTROPHIC FLAGELLATES

167 **Schreier, J.**; Smith, C.; Fu, H.; Suggs, E.; Sharma, S.; Moran, M.: GENETIC BASIS OF MICROBIAL SUCCESS DURING MARINE MICROBIOME ASSEMBLY

168 **Zhao, Z.**; Deng, H.; Gong, J.: THE TOP-DOWN AND BOTTOM-UP CONTROLS OF SYNECHOCOCCUS IN COASTAL OCEAN

169 **Campbell, C.**; Campbell, C.; Lavery, P.; Huggett, M.; McMahon, K.: IMPACT OF NUTRIENTS AND STORMWATER ON PERIPHYTON IN A SHALLOW EMBAYMENT

CS025 PHYTOPLANKTON ECOLOGY AND PHYSIOLOGY

170 **Hillhouse, J.**; Genzer, J.; Kamalanathan, M.; Quigg, A.: ROLE OF NITROGEN STRESS IN DEFINING THE RESPONSE OF DIATOM THALASSIOSIRA PSEUDONANA TO CONTAMINANTS

171 **Acosta Caraballo, Y.**; Wu, M.: PHYTOPLANKTON ASSEMBLAGES IN SELECTED FRESHWATERS OF NEW JERSEY

172 **Montgomery, Q.**; Johnson, R.; Bates, N.: INTERANNUAL VARIABILITY IN COCCOLITHOPHORE PIGMENTS ACROSS THE NORTH ATLANTIC SUBTROPICAL GYRE

173 **Santos, M.**; Beaulieu, S.; Peacock, E.; Crockford, T.; Shalapyonok, A.; Futrelle, J.; Fowler, B.; Archibald, K.: A GRADIENT IN PHYTOPLANKTON SIZE DISTRIBUTION ACROSS THE NORTHEAST U.S. SHELF

174 **Bishop, I.**; Rynearson, T.: WHO CAN BEAT THE HEAT? THERMAL TOLERANCE AND GROWTH RATE VARIABILITY IN COLD-ADAPTED DIATOMS

CS026 PRIMARY PRODUCTION

17 **Mescioglu, E.**; Rahav, E.; Frada, M.; Galletti, Y.; Rosenfeld, S.; Rahev, O.; Santinelli, C.; Ho, T.; Herut, B.; Paytan, A.: BIOAEROSOL IMPACTS ON PICO-PHYTOPLANKTON AND PRIMARY PRODUCTION IN THE NORTHERN RED SEA; A MESOCOSM APPROACH

18 **Hollingshead, K.**; Lee, S.; Moser, K.: INVESTIGATING THE COMPOUNDING EFFECTS OF LAND USE CHANGE AND WARMING TEMPERATURES ON ALGAL PRODUCTION USING PALEOLIMNOLOGY IN CHANDOS LAKE, ONTARIO (CANADA)

19 Savelli, R.; Cugier, P.; Polsenaere, P.; Méléder, V.; Lavaud, J.; Barnett, A.; Dupuy, C.; **Le Fouest, V.**: PHYSICAL-BIOLOGICAL COUPLED MODELLING FOR MAPPING MICROPHYTOBENTHOS PRIMARY PRODUCTION ON A TEMPERATE INTERTIDAL MUDFLAT

CS033 CORAL REEF ECOSYSTEMS

175 **Grier, S.**; Fong, P.: BOTH SPECIES INTERACTIONS AND NUTRIENTS MAY REDUCE THE NEGATIVE AFFECTS OF DENSITY ON GROWTH OF A DOMINANT MACROALGA ON A HUMAN-IMPACTED FRINGING CORAL REEF IN THE SOUTH PACIFIC.

176 Veglia, A.; Hammerman, N.; **Rivera, C.**; Lucas, M.; Galindo, A.; Schizas, N.: CHARACTERIZING POPULATION STRUCTURE OF CORAL-ASSOCIATED FAUNA FROM MESOPHOTIC AND SHALLOW HABITATS IN THE CARIBBEAN

177 **Packard, A.**; Hendrick, G.; Nicholson, M.; Sikkel, P.: CLOSING THE LOOP: INTEGRATING PARASITES IN TROPHIC CONNECTIVITY OF DIEL MIGRATORY FISHES

178 **Torres-Perez, J.**; Detres, Y.; Armstrong, R.; Aragones-Fred, C.; Cruz Motta, J.: CORALBASICS: A MULTI-LEVEL CITIZEN SCIENCE APPROACH FOR CORAL REEF ASSESSMENT IN SOUTHWEST, PUERTO RICO

179 **Olmeda-Saldana, M.**; Williams, S.; Cruz-Motta, J.: DIADEMA ANTILLARUM GRAZING EFFECTS ON ALGAL/BENTHIC COVER AND DIVERSITY IN LA PARGUERA NATURAL RESERVE

180 **Hoch, M.**; McNab, M.; May, H.; Tran, A.; Wegley Kelly, L.; Nelson, C.: EFFECT OF SEA SURFACE TEMPERATURE ON MICROBIOMES OF *MANICINA AREOLATA* (ROSE CORAL) FROM SEAGRASS HABITATS OF BELIZE

181 **Barry, J.**; Sikkel, P.; Jerris, K.: EFFECTS OF THE PARASITIC ISOPOD ANILOCRA CHROMIS (CYMOTHOIDAE) ON ITS HOST THE BROWN CHROMIS, CHROMIS MULTILLINEATA (POMACENTRIDAE)

182 **Gomez, A.**; McDonald, K.; Shein, K.; DeVries, S.; Armstrong, R.; Hernandez, W.; Carlo, M.: EVALUATING CORAL RESILIENCY IN PUERTO RICO BY COMPARING IN SITU AND SATELLITE-BASED OCEAN TEMPERATURES AND CORAL-ALGAE ASSOCATION DYNAMICS

183 **Jarrett, J.**: POPULATION STRUCTURE OF *HOLOTHURIA MEXICANA* IN THE WESTERN CARIBBEAN

184 **Eckert, R.**; Studivan, M.; Voss, J.: POPULATIONS OF THE CORAL SPECIES *MONTASTRAEA CAVERNOSA* LACK VERTICAL CONNECTIVITY ON THE BELIZE BARRIER REEF

185 **Morales Ruiz, C.**; Weil, E.: RECOLONIZATION OF ORBICELLA FAVEOLATA COLONIES KILLED DURING THE THERMAL ANOMALY OF 2005: STATUS OF THE NEW COMMUNITIES AND JUVENILE DYNAMICS.

186 **Banon, Y.**; Eakin, M.; Devotta, D.; Liu, G.; McDonald, K.: HEAT STRESS AND BLEACHING IN CORAL REEF COMMUNITIES DURING THE 2014-2016 CARIBBEAN BLEACHING EVENT

CS034 ESTUARINE ECOSYSTEMS

187 **Ceroli, A.**; Schmitt, J.: COMPARISON OF COASTAL WETLANDS' VULNERABILITIES TO SEA LEVEL RISE

188 **Slaughter, A.**; Kimmerer, W.; Loboschefsky, E.; Takekawa, J.: DOES HIGHER ZOOPLANKTON ABUNDANCE IN MANAGED AND TIDAL WETLANDS TRANSLATE TO ENHANCED FOOD SUPPLY FOR FISH IN THE OPEN WATERS OF SAN FRANCISCO ESTUARY?

189 **Politi, T.**; Zilius, M.; Ruginis, T.; Castaldelli, G.; Bartoli, M.; Daunys, D.: ESTUARINE MACROFAUNA BIODIVERSITY AFFECTS BENTHIC BIOGEOCHEMISTRY ALONG A SALINITY GRADIENT

190 **Stanganelli, J.**; Jeppesen, R.; Wasson, K.; Mize, Z.; Gaskins, L.: FACTORS INFLUENCING THE ABUNDANCE AND DECLINE OF AN INVASIVE JAPANESE MUDSNAIL, BATILLARIA ATTRAMENTARIA, AT ELKHORN SLOUGH ESTUARY

191 **Gonzalez-Garcia, M.**; Fontaine, D.; Davidson, I.: MANIPULATING PROPAGULE PRESSURE IN A MIMIC INVASION: CHAETECEROS MUELLARI IN A CHESAPEAKE BAY PLANKTON COMMUNITY.

192 **Jagoe, C.**; Marquez, M.; Chauhan, A.; Pathak, A.: MICROBIOME OF CULTURED OYSTERS (CRASSOSTREA VIRGINICA) FROM THE NORTHERN GULF OF MEXICO

CS035 CHANGING BIOGEOCHEMISTRY AND ECOLOGY ACROSS POLAR AQUATIC SYSTEMS IN THE 21ST CENTURY

402 **Dasarathy, S.**; Bowman, J.: DETERMINING THE RELATIONSHIP BETWEEN OCEANOGRAPHIC PARAMETERS AND AEROSOL FORMATION IN THE PALMER LTER WITH NASA CALIPSO

403 **Pretty, T.**: IMPACT OF FOREST FIRE ON AQUATIC INVERTEBRATES IN SUBARCTIC LAKES OF THE TAIGA PLAINS

404 **Teillet, M.**; Gray, D.; Latta, A.: IMPACTS OF GRANULAR MATERIAL ON WATER QUALITY AND FISHING OPPORTUNITIES IN THE MACKENZIE RIVER SYSTEM, NORTHWEST TERRITORIES

405 **DiTullio, G.**; Schanke, N.; Bolinesi, F.; Mangoni, O.; Lee, P.: LIGHT, NITROGEN AND TRACE METAL CO-LIMITATION IN THE HIGH ARCTIC OCEAN

CS037 COASTAL ECOSYSTEMS

193 **Ishaque, A.**; Elfadul, R.; Jesien, R.; Elnabawi, A.; Chigbu, P.: ANALYSIS OF EMERGING CONTAMINANTS IN MARYLAND COASTAL BAYS USING IN VITRO BIOASSAYS AS BIOLOGICAL SCREENING TOOLS

194 **Colon, D.**; Santos Flores, C.; Ramirez, W.: CHANGES IN SHORELINE AND THE REALTION IN SHORELINE AND THE RELATION BETWEEN DIVERSITY OF SHELL-MOLLUSKS AND SEDIMENT SORTING OF THE WEST COAST OF PUERTO RICO

195 **Sievers, M.**; Brown, C.; Tulloch, V.; Pearson, R.; Haig, J.; Turschwell, M.; Connolly, R.: COASTAL WETLANDS IN THE CONSERVATION OF MARINE MEGAFAUNA

196 **Lee, J.**; Kim, S.; Long, M.; Park, J.; Lee, T.; Yoo, S.; Son, Y.: ESTIMATION OF THE INFLUENCE OF BARREN GROUND ON COASTAL ROCKY BOTTOM ECOSYSTEM METABOLISM USING A NONINVASIVE EDDY COVARIANCE TECHNIQUE: APPLICATION TO SUBTROPICAL JEJU ISLAND

197 **Santos, A.**; Jesus, R.; Andrade, R.; de Andrade, J.; Masque, P.; Hatje, V.: HG SPATIAL DISTRIBUTIONS AND CHRONOLOGY OF CONTAMINATION FROM PRISTINE TO HIGHLY CONTAMINATED SITES IN TODOS OS SANTOS BAY (NORTHEAST BRAZIL)

198 **Folger, C.**; Lee, H.; Reusser, B.: PATTERNS OF RISK TO BIVALVES AND THE SERVICES THEY PROVIDE ALONG THE WEST COAST OF THE U.S. FROM THE ARCTIC TO SOUTHERN CALIFORNIA

199 **Juliussen, C.**; Smith, J.; Carr, M.: SIZE STRUCTURE OF PURPLE SEA URCHINS (STRONGYLOCENTROTUS PURPURATUS) BETWEEN KELP FORESTS AND URCHIN BARRENS HABITATS

200 **Hanson, A.**; Oczkowski, A.; Santos, E.; Martin, R.; Buffum, H.; Huertas, E.; Watson, E.; Wigand, C.: SYNTHESIS OF LONG-TERM WATER QUALITY AND MODERN BIOGEOCHEMICAL SEDIMENT DATA IN THE SAN JUAN BAY ESTUARY

201 **Morales-Nunez, A.**; Chigbu, P.: THE COMMUNITY STRUCTURE OF SOFT-BOTTOM MACROBENTHIC FAUNA IN DEAD-END CANALS IN THE MARYLAND COASTAL BAYS, USA

CS041 UNDERGRADUATE RESEARCH PROJECTS

209 **Leopo, D.**; Beinart, R.: COMPARING MORPHOLOGY AND GENOTYPES IN PUTATIVELY CRYPTIC ALVINICONCHA SPECIES FROM HYDROTHERMAL VENTS IN THE LAU BASIN, TONGA

210 **Bantle, A.**; Trainer, V.; Adams, N.: DETECTION AND ENUMERATION OF AZADINIUM SPECIES IN WASHINGTON STATE, USA

211 **Froney, M.**; EFFECTS OF OIL EXPOSURE ON THE EXPRESSION OF IGF GENES IN PACIFIC HERRING

212 **Bredehoeft, M.**; Bredehoeft, M.; Wu, J.; Rogers, S.: IMPACT OF ANAEROBIC CO-DIGESTION OF CULTIVATED SEAWEED ON NITROGEN BIOEXTRACTION AND RECOVERY POTENTIAL

213 **Togami, H.**; Rielly, E.: ARTIFICIAL SEAGRASS HABITATS MODEL THE INFLUENCE OF HABITAT LOSS AND FRAGMENTATION ON EPIBIONT COMMUNITIES

214 **Wilson, J.**; Deheyn, D.: ASSESSING MULTI-FUNCTIONAL ADAPTATIONS FOR THRIVING IN COASTAL ENVIRONMENTS: THE CASE OF THE COMMON ICE PLANT, MESEMBRYANTHEMUM CRYSTALLINUM

215 **Sablan, L.**; Wiegner, T.: ASSESSING SEWAGE POLLUTION USING FECAL INDICATORS AT PUAKŌ AND ADJACENT RESORTS

216 **Cleary, N.**; Caracappa, J.; Daw, J.; Munroe, D.: ATLANTIC HORSESHOE CRAB (LIMULUS POLYHEMUS) MOVEMENT WITHIN OYSTER FARMS ON THE DELAWARE BAY SHORELINE

217 **MacVittie, S.**; Krediet, C.: BACTERIAL COMMUNITY STABILITY AND HEAT STRESS RESPONSE IN AIPTASIA

218 **Scott, N.**; Taylor, J.: CHARACTERISTICS OF MANTIS SHRIMP TELSON ARMOR

219 **O'Hare, M.**; Womack, N.; Raigoza, S.; Walters, T.; Wang, Y.; Frischer, M.: DEVELOPMENT OF NOVEL SENTINEL ASSAY TO INVESTIGATE THE WATERBORNE TRANSMISSION OF SHRIMP BLACK GILL

[1] REPRESENTS INVITED PRESENTATIONS

220 **Gimpelevich, M.**; Sutherland, K.: DISTRIBUTION OF THE TROPICAL PYROSOMA ATLANTICUM IN THE NORTHERN CALIFORNIA CURRENT FROM VERTICAL CAMERA TOWS

221 **Laroche, R.**; Benedict, C.; Titus, B.; Rodriguez, E.; Meyer, C.: FIRST CHARACTERIZATION OF THE CLOWNFISH-HOSTING SEA ANEMONES MICROBIOME ACROSS HOST AND HABITAT

222 **Schlink, C.**; Granger, J.; Tobias, C.; Rollinson, V.: NO DETECTABLE DENITRIFICATION TO N2 IN NEAR SHORE NEW ENGLAND SEDIMENTS OVERLAIN BY INVASIVE CLADOPHORA MATS

223 **MacMillan, J.**; Vokhshoori, N.; McCarthy, M.: NOVEL APPROACH FOR INVESTIGATING COASTAL FOOD WEB RESPONSES TO 2015-2016 EL NINO IN THE MONTEREY BAY FROM δ13C AND δ15N OF CALIFORNIA MUSSEL SHELL PERIOSTRACUM

224 **Leslie, J.**; Perrtree, R.; Ylitalo, G.; Balmer, B.; Bolton, J.; Boyd, D.; Cox, T.: PERSISTENT ORGANIC POLLUTANT CONCENTRATIONS IN COMMON BOTTLENOSE DOLPHINS (TURSIOPS TRUNCATUS) AND ASSOCIATED SPATIAL PATTERNS NEAR SAVANNAH, GA (USA)

225 **Benedict, C.**; Laroche, R.; Titus, B.; Gusmão, L.; Meyer, C.; Rodriguez, E.: PHYLOGENETIC RELATIONSHIPS AMONG THE CLOWNFISH-HOSTING SEA ANEMONES REVEALS AT LEAST FOUR INDEPENDENT ORIGINS OF THE SYMBIOSIS

226 **Dailey, E.**; Knapp, A.; Goodwin, D.: SETTING A BASELINE: INTERANNUAL VARIABILITY OF CARBONATE PARAMETERS IN THE WESTERN SOUTH PACIFIC

227 **Anderson, C.**: SIMULATIONS OF THE NASHUA RIVER FLOODING IN FITCHBURG, MASSACHUSETTS

228 **Harris, J.**; McCormick, A.; Phillips, J.; Botsch, J.; McCary, M.; Ives, A.: SPATIAL AND TEMPORAL VARIATION IN NUTRIENT LIMITATION OF BENTHIC ALGAE IN LAKE MYVATN, ICELAND

229 **Cook, L.**; Rykaczewski, R.: STAGE-BASED DEMOGRAPHIC MATRIX MODELING AS AN EXPLORATORY TOOL OF MARINE POPULATION SENSITIVITY TO ENVIRONMENTAL STRESS

230 **Gallagher, S.**: THE ANALYSIS OF THE LAND USE AND LAND COVER OF THE NASHUA RIVER WATERSHED

231 **Esbrí, L.**; Cáceres-Charneco, R.; Ortiz-Zayas, J.: THE WATER CHEMISTRY OF THE ARTIFICIAL REINTRODUCTION PONDS FOR THE PUERTO RICAN CRESTED TOAD.

232 **Diaz Santana, G.**; Santos, H.: TITLE: PALEO-ENVIRONMENT, SEQUENCE STRATIGRAPHY AND DEPOSITIONAL HISTORY OF THE JUANA DIAZ FORMATION, ALONG THE JACAGUAS RIVER NORTH OF HACIENDA LUCIANA AND SOUTH OF PR-52 EXPOSURE.

233 **McDermith, E.**; Whitney, L.; Lomas, M.: UNSUSPECTING PLAYERS IN THE DISSOLVED ORGANIC PHOSPHORUS POOL: PHOSPHONATES AND EUKARYOTIC PHYTOPLANKTON

CS043 GLOBAL OCEANOGRAPHY AND LIMNOLOGY

355 **Anderson, Z.**; Lopez, P.; Johnson, R.; Lomas, M.; Bates, N.: THREE DECADES OF SUSTAINED OBSERVATIONS OF OPEN OCEAN PHYSICS AND BIOGEOCHEMISTRY: AN OVERVIEW OF THE BERMUDA ATLANTIC TIME-SERIES STUDY.

CS045 PHYSICAL DYNAMICS

406 **Wewerka, S.**; Walter, R.; Bockmon, E.; O'Leary, J.: THERMOHALINE STRUCTURE AND EELGRASS RESILIENCE IN A LOW-INFLOW ESTUARY

407 **Evans, C.**; Canals, M.: TOWARDS IMPLEMENTING THE OPERATIONAL USE OF HIGH FREQUENCY RADAR WAVE AND WIND MEASUREMENTS

408 **Coles, V.**; Brooks, M.; Coles, W.: INERTIAL EFFECTS ON THE ADVECTION AND DISTRIBUTION OF PELAGIC SARGASSUM

409 **Perez, L.**; Gon, C.; Gough, M.; Freismuth, T.; MacMahan, J.: OBSERVATIONS OF INFRAGRAVITY (30 S – 5 MIN) SURGES (WAVES) IN INTERTIDAL ROCKY CHANNELS

SS006 PLASTICS IN THE AQUATIC ENVIRONMENT

447 **Lecher, A.**; Harris, E.; Doctor, E.: CONCENTRATIONS OF SMALL PLASTIC PIECES ON FLORIDA BEACHES: EAST AND WEST COAST COMPARISONS

448 **Dodson, G.**; Shotorban, K.; Hatcher, P.; Waggoner, D.; Wagner, J.; Ghosal, S.; Noffke, N.: MICROPLASTIC PARTICLES AND MICROFIBERS IN BEACH SEDIMENTS FROM SELECTED SITES IN VIRGINIA AND NORTH CAROLINA, USA

449 **Zhang, F.**; Xu, J.; Zhu, L.; Peng, G.; Wang, X.; Li, D.: INGESTION AND TROPHIC TRANSFER OF MICROPLASTICS BY COMMERCIAL FISH OFF THE EAST CHINA SEA COAST

450 **Mabry, L.**; Urban-Rich, J.: MICROPLASTICS ABUNDANCE IN THE CHARLES RIVER, MASSACHUSETTS

451 **García - Morales, C.**; Ortiz Zayas, D.: IDENTIFYING THE OCCURRENCE AND DISTRIBUTION OF MICROPLASTIC PARTICLES IN WATER AND SEDIMENTS IN THE SAN JUAN BAY ESTUARY

452 Zhao, S.; **Zhu, L.**; Gao, L.; Li, D.: METHODOLOGICAL LIMITATIONS FOR MICROPLASTIC QUANTIFICATION IN THE OCEAN: RECOMMENDATIONS FOR KNOCKING DOWN THE BARRIERS

453 **Foster, T.**; Stauffer, B.: THE BIG PROBLEM OF SMALL PLASTICS: OPTIMIZING STAINING METHODS TO QUANTIFY MICROPLASTICS IN LOUISIANA ESTUARIES

454 **Nakajima, R.**; Tsuchiya, M.; Kitahashi, T.; Lindsay, D.; Fujikura, K.; Fukushima, T.: A NEW SMALL DEVICE MADE OF GLASS FOR SEPARATING MICROPLASTICS FROM MARINE SEDIMENTS

455 **Morrison, A.**; Colson, B.; Marx, C.; Dhoonmoon, C.; Anderson, C.; Batterton, J.; White, H.; Michel, A.: INVESTIGATING METHODS FOR THE RAPID IDENTIFICATION AND ANALYSIS OF MARINE PLASTICS

456 **Medina, L.**; Zaliznyak, T.; Taylor, G.: A NOVEL ANALYTICAL METHOD TO ASSESS MICROPLASTIC DIVERSITY, ABUNDANCE AND MASS IN THE MARINE ENVIRONMENT SAMPLES

457 **Poulton, N.**; Nelson, H.; Wright, H.; Lubelczyk, L.; Cobb, C.: DETECTION AND ENUMERATION OF MICROPLASTICS USING IMAGING FLOW CYTOMETRY (FLOWCAM)

458 **Tonti, N.**; Urban-RIch, J.; Tobin, C.: MICROFIBERS WITHIN OYSTERS AT 3 SITES ALONG THE MASSACHUSETTS COAST

459 **Roper, Z.**; Wilson Grimes, K.; Habtes, S.; Lavender Law, K.: SPATIAL AND TEMPORAL TRENDS IN MARINE DEBRIS FROM A TERRITORY-WIDE, CITIZEN SCIENCE-COLLECTED DATASET IN THE UNITED STATES VIRGIN ISLANDS (USVI)

460 Sorokopud-Jones, M.; Wuerz, M.; Amjad, H.; Delacruz, B.; Wiegand, M.; **Huebner, J.**: MICROPLASTICS HAVE LIMITED EFFECTS ON BASIC LIFE CYCLE PARAMETERS OF MEDAKA (*ORYZIAS LATIPES*)

THURSDAY

[T] REPRESENTS TUTORIAL PRESENTATIONS

461 **Jovanovic, B.**; Stankovic, J.; Milosevic, D.; Savic-Zdravkovic, D.; Savic, A.; Yalçın, G.; Çetinbag, D.; Öztürk, D.; Vebrová, L.; Boukal, D.; Richardson, J.; Feuchtmayr, H.; Beklioğlu, M.: COMPARATIVE INDOOR AND OUTDOOR STUDY OF MICROPLASTIC EFFECTS ON NON-BITING MIDGES (DIPTERA: CHIRONOMIDAE)

462 Catching, K.; **Smith, E.**; Borel, A.; Henslee, H.; McNab, M.; Tran, A.; Nelson, C.; Kelly, L.; Hoch, M.: THE ROLE OF PLASTIC WASTE IN CAUSING DISEASE IN THE CORAL PORITES ASTREOIDES: DIRECT PATHOGEN TRANSMISSION OR INDIRECT FACILITATION?

463 **Lane, B.**; Corbau, C.; Malkin, S.; Nardin, W.: COUPLING PLASTIC DEGRADATION WITH BIO-COASTAL PROCESSES

464 **Dudek, K.**; Polidoro, B.; Neuer, S.: DEGRADATION AND DEPOSITION: THE ROLE OF BIOTIC AND ABIOTIC PROCESSES IN MICROPLASTICS' FATE IN THE OCEAN

465 **Long, R.**; Mays, J.; Hollis, B.; Rolle, S.: IMPACT OF BISPHENOL A UPON AQUATIC BACTERIAL ORGANIC MATTER UTILIZATION AND ECOSYSTEM SERVICE

SS007 HALOPHILA STIPULACEA: ECOLOGY AND MANAGEMENT OF THE GLOBALLY INVASIVE SEAGRASS

378 **Farchette, A.**; Jobsis, P.; Wyllie-Echeverria, S.; Tussenbroek, B.; Willette, D.; Kenworthy, J.; Stanford, A.; Turner, T.: HALOPHILA STIPULACEA WITHIN THE EASTERN CARIBBEAN; DECEMBER 2018 WEBINAR SCHEDULED

SS012 TRACING THE NATURAL AND ANTHROPOGENIC CARBON CYCLE ACROSS AQUATIC ENVIRONMENTS

113 **Rodríguez-Cardona, B.**; Wymore, A.; Kortelainen, P.; Argerich, A.; Johnson, S.; McDowell, W.: LONG-TERM TRENDS IN DISSOLVED ORGANIC CARBON FROM FLUVIAL SYSTEMS ACROSS BIOMES

114 **Goodman, K.**; Cawley, K.: THE NATIONAL ECOLOGICAL OBSERVATORY NETWORK AQUATIC SAMPLING: INTEGRATING HIGH FREQUENCY SENSOR DATA WITH OBSERVATIONAL SAMPLING FOR CARBON FLUXES

SS014 ECOSYSTEM BASED MANAGEMENT: HOLISTIC APPROACHES TO EFFECTIVE MANAGEMENT OF REGIONAL ECOSYSTEM

379 **McNally, S.**; Schillaci, C.; Tlusty, M.; Ford, K.: THE DEVELOPMENT OF A STATE-WIDE AQUACULTURE PERMITTING PLAN FOR MASSACHUSETTS USING AN ECOSYSTEM BASED MANAGEMENT FRAMEWORK

380 **Ilia, A.**; O'Donnell, J.: THE POTENTIAL ROLE OF SALT MARSH ON COASTAL PROTECTION IN LONG ISLAND SOUND

381 **Cabrera, L.**; Ortiz, J.: SURFACE-GROUNDWATER INTERACTIONS IN THE RIO LOCO ALLUVIAL AQUIFER, SOUTHWESTERN, PUERTO RICO

382 **Castejon Brignoni, R.**; Melendez Diaz, J.; Ortiz Zayas, J.: IMPACTS IN THE OPERATIONAL BUDGET OF THE PUERTO RICO ELECTRICAL POWER AUTHORITY (PREPA) BASED ON THE SEDIMENT THE WATER BRINGS INTO YAUCO II HYDROELECTRIC PLANT

SS018 METABOLIC DIVERSITY IN MARINE BIOGEOCHEMICAL CYCLES IN PRESENT AND FUTURE OCEAN

34 **Carroll, J.**; Juescher, E.; Van Oostende, N.; Ward, B.: GENOME-BASED GROWTH RATE ESTIMATES ROBUST IN SYNCHRONIZED SYNECHOCOCCUS CULTURES

35 **Rivera García, L.**; Rivera Vicéns, R.; Veglia, A.; Schizas, N.: DE NOVO TRANSCRIPTOME ASSEMBLY OF THE SOFT OCTOCORAL BRIAREUM ASBESTINUM (DIGITATE MORPHOLOGY) FROM SOUTHWEST CONTINENTAL SHELF OF PUERTO RICO

SS019 BLUE CARBON: FROM THE ECOSYSTEM TO THE MARKETS

383 **Faria Patire, V.**; Santos Andrade, M.; Almeida Silva, L.; Hatje, V.: ASSESSING SPATIAL AND VERTICAL VARIATION OF ORGANIC MATTER RECALCITRANCE AND δ13C IN TROPICAL MANGROVE SOILS

384 Dórea Filho, A.; **Hatje, V.**; Barros, F.: BLUE CARBON STOCKS IN MANGROVE SOILS VARY WITHIN ESTUARINE GRADIENTS

SS020 LARGE LAKES OF THE WORLD: DETECTING CHANGES USING LONG-TERM MONITORING

356 **Bockwoldt, K.**; Collingsworth, P.; Hood, J.; Watkins, J.; Rudstam, L.; Barbiero, R.; Hinchey-Malloy, E.: LONG-TERM COMPARISONS OF NEARSHORE AND OFFSHORE LAKE ERIE ZOOPLANKTON COMMUNITIES

SS021 TRANSITIONING SCIENTIFIC RESEARCH INTO MEANINGFUL APPLICATIONS

385 **Maranda, L.**; Ullman, D.; Rines, J.; LaGarenne, R.; Wilson, P.; Barnes, J.; Gomez-Chiarri, M.; McManus, M.: MONITORING TOXIC PHYTOPLANKTON IN LOWER NARRAGANSETT BAY: DETERMINING THE SOURCE OF PSEUDO-NITZSCHIA BLOOMS

SS022 PHYTOPLANKTON-BACTERIA INTERACTIONS: MOLECULAR INSIGHTS, CHEMICAL DRIVERS, AND BEHAVIORAL DYNAMICS

36 Chebli, G.; Patin, N.; Brown, E.; Kubanek, J.; Stewart, F.; **Garfield, C.**: METABOLITE AND MICROBIAL COMPOSITION DRIVERS DURING A FLORIDA RED TIDE BLOOM

37 **Parsons, R.**; Stead, Z.; McLeod, K.; Bulski, K.; Bittar, T.; Carlson, C.; Harvey, E.: LINEAGE SPECIFIC MORTALITY RATES OF BACTERIOPLANKTON IN THE SARGASSO SEA

38 **Rosario, M.**; Diou-Cass, Q.; Pollara, S.; Perkins, Z.; Harvey, E.; Whalen, K.; Rowley, D.: CHEMICAL INTERACTIONS BETWEEN EMILIANI HUXLEYI AND PSEUDOALTEROMONAS PISCICIDA: THE INFLUENCE OF THE QUORUM SENSING MOLECULE HHQ ON EMILIANI HUXLEYI GROWTH

39 **Matrai, P.**; Countway, P.; Rauschenberg, C.; Romano-Young, K.; Moore, K.; Peart, S.: SEASONAL IMPACT OF CONTINUOUS DMSP SUPPLY RATES ON THE DIVERSITY AND FUNCTION OF ANTARCTIC MICROBIAL COMMUNITIES

40 **Hays, M.**; Carlson, M.; Rathwell, C.; Rocap, G.: HOST ASSOCIATED MICROBIAL COMMUNITY INFLUENCES RESISTANCE TO VIRAL INFECTION IN THE DIATOM THALASSIOSIRA PSEUDONANA

41 **Pollara, S.**; Becker, J.; Nunn, B.; Whalen, K.: INVESTIGATING CHEMICALLY INDUCED STASIS IN MARINE COCCOLITHOPHORE *EMILIANIA HUXLEYI* FOLLOWING EXPOSURE TO THE BACTERIAL SIGNALING MOLECULE, HHQ

SS023 CARBON CYCLING ACROSS GRADIENTS IN THE LAND-OCEAN-CONTINUUM

357 **Mejias-Rivera, C.**; Armstrong, R.; Morell, J.: PARTICULATE ORGANIC CARBON TEMPORAL AND SPATIAL VARIABILITY IN SOUTHERN PUERTO RICO: AN ASSESSMENT OF SOURCES, CONNECTIVITY, ECOLOGICAL IMPLICATIONS AND BIO-OPTICAL PROPERTIES

[1] REPRESENTS INVITED PRESENTATIONS

358 Lee, J.; Hwan, Y.; **Kim, S.**: ORGANIC CARBON BUDGETS AND NUTRIENT FLUXES OF CONTINENTAL SHELF SEDIMENTS IN THE EAST CHINA SEA

359 **Medeiros, P.**; Letourneau, M.: SEASONAL CHANGES IN DISSOLVED ORGANIC MATTER COMPOSITION IN THE ALTAMAHA RIVER, GEORGIA, USA

361 **Schwartz, J.**; Peng, S.; Schlesinger, D.; Jiao, Y.; Rhew, R.; Myneni, S.: SALINITY GRADIENT EFFECTS ON HALOGENATION OF AND HALOCARBON EMISSIONS FROM SOIL ORGANIC MATTER

362 **Davis, S.**; Chen, R.; Peri, F.: THE EFFECT OF EPISODIC RAIN EVENTS ON DISSOLVED ORGANIC CARBON (DOC) FLUXES IN A SMALL URBAN WATERSHED

363 **Travis, C.**; Schlesinger, D.; Myneni, S.: BROMINATION OF COASTAL ORGANIC MATTER IN THE PRESENCE OF FE

364 **Logozzo, L.**; Raymond, P.: THE COUPLED CYCLING OF DISSOLVED IRON AND DISSOLVED ORGANIC MATTER IN THE CONNECTICUT RIVER

365 **Taveras, Z.**; McMahon, R.; Harvey, H.: CHANGING ORGANIC CARBON SOURCES ACROSS THE CANADIAN ARCTIC SHELF AND ITS RELATIONSHIP WITH SEDIMENT MICROBENTHOS

366 **Menendez, A.**; Tzortziou, M.; Grunert, B.: CDOM DYNAMICS IN THE LONG ISLAND SOUND

367 **Gelinas, Y.**; Leone, F.; Harutyunyan, R.; Imfeld, A.; Rossi, E.: 13C/12C ANALYSIS OF BACTERIA-SPECIFIC FATTY ACIDS TO ASSESS THE LABILITY OF TERRESTRIAL AND MARINE ORGANIC MATTER IN SEDIMENTS OF THE ST. LAWRENCE ESTUARY

SS027 SMALL SCALE SPATIAL AND TEMPORAL PATTERNS IN PARTICLES, PLANKTON, AND OTHER ORGANISMS

410 **Stoer, A.**; Davis, R.; Comeau, A.; MacIntyre, H.: ESTIMATING PARTICULATE CARBON ON THE SCOTIAN SHELF FROM REMOTELY SENSED MEASUREMENTS OF PARTICLE BACKSCATTER

411 **Kumler, W.**; Fredricks, H.; Ossolinski, J.; Allen, A.; Thamatrakoln, K.; Bidle, K.; Van Mooy, B.; Edwards, B.: SIGN OF THE TIMES: THE LIPID SIGNATURE OF A COLLAPSING PHYTOPLANKTON BLOOM

412 **Yelton, R.**; Slaughter, A.; Kimmerer, W.: SOURCE OR SINK? TIDAL FLUX OF A CALANOID COPEPOD AT A RESTORED MARSH IN SAN FRANCISCO ESTUARY

413 **Chapina, R.**; O'Malley, B.; Stockwell, J.: TO STAY OR NOT TO STAY? A MODEL TO TEST DECISION MAKING OF MYSIS DILUVIANA TO VERTICALLY MIGRATE

SS030 BIOLUMINESCENT BAYS OF THE CARIBBEAN: SCIENCE, MANAGEMENT, OUTREACH, AND RECOVERY

386 **Cabral Guadalupe, A.**; Martin Bras, M.; Fernandez Porto, J.: ANALISYS OF HISTORICAL DATA FOR PUERTO MOSQUITO, VIEQUES, SUMMARY, COMPARISON OF PAST AND CURRENT RESULTS, WITH IMPLICATIONS FOR INDICATORS OF ECOSYSTEM HEALTH

387 **Yen, J.**: CITIZEN SCIENCE: KEEPING PYROCYSTIS FUSIFORMIS ALIVE IN YOUR HOME.

388 **Ayala-Torres, R.**; Ayala, R.; Otero, E.: DISSOLVED OXYGEN PATTERNS ACROSS THE WATER COLUMN INFLUENCED BY TIDES, AND THE IMPACT ON BIOLUMINESCENCE AT BAHÍA FOSFOSRESCENTE, PUERTO RICO

389 **Rivera Oquendo, E.**; Martinez, K.; Govender, Y.: IMPACTS OF CLIMATE ON BIOLUMINESCENCE IN THE LAGUNA GRANDE AT FAJARDO

390 **Flores, J.**; Ayala Rivera, F.; Govender, Y.; Siebens Morales, Y.: PATTERNS OF SEDIMENTATION AND DINOFLAGELLATE DENSITY IN THE LAGUNA GRANDE, FAJARDO, PUERTO RICO

391 Nevarez, J.; Cadilla, J.; Torres, M.; Roldán, Z.; Vélez, S.; **Sastre, M.**: PLANKTONIC TAXA COMPOSITION IN LAGUNA GRANDE BASIN, PUERTO RICO

392 **Ramos, J.**; Martinez, R.; Govender, Y.: TEMPORAL DYNAMICS OF DINOFLAGELLATES IN THE CARIBBEAN

SS031 IMPACTS OF (SUB)MESOSCALE OCEAN DYNAMICS ON MARINE ECOSYSTEMS

415 **Sasai, Y.**: IMPACT OF OCEAN PHYSICS ON MARINE ECOSYSTEMS IN THE KUROSHIO AND KUROSHIO EXTENSION REGIONS

416 **Pacheco, F.**; Ferreira de Pinho, U.; Lethaby, P.; Johnson, R.; Bates, N.: AN ASSESSMENT OF CURRENTS IN THE UPPER OCEAN AT THE BERMUDA ATLANTIC TIME-SERIES STUDY SITE USING A VESSEL-MOUNTED ACOUSTIC DOPPLER CURRENT PROFILER

417 **Abbott, K.**; Lindo Atichati, D.: MESOSCALE AND SUBMESOSCALE EDDY DYNAMICS IN THE NORTHWESTERN ATLANTIC OCEAN AND IMPLICATIONS FOR EDDY-INDUCED CARBON TRANSPORT IN THE SARGASSO SEA

418 **Mukherjee, S.**; Habtes, S.: ANALYSIS OF INPUT CLIMATOLOGIES FOR THE DEVELOPMENT OF A COASTAL OCEAN CIRCULATION MODEL FOR THE US VIRGIN ISLANDS AND PUERTO RICO

419 **Penna, A.**; Gaube, P.: MESOSCALE VARIABILITY OF ACOUSTIC BACKSCATTERING IN THE NORTH ATLANTIC

420 **Arzeno, I.**; Giddings, S.; Pawlak, G.: COASTAL NEAR-INERTIAL TEMPERATURE ANOMALIES AND THEIR POTENTIAL BIOLOGICAL RESPONSE

421 **Chaichitehrani, N.**; Coles, V.; Shropshire, T.; Stukel, M.; Silsbe, G.: MICRO- AND MESOZOOPLANKTON GRAZING RATE IN THE GULF OF MEXICO USING PHYSICAL-BIOLOGICAL NUMERICAL MODELS

422 **Le Gal, M.**; Mitarai, S.: MODELING MANGROVE PROPAGULE DISPERSION AROUND THE BAJA CALIFORNIA PENINSULA THROUGH NUMERICAL HYDRODYNAMICS MODEL

SS041 THE NEXT GENERATION: UNDERGRADUATE RESEARCH IN PUERTO RICO AND THE US VIRGIN ISLANDS

341 **Rodríguez, J.**; Ríos, J.; Rodríguez, C.: ABUNDANCE OF FISH LARVAE SPECIES IN A TROPICAL ESTUARY, CANO BOQUILLA, MANATI, PUERTO RICO.

342 **Barreto Velez, T.**; Smoak, J.; Chappel, A.; Ross, M.; Meeder, J.; Kominoski, J.; Fourqurean, J.; Crowl, T.; Ogurcak, D.: DOES CLIMATE DRIVE CHANGES IN NUTRIENT CONCENTRATIONS OF MANGROVE SEDIMENTS? A COMPARISON BETWEEN BASIN AND FRINGE FORESTS IN LA PARGUERA, PUERTO RICO

343 **Papotto, J.**; Rodrigues, L.; Ramos Soler, I.; Padilla Plaza, G.; Wagner, E.; Goldsmith, S.: GEOCHEMICAL ANALYSIS OF STREAM SEDIMENTS COLLECTED FROM SOUTHERN PUERTO RICO FOLLOWING HURRICANE MARIA

344 **Rivera Cruz, J.**; Barreto Velez, T.; Santos-Crespo, M.; Ogurcak, D.; Crowl, T.; Fourqurean, J.; Ross, M.; Meeder, J.; Charles, S.; Kominoski, J.; Smoak, J.; Chappel, A.; Yu, M.; Lugo, A.: VARIATION IN SEDIMENT NUTRIENT CONCENTRATIONS IN AN URBAN-MANGROVE ECOSYSTEM, PINONES, PUERTO RICO

THURSDAY

T REPRESENTS TUTORIAL PRESENTATIONS

345 **Santos-Crespo, M.**; Barreto-Vélez, T.; Rivera-Cruz, J.; Ogurcak, D.; Crowl, T.; Smoak, J.; Chappel, A.; Fourqurean, J.; Ross, M.; Kominoski, J.; Meeder, J.: VARIATION IN THE NUTRIENT CONTENT OF SEDIMENTS DOWN CORE IN THE BASIN AND FRINGE MANGROVE FORESTS OF JOBOS BAY, PUERTO RICO

SS042 LONG-TERM ECOLOGICAL STUDIES OF FRESHWATER ECOSYSTEMS IN THE GREATER CARIBBEAN BIOGEOGRAPHIC REGION

205 **Santana-Cruz, A.**; Behra, M.: EFFECTS OF HERBICIDE RUN-OFFS, GLYPHOSATE AND ATRAZINE, ON DEVELOPMENT AND FUNCTION OF THE ZEBRAFISH LARVAL NERVOUS SYSTEM

206 **Santos, C.**; Martínez, G.: LONG-TERM CHANGES IN THE PHYTOPLANKTON COMPOSITION OF EIGHT RESERVOIRS OF PUERTO RICO

SS044 CONNECTING WATERSHED CHARACTERISTICS TO FLUVIAL EXPORTS AND IN-STREAM BIOGEOCHEMICAL PROCESSES

47 **Herreid, A.**; Wymore, A.; Varner, R.; McDowell, W.: CONTROLS ON GREENHOUSE GAS PRODUCTION IN STREAMS ACROSS A LAND USE GRADIENT

48 **Ortega Pieck, A.**; Brooks, E.; Roley, S.; Boll, J.; Huggins, D.: LINKING SEASONAL GROUNDWATER AND SURFACE NUTRIENT DYNAMICS TO STREAM FUNCTION IN INTENSIVELY MANAGED SEMI-ARID LANDSCAPES

49 **Jaimes, R.**; Aguilar, C.; Brown, B.; Cuhel, R.: RIVERINE GRADIENTS OF KEY BIOGEOCHEMICALS THROUGH AGRICULTURAL, (SUB) URBAN, HARBOR, AND COASTAL ECOTOPES DEMONSTRATE VALUE OF WATERSHED MANAGEMENT

SS045 DRONE REMOTE SENSING FOR AQUATIC SCIENCES: CHALLENGES AND SUCCESSES

437 **Lopez, O.**: HIGH RESOLUTION DRONE-BASED AERIAL IMAGERY FOR REMOTE SENSING OF TEMPORAL CHANGES ON SEAGRASS MEADOWS IN CAYO CARACOLES, LA PARGUERA, PUERTO RICO

438 **Hsu, A.**; Lo, E.; Costa, M.; Aburto, O.: MONITORING BAJA'S MANGROVES: REMOTE SENSING TECHNIQUES ESTIMATING MANGROVE COVERAGE IN REAL-TIME

439 Monks, L.; Russell, J.; Lee, T.; Hargrave, K.; Fox, P.; Myint, S.; **Neuer, S.**: DRONE AND SATELLITE BASED ESTIMATES OF CHLOROPHYLL A AND TAXON-SPECIFIC BIOMASS: A COMPARATIVE APPROACH TO MONITORING ALGAL BLOOMS

440 **Montanez, A.**; Grafals-Soto, R.; Castro-Jimenez, A.; Barreto, M.: UNMANNED AIRCRAFT SYSTEMS (UAS) AS BEACH AND DUNE MONITORING PLANNING TOOL FOR COASTAL ZONE MANAGEMENT IN PUERTO RICO

SS048 BASIC AND APPLIED RESEARCH FOR RESOURCE MANAGEMENT AND ENVIRONMENTAL COMPLIANCE

393 **Jickling, N.**; Strong, A.: THE ROLE OF AREA BASED MANAGEMENT TOOLS AS GENERATORS OF COLUMN CARBON MITIGATION ECOSYSTEM SERVICES

394 **Thrift, E.**: EVALUATING THE EFFECTS OF MESO-PREDATOR SCREENING ON GHOST CRAB PREDATION OF LOGGERHEAD SEA TURTLE NESTS

395 **Xu, H.**; Canals, M.; Capella Hernández, J.; Garcia Moreno, F.; Evans, C.: OCEANOGRAPHIC PATHWAYS AND HYDRODYNAMIC CONNECTIVITY BETWEEN MARINE PROTECTED AREAS IN THE US VIRGIN ISLANDS AND EASTERN PUERTO RICO

SS052 CARBON CYCLING WITHIN COASTAL WETLANDS AND WATER BODIES

369 **Hamersley, M.**; Tan, C.; Kanaoka, K.: CO-INHIBITION OF METHANOGENESIS BY NITRATE AND SULFATE: CONNECTIONS WITH DENITRIFICATION

370 **Richardson, C.**; Fackrell, J.; Paytan, A.; Glenney, K.: SOURCES AND MAGNITUDE OF LATERAL C LOSSES FROM DRAINED PEATLANDS

371 Salgado-Hernanz, P.; **Regaudie de Gioux, A.**; Antoine, D.; Basterretxea, G.: PRIMARY PRODUCTION IN COASTAL WATERS OF THE MEDITERRANEAN SEA: VARIABILITY, TRENDS AND CONTRIBUTION TO BASIN SCALE BUDGETS

372 **Bowles, M.**; Segarra, K.; Samarkin, V.; Porubsky, W.; Weston, N.; Joye, S.: DECOUPLING SULFATE REDUCTION AND ANAEROBIC METHANE OXIDATION IN SALT MARSH SEDIMENTS

SS058 BRAVE NEW WORLD: THE ECOLOGY OF HIGHLY IMPACTED WATERBODIES

207 **King, L.**; Brothers, S.; Brahney, J.: A PALEOLIMNOLOGICAL HISTORY OF EUTROPHICATION IN UTAH LAKE

208 **Wikström, J.**; Bonaglia, S.; Renman, G.; Walve, J.; Hedberg, J.; Thillman, T.; Gunnarsson, J.: THE EFFECTS ON BENTHIC FLUXES OF GASES, NUTRIENTS AND CONTAMINANTS FROM LAKE SEDIMENTS AFTER THIN LAYER CAPPING AND ALUMINIUM INJECTION

234 **Hernandez Figueroa, E.**; Pinto Pacheco, S.; Cuevas, E.: DO TIDES MATTER IN WATER TABLE LEVELS AND SPATIO-TEMPORAL SALINITY IN A COASTAL URBAN WETLAND? THE CASE OF CIÉNAGA LAS CUCHARILLAS, CATAÑO PUERTO RICO.

SS062 MIXOTROPHIC PROTISTS: AN UNDERRATED MAJORITY IN MARINE AND FRESHWATER ECOSYSTEMS?

Location: Exhibit Hall B

50 **Mars Brisbin, M.**; Mitarai, S.: SEEING IS BELIEVING: PAIRING HIGH-THROUGHPUT SEQUENCING WITH HIGH-THROUGHPUT IMAGING TO EVALUATE ACANTHARIAN ABUNDANCE AND LIFE-HISTORY

51 **Luo, H.**; Goes, J.; Jiang, X.; Gomes, H.; Mckee, K.; Lin, S.: DYNAMICS OF NUTRIENT CYCLING AND PHYTOPLANKTON SPECIES SUCCESSION DURING THE EVOLUTION OF NOCTILUCA BLOOMS ALONG THE COAST OF OMAN

52 **Florenza, J.**; Dietzel, K.; Bertilsson, S.: FLUORESCENTLY LABELLED BACTERIA IN COMBINATION WITH FACS TO IDENTIFY MIXOTROPHS IN LAKE WATER SAMPLES

53 **Ivanković, M.**; Bengtsson, M.; Ptacnik, R.: PREY SELECTIVITY OF FRESHWATER MIXOTROPHIC AND HETEROTROPHIC FLAGELLATES

54 **Holen, D.**: STOMATOCYST MORPHOTYPE AND ENCYSTMENT RATES IN OCHROMONAS PINGUIS, A MIXOTROPHIC CHRYSOPHYTE.

SS065 TURNING THE LIGHTS ON FOR DEEP-SEA ECOSYSTEMS IN THE CARIBBEAN, GULF OF MEXICO, AND US SE ATLANTIC

446 **Cook, L.**; Palter, J.; Gonçalves, A.; Bianchi, D.: TEMPORAL AND SPATIAL VARIABILITY OF MESOPELAGIC MIGRATORS IN THREE ATLANTIC LOCALES FROM A DECADE OF WEEKLY SURVEYS

[1] REPRESENTS INVITED PRESENTATIONS

SS066 LARGE RIVERS OF THE WORLD AS PIPES, CHIMNEYS AND REACTORS

466 **Rodríguez-Ortiz, N.**; Ramírez, A.: EFFECTS OF STREAM CHANNELIZATION ON HABITAT CONDITION AND MACROINVERTEBRATES IN URBAN STREAMS IN PUERTO RICO

467 **Colin, N.**; Górski, K.; Manosalva, A.; Habit, E.: EVALUATION OF THE EFFECTS OF MULTIPLE STRESSORS ON FISH IN TEMPERATE RIVER SYSTEMS OF CENTRAL-SOUTHERN CHILE: POPULATION AND COMMUNITY RESPONSES

SS069 CLIMATE ADAPTATION IN AQUATIC AND HUMAN SYSTEMS

70 **Torres Díaz, M.**; Ruiz Díaz, C.; Toledo Hernández, C.; Ramírez Lugo, J.; Díaz Vázquez, L.: IMMUNE RESPONSE OF ACROPORA CERVICORNIS TO SOLAR RADIATION AND SEA SURFACE TEMPERATURES

SS072 METHODS IN AQUATIC SCIENCE EDUCATION

348 **MacMillan, G.**; Gérin-Lajoie, J.; Snowball, H.; Rowell, J.; Monfette, M.; Hébert-Houle, E.; Townley, E.; Amyot, M.; Franssen, J.; Herrmann, T.; Lévesque, E.; Dedieu, J.: COMMUNITY-BASED ENVIRONMENTAL MONITORING IN NUNAVIK (NORTHERN QUEBEC, CANADA) FROM THE PERSPECTIVE OF AN EARLY CAREER RESEARCHER

349 **Reyes-Maldonado, R.**; Ramírez, A.: CHIRONOMUS SP. FLORIDA AS EDUCATIONAL TOOL: POTENTIAL USES OF THE SPECIES IN PUERTO RICO

350 **Venn, C.**; Hallen, C.: EVOLUTION OF AN UNDERGRADUATE COURSE IN AQUEOUS GEOCHEMISTRY: SUCCESSES AND CHALLENGES

351 **Santos, C.**: PROMOTING INTEREST IN FRESHWATER BIODIVERSITY THROUGH DRAWING EXPERIENCES: A STUDENT-ASSISTED ILLUSTRATED GUIDE TO PUERTO RICO'S BIOTA

352 **Wolfe, W.**; Jones-Cervantes, S.; Mullane, K.; Nesbit, K.; Rasina, B.: SCOPE: A VOLUNTEER STEM OUTREACH PROGRAM CONNECTING COMMUNITIES TO RESEARCH AT SCRIPPS INSTITUTION OF OCEANOGRAPHY

353 **Dubickas, K.**; Ross, B.; Amergian, K.: THE OCEANOGRAPHY CAMP FOR GIRLS: 28 YEARS OF EXPERIENTIAL LEARNING TO ENGAGE TEENS IN OCEAN RESEARCH AND CAREERS

354 **O'Neil, J.**; Birney, L.; Newton, R.; Fraioli, A.; Bone, E.; Moore, G.; Green, A.; Goodwin-Segal, T.; Dennison, W.: USING URBAN HARBORS FOR EXPERIENTIAL, ENVIRONMENTAL LITERACY: CASE STUDIES OF NEW YORK AND CHESAPEAKE BAY

220 papers

THURSDAY

FRIDAY ORALS

CS012 CARBON FLUXES IN FW AND MARINE ENVIRONMENT

Chair(s): Rob Striegl, US Geological Survey (rstriegl@usgs.gov)
Sarah Ellen Johnston, Florida State University (sarahellenjohnston@gmail.com)
Kimberly Wickland, USGS (kpwick@usgs.gov)

Location: Room 101 A/B

14:00 **Conrad, R.**; Klose, M.; Claus, P.; Enrich-Prast, A.: ACETATE TURNOVER AND METHANOGENIC PATHWAYS IN LAKE SEDIMENTS OF AMAZONIA

14:15 **Lambrecht, N.**; Katsev, S.; Wittkop, C.; Hall, S.; Fakhraee, M.; Swanner, E.: BIO-PHYSICAL CONTROLS ON THE METHANE FLUX FROM FERRUGINOUS MEROMICTIC LAKES IN THE MIDWEST

14:30 **DelVecchia, A.**; Balik, J.; Campbell, S.; West, D.; Taylor, B.; Wissinger, S.: CARBON DIOXIDE CONCENTRATIONS AND EFFLUX FROM PERMANENT, SEMI-PERMANENT, AND TEMPORARY SUBALPINE PONDS

14:45 **Torrens, C.**; Lyons, W.; Welch, K.; Gooseff, M.: SPATIOTEMPORAL ANALYSIS OF CARBON FLUXES IN GLACIAL MELTWATER STREAMS, ANTARCTICA

15:00 **Athukorala, D.**; Ziegler, S.; Prestegaard, K.; Bowering, K.; Skinner, A.: SNOWPACK FORMATION AND MELT CHARACTERISTICS REPRESENT KEY DETERMINANTS OF DISSOLVED ORGANIC CARBON EXPORT IN A HEADWATER BOREAL FOREST WATERSHED

15:15 **Ziegler, S.**; Bowering, K.; Athukorala, D.; Prestegaard, K.; Edwards, K.: CLIMATE EFFECTS ON SOIL DISSOLVED ORGANIC CARBON FLUXES AND EXPORT IN SMALL MESIC BOREAL FOREST WATERSHEDS

15:45 **Hall, C.**: WINDS OF CHANGE: EFFECTS OF WIND STRESS ON PHYTOPLANKTON PARTICULATE ORGANIC CARBON FLUX IN THE CALIFORNIA CURRENT SYSTEM

16:00 **Toullec, J.**; Vincent, D.; Frohn, L.; Miner, P.; Le Goff, M.; Devesa, J.; Moriceau, B.: COPEPOD GRAZING ON DIATOM AGGREGATES: INFLUENCES ON PARTICLES SIZE, SINKING VELOCITIES AND CARBON EXPORT

16:15 **Hemsley, V.**; Trimmer, M.; Fuessel, J.; Lam, P.; Duret, M.: RAPID REMINERALISATION CONSTRAINS THE BIOLOGICAL CARBON PUMP

16:45 Reygondeau, G.; Guidi, L.; Prieur, L.; **Legendre, L.**: EFFECTS OF THE DEPTH OF THE PERMANENT PYCNOCLINE ON THE ESTIMATES OF BIOGENIC CARBON SEQUESTRATION IN THE WORLD OCEAN

17:00 **Carrion Banuchi, C.**; Howard, J.; Lopes, C.; Wilson, S.; Fourqurean, J.: SEAGRASS PRESENCE AND SEDIMENT GRAIN SIZE CONTROL ORGANIC MATTER BREAKDOWN RATES WITHIN FLORIDA KEYS NATIONAL MARINE SANCTUARY

17:15 **Johnson, R.**; Gulick, A.; Constant, N.; Bolten, A.; Smulders, F.; Christianene, M.; Nava, M.; Kolasa, K.; Bjorndal, K.: CONSISTENT RESPONSE IN SEAGRASS ECOSYSTEM METABOLIC CARBON FLUX TO GREEN TURTLE GRAZING ACROSS CARIBBEAN MEADOWS

CS015 ENVIRONMENTAL CHANGE

Chair(s): John P. Smol, Queen's University (smolj@queensu.ca)

Location: Room 103 A

15:45 **Coldsnow, K.**; Relyea, R.: AQUATIC MACROPHYTE MITIGATE SOME, BUT NOT ALL, EFFECTS OF VARIOUS ROAD SALTS ON FRESHWATER COMMUNITIES

16:00 **Smith, S.**; Fox, S.; Plaisted, H.; Lee, K.: CHANGES IN THE THERMAL STRUCTURE OF FRESHWATER LAKES WITHIN CAPE COD NATIONAL SEASHORE (MASSACHUSETTS, USA) FROM 1996 TO 2014

16:15 **Cantwell, H.**; Perkins, R.: DIRECT AND INDIRECT IMPLICATIONS OF CLIMATE CHANGE UPON THE WATER QUALITY OF A SUPPLY-CRITICAL UK RESERVOIR.

16:30 Duda, M.; Robertson, G.; Blais, J.; Kimpe, L.; **Smol, J.**: A NOVEL APPROACH TO CONSERVATION BIOLOGY: USING PALEOLIMNOLOGY TO RECONSTRUCT THE LONG-TERM POPULATION TRENDS OF A VULNERABLE SEABIRD

16:45 **Albergaria-Barbosa, A.**; Guimarão, L.; Santana, J.; Souza, J.; França, E.: TEMPORAL AND SPATIAL EVALUATION CHANGES IN THE ORGANIC MATTER SOURCES DEPOSITED IN A PRISTINE TROPICAL ESTUARY (ESTUARY OF ITAPICURU RIVER, BAHIA, BRASIL)

17:00 **Klein, S.**; Pitt, K.; Lucas, C.; Hung, S.; Schmidt-Roach, S.; Aranda, M.; Duarte, C.: TEMPERATURE OSCILLATIONS ENHANCE THE THERMAL TOLERANCE OF A RESISTANT NON-CALCIFYING CNIDARIAN HOLOBIONT

CS021 AQUATIC FOOD WEBS

Chair(s): Albert Calbet, Institut de Ciències del Mar-CSIC (acalbet@icm.csic.es)

Location: Room 103 B

09:00 **Kimmerer, W.**: THE DYNAMICS OF ESTUARINE PLANKTONIC FOOD WEBS[1]

09:15 **Gearty, A.**; Ignoffo, T.; Slaughter, A.; Kimmerer, W.: FOOD QUANTITY AND QUALITY, OR FEMALE MORTALITY: WHAT CONTROLS COPEPOD EGG PRODUCTION RATES?

09:30 **Haraguchi, L.**; Jakobsen, H.; Carstensen, J.: ANNUAL CYCLES IN PHYTOPLANKTON: SEASONALITY OR SUCCESSION?

09:45 **Harvey, E.**: A MATTER OF TIME: DIEL RATES OF PHYTOPLANKTON GROWTH AND MORTALITY IN THE OCEAN

10:00 **Leitao de Oliveira Junior, E.**; Ali Ger, K.; de Fátima Panosso, R.: CONTRASTING TOP-DOWN EFFECT OF A CYCLOPOID VS. CALANOID COPEPOD ON FILAMENTOUS CYANOBACTERIA DOMINANCE

10:15 **Pierson, J.**; Lavrentyev, P.; Bastianini, M.: PLANKTON TROPHIC DYNAMICS DRIVEN BY DIATOM CYTOTOXINS

14:00 **Novotny, A.**; Zamora-Terol, S.; Winder, M.: KEY TROPHIC LINKS OF MICRO- AND MESOZOOPLANKTON HIGHLIGHTED BY SELECTIVE DNA METABARCODING

14:15 **Helenius, L.**; Johnson, C.; Budge, S.: LIPIDS AT THE PLANT-ANIMAL INTERFACE: USING A STABLE ISOTOPE LABELING METHOD TO EVALUATE TRANSFER AND ASSIMILATION OF ESSENTIAL FATTY ACIDS IN MARINE PLANKTON

14:30 **Fridolfsson, E.**; Bunse, C.; Legrand, C.; Lindehoff, E.; Majaneva, S.; Hylander, S.: SEASONAL VARIATION AND SPECIES-SPECIFIC CONCENTRATIONS OF THE ESSENTIAL VITAMIN B1 (THIAMIN) IN ZOOPLANKTON AND SESTON

14:45 **Parrish, C.**; Pazanini, C.; Connelly, T.; Carreón-Palau, L.: TROPHIC TRANSFER OF ESSENTIAL POLYUNSATURATED FATTY ACIDS IN MARINE FOOD WEBS IN THREE VERY DIFFERENT ENVIRONMENTS

[1] REPRESENTS INVITED PRESENTATIONS

15:00 **Strandberg, U.**; Hiltunen, M.; Rissanen, N.; Taipale, S.; Kankaala, P.: IMPAIRED TRANSFER OF ESSENTIAL FATTY ACIDS AT THE ALGAE-ZOOPLANKTON INTERFACE IN HUMIC LAKES

15:15 **Lukić, D.**; Horváth, Z.; Vad, C.; Póda, C.; Ptacnik, R.: ENVIRONMENTAL CONSTRAINT OF INTRAGUILD PREDATION: INORGANIC TURBIDITY SHAPES OMNIVORY IN FAIRY SHRIMPS

15:45 **Dillon, K.**; Premo, Z.: FOOD WEB STRUCTURE OF SARGASSUM COMMUNITIES IN THE NORTHERN GULF OF MEXICO.

16:00 **Morley, S.**; Duda, J.; Pess, G.; Johnson, R.; McHenry, M.: FOODWEB RESPONSE DURING AND AFTER DAM REMOVAL

16:15 **Dubois, M.**; Grabowski, R.; Gill, A.: FISH COMMUNITY RESPONSES TO FLOW AND PHYSICAL HABITAT RESTORATION IN RIVERS

16:30 **van Dorst, R.**; Gårdmark, A.; Svanbäck, R.; Huss, M.: THE EFFECT OF BROWNING ON FRESHWATER PELAGIC FOOD WEBS: TOP-DOWN VS. BOTTOM-UP EFFECTS

16:45 **Vesterinen, J.**; Taipale, S.; Kahilainen, K.; Strandberg, U.; Kankaala, P.: LITTORAL BENTHIC INVERTEBRATES CONSTITUTE A HIGH-QUALITY FOOD RESOURCE FOR FISH IN SUBARCTIC LAKES

17:00 **Mercado-Molina, A.**; Trexler, J.: RESOURCE-CONSUMER RELATIONSHIP ALONG A GRADIENT OF WATER FLOW

17:15 **Rhude, K.**; Sterner, R.: DIPOREIA SEDIMENT PREFERENCE ALONG A LAKE SUPERIOR TRANSECT

CS025 PHYTOPLANKTON ECOLOGY AND PHYSIOLOGY

Chair(s): Michael Behrenfeld, Oregon State University (mjb@science.oregonstate.edu)

Location: Room 102 A/B/C

09:00 **Kellogg, M.**; McIlvin, M.; Moran, D.; Marchetti, A.; Moffett, J.: EFFICIENT CO SUBSTITUTION FOR ZN IN THE NORTHEAST PACIFIC OCEAN DIATOMS PSEUDO-NITZSCHIA DELICATISSIMA AND THALASSIOSIRA SP.

09:15 **Shire, D.**; Kustka, A.: PROTEOMIC APPROACHES TOWARDS UNDERSTANDING ZN, CD, AND CO SUBSTITUTION IN THE COSMOPOLITAN COCCOLITHOPHORE EMILIANIA HUXLEYI

09:30 **Bornhorst, R.**; Shrestha, R.; Hildebrand, M.; Kustka, A.: A REVERSE GENETIC AND KINETIC INVESTIGATION OF THE REDUCTIVE-OXIDATIVE IRON UPTAKE PATHWAY IN THALASSIOSIRA PSEUDONANA

09:45 **Berman-Frank, I.**; Tzubari, Y.; Magnezi, L.; Be'er, A.: IRON AND PHOSPHORUS DEPRIVATION INDUCE SOCIALITY IN THE MARINE BLOOM-FORMING CYANOBACTERIUM TRICHODESMIUM

10:00 **Liefer, J.**; Fyfe, M.; Faulkner, S.; Irwin, A.; Follows, M.; Finkel, Z.: NITROGEN AND PHOSPHORUS STARVATION HAVE DISTINCT EFFECTS ON PROCHLOROCOCCUS C:N:P AND ITS MACROMOLECULAR BASIS

10:15 **Sheward, R.**; Liefer, J.; Irwin, A.; Finkel, Z.: INFLUENCE OF ENVIRONMENTAL CONDITIONS ON THE C:N:P OF THE DOMINANT MARINE CALCIFIER EMILIANIA HUXLEYI

14:00 **Anderson, S.**; Kling, J.; Kremer, C.; Franzè, G.; Hutchins, D.; Litchman, E.; Menden-Deuer, S.; Rynearson, T.: WINNERS AND LOSERS IN A CHANGING TIDE: TEMPERATURE-NUTRIENT IMPACT ON PHYTOPLANKTON COMMUNITY DYNAMICS

14:15 **Kling, J.**; Phan, M.; Fu, F.; Anderson, S.; Franzè, G.; Wilburn, P.; Kremer, C.; Litchman, E.; Rynearson, T.; Hutchins, D.: THERMAL DIVERSITY IN A COASTAL MARINE *SYNECHOCOCCUS* COMMUNITY SELECTED UNDER LOW- AND HIGH-TEMPERATURES

14:30 **Trombetta, T.**; Vidussi, F.; Mas, S.; Parin, D.; Simier, M.: WATER TEMPERATURE MAJOR DRIVER OF PHYTOPLANKTON BLOOMS IN COASTAL WATERS

14:45 **Wierenga, J.**; Ibelings, B.: COLD THERMAL REFUGIUM FOR PLANKTOTHRIX RUBESCENS FROM CHYTRID PARASITES.

15:00 **Kärcher, O.**; Filstrup, C.; Brauns, M.; Tasevska, O.; Patceva, S.; Walz, A.; Frank, K.; Markovic, D.: CHLOROPHYLL A – NUTRIENT AND TEMPERATURE RELATIONSHIPS, AND PREDICTIONS FOR LAKES ACROSS MOUNTAIN REGIONS

15:15 **Erratt, K.**; Creed, I.; Trick, C.: DO CYANOBACTERIA LIKE IT HOT AND DIRTY: THE SYNERGETIC EFFECT OF INCREASED LAKE TEMPERATURE AND BROWNING ON CELLULAR PERFORMANCE AND TOXICITY.

15:45 **Yin, J.**; Wilkerson, F.; Blaser, S.; Frantzich, J.: PHYTOPLANKTON PRODUCTIVITY AND NUTRIENT UPTAKE IN THE NORTHERN SAN FRANCISCO ESTUARY

16:00 **Filstrup, C.**; Bramburger, A.: DO DIFFERENCES IN STREAM OPTICAL PROPERTIES DETERMINE PHYTOPLANKTON STRUCTURING IN LARGE LAKES?

16:15 **Heinle, M.**; Flandez, A.; Clardy, T.; Qurban, M.; Prihartato, P.; Hikmawan, T.; Abdulkader, K.: THE ROLE OF NANO- AND PICOPHYTOPLANKTON IN COASTAL PHYTOPLANKTON COMMUNITIES OF THE ARABIAN GULF

16:30 **Li, X.**; Yan, M.: RESPONSES OF TOXIC ALGAE TO ENVIRONMENTAL VARIABLES IN THE PEARL RIVER ESTUARY AND HONG KONG WATERS

16:45 **Kamalanathan, M.**; Mapes, S.; Faulkner, P.; Hillhouse, J.; Hala, D.; Campbell, D.; Beardall, J.; Raven, J.; Quigg, A.: BIOCHEMICAL PATHWAYS ORCHESTRATING THE DIVERSE RESPONSE OF DIFFERENT PHYTOPLANKTON SPECIES TO OIL EXPOSURE

17:00 **Sprecher, B.**; Zhang, H.; Lin, S.: GENETIC TRANSFORMATION EFFORTS IN THE DINOFLAGELLATE OXYRRHIS MARINA

17:15 **Zhou, L.**; Lin, S.: ESTIMATING PHYTOPLANKTON CARBON BIOMASS USING DNA

CS033 CORAL REEF ECOSYSTEMS

Chair(s): Justin Baumann, University of North Carolina at Chapel Hill (baumannj@live.unc.edu)
Travis Courtney, Scripps Institution of Oceanography (traviscourtney@gmail.com)

Location: Room 104

09:00 **Baumann, J.**; Bove, C.; Carne, L.; Castillo, K.: ACCLIMATIZATION TO ENVIRONMENTAL HETEROGENEITY LIMITED BY LOCAL ADAPTATION IN SIDERASTREA SIDEREA BUT NOT PSEUDODIPLORIA STRIGOSA CORALS IN BELIZE

09:15 **Beatty, D.**; Clements, C.; Mathew Valayil, J.; Jarvis, S.; Ritchie, K.; Stewart, F.; Hay, M.: ACCLIMATIZATION IN CORAL ANTI-VIBRIO CHEMICAL DEFENSE AFTER RAPID SHIFTS IN REEF STATE

09:30 **Townsend, J.**; Brandt, M.; Mukherjee, S.; Medina, M.; Smith, T.: LESION RECOVERY OF CORALS ALONG A SHALLOW TO MESOPHOTIC DEPTH GRADIENT

FRIDAY

09:45 **Silva, D.**; Peixoto, R.; Rachid, C.; Soriano, A.; Fragoso, H.; Villela, h.; Duarte, G.; Rosado, P.; Martins, E.; Ximenes, D.: EVALUATION OF OIL SPILL REMEDIATION STRATEGIES FOR PROTECTING BRAZILIAN CORALS AND THEIR ASSOCIATED MICROBIOME

10:15 **Ros, M.**; Camp, E.; Leggat, B.; Suggett, D.: UNLOCKING THE BLACK-BOX OF INORGANIC CARBON-UPTAKE AND UTILISATION STRATEGIES AMONGST CORAL ENDOSYMBIONTS (SYMBIODINIACEAE)

14:00 **Lin, S.**; Yu, L.; Zhang, H.: TRANSCRIPTOMIC RESPONSES TO THERMAL STRESS AND VARIED PHOSPHORUS CONDITIONS IN SYMBIODINIUM KAWAGUTII

14:15 **Brown, K.**; Bender-Champ, D.; Achlatis, M.; van der Zande, R.; Kubicek, A.; Mello-Athayde, M.; Hoegh-Guldberg, O.; Dove, S.: DIFFERENTIAL CARBONATE BUDGETS SHAPE GEOMORPHOLOGICAL ZONES ON A PLATFORM CORAL REEF

14:30 **Navarro, B.**; Santano, J.; Milton, I.; Warren, R.: STRUCTURAL COMPLEXITY IN CORAL REEF ECOSYSTEMS

14:45 **Courtney, T.**; Andersson, A.: EVALUATING MEASUREMENTS OF ECOSYSTEM-SCALE CORAL REEF CALCIFICATION UNDER GLOBAL ENVIRONMENTAL CHANGE

15:00 **Long, M.**; Rheuban, J.; Conner, K.; Hopkinson, B.; McCorkle, D.: ECOSYSTEM PRODUCTIVITY AND CALCIFICATION AT MARGINAL CORAL REEF SITES USING TURBULENT BOUNDARY LAYER TECHNIQUES

15:15 **Gonzalez, F.**: ASSESSING METABOLIC CHANGES OF THE CARIBBEAN CORAL ORBICELLA FAVEOLATA DURING ITS GAMETOGENESIS USING THE CISME (CORAL IN SITU METABOLISM) DEVICE.

16:00 **Hollister, K.**; Ennis, R.; Spalding, H.; Smith, T.: CORAL-ALGAL COMPETITION AND OVERGROWTH DYNAMICS OF A RAPIDLY EMERGING RED ALGA (*RAMICRUSTA* SP.) IN ST. THOMAS, US VIRGIN ISLANDS

16:30 **Garcia-Hernandez, J.**; Sanchez, P.; Hammerman, N.; Schizas, N.: FISH, CORAL, AND SPONGE ASSEMBLAGES ASSOCIATED WITH ALTIPHOTIC AND MESOPHOTIC REEFS ALONG THE GUANICA BIOSPHERE RESERVE CONTINENTAL SHELF EDGE, SOUTHWEST PUERTO RICO

16:45 **Sturm, A.**; González Méndez, J.; Eckert, R.; Chei, E.; Dodge, D.; Studivan, M.; González Díaz, P.: ASSESSING POPULATION CONNECTIVITY OF THE CORAL SPECIES, *MONTASTRAEA CAVERNOSA*, ACROSS LOCAL AND REGIONAL SCALES IN CUBA AND THE TROPICAL WESTERN ATLANTIC

17:00 **Schmidt-Roach, S.**; Muniz Baretto, M.; Wang, X.; Cziesielski, M.; Prasanna, A.; Aranda, M.: UNDERSTANDING POPULATION STRUCTURE AND THERMAL TOLERANCE CAPACITIES OF THE COMMON BRAIN CORAL *PLATYGYRA* IN SAUDI ARABIAN WATERS

17:15 **Ruiz-Jones, L.**; Caruso, C.; Rocha de Souza, M.; Gates, R.: USING RADSEQ TO INVESTIGATE PATTERNS OF CORAL GENETIC DIVERSITY ACROSS AN ENVIRONMENTAL MOSAIC

CS035 CHANGING BIOGEOCHEMISTRY AND ECOLOGY ACROSS POLAR AQUATIC SYSTEMS IN THE 21ST CENTURY

Chair(s): Jeff Bowman, University of California at San Diego (jsbowman@ucsd.edu)
Jim McClelland, University of Texas (jimm@utexas.edu)
Peter Hernes, University of California (pjhernes@ucdavis.edu)
Ronnie Glud, University of Southern Denmark (rnglud@biology.sdu.dk)
Suzanne Tank, University of Alberta (suzanne.tank@ualberta.ca)
Marguirite Xenopoulos, Trent University (mxenopoulos@trentu.ca)
Hyewon Kim, University of Virginia (hk8m@virginia.edu)
Andrew Thurber, Oregon State University (athurber@coas.oregonstate.edu)

Location: Room 202

09:00 **Kim, H.**; Ducklow, H.; Schofield, O.; Steinberg, D.; Doney, S.: DATA ASSIMILATIVE MODELING OF MARINE ECOSYSTEM AND MICROBIAL DYNAMICS IN THE COASTAL WEST ANTARCTIC PENINSULA

09:15 **Capelle, D.**; Kuzyk, Z.; Papakyriakou, T.; Miller, L.; Guéguen, C.; Macdonald, R.: CARBON BUDGET FOR HUDSON BAY

09:30 **Rao, D.**; Saito, M.; DiTullio, G.; Follows, M.: *IN SITU* UPTAKE RATES OF VITAMIN B_{12} AND COBALT ALONG AN ANTARCTIC COSTAL WATER TRANSECT FROM THE AMUNDSEN SEA TO ROSS SEA

09:45 **Yoshida, K.**; McMinn, A.; Seger, A.; Karsh, K.; Suzuki, K.: PHYSIOLOGICAL RESPONSES OF ICE ALGAE TO FE AND LIGHT AVAILABILITY: EX SITU INCUBATION OF ICE ALGAE USING A LOW-FE ICE TANK

10:00 **MacMillan, G.**; Chételat, J.; Richardson, M.; Amyot, M.: ENVIRONMENTAL DRIVERS OF MERCURY BIOACCUMULATION IN ARCTIC FRESHWATER ZOOPLANKTON

10:15 **Vucic, J.**; Gray, D.: ABIOTIC FACTORS INFLUENCING ZOOPLANKTON COMMUNITY STRUCTURE IN SMALL ARCTIC LAKES

14:00 **Johnston, S.**; Bogard, M.; Striegl, R.; Dornblaser, M.; Podgorski, D.; Butman, D.; Spencer, R.: DRIVERS OF DISSOLVED ORGANIC MATTER COMPOSITION IN NORTHERN HIGH LATITUDE LAKES

14:15 **Laney, S.**; Okkonen, S.; Longnecker, K.; Stramski, D.; Koestner, D.: ASSESSING THE CONTRIBUTION OF ORGANIC CARBON TO THE COASTAL BEAUFORT SHELF DURING THE SPRING FRESHET OF THE KUPARUK AND SAGAVANIRKTOK RIVERS

14:30 **McClelland, J.**; Connolly, C.; Rawlins, M.: FLUVIAL NITROGEN EXPORT FROM THE NORTH SLOPE OF ALASKA TO THE BEAUFORT SEA: SPATIAL PATTERNS AND IMPLICATIONS FOR BIOLOGICAL PRODUCTION IN COASTAL WATERS

14:45 **Bristol, E.**; Connolly, C.; McClelland, J.; Lorenson, T.; Richmond, B.; Ilgen, A.; Herrick, C.; Bull, D.; Kanevskiy, M.; Iwahana, G.; Jones, B.: BIOGEOCHEMISTRY OF ERODING SOILS ALONG THE BEAUFORT SEA COASTLINE: RESULTS FROM PERMAFROST CORES COLLECTED AT DREW POINT, AK

15:00 **Shakil, S.**; Tank, S.; Vonk, J.: MINERALIZATION POTENTIAL OF PARTICULATE ORGANIC CARBON MOBILIZED TO STREAMS BY PERMAFROST THAW SLUMPS ON THE PEEL PLATEAU (NT, CANADA)

[1] REPRESENTS INVITED PRESENTATIONS

CS043 GLOBAL OCEANOGRAPHY AND LIMNOLOGY

Chair(s): Adam Heathcote, Science Museum of Minnesota (adam.heathcote@gmail.com)
Jean-Francois Lapierre Université de Montréal (jfrancoislapierre@gmail.com)

Location: Room 208 C

09:00 **Downing, J.**: THE NEED FOR MORE GLOBAL INFERENCE IN OCEANOGRAPHY AND LIMNOLOGY: WHAT'S DONE, HOW IT'S DONE, AND HOW WE CAN TO DO BETTER[T]

09:30 **Plueddemann, A.**; Trowbridge, J.; Dever, E.; Kelley, D.; Brennan-Tonetta, M.: THE OCEAN OBSERVATORIES INITIATIVE (OOI)

09:45 **Hill, R.**; Leibowitz, S.; Weber, M.; Iiames, J.; Schaeffer, B.; Salls, W.: USING THE LAKE-CATCHMENT (LAKECAT) DATASET TO CHARACTERIZE, MODEL, AND PREDICT LAKE CONDITIONS ACROSS THE CONTERMINOUS US

10:00 **Newton, J.**; Jewett, L.; Tilbrook, B.; Isensee, K.; Feely, R.: THE GLOBAL OCEAN ACIDIFICATION OBSERVING NETWORK, GOA-ON: OBSERVING ON LOCAL SCALES GLOBALLY

10:15 **Lapierre, J.**; Collins, S.; Oliver, S.; Stanley, E.: INCONSISTENT BROWNING OF NORTHEASTERN US LAKES DESPITE INCREASED PRECIPITATION AND RECOVERY FROM ACIDIFICATION

14:00 **Lomas, M.**; Baer, S.; Johnson, R.; Lomas, D.; Martiny, A.; Bates, N.: PARTICULATE STOICHIOMETRY OF THE WESTERN NORTH ATLANTIC OCEAN

14:15 **Shahabinia, A.**; Bogard, M.; del Giorgio, P.: PATTERNS AND REGULATION OF ECOSYSTEM METABOLISM IN LAKES ACROSS CANADA

14:30 **Fergus, C.**; Hill, R.; Brooks, J.; Kaufmann, P.; Pollard, A.; Weber, M.; Herlihy, A.; Paulsen, S.: WATER-LEVEL DRAWDOWN IN LAKES: WHAT FACTORS INCREASE THE RISK OF SEVERE DRAWDOWN ACROSS THE CONTERMINOUS U.S.?

14:45 **Rivera, J.**; Lumpkin, R.; Goni, G.; Rivero, U.; Dolk, S.; Olascoaga, M.; Covelli, M.; Torres Garcia, M.: COASTAL CURRENT ESTIMATION BY LAGRANGIAN METHODS IN COSTA RICA AND BAHIA DE MOSQUITOS PANAMA IN THE CARIBBEAN SEA

15:00 **Kovalenko, K.**; Reavie, E.; Edlund, M.: REGIONAL PALEOLIMNOLOGY REVEALS WIDESPREAD ANTHROPOGENIC IMPACTS

15:15 **Paquette, C.**; Gregory-Eaves, I.; Beisner, B.: ZOOPLANKTON SUBFOSSIL ASSEMBLAGE SHIFTS SINCE PREINDUSTRIAL TIMES ACROSS CANADIAN ECOZONES AND IN RELATION TO HUMAN ACTIVITIES

SPOT001: PUBLICATIONS SPOTLIGHT SESSION

Chair(s): Marguerite A. Xenopoulos, Trent University (mxenopoulos@trentu.ca)
Christopher T. Filstrup, University of Minnesota Duluth (filstrup@d.umn.edu)
Paul Kemp, Editor, Limnology and Oceanography: Methods (lomethods-editor@aslo.org)
Patricia A. Soranno, Michigan State University (soranno@msu.edu)
Adrienne J. Sponberg, ASLO Director of Communications and Science (asponberg@aslo.org)

Location: Room 209 A/B

09:00 **Hundey, B.**; Wood-Charlson, E.; Tolar, B.; Shirey, P.; Olker, J.: TIDAL MOMENTUM: SCIENCE COMMUNICATION TRAINING OPPORTUNITIES THAT MEET EARLY-CAREER SCIENTIST RECOMMENDATIONS

09:15 **Kujawinski, E.**; Kido Soule, M.; Swarr, G.; Longnecker, K.; Widner, B.; Frank, A.: DISSOLVED MARINE METABOLOMICS

09:30 Bonnet, S.; **Benavides, M.**; Berthelot, H.; Caffin, M.; Van Wambeke, F.; Moutin, T.; Berman-Frank, I.; Barthaux, V.; Grosso, O.; Grégori, G.; Heliás-Nunige, S.; Hunt, B.; Carlotti, F.: HOW DOES DIAZOTROPH DERIVED NITROGEN TRANSFER IN THE PLANKTONIC FOOD WEB?

09:45 **Gradoville, M.**; Bombar, D.; Crump, B.; Letelier, R.; Zehr, J.; White, A.: DIVERSITY AND ACTIVITY OF NITROGEN-FIXING COMMUNITIES ACROSS OCEAN BASINS: OVERVIEW AND RECENT DEVELOPMENTS

10:00 **Glibert, P.**; Wilkerson, F.; Dugdale, R.; Raven, J.; Dupont, C.; Leavitt, P.; Parker, A.; Burkholder, J.; Kana, T.: PLUSES AND MINUSES OF AMMONIUM AND NITRATE UPTAKE AND ASSIMILATION BY PHYTOPLANKTON AND IMPLICATIONS FOR PRODUCTIVITY AND COMMUNITY COMPOSITION, WITH EMPHASIS ON NITROGEN-ENRICHED CONDITIONS

14:00 **Mette, M.**: APPLICATIONS OF BIVALVE SCLEROCHRONOLOGY TO ECOLOGICAL AND CLIMATOLOGICAL STUDIES

14:15 **Tzadik, O.**; Curtis, J.: CHEMICAL ARCHIVES IN FISHES BEYOND OTOLITHS: A REVIEW ON THE USE OF OTHER BODY PARTS AS CHRONOLOGICAL RECORDERS OF MICROCHEMICAL CONSTITUENTS FOR EXPANDING INTERPRETATIONS OF ENVIRONMENTAL, ECOLOGICAL, AND LIFE-HISTORY CHANGES

14:30 **Casas-Ruiz, J.**: A TALE OF PIPES AND REACTORS: CONTROLS ON THE IN-STREAM DYNAMICS OF DISSOLVED ORGANIC MATTER IN RIVERS

14:45 **Kraus, T.**; Hansen, A.; Downing, B.; Pellerin, B.; Fleck, J.; Bergamaschi, B.: USING OPTICAL PROPERTIES TO LOOK BACKWARDS AND FORWARDS IN TIME: DISSOLVED ORGANIC MATTER SOURCE, PROCESSING, REACTIVITY AND FATE

15:00 **Deemer, B.**; Hayes, N.; Strock, K.; Corman, J.; Razavi, R.; Dibble, K.; Yackulic, C.: CATCHMENT AND MANAGEMENT CHARACTERISTICS ARE KEY IN DETERMINING RESERVOIR RESPONSE TO CLIMATE CHANGE

15:15 **Fasching, C.**; Yao, H.; Rusak, J.; Xenopoulos, M.: HYDROLOGICAL EFFECTS ON STREAM-LAKE COUPLING AND DECOUPLING C:N:P STOICHIOMETRY, BUDGETS AND EXPORT FLUXES

15:45 **McCrackin, M.**: MANAGING EUTROPHICATION: THE CASE OF THE BALTIC SEA

16:00 **Kohlbach, D.**; Graeve, M.; Lange, B.; Schaafsma, F.; Lebreton, B.; David, C.; Vortkamp, M.; Peeken, I.; Ferguson, S.; Michel, C.; Flores, H.: CRITICAL IMPORTANCE OF SEA ICE-DERIVED CARBON IN ARCTIC AND ANTARCTIC FOOD WEB PROCESSES

16:15 **Burson, A.**; Stomp, M.; Akil, L.; Brussaard, C.; Huisman, J.: UNBALANCED REDUCTIONS OF NUTRIENT LOADS HAS CREATED AN OFFSHORE GRADIENT FROM PHOSPHORUS TO NITROGEN LIMITATION IN THE NORTH SEA

16:30 **Darnis, G.**; Geoffroy, M.; Daase, M.; Søreide, J.; Swoboda, S.; Cottier, F.; Renaud, P.; Berge, J.: THE KEY ROLE OF ZOOPLANKTON PROCESSES IN THE EXPORT OF BIOGENIC MATTER DURING THE TRANSITION FROM POLAR NIGHT TO SPRING IN A HIGH-ARCTIC MARINE SYSTEM

16:45 **Yokota, K.**; Mehlrose, M.; Hastings, C.; Davidson Parry, E.; Waterfield, H.; Kwietniewski, E.; Wells, B.: AQUATIC MICROPLASTICS AFFECT PRIMARY PRODUCERS, TOO!

FRIDAY

[T] REPRESENTS TUTORIAL PRESENTATIONS

SS002 ENVIRONMENTAL EPIGENETICS: KEY MECHANISMS OF ACCLIMATIZATION AND ADAPTATION TO GLOBAL OCEAN CHANGE

Chair(s): Jose Eirin-Lopez, Florida International University (jeirinlo@fiu.edu)
Hollie Putnam, University Of Rhode Island (hputnam@uri.edu)

Location: Room 103 A

09:00 **Rosado Borrero, G.**; Schizas, N.: TELOMERE LENGTH VARIATION AMONG ENDANGERED CARIBBEAN SCLERACTINIAN CORAL ACROPORA CERVICORNIS

09:15 **Cziesielski, M.**; Mariappan, K.; Aranda, M.: HISTONE MODIFICATIONS REGULATE AND MAINTAIN CNIDARIAN-ALGAE SYMBIOSIS

09:30 **Rodriguez-Casariego, J.**; Mercado-Molina, A.; Baums, I.; Sabat, A.; Eirin-Lopez, J.: NON-GENETIC MECHANISMS OF CORAL RESPONSE TO GLOBAL CHANGE: PRELIMINARY EPIGENETIC ANALYSES IN CORALS IMPACTED BY HURRICANES IRMA AND MARIA IN PUERTO RICO

09:45 **Roberts, S.**: COMPARATIVE ANALYSES OF DNA METHYLATION PATTERNS IN BIVALVES

10:00 **Nguyen, N.**; Dimond, J.: DNA METHYLATION PROFILING OF ANTHOPLEURA ELEGANTISSIMA USING NANOPORE SEQUENCING

10:15 **Beal, A.**; Wells, R.; Kiszka, J.; Eirin-Lopez, J.: THE BOTTLENOSE EPIGENETIC AGING TOOL (BEAT): A RELIABLE MOLECULAR TOOL FOR AGING WILD SMALL CETACEANS

14:00 **Prasanna, A.**; Schmidt-Roach, S.; Liew, Y.; Cziesielski, M.; Zahran, N.; Aranda Lastra, M.: A STUDY ON DNA METHYLATION CHANGES IN EXAIPTASIA PALLIDA IN RESPONSE TO LONG TERM HEAT ACCLIMATION

14:15 **Putnam, H.**; Spencer, L.; Roberts, S.; Goetz, F.: THE POTENTIAL FOR ACCLIMATIZATION TO OCEAN ACIDIFICATION THROUGH EPIGENETIC MEMORY IN GEODUCK CLAMS

14:30 **Fellous, A.**; Shama, L.: DNA METHYLATION DURING GAMETOGENESIS AND EMBRYOGENESIS OF THREESPINE STICKLEBACK : A SOURCE OF HERITABLE PHENOTYPIC DIVERSITY FOR ADAPTATION TO CLIMATE CHANGE ?

14:45 **Leach, T.**; Strader, M.; Hofmann, G.: INVESTIGATING THE ROLE OF MATERNAL CONDITIONING ON OFFSPRING PERFORMANCE AND DNA METHYLATION PATTERNS IN THE PURPLE SEA URCHIN

15:00 **Strader, M.**; Hofmann, G.: INTRA- AND TRANSGENERATIONAL PLASTICITY OF DNA METHYLATION IN THE PURPLE SEA URCHIN, STRONGYLOCENTROTUS PURPURATUS

15:15 Liew, Y.; Howells, E.; Wang, X.; Michell, C.; Burt, J.; Idaghdour, Y.; **Aranda Lastra, M.**: INTERGENERATIONAL EPIGENETIC INHERITANCE IN REEF-BUILDING CORALS

SS006 PLASTICS IN THE AQUATIC ENVIRONMENT

Chair(s): Aron Stubbins, Northeastern University, Boston (aron.stubbins@northeastern.edu)
Cristina Romera-Castillo, Institut de Ciencias del Mar, CSIC, Barcelona (crisrc@icm.csic.es)
Kara Lavender Law, Sea Education Association, Woods Hole (klavender@sea.edu)
Daoji Li, East China Normal University, Shanghai (daojili@sklec.ecnu.edu.cn)

Location: Room 201

09:00 Arias, M.; **Grossart, H.**: MICROPLASTIC PARTICLES: A NEWLY EMERGING MICROBIAL HABITAT IN AQUATIC ECOSYSTEMS WITH MULTIPLE AND OFTEN UNFORESEEN CONSEQUENCES FOR ECOLOGY AND HUMAN HEALTH

09:30 **Koski, M.**; Christensen, A.; Nielsen, T.: EFFECT OF CAR TIRE AND RUBBER GRANULES ON COASTAL COPEPODS

09:45 **Pedersen, A.**; Boegehold, A.; Gopalakrishnan, K.; Kashian, D.: SUBLETHAL EFFECTS IN DREISSENA BUGENSIS FOLLOWING EXPOSURE TO MICROPLASTICS

10:00 **Schmidt, N.**; Castro-Jimenez, J.; Oursel, B.; Sempéré, R.: PLASTIC ADDITIVES IN SEAWATER, SEDIMENT AND ZOOPLANKTON SAMPLES FROM THE BAY OF MARSEILLE, SOUTHERN FRANCE

10:15 **Carrasquillo-Rosa, G.**; Zayas, B.: CELLULAR EVENTS RESULTING FROM CONTINUES EXPOSURE OF AQUATIC SPECIES TO THE PLASTICIZER DIBUTYL PHTHALATE, AN ENVIRONMENTAL CONTAMINANT

14:00 **Urban-Rich, J.**; Carilli, J.; Baumann, J.; Rotjan, R.: ARE MICROPLASTICS INGESTED OR INCORPORATED WITHIN REEF-FORMING CORALS?

14:15 **Seeley, M.**; Song, B.; Hale, R.: EFFECT OF DIFFERENT MICROPLASTICS ON SEDIMENT MICROBIAL COMMUNITIES, DENITRIFICATION AND NUTRIENT FLUXES

14:30 **Beklioğlu, M.**; Yalçın, G.; Çetinbağ, D.; Jovanovic, B.; Öztürk, D.; Vebrová, L.; Boukal, D.; Milošević, D.; Savić, D.; Stanković, J.; Feuchtmayr, H.: IN-SITU MESOCOSMS EXPERIMENT FOR INVESTIGATING IMPACTS OF MICROPLASTICS ON LITTORAL FOOD WEB

15:00 **Wiggin, K.**; Holland, E.: MICROPLASTIC POLLUTION SIZED 500-3UM IN AQUATIC ENVIRONMENTS NEAR HIGHLY URBANIZED AREAS

15:15 **Lasseigne, D.**; Masura, J.; Wilson, K.; Brandt, M.: MICROPLASTIC AND MICROFIBER ABUNDANCES IN COASTAL ENVIRONMENTS OF ST. THOMAS, USVI.

15:45 **Miller, M.**; Steele, C.; Diego, R.: FACTORS INFLUENCING THE DISTRIBUTION OF MICROPLASTICS AND MARINE DEBRIS ON CALIFORNIA'S CHANNEL ISLANDS

16:00 **Martin, C.**; Al Nahdi, R.; Duarte, C.: THE HISTORY OF MARINE PLASTIC POLLUTION IN A HANDFUL OF SEDIMENT

16:15 **Crew, A.**; Gregory-Eaves, I.; Ricciardi, A.: ASSESSING THE DIVERSITY, ABUNDANCE AND DISTRIBUTION OF MICROPLASTICS IN THE SEDIMENTS OF THE ST. LAWRENCE RIVER

16:30 **Xiong, X.**; Wu, C.; Elser, J.: MICROPLASTICS IN HIGH-ALTITUDE LAKES – A COMPARATIVE STUDY BETWEEN CHINA AND U.S.

16:45 **Ward, J.**; Zhao, S.; Mladinich, K.; Griffin, T.; Holohan, B.; Shumway, S.: EXAMINING THE USE OF BIVALVE MOLLUSCS AS INDICATORS OF MICROPLASTICS IN THE ENVIRONMENT

17:00 **Lascelles, N.**; Jagoe, C.; Deshpande, A.; Freeman, D.; Drayton, D.: ANALYSIS AND CHEMICAL CHARACTERIZATION OF MICROPLASTIC POLLUTION IN COASTAL WATERS

17:15 **Li, D.**: PROGRESS AND PROSPECTS OF MARINE MICROPLASTIC RESEARCH IN CHINA

[1] REPRESENTS INVITED PRESENTATIONS

SS019 BLUE CARBON: FROM THE ECOSYSTEM TO THE MARKETS

Chair(s): Miguel-Angel Mateo, Spanish Council for Scientific Research (CSIC) (mateo@ceab.csic.es)
Paul Lavery, Edith Cowan University (p.lavery@ecu.edu.au)
Oscar Serrano, Edith Cowan University (o.serranogras@ecu.edu.au)
Martin Skov, Bangor University (mwskov@bangor.ac.uk)

Location: Room 204

09:00 **Chmura, G.**: IDENTIFYING PREDICTORS OF BLUE CARBON (SALT MARSH SOIL ORGANIC CARBON) ACCUMULATION RATES - EXAMPLES FROM EASTERN CANADA

09:15 **Schiebel, H.**; Lawlor, K.; Templeton, T.: CREATING A NEW ENGLAND SALT MARSH SEDIMENT PROFILE: METHODS AND IMPLICATIONS FOR CLIMATE CHANGE MITIGATION STRATEGIES

09:30 **Gorham, C.**; Lavery, P.; Kelleway, J.; Salinas, C.; Serrano, O.: CARBON STORAGE BY TIDAL MARSH ECOSYSTEMS IN WESTERN AUSTRALIA

09:45 **Piñeiro Juncal, N.**; Diaz-Almela, E.; Leiva-Dueñas, C.; Deulofeu, O.; Soler, M.; Benjumea, D.; Giralt, S.; Frigola, J.; Martinez-Cortizas, A.; Serrano, O.; Lavery, P.; Mateo, M.: SEDIMENTARY PROCESSES AS A CONTROL OF BLUE CARBON STOCKS AND FLUXES: A STUDY CASE FROM THE SPANISH ANDALUSIAN COASTS

10:00 **Bjork, M.**; Gullström, M.; Lyimo, L.; George, R.; Lyimo, T.; Hamisi, M.; Mangora, M.; Mtolera, M.; Deyanova, D.; Dahl, M.; Asplund, M.: POSSIBLE DRIVERS OF METHANE (CH4) AND NITROUS OXIDE (N2O) EMISSIONS IN SEAGRASS MEADOWS

10:15 **Lopes, C.**: TRENDS IN INORGANIC CARBON DYNAMICS OF SEAGRASS THALASSIA TESTUDINUM

14:00 **Leiva Dueñas, C.**; Piñeiro-Juncal, N.; Díaz-Almela, E.; Serrano, O.; Belshe, F.; Lavery, P.; Martínez-Cortizas, A.; Mateo, M.: FACTORS INFLUENCING LONG-TERM BLUE CARBON ACCUMULATION IN SEAGRASSES MEADOWS ALONG ANDALUSIAN COAST: A PALEO-PERSPECTIVE.

14:30 **Costa, M.**; Ezcurra, E.; Ezcurra, P.; Salinas-de-León, P.; Turner, B.; Leichter, J.; Aburto-Oropeza, O.: SEDIMENT DEPTH AND ACCUMLATION AGE CONSTRAIN MANGROVE CARBON STOCKS, DEPENDING ON FOREST TYPE

14:45 **Ortega, A.**: METAGENOMES REVEAL WORLDWIDE OCCURRENCE OF MACROALGAE IN THE OCEAN

15:00 **Serrano, O.**; Lavery, P.; Lovelock, C.: THE POTENTIAL OF AUSTRALIAN BLUE CARBON ECOSYSTEMS FOR CLIMATE CHANGE MITIGATION - A NATIONAL ASSESSMENT.

15:15 **Diaz Almela, E.**; Piñeiro Juncal, N.; Leiva Dueñas, C.; García-Orellana, J.; Frigola, J.; Soler, M.; Benjumea-Novalbos, D.; Bruach-Menchén, J.; Serrano, O.; Belshe, E.; Lavery, P.; Vasquez-Loarte, A.; Pagès Fauria, J.; Giralt, S.; Hernández Carrero, I.; Masque, P.; Mateo, M.: A BLUE CARBON INVENTORY: STOCKS AND FLUXES FOR THE REGION OF ANDALUSIA (SW MEDITERRANEAN)

15:45 **Herrera-Silveira, J.**; Pech, M.; Morales, S.; Caamal, J.; Cinco, S.; Pech, E.; Camacho, A.; Mendoza, J.; Ramírez, J.; Pérez, O.; Teutli, C.: BLUE CARBON OF MEXICO: CARBON STOCKS AND FLUXES IN MANGROVES AND SEAGRASSES, NATIONAL AND REGIONAL SYNTHESIS

16:00 **Rovai, A.**; Twilley, R.; Castaneda-Moya, E.; Pagliosa, P.; Fonseca, A.; Riul, P.: ADJUSTING THE CONTRIBUTION OF MANGROVES TO GLOBAL CARBON STOCKS

16:15 **Kumagai, J.**; Costa, M.; Aburto, O.: MANGROVES, MONEY, AND, MANAGEMENT: ADRESSING THE DISCONNECT BETWEEN BLUE CARBON RESEARCH AND GOVERNMENT ACTION

16:30 **Mateo, M.**; Diaz Almela, E.; Mendoza, R.; Vivas, S.; Piñeiro-Juncal, N.; Leiva Dueñas, C.; Otero, M.; Peñalver, P.; León, D.; Alhama, M.; Gómez, A.; Moreno, D.; Barrajón, A.; de la Rosa, J.; Remón, J.; Fernández, M.; Arroyo, M.; Giralt, S.; Serrano, O.; Belshe, E.; Lavery, P.; Sans, A.; Alaman, M.; Monnier, B.: THE PROJECT "LIFE BLUE NATURA": GETTING MUDDY TO CLEAR THE PATH OF BLUE CARBON TO THE EMISSION OFFSET SYSTEMS

16:45 McGlathery, K.; **Aoki, L.**; Oreska, M.; Berger, A.; Wiberg, P.; Berg, P.: SEAGRASS RESTORATION AND RESILIENCE DRIVES LONG-TERM BLUE CARBON ACCUMULATION

17:00 **Watson, E.**; Powell, E.; Champlin, L.; Fountain, M.; Wigand, C.: CAP AND TRADE FUNDED COASTAL WETLAND RESTORATION-CARBON SEQUESTRATION PROJECTS: EXAMPLES FROM CALIFORNIA AND THE US NORTHEAST

17:15 Howard, J.; **Diazgranados, M.**; Sierra, P.; Espinosa, R.; Ramos, J.; Pidgeon, E.: DEVELOPING A BLUE CARBON MANGROVE CONSERVATION PROJECT IN CISPATA, COLOMBIA

SS030 BIOLUMINESCENT BAYS OF THE CARIBBEAN: SCIENCE, MANAGEMENT, OUTREACH, AND RECOVERY

Chair(s): Michael Latz, Scripps Institution of Oceanography (mlatz@ucsd.edu)
Mark Martin Bras, Vieques Conservation and Historical Trust (biobaypatrick@gmail.com)

Location: Room 101 A/B

09:00 **Martin Bras, M.**; Cabral Guadalupe, A.; Fernandez Porto, J.: VARIATION IN ABUNDANCE *PYRODINIUM BAHAMENSE* BEFORE AND AFTER HURRICANE MARIA IN PUERTO MOSQUITO VIEQUES BIOLUMINESCENT BAY, VIEQUES

09:15 **Carter, M.**; Martin Bras, M.; Cabral Guadalupe, A.; Fernández Porto, J.; Latz, M.: DRIVERS OF *PYRODINIUM BAHAMENSE* ABUNDANCE OVER MULTI-YEAR TIMESCALES: SEARCHING FOR THE COMPLETE STORY

09:45 Reale-Munroe, K.; **Castillo, B.**; Pinckney, J.; Munroe, M.: INFLUENCES OF WATERSHED INPUTS ON WATER QUALITY AND BIOLUMINESCENT DINOFLAGELLATES IN MANGROVE LAGOON, ST. CROIX, USVI

10:00 **Reale-Munroe, K.**; Pinckney, J.; Greenfield, D.; Tomas, C.; Lane, C.; Castillo, B.; Latz, M.; Zimberlin, M.; Reidhaar, P.; Hillis-Starr, Z.: HIGHLIGHTS - BIOLUMINESCENT MANGROVE LAGOON, ST. CROIX, USVI

10:15 **Latz, M.**; Carter, M.: FIRST BIOLUMINESCENCE MEASUREMENTS OF A 'NEW' BIOLUMINESCENT LAGOON IN SALT RIVER BAY, ST. CROIX, USVI

SS031 IMPACTS OF (SUB)MESOSCALE OCEAN DYNAMICS ON MARINE ECOSYSTEMS

Chair(s): Alice Della Penna, Univ. of Washington - Applied Physics Laboratory (alice.dellapenna@gmail.com)
Alison Chase, University of Maine (alison.p.chase@maine.edu)
Peter Gaube, Univ. of Washington - Applied Physics Laboratory (pgaube@apl.uw.edu)

Location: Room 209 C

09:00 **Gaube, P.**: THE STRUCTURING OF OPEN-OCEAN ECOSYSTEMS BY (SUB)MESOSCALE DYNAMICS

09:15 **Lehahn, Y.**; Koren, I.; Sharoni, S.; d'Ovidio, F.; Vardi, A.; Boss, E.: ENHANCEMENT AND PROLONGATION OF (SUB)MESOSCALE PHYTOPLANKTON BLOOMS BY HORIZONTAL STIRRING AND MIXING

FRIDAY

09:30 **Clayton, S.**; Palevsky, H.; Thompson, L.; Quay, P.: MULTI-SCALE CONTROLS ON NET COMMUNITY PRODUCTION AND CHLOROPHYLL IN THE KUROSHIO EXTENSION

09:45 Lévy, M.; Dutkiewicz, S.; Jahn, O.; Clayton, S.; **Mangolte, I.**: IMPACT OF SUBMESOSCALE PROCESSES ON PHYTOPLANKTON COMMUNITY STRUCTURE ACROSS OCEANIC FRONTS

10:00 **Chase, A.**; Della Penna, A.; Gaube, P.: INFLUENCE OF SMALL-SCALE FRONTAL FEATURES ON PHYTOPLANKTON COMMUNITY COMPOSITION IN THE NORTH ATLANTIC OCEAN

10:15 **de Verneil, A.**; Lachkar, Z.; Smith, S.; Lévy, M.: BALANCE OF CARBON FLUXES IN A SUBMESOSCALE RESOLVING MODEL OF THE ARABIAN SEA UPWELLING REGION

14:00 **Wear, E.**; Carlson, C.; Church, M.: BACTERIOPLANKTON METABOLIC RESPONSE TO PHYTOPLANKTON LYSATE VARIES ACROSS A MESOSCALE EDDY DIPOLE IN THE NORTH PACIFIC SUBTROPICAL GYRE

14:15 **Sutula, M.**; Kessouri, F.; McWilliams, J.; Bianchi, D.; Renault, L.; Frenzel, H.; Deutsch, C.: SEASONAL MODULATION OF NUTRIENTS CYCLES AND PHYTOPLANKTON BIOMASS BY SUBMESOSCALE CIRCULATION IN THE CALIFORNIA CURRENT SYSTEM

14:30 **Lowe, A.**; Edwards, C.; Drake, P.: A MODEL INVESTIGATION OF THE INFLUENCE OF SUBMESOSCALE FRONTAL DYNAMICS ON CONNECTIVITY OF NEARSHORE ROCKFISH POPULATIONS AROUND THE MONTEREY PENINSULA

15:00 **Hanz, U.**; Rapp, H.; Duineveld, G.; Roberts, E.; Reichart, G.; Mienis, F.: OCEAN DYNAMICS AT AN ARCTIC SEAMOUNT INFLUENCING A DEEP-SEA SPONGE REEF

15:15 **Ewen, K.**; Habtes, S.; Nemeth, R.; Holstein, D.; Gerard, T.: THE USE OF LARVAL DISPERSAL MODELS TO IDENTIFY POSSIBLE SPAWNING SITES FOR PARROTFISH (FAMILY SCARIDAE) IN THE U.S. VIRGIN ISLANDS

SS033 GOING DEEPER: LINKAGES ACROSS THE OCEANIC WATER COLUMN

Chair(s): Sergio Morales, University of Otago (sergio.morales@otago.ac.nz)
Federico Baltar, University of Vienna (federico.baltar@otago.ac.nz)

Location: Room 209 C

15:45 Wenley, J.; **Baltar, F.**; Morales, S.: SEASONALITY AND CONNECTIVITY OF THE SURFACE AND DEEP OCEAN'S PROKARYOTIC COMMUNITIES

16:00 **Cruz, B.**; Brozak, S.; Neuer, S.: OPTICAL AND DNA-BASED CHARACTERIZATION OF SINKING PARTICLES AT THE BERMUDA ATLANTIC TIME-SERIES STUDY STATION (BATS)

16:15 **Huete-Stauffer, T.**; Logares, R.; Ansari, M.; Calleja, M.; Morán, X.: THE RED SEA MICROBIOME: SEASONAL CONNECTION BETWEEN THE SURFACE AND THE DEEP SCATTERING LAYER?

16:30 **Rigonato, J.**; Budinich, M.; Brandão, M.; Murillo, A.; Pierella, J.; Soviadan, D.; Gregory, A.; Sullivan, M.; Vargas, C.; Eveillard, D.; Bowler, C.; Bork, P.; Lombard, F.; Wincker, P.; Stemmann, L.; Jaillon, O.; Ocean, T.: UNVEILING COMMUNITIES STRUCTURE OF GLOBAL OCEAN MESOPELAGIC PLANKTON

16:45 **Morales, S.**; Lockwood, S.; Baltar, F.: THE MARINE METHANE PARADOX: PLAYERS AND RELEVANCE OF AN UNACCOUNTED METHANE SOURCE IN THE AEROBIC OCEAN ALONG THE WATER COLUMN

17:00 **Reinthaler, T.**; Herndl, G.: DEEP-SEA HETEROTROPHIC MICROBIAL ACTIVITY IN THE WORLD'S OCEANS

17:15 **Mara, P.**; Sylvan, J.; Schubotz, F.; Burgaud, G.; Klein, F.; Beaudoin, D.; Wee, S.; Dick, H.; Lott, S.; Cox, R.; Meyer, L.; Edgcomb, V.: EXPLORATION OF LIFE IN THE LOWER OCEANIC CRUST: STRATEGIES FOR MICROBIAL SURVIVAL

SS054 REMOTE SENSING OF WATER QUALITY AND QUANTITY

Chair(s): Kiana Zolfaghari, University of Waterloo (kzolfagh@uwaterloo.ca)
Claude R. Duguay, University of Waterloo (crduguay@uwaterloo.ca)
Daniel Odermatt, EAWAG (daniel.odermatt@eawag.ch)

Location: Room 202

15:45 **Yousef, F.**: LONG-TERM THERMAL DYNAMICS IN ALPINE LAKES OF WESTERN ROCKY MOUNTAIN

16:00 **Zhang, Y.**: PERMANENT WATER TRANSPARENT ESTIMATION IN CHINA USING LANDSAT SERIES IMAGES

16:15 **Schalles, J.**: SATELLITE CLASSIFICATION AND MAPPING OF CENTRAL SOUTH ATLANTIC BIGHT WATERS USING SPECTRAL LIBRARY DERIVED CLUSTER ANALYSIS

16:30 **Uudeberg, K.**; Arikas, A.; Soomets, T.; Randla, M.; Ansper, A.; Reinart, A.: WATER QUALITY CHANGES IN INLAND AND COASTAL WATERS CAPTURED BY SATELLITES USING OPTICAL WATER TYPE GUIDED APPROACH

16:45 **Hooker, K.**; Beyer, J.; Xiao, X.; Hambright, K.: IMPROVING PREDICTIONS OF HARMFUL ALGAL BLOOMS USING REMOTE SENSING: IS MORE DATA THE ANSWER?

17:00 **Sørensen, K.**; Norli, M.; Hjermann, D.; Kristiansen, T.; Marty, S.; King, A.; Jaccard, P.; Mengeot, C.; Protsenko, E.; Eikrem, W.: USE OF FERRYBOX SHIPS OF OPPORTUNITY SYSTEMS FOR SATELLITE PRODUCT VALIDATION

17:15 **Duguay, C.**; Crétaux, J.; Simis, S.; Merchant, C.; Giardino, C.: DELIVERING THE LAKE ESSENTIAL CLIMATE VARIABLES - AN UPDATE FROM ESA CCI LAKES

SS058 BRAVE NEW WORLD: THE ECOLOGY OF HIGHLY IMPACTED WATERBODIES

Chair(s): Grace Wilkinson, Iowa State University (wilkinso@iastate.edu)
Eric Moody, Iowa State University (ekmoody@iastate.edu)

Location: Room 208 A/B

09:00 **Uye, S.**: FORMERLY PRODUCTIVE FISHING GROUNDS GET IN TROUBLE WITH JELLYFISH: TEMPORAL CHANGE IN ANTHROPOGENIC IMPACTS IN TWO JAPANESE COASTAL ECOSYSTEMS, THE SETO INLAND SEA AND THE ARIAKE SEA

09:15 **Durdall, A.**; Nick, S.; Pott, C.; Nemeth, R.; Grimes, K.: DECLINE OF A CRITICAL FISH NURSERY HABITAT IN GREAT POND, ST. CROIX, USVI: INVASIVE TILAPIA, MANGROVE DIE-OFF, FISH KILLS AND HURRICANE MARIA

09:30 **Broman, E.**; Raymond, C.; Sommer, C.; Gunnarsson, J.; Creer, S.; Nascimento, F.: BALTIC MEIOFAUNAL DIVERSITY AND COMMUNITY COMPOSITION ARE DRIVEN BY SALINITY AND INTERACTIONS WITH MACROFAUNA

10:00 **Tolar, B.**; Boye, K.; Bobb, C.; Spielman, L.; Bargar, J.; Francis, C.: MICROBIAL SIGNATURES OF SEASONAL REDOX TRANSITIONS IN A URANIUM-CONTAMINATED FLOODPLAIN

10:15 **Hébert, M.**; Fugère, V.; Beisner, B.; Fussmann, G.; Gonzalez, A.: THE IMPACTS OF RISING GLYPHOSATE USE ON PHOSPHORUS LOADING AND PLANKTON COMMUNITIES IN AGRICULTURAL WATERSHEDS

14:00 **Hoffman, D.**; McCarthy, M.; Davis, T.; Gossiaux, D.; Burtner, A.; Johengen, T.; Palladino, D.; Gardner, W.; Myers, J.; Newell, S.: SPATIAL AND TEMPORAL DYNAMICS OF WATER COLUMN AMMONIUM IN WESTERN LAKE ERIE: UPTAKE, INTERNAL RECYCLING, AND CYANOHABS

FRIDAY

[1] REPRESENTS INVITED PRESENTATIONS

14:15 **Evans, M.**; Nevers, M.: IMPACTS OF MULTIPLE ECOSYSTEM STRESSORS ON NUISANCE CLADOPHORA GROWTH

14:30 **Fleck, R.**; Albright, E.; Grigel, H.; Moody, E.; Ortiz, D.; Walter, J.; Wilkinson, G.: PREDICTING TOXIC ALGAL BLOOMS IN LAKES LOCATED IN A HIGHLY EUTROPHIC LANDSCAPE

14:45 **Moody, E.**; Butts, T.; Fleck, R.; Jeyasingh, P.; Wilkinson, G.: FROM GLYCOLYSIS TO GREEN LAKES: EVOLUTION OF DAPHNIA ALTERS ECOSYSTEM FUNCTION IN HYPEREUTROPHIC LAKES

15:00 **Scott, T.**: COMMUNITY, IDENTITY, STABILITY – REVISITING THE NITROGEN : PHOSPHORUS RELATIONSHIP IN LAKES[T]

15:15 **Stanley, B.**; Sipler, R.; Roberts, Q.; Bott, C.; Song, B.; Bronk, D.: ESTUARINE WATER MICROBIOME RESPONSES TO DIFFERENTLY TREATED WASTEWATER EFFLUENTS

15:45 **Taylor, J.**; Nifong, R.; Moore, M.; Farris, J.: DISCONTINUITY BETWEEN PATTERN AND PROCESS IN SIMULATED AGRICULTURAL DITCHES YIELD ECOLOGICAL AND METHODOLOGICAL QUESTIONS REGARDING DIEL DENITRIFICATION MODELS[T]

16:00 **Nick, S.**; Grimes, K.: DO YOU REALLY WANT TO SWIM THERE? AN INVESTIGATION OF ENTEROCOCCUS BACTERIA AT BEACHES OF THE UNITED STATES VIRGIN ISLANDS

16:15 **Bonaglia, S.**; Rämö, R.; Nybom, I.; Wikström, J.; Marzocchi, U.; Leermakers, M.; Walve, J.; Nascimento, F.; Sobek, A.; Gunnarsson, J.: SEDIMENT REMEDIATION WITH ACTIVE SORBENTS: EFFECTS ON CONTAMINANT FLUXES, BIOGEOCHEMICAL PROCESSES AND MEIOFAUNA

16:30 **Matsuzaki, S.**; Shinohara, R.; Uchida, K.; Sasaki, T.: CATCH DIVERSIFICATION PROVIDES MULTIPLE BENEFITS IN INLAND FISHERIES

16:45 **Dexter, E.**; Bollens, S.; Cordell, J.; Rollwagen-Bollens, G.: INSIGHTS FROM A 20-YEAR ZOOPLANKTON TIME-SERIES ENCOMPASSING A WIDE RANGE OF NORTHEAST PACIFIC ESTUARIES.

203 papers

FRIDAY

[T] REPRESENTS TUTORIAL PRESENTATIONS

AUTHOR INDEX

A

Abad, J. 39
Abaya, L. 67
Abbott, K. 80
Abdala, Z. 41
Abdulkader, K. 45, 84
Abdullah, H. 71
Abdul-Samad, H. 62
Abe, O. 57
Abernethy, E. 48
Abonyi, A. 60
Abuel-Regal, M. 58
Aburto, O. 81, 88
Aburto-Oropeza, O. 88
Abuyen, K. 55
Achlatis, M. 85
Acker, M. 41, 75
Ackermann, M. 71
Acosta, A. 60
Acosta Caraballo, Y. 76
Adair, E. 75
Adame, M. 56
Adams, H. 46
Adams, N. 77
Adam, T. 68
Addy, K. 72
Adelson, A. 51
Adeola, U. 52
Aeby, G. 61
Aeppli, C. 53
Afshar, M. 56
Agosto Shaw, K. 76
Agrawal, A. 54
Aguiar, A. 39
Aguiar, D. 67
Aguilar, C. 48, 67, 69, 74, 81
Aguilar-Islas, A. 42
Agustí, S. 46, 61, 62
Ahmed, Q. 51
Ahmerkamp, S. 56, 71
Ahrens, J. 56
Aijaz, S. 67
Akil, L. 86
Aksnes, D. 40
Alaman, M. 88
Alanis, B. 49, 73
Albergaria-Barbosa, A. 75, 83
Albert, S. 40, 59, 72
Albertson, T. 45
Albino, D. 55
Albright, E. 68, 90
Albuquerque, F. 74
Alderman, D. 37, 54, 55, 60
Alegria, H. 43
Aleman, J. 38
Aleman, M. 53
Alexander, M. 39, 73
Algar, C. 51, 71
Alhama, M. 88
Alharbi, A. 75
Alhgren, N. 36
Ali Ger, K. 83
Alin, S. 49, 59
Allam, A. 70
Allen, A. 41, 64, 80
Allen, G. 65
Allen, M. 64
Alleosfour, A. 73
Aller, J. 69
Aller, R. 69
Almeda, R. 47
Almeida, R. 39
Almeida Silva, L. 79
Al-Moosawi, L. 35
Al Nahdi, R. 87
Al-Najjar, T. 36
Alneberg, J. 61
Al-Nuwairah, M. 45
Al-Otaibi, N. 44
Altabet, M. 42, 51
Althoen, C. 53
Altieri, A. 51
Altiok, H. 36
Aluwihare, L. 54
Alvarado, E. 54
Alvarez, B. 55
Alvarez Caraveo, B. 55
Alvarez, J. 44
Alvarez-Rosario, J. 64
Amador, A. 62
Amaro, C. 60
Amazon, D. 39
Amergian, K. 82
Amin, S. 63
Amjad, H. 78
Amod, R. 72
Amon, R. 65
Amyot, M. 43, 65, 82, 85
Andersen, G. 61
Andersen, K. 39
Anderson, C. 68, 70, 78
Anderson, D. 46
Anderson, I. 43, 60
Anderson, K. 54
Anderson, R. 42, 71
Anderson, S. 45, 49, 71, 84
Anderson, Z. 78
Andersson, A. 59, 61, 85
Andrade, R. 77
Angarita, H. 39
Angelini, C. 39
Anggoro, A. 36
Anglès, S. 48
Anlauf, H. 60
Ansari, M. 89
Ansper, A. 89
Antczak, O. 54
Anthony, K. 41
Antoine, D. 81
Antoine, J. 54
Anton, A. 62
Antoniou, E. 47
Aoki, L. 88
Apostolaki, E. 62
Aquino-Pereira, A. 64
Aragones-Fred, C. 76
Aragón-González, J. 71
Aranda Lastra, M. 87
Aranda, M. 83, 85, 87
Archer, S. 46, 49, 53, 54
Archibald, K. 76
Ardelan, M. 35
Ardón, M. 72
Ares, A. 44
Arevalo-Martinez, D. 42
Argerich, A. 79
Arias, M. 87
Arikas, A. 89
Arismendi, I. 48
Arístegui, J. 43
Arjona, A. 46, 54
Armbrust, E. 69, 73
Armengol, L. 64
Armenteros-Almanza, M. 57, 65
Armitage, A. 48
Armstrong, A. 48
Armstrong, R. 72, 76, 79
Arnaud, M. 41
Arnold, E. 48
Arnold, R. 66, 70
Arnosti, C. 69
Arocho Hernández, N. 60
Arrington, E. 70
Arroyo, M. 88
Arteaga, M. 52, 65
Arula, T. 61
Arzeno, I. 80
Arzola, N. 45
Asch, R. 39, 55, 56
Ashford, O. 55
Askin, S. 55
Asplund, M. 88
Aßmann, S. 42
Astor, Y. 71
Atakpa, D. 49
Athukorala, D. 83
Attermeyer, K. 40
Auger, P. 42, 57
Aulenbach, B. 72
Aumont, O. 42
Auscavitch, S. 65
Austnes, K. 40
Avarachen, M. 35
Awad, M. 58
Axler, R. 35
Ayala Crespo, C. 43
Ayala, R. 80
Ayala Rivera, F. 80
Aydin, K. 39
Aylagas, E. 60
Aziz, T. 47

B

Babcock-Adams, L. 41, 57
Babin, M. 38
Baer, E. 48
Baer, S. 86
Baeza, M. 38
Bailey, S. 68
Bailey, V. 70
Bailleul, B. 37
Baker, A. 41
Baker, S. 37
Balaguru, K. 63
Balbín, R. 69
Balch, W. 54, 74
Balik, J. 83
Ballantyne, F. 72
Balmagia, J. 53
Balmer, B. 78
Balmonte, J. 69
Balser, A. 67
Baltar, F. 69, 89
Balz, D. 59
Bam, W. 41
Bange, H. 42
Banister, R. 54
Banon, Y. 77
Bantle, A. 77
Baracchini, T. 47
Barak, S. 62
Barbeau, K. 41, 57
Barbera, P. 60
Barberena K. 64
Barberena-Arias, M. 48, 64
Barbero, L. 65
Barber, P. 36, 38, 55, 62
Barbiero, R. 37, 45, 69, 70, 79
Barbosa, L. 63
Barby, C. 39
Barco, R. 55
Bargar, J. 89
Barnes, J. 79
Barnes, T. 40
Barnett, A. 76
Barnett, W. 59
Barouillet, C. 67
Barrajón, A. 88
Barral, A. 66
Barreto, M. 81
Barreto-Vélez, T. 39, 80, 81
Barrett, L. 48
Barrett, R. 38
Barros, F. 79
Barros, M. 76
Barry, J. 76
Barthaux, V. 86
Bartlett, B. 56
Bartley, T. 38
Bartoli, M. 43, 77
Bartoloni, S. 57
Barton, A. 53
Bartsch, W. 35
Barufi, J. 76
Baskaran, M. 41
Baskett, M. 41
Basterretxea, G. 81
Bastianini, M. 83
Bastidas, C. 42
Bastviken, D. 59
Basu, N. 68
Batchelor, R. 38, 48
Bates, N. 35, 73, 76, 78, 80, 86
Bateson, D. 63
Batterbee, C. 53
Batterton, J. 78
Battin, T. 59
Baudoux, A. 69
Baulch, H. 37, 44, 59
Baumann, H. 38
Baumann, J. 84, 87
Baumann, U. 68
Baumgartner, M. 53
Baumiller, T. 65
Baums, I. 45, 54, 65, 87
Baxter, C. 59
Bayer, B. 35
Beal, A. 87
Beal, J. 57
Beane, R. 48
Beardall, J. 84
Beardsall, B. 45
Beatty, D. 84
Beaudoin, D. 53, 89
Beaulieu, J. 59
Beaulieu, S. 76
Becker, J. 63, 69, 79
Beck, M. 56
Bednarsek, N. 49, 59, 67
Be'er, A. 84
Beets, J. 67
Beghoura, H. 42
Begin, C. 41, 69
Behra, M. 81
Behrends, T. 68
Behrenfeld, M. 45, 46
Beinart, R. 77
Beirne, E. 53
Beisner, B. 38, 47, 71, 86, 89
Beja, O. 73
Beklioğlu, M. 79, 87
Bélanger, S. 46
Beldowski, J. 67
Bellerby, R. 63
Bell, G. 38
Belshe, E. 88
Belshe, F. 88
Beman, J. 71
Benavides, M. 43, 86
Bender-Champ, D. 85
Bender, J. 36
Benedict, C. 78
Benes, K. 52
Bengtsson, M. 60, 65, 81
Benitez-Nelson, C. 51, 57
Benitez-Requena, L. 47
Benito-Granell, X. 39
Benjumea, D. 88
Benjumea-Novalbos, D. 88
Benk, S. 44, 63
Bennett, M. 35
Bento, J. 57
Bento Netto Junior, J. 42
Beraki, S. 55
Beraldo Bittar, T. 51
Bercovici, S. 52
Bergamaschi, B. 51, 52, 86
Bergbusch, N. 66
Berge, J. 86
Berger, A. 88
Berger Muñoz, T. 42
Berger, S. 47, 65
Berger, T. 57
Berg, P. 88
Bergvall, C. 73
Berman-Frank, I. 45, 84, 86
Bernát, G. 45
Bernus, J. 62
Berrio, C. 75
Berry, N. 37
Berthelot, H. 86
Bertilsson, S. 47, 61, 81
Bertin, A. 67
Bertin, M. 75

Bertin, X. 62
Berumen, M. 60
Best, B. 36
Besterman, A. 71
Betancourt, A. 54
Betancourt Gomez, N. 64
Beyer, J. 89
Bhattacharjee, A. 46
Bhattacharya, D. 41
Bhattacharya, R. 43
Białk-Bielińska, A. 61
Bianchi, D. 67, 81, 89
Bianucci, L. 63
Bibby, T. 35
Biddanda, B. 73
Biddle, J. 52, 55, 73
Bideau, S. 54
Bidlack, A. 68
Bidle, K. 46, 80
Biederbrick, J. 37
Bielmyer-Fraser, G. 70
Bienhold, C. 71
Bigatti, G. 36
Bigler, L. 45
Billick, I. 38
Bilodeau, F. 43
Binder, B. 49
Binder, F. 37
Birch, J. 49
Bird, M. 65
Birgand, F. 72
Birgul, A. 43
Birney, L. 82
Bishop, I. 76
Bittar, T. 79
Bitterwolf, K. 36, 54
Bitterwolf, S. 36
Bjork, M. 88
Bjorndal, K. 83
Black, D. 51
Black, E. 42
Blackford, J. 64
Blais, J. 83
Blanc-Mathieu, R. 36
Blanco Bercial, L. 53
Blaser, S. 84
Blenkinsopp, C. 47
Bliem, C. 69
Blough, N. 44
Blount, K. 70
Bobb, C. 89
Bobo-Shisler, C. 53
Bockmon, E. 51, 57, 78
Bockwoldt, K. 79
Bodker, E. 70
Boedecker, A. 45, 60, 67, 68
Boegehold, A. 48, 68, 87
Boehm, A. 42
Boehrer, B. 59
Boersma, M. 47, 67
Boetius, A. 71
Bogard, M. 51, 85, 86
Bograd, S. 60
Bohnenstiehl, D. 48
Boisen, O. 51
Boiteau, R. 36, 57
Bolanos, L. 46, 52
Bolinesi, F. 77
Bollens, S. 66, 90
Boll, J. 81
Bolt, C. 42
Bolten, A. 83
Bolton, J. 78
Bombar, D. 43, 86
Bonaglia, S. 43, 81, 90
Bond-Lamberty, B. 70
Bone, E. 82
Bonin, C. 37
Bonisteel, E. 45
Bonnet, S. 86
Bonomi Barufi, J. 45
Borchardt, D. 47
Borecka, M. 61
Borel, A. 54, 79
Borgen, M. 54
Bork, P. 89
Bornhorst, R. 57, 84
Bornman, T. 35
Borrelli, M. 49
Boshers, D. 51
Bosma, K. 49
Bossa, M. 65
Boss, E. 46, 88
Bostick, B. 75
Boswell, K. 65
Botsch, J. 61, 78
Bott, C. 35, 90
Bottino, F. 39
Bouchard, C. 51
Bouffard, D. 47
Boughman, J. 61
Bouissou, A. 60
Boukal, D. 79, 87
Boulet, B. 41
Bour, A. 44
Bourgeois, A. 46
Bourne, D. 41
Bourque, J. 65
Bove, C. 84
Bowden, B. 72
Bowering, K. 43, 83
Bowler, C. 89
Bowles, M. 81
Bowman, J. 55, 61, 66, 73, 77
Boyce, D. 49
Boyd, D. 78
Boye, K. 89
Boyko, V. 52
Braakman, R. 69
Bracher, A. 42
Bradt, S. 35
Brady, C. 54
Brady, D. 59
Brady, P. 63
Bragg, D. 48
Brahmstedt, E. 43, 66
Brahney, J. 81
Brambilla, D. 61
Bramburger, A. 67, 84
Brandão, M. 89
Brander, S. 54
Brandon, J. 48
Brandt, M. 48, 50, 54, 61, 62, 66, 84, 87
Brandtneris, V. 48
Brandt, S. 49
Branstrator, D. 76
Brauns, M. 44, 84
Bredehoeft, M. 77
Breier, C. 73
Breier, J. 49, 72
Breitbart, M. 42
Brendonck, L. 39
Brennan, R. 38
Brennan-Tonetta, M. 86
Brenneis, T. 41
Bretherton, L. 40
Brickman, D. 70
Bridgham, S. 70
Brignac, K. 55
Brill, R. 36, 53, 54
Brinker, A. 67
Brisson, S. 36
Bristol, E. 85
Bristow, L. 35, 71
Broach, K. 36
Broder, T. 52
Brodie, S. 60
Broek, T. 44
Broman, E. 43, 89
Bronk, D. 35, 90
Brooke, S. 65
Brooks, A. 68
Brooks, E. 81
Brooks, G. 40, 41, 57, 69
Brookshire, J. 72
Brooks, J. 86
Brooks, M. 62, 78
Broomell, S. 55
Brosnahan, M. 46
Brothers, S. 81
Brovold, S. 68
Browder, J. 54, 68
Brown, A. 44
Brown, B. 81
Brown, C. 40, 59, 62, 67, 73, 77
Brown, E. 79
Browne, K. 74
Brown, I. 35
Browning, T. 40, 41
Brown, K. 85
Brown, L. 64
Brownlee, E. 53
Brown, M. 38
Brown, P. 38, 53
Brozak, S. 89
Bruach-Menchén, J. 88
Bruggeman, J. 64
Bruland, K. 57
Brumsack, H. 56, 57
Bruns, P. 38
Brussaard, C. 86
Brycki, J. 54
Brzezinski, M. 57
Brzeziński, T. 67
Bubley, W. 60
Bucheit, J. 54
Buck, C. 42
Buck, K. 36, 41, 42
Bucklin, C. 50, 66, 74
Budge, S. 83
Budinich, M. 89
Buelo, C. 68
Buesseler, K. 42, 57
Buffum, H. 77
Bukaveckas, P. 43
Bull, D. 85
Bullister, J. 51
Bullock, L. 45, 55
Bulski, K. 79
Bundy, M. 75
Bundy, R. 41, 42, 57
Bunnell, D. 37
Bunse, C. 69, 83
Burgaud, G. 89
Burger, J. 42
Burgin, A. 37
Burgos Rodriguez, J. 50
Burian, S. 68
Burke, A. 45
Burkholder, J. 86
Burlakova, L. 69, 70
Burnes, R. 73
Burnet, S. 68
Burnham, T. 39
Burns, J. 67
Burns, S. 41
Burns, W. 54, 56
Burson, A. 86
Burt, J. 87
Burtner, A. 89
Burt, W. 57
Bushnell, M. 59
Bushnell, P. 54
Buskey, E. 41
Butman, D. 51, 85
Butts, T. 90
Byrne, R. 51, 57, 59

C

Caamal, J. 88
Cabana, G. 66
Cabral Guadalupe, A. 63, 80, 88
Cabrera, J. 71
Cabrera, L. 79
Cabrera Rivera, L. 39
Cabrera-Rivera, L. 39
Cáceres-Charneco, R. 78
Cade-Menun, B. 75
Cadilla, J. 80
Cael, B. 71
Caffin, M. 86
Cagnat, X. 41
Cahir, C. 69
Cahyani, D. 36
Cai, M. 61
Cai, W. 49, 57
Calbet, A. 76
Calleja, M. 44, 68, 89
Callwood, V. 64
Calosi, P. 38
Camacho, A. 88
Camacho Ibar, V. 65
Camara, M. 59
Cambazoglu, K. 51
Campbell, B. 61
Campbell, C. 76
Campbell, D. 45, 84
Campbell, L. 48, 76
Campbell, S. 83
Camp, E. 85
Canals-Silander, M. 40, 48, 56, 66, 67, 70, 78, 81
Canava, K. 37, 54, 55, 60
Cancel-Villamil, J. 39
Candela, J. 65
Candelas, F. 35
Canela, R. 55
Caner, H. 36
Cannady, A. 55, 73
Canning, A. 70
Canonico, G. 36
Cantwell, H. 83
Cao, Z. 70
Capella Hernández, J. 81
Capella, J. 70
Capelle, D. 85
Capone, D. 55, 73
Caprara, S. 36
Caracappa, J. 77
Carbonell-Moore, C. 46
Cardenas, A. 63
Carey, C. 47, 71, 74
Carilli, J. 87
Carlisle, A. 36
Carlo, M. 76
Carlotti, F. 86
Carlson, A. 75
Carlson, C. 44, 52, 61, 69, 70, 79, 89
Carlson, J. 54
Carlson, L. 69
Carlson, M. 71, 79
Carne, L. 84
Carney, K. 73
Carpenter, R. 59
Carpenter, S. 68
Carrasquillo-Rosa, G. 87
Carreón-Palau, L. 83
Carrière-Garwood, J. 55
Carrion Banuchi, C. 83
Carr, M. 77
Carroll, G. 60
Carroll, J. 69, 79
Carstensen, J. 83
Carter, B. 49
Carter, M. 57, 63, 73, 88
Caruso, C. 85
Carvalho, S. 60, 68
Carvallo, F. 48
Cary, C. 49
Casas-Ruiz, J. 86
Casciotti, K. 43
Casillas, P. 40
Cassell, J. 62
Castaldelli, G. 77
Castaneda-Moya, E. 88
Castañeda-Moya, E. 39
Castaño, J. 75
Castejón-Brignoni, R. 39, 79
Castelao, R. 42
Castillo, B. 88
Castillo, K. 84
Castro-Jimenez, A. 81
Castro-Jimenez, J. 87
Catalan, E. 55
Catalán, N. 63
Catching, K. 79
Cavicchioli, R. 46

Cawley, K. 79
Cerderia, S. 36
Cernadas-Martin, S. 71
Ceroli, A. 77
Cerveny, J. 45
Çetinbag, D. 79
Çetinbağ, D. 87
Ceulemans, R. 46
Chacagueda, F. 56
Chaffin, J. 60, 67
Chaguaceda Borjabad, F. 47
Chaichitehrani, N. 80
Chaiharn, C. 73, 75
Chambers, R. 49
Champlin, L. 88
Chandra, S. 72
Chan, E. 49, 72, 73
Chang, B. 51
Chang, J. 41
Chanton, J. 43
Chant, R. 55
Chapina, R. 80
Chapman, J. 53
Chapman, L. 38
Chapman, P. 43
Chappel, A. 80, 81
Chappell, J. 39
Chappell, P. 36, 41
Chardón Maldonado, P. 40, 56, 70
Charlebois, P. 70
Charles, M. 64
Charles, S. 48, 80
Charmasson, S. 41
Chase, A. 46, 89
Chase, D. 63
Chase, J. 74
Chauhan, A. 77
Chaves-Fonnegra, A. 48
Chavez, F. 36, 42
Chebli, G. 79
Checkley, D. 39
Chei, E. 85
Chen, C. 74
Cheng, H. 71
Chen, R. 36, 49, 80
Chen, T. 45
Cherepanova, O. 36
Cherrier, J. 35
Chételat, J. 85
Cheung, I. 55
Chiang, K. 55
Chiappa, X. 71
Chien, C. 36, 39
Chigbu, P. 44, 45, 61, 77
Chikaraishi, Y. 75
Chiquillo, K. 62
Chisholm, S. 69
Chmiel, R. 57
Chmura, G. 88
Choi, B. 75
Choi, C. 46
Choi, F. 54
Cholewiak, D. 54
Chrétien, E. 38
Christensen, A. 47, 87
Christian, A. 55
Christianene, M. 83
Christian, R. 48
Chu, C. 44
Chu, K. 40
Chu, P. 47
Chu, R. 40
Church, I. 73
Church, M. 41, 89
Chu, S. 62
Cinco, S. 88
Ciobanu, M. 45
Cipolla, K. 55
Cira, E. 68
Cisneros, A. 36
Clapham, M. 36
Claquin, P. 45
Clardy, T. 45, 84
Clarke, E. 49
Clark, F. 45
Clark, R. 53
Claus, P. 83
Clayton, S. 89
Cleary, N. 77
Clements, C. 84
Clevenger, S. 57
Close, H. 65
Closek, C. 42
Closset, I. 57
Clow, D. 72
Clower, P. 56, 62
Coale, T. 41
Cobb, C. 78
Cobleigh, K. 56, 61
Cockshutt, A. 45
Codden, C. 66
Coello-Camba, A. 36
Coesel, S. 73
Coffey, K. 68
Cohen, J. 49
Cohen, N. 64
Cohen, R. 44
Cohen, S. 38, 76
Coker, D. 60
Colbert, S. 67
Coldsnow, K. 38, 83
Coleman, H. 55
Coles, D. 54
Coles, V. 69, 78, 80
Coles, W. 78
Colin, N. 82
Colin, S. 53
Collingsworth, P. 37, 70, 79
Collin, R. 51
Collins, A. 35
Collins, C. 36
Collins, S. 86
Colombo, R. 52
Colom, W. 47
Colon, D. 77
Colón, F. 64
Colon-Gaud, C. 48
Colon Molina, D. 52
Colson, B. 53, 78
Combes, V. 57
Combs, I. 57
Comeau, A. 80
Comstock, J. 52, 61
Congdon, V. 48
Connell, J. 75
Connelly, T. 83
Conner, K. 53, 85
Connolly, C. 85
Connolly, R. 67, 77
Conrad, D. 44
Conrad, R. 83
Constant, N. 83
Contreras, P. 52
Cook, L. 78, 81
Cook, P. 69
Cooney, E. 67
Cooper, A. 36
Cooper, W. 43, 53
Cope, J. 53
Coppola, L. 45
Corbau, C. 79
Cordell, J. 90
Cordero Quiros, N. 56
Cordes, E. 44, 65, 66
Corilo, Y. 36
Corman, J. 36, 67, 86
Cornwell, J. 60
Correa, A. 48
Cortez, D. 52
Cosca-Baresh, A. 53
Cossu, R. 40
Costa, M. 81, 88
Costa, N. 38
Costanza, C. 37, 54, 55, 60
Costello, D. 68
Costello, J. 53, 71
Cotner, J. 60
Cottier, F. 86
Couch, C. 67
Coulombier, T. 62
Countway, P. 53, 79
Courtney, S. 56
Courtney, T. 85
Couto, T. 38
Covelli, M. 86
Coveney, M. 73
Cover, M. 48
Covich, A. 39, 72
Cowles, G. 62
Cox, R. 89
Cox, T. 78
Coyne, K. 68
Cozy, M. 55
Craig, C. 76
Crawford, J. 68
Creed, I. 40, 84
Creer, S. 89
Crespo-Medina, M. 48
Crétaux, J. 89
Cretoiu, S. 46
Crew, A. 87
Cristescu, M. 38
Crockett, J. 48
Crockford, T. 76
Cross, C. 62
Crowder, L. 54
Crowl, T. 38, 39, 64, 72, 80, 81
Crump, B. 66, 86
Crumrine, P. 38
Cruz, B. 89
Cruz, K. 37
Cruz Motta, J. 76
Cruz-Motta, J. 76
Cruz Reyes, R. 45, 54, 55
Cruz-Rivera, E. 62
Cuevas, E. 66, 73, 81
Cugier, P. 76
Cuhel, R. 48, 67, 69, 74, 81
Cumming, B. 67
Cunha, D. 39
Curchitser, E. 39
Curd, E. 60
Curdia, J. 60
Curtis, B. 73
Curtis, J. 86
Cuzin, P. 46
Cyronak, T. 59
Cyterski, M. 35
Cziesielski, M. 85, 87
Czub, M. 67

D

Daase, M. 86
Dabney, B. 75
Daehnke, K. 43
Dahl, A. 69
Dahlenburg, C. 70
Dahl, M. 88
Dailey, E. 78
Dai, M. 70
Daims, H. 35
Dale, A. 62
Dalgleish, F. 65
Dalman, E. 47, 56
Daly, K. 45
Damashek, J. 35
Dam, H. 38, 53
D'Andrilli, J. 51, 63
Daneshmand, V. 47
Daniel, I. 64
Daniel, S. 70
Danylchuk, A. 62
Darnis, G. 86
Dasarathy, S. 77
Da Silva, L. 61
Das, N. 45
Data, B. 43
Daunys, D. 77
Davenport, J. 73
David, C. 86
Davidson, I. 77
Davidson Parry, E. 86
Davies, A. 65
Davies, I. 49
Davies, K. 74
Davies, S. 48
Dávila, Y. 71
Davis, A. 51
Davis, C. 51
Davis, G. 53, 54
Davis, H. 37, 54, 55, 60
Davis, K. 40, 51
Davis, R. 80
Davis, S. 68, 80
Davis, T. 89
Daw, J. 77
Dawson, M. 38, 60
Dawydiak, W. 53
Deacy, B. 54
de Andrade, J. 77
Dearden, E. 51
de Arruda, E. 39
De Bonville, J. 43
Dedieu, J. 82
Dedrick, J. 54
Deemer, B. 86
Deering, N. 40, 59, 72
de Fátima Panosso, R. 83
Defeo, O. 36
DeForce, E. 66
DeGrandpre, M. 57
Deheyn, D. 77
Deininger, A. 40, 61
De Jesus Roman, A. 39
De La Cour, J. 69
Delacruz, B. 78
De-La-Cruz, L. 71
de la Rosa, J. 88
Delebecq, G. 46
DeLeon Sanchez, E. 55
Delgado-Huertas, A. 61
del Giorgio, P. 43, 44, 59, 65, 66, 86
Del Giudice, D. 47
Dellaert, Z. 53
Della Penna, A. 89
Delorme, N. 59
Delpin Sosa, M. 54
del Valle, C. 64
Del Valle, T. 64
DelVecchia, A. 83
Del Vecchio, R. 44
deMayo, J. 38
Demayo, J. 38
Demirel, N. 36
Demir, F. 68
Demir, V. 36
Demopoulos, A. 65
Deng, H. 76
Denis, V. 66
Denney, B. 73
Dennison, W. 63, 82
Denny, R. 49
deNoyelles, J. 37
De Pol-Holz, R. 71
Dergam, J. 39
DeRosia-Banick, K. 42
Desai, D. 60
de Senerpont Domis, L. 35
Deshpande, A. 55, 87
Dessu, S. 38
Detres, Y. 76
Deulofeu, O. 88
Deutsch, C. 39, 67, 89
Dever, E. 86
de Verneil, A. 89
Devesa, J. 83
Devito, K. 72
Devol, A. 71
Devotta, D. 69, 77
DeVries, S. 76
DeWeese, N. 76
Dexter, E. 90
Deyanova, D. 88
Dhoonmoon, C. 78
Diamond, S. 59
Diaz Almela, E. 88
Diaz-Almela, E. 88

Díaz-Almela, E. 88
Diaz-Asencio, M. 57
Diaz, F. 42
Diazgranados, M. 88
Diaz, H. 38
Diaz, J. 39, 44
Diaz, K. 64
Diaz Santana, G. 78
Díaz Vázquez, L. 82
Díaz-Velázquez, E. 35
Dibble, K. 86
Dick, H. 89
Dickson, A. 51, 57, 62
Di Domenico, M. 36
Diego, R. 87
Diercks, A. 41
Dierking, J. 53
Dierssen, H. 37
Dietrich, D. 75
Dietzel, K. 81
Dietze, M. 47
Dietz, M. 75
Dieunel, D. 52
Dijkstra, N. 68
Dilewski, C. 47
Dillon, K. 84
Dillon, P. 37
Di Lorenzo, E. 42, 57
DiMarco, S. 40, 43
Dimino, T. 55
Dimond, J. 87
Dingmon, T. 55
Dinh, K. 47
DiNicola, G. 45
Dinniman, M. 51
Diou-Cass, Q. 79
Dittmar, T. 39, 44
DiTullio, G. 57, 64, 77, 85
Dixon, K. 40
Dixson, D. 52
Djeghri, N. 45
Djurhuus, A. 42
Doall, M. 42
Doctor, E. 78
Dodds, W. 72
Dodge, D. 85
Dodson, G. 78
Dodson, S. 60
Doherty, S. 65
Dolan, J. 45
Dolk, S. 86
Domaizon, I. 46
Domingo, C. 38
Doney, S. 46, 85
Donner, S. 69
Dooley, K. 69
Doo, S. 59
Dopson, M. 43, 61
Dórea Filho, A. 79
Dornback, M. 55
Dornblaser, M. 51, 85
Dornelas, M. 41
Dorrington, R. 35
Dorsey, K. 53
dos Santos, A. 39
Douglas, A. 41, 57
Dove, S. 85
d'Ovidio, F. 88
Dowd, M. 70
Downing, B. 51, 52, 86
Downing, G. 63
Downing, J. 86
Downs, E. 37
Drake, P. 89
Drake, T. 43
Drayton, D. 55, 87
Drenkard, E. 39
Dronova, I. 72
Druffel, E. 71
Druschel, G. 75
Drysdale, J. 57
D'souza, N. 45, 56
Duarte, C. 61, 62, 83, 87
Duarte Ferreira, G. 76
Duarte, G. 85
Duarte Moreno, H. 47, 67
Dubickas, K. 45, 82
Dubois, M. 84
Ducklow, H. 85
Duda, J. 84
Duda, M. 83
Dudek, K. 79
Duff, N. 45
Duff, R. 73
Duffy, E. 36
Dugdale, R. 86
Duguay, C. 89
Duineveld, G. 65, 89
Duke, J. 37, 38, 54, 55, 60
Dulai, H. 41
Du, M. 46
Dumbauld, B. 53
Dunbabin, M. 59
Dunham, A. 49
Dunlap-Smith, A. 48
Dunne, J. 39
Dunton, K. 48
Dupont, C. 41, 57, 64, 71, 86
Dupuy, C. 62, 76
Duran, A. 60
Durand, P. 46
Durdall, A. 89
Duret, M. 43, 83
Durgadeen, N. 54
Durham, B. 69, 73
Duris, J. 68
Durkin, A. 65
Dursun, F. 36, 71
Dutkiewicz, S. 39, 49, 89
Dutz, J. 53
Duval, T. 51
Duvert, C. 65
Du, Y. 51
Dyhrman, S. 69
Dzieweczynski, T. 38, 54
Dziewit, M. 41

E

Eakin, C. 69
Eakin, M. 77
East, T. 41
Eberhard, E. 56, 59, 72
Eckard, R. 63
Eckert, R. 77, 85
Eddy, T. 37
Edgcomb, V. 61, 67, 71, 89
Ediger, D. 36, 71
Edlund, M. 68, 86
Edman, R. 60
Edmonds, D. 70
Edmunds, P. 59
Edwards, B. 80
Edwards, C. 89
Edwards, K. 43, 64, 83
Edwards, R. 71
Effertz, C. 45
Egger, M. 68
Eggers, L. 60
Eggleston, D. 48
Eggleston, E. 43
Ehrlich, E. 46
Eichner, M. 45
Eikrem, W. 89
Eirin-Lopez, J. 87
Eisenstadt, A. 51
Eklöv, P. 47, 56
Elabd, I. 36
Elfadul, R. 77
Elgin, A. 69
Ellegaard, M. 56
Ellen, J. 71
Elliott, A. 47
Ellis, G. 69
Elmarsafy, M. 56
Elmi, D. 71
Elnabawi, A. 77
Elser, J. 67, 87
El-Wazzan, E. 58
Emanuele, G. 62
Emeis, K. 43
Emerson, N. 48
Encinas Fernandez, J. 59
Endo, H. 36
Endo, Y. 56, 70
English, C. 52, 70
Engman, A. 44
Engstrom, D. 68
Ennis, R. 48, 85
Enrich-Prast, A. 83
Enriquez, C. 71
Eppinga, M. 39
Erazo, N. 55, 61
Erdner, D. 63
Erisman, B. 56
Erratt, K. 40, 84
Esbrí, L. 78
Escobar-Briones, E. 36
Esparra-Escalera, H. 52
Espinosa, R. 88
Estrada, C. 55
Estrada, S. 55
Evans, C. 67, 78, 81
Evans, M. 35, 67, 68, 90
Eveillard, D. 89
Eveleth, R. 46
Everhart, C. 57
Ewanchuk, P. 44, 55
Ewen, K. 89
Eyre, B. 59
Ezcurra, E. 88
Ezcurra, P. 88

F

Fabbri, E. 47
Faber, J. 67
Fackrell, J. 67, 81
Fadeev, E. 71
Fadum, J. 39
Fakhraee, M. 83
Falkenberg, L. 49
Falkowski, P. 37
Fan, R. 75
Farchette, A. 56, 79
Faria Patire, V. 79
Farrell, K. 74
Farris, J. 90
Fasching, C. 63, 86
Faulkner, P. 36, 84
Faulkner, S. 84
Fawcett, S. 42, 43, 51
Fazekas, H. 72
Feddersen, F. 62
Federal, D. 49
Feely, R. 49, 59, 67, 86
Fei, C. 63
Feldmesser, E. 46
Felipe, Y. 71
Felker-Quinn, E. 70
Fellous, A. 87
Feng, Z. 74
Fenical, W. 54
Fennel, K. 70
Fenwick, I. 54
Fergus, C. 86
Ferguson, M. 45
Ferguson, S. 86
Fernandez-Figueroa, E. 38
Fernandez, M. 55
Fernández, M. 88
Fernandez Porto, J. 63, 80, 88
Fernández Porto, J. 88
Ferner, M. 52
Ferreira de Pinho, U. 80
Ferrer-Gonzalez, F. 69
Ferrer-González, F. 69
Feuchtmayr, H. 79, 87
Few, C. 55
Fidalgo De Souza, L. 52
Fiechter, J. 39
Fielder, C. 59
Field, J. 60
Fields, D. 45, 46, 53, 71
Fields, L. 67
Fies, J. 53
Fietzek, P. 70
Figueiredo, R. 47
Figueredo, N. 54
Figueroa, J. 38
Figueroa, Y. 39, 67
Filstrup, C. 84
Finiguerra, M. 38
Finkel, Z. 84
Finkler, N. 39
Fink-Mercier, C. 65
Fink, P. 37, 47, 66
Finlay, K. 59
Finnie, B. 59
Fiorendino, J. 48
Fiori, E. 70
Fischer, A. 46
Fischer, R. 65
Fisher, C. 45, 54, 55, 65
Fisher, M. 48
Fisher, N. 45
Fisher, P. 40
Fisher, R. 55
Fiss, M. 52
Fitch, A. 70
Fitzenreiter, K. 62
Fitzgibbon, A. 68
Fitz, K. 49
Fitzpatrick, P. 51
Fitzsimmonds, J. 42
Fitzsimmons, J. 40, 46
Flandez, A. 84
Flecker, A. 39
Fleck, J. 86
Fleck, R. 40, 68, 90
Fleisher, M. 71
Flesher, D. 69
Flick, R. 57
Flood, R. 65
Florea, K. 43
Florenza, J. 81
Flores, A. 36
Flores, H. 86
Flores-Hernandez, M. 38, 39
Flores, J. 80
Fluggen, M. 72
Flynn, K. 64
Flynn, R. 42
Fogarty, L. 68
Foks, S. 65
Foley, C. 70
Folger, C. 77
Follett, C. 57
Follows, M. 49, 61, 84, 85
Fong, C. 38, 55
Fong, P. 38, 55, 61, 62, 76
Fonseca, A. 88
Fontaine, D. 73, 77
Forbes, H. 50, 74
Forbes, Jr, H. 64
Forbes, Jr., H. 66
Forbes, K. 62
Forde, A. 57
Ford, K. 79
Ford, M. 65
Forget, G. 49
Forrer, H. 51
Forsberg, B. 39
Forsch, K. 41, 57
Forsman, A. 43
Fortin, S. 43
Fortin Saint-Gelais, N. 35, 49
Foster, S. 67
Foster, T. 78
Fountain, M. 88
Fourqurean, J. 48, 80, 81, 83
Fowler, B. 76
Fox, P. 81
Fox, S. 83
Frada, M. 76
Fragoso, H. 85
Fraioli, A. 82
França, E. 83
França, M. 75

France, S. 65
Francis, C. 89
Francis, G. 64
Francois, R. 57
Frank, A. 69, 86
Frank, K. 84
Franklin, B. 73
Frankovich, T. 68
Franks, D. 54
Franqui Rivera, G. 52
Franssen, J. 49, 82
Frantzich, J. 84
Franzè, G. 46, 53, 84
Franzellitti, S. 47
Frawley, T. 54
Frazer, T. 61
Frazier, B. 60
Fredricks, H. 80
Freeman, D. 55, 87
Freeman, M. 39
Freismuth, T. 78
Frenzel, H. 89
Freyria, N. 73
Frías, M. 47
Frickenhaus, S. 44
Fridolfsson, E. 83
Frigola, J. 88
Frigstad, H. 61, 63
Frischer, M. 77
Fritz, S. 39
Frková, Z. 47
Frohn, L. 83
Froney, M. 77
Frontera, T. 48
Frost, E. 59
Frost, P. 45
Fu, B. 51
Fuchsman, C. 71
Fuessel, J. 35, 43, 83
Fu, F. 41, 84
Fugere, V. 38
Fugère, V. 38, 89
Fu, H. 76
Fuhrman, J. 36, 53, 61, 73
Fujii, M. 70
Fujiki, T. 51, 57
Fujikura, K. 78
Fujimura, A. 62
Fukushima, T. 78
Fulton, K. 41, 57
Fulweiler, R. 67
Fumo, J. 57
Funk, K. 56
Furman, B. 48
Fussmann, G. 38, 44, 89
Futrelle, J. 76
Fyfe, M. 84

G

Gade, A. 45
Gaedke, U. 46
Gaiser, E. 39
Gaja, S. 67
Gale, D. 40
Gale, S. 59
Galindo, A. 44, 76
Galindo-Sánchez, C. 52, 65
Gallagher, S. 78
Gallardo-Escarate, C. 42
Galletti, Y. 76
Galvez, B. 60
Gangrade, S. 55
Gao, L. 52, 78
Gao, Y. 49, 52, 68
Garcia-Bermudez, M. 44, 72
Garcia, E. 56
Garcia-Hernandez, J. 85
García - Morales, C. 78
Garcia Moreno, F. 81
García-Orellana, J. 88
Garcia Rodriguez, B. 39, 46
Garcia, S. 73
García-Troche, E. 58
García-Villacorta, R. 39
Garczarek, L. 42, 60
Gårdmark, A. 84
Gardner, W. 67, 89
Garfield, C. 79
Gargur-Leal, I. 75
Garza, C. 37, 48, 50
Gasbarro, R. 65
Gasca, J. 65
Gasca-Pineda, J. 52
Gaskill, J. 37
Gaskins, L. 77
Gassett, P. 42
Gast, R. 53, 61
Gates, R. 85
Gaube, P. 46, 80, 88, 89
Gaul, A. 51
Gauthier, J. 46
Gazzola, V. 73
Gearty, A. 83
Geiger, E. 69
Geist, S. 48, 56
Gelber, A. 51
Gelinas, Y. 80
Geller, H. 38
Ge, M. 43
Genzer, J. 76
Geoffroy, M. 86
Georgacopoulos, O. 35
George, A. 50
George Host Ce, D. 35
George, R. 88
Geraldes, M. 51
Gerardin, M. 66
Gerard, T. 89
Gérin-Lajoie, J. 82
Gernez, P. 62
Geuer, J. 41
Ghosal, S. 78
Giandonato, T. 55
Giardino, C. 89
Gibbes, B. 40, 72
Gibbons, K. 35, 67
Gibbs, A. 74
Giblin, A. 51
Gibson, D. 50
Giddings, S. 51, 61, 62, 80
Giebel, H. 69
Gilbert, S. 74
Gilg, M. 55
Gill, A. 84
Gilmartin, J. 41
Gimpelevich, M. 78
Gioli, B. 59
Giordano, M. 45
Giovannoni, S. 46, 52, 61, 69
Giralt, S. 88
Girard, F. 45
Givens, C. 68
Gleixner, G. 44, 63
Glenney, K. 81
Glenn, S. 49
Glibert, P. 86
Glickman, O. 45, 54
Glover, D. 46
G. Morán, X. 68
Gobler, C. 42, 59
Gochfeld, D. 48
Goebel, M. 37
Goehring, L. 37, 54, 55, 60
Goeltz, J. 51
Goericke, R. 57
Goes, J. 81
Goetz, F. 87
Gold, A. 72
Gold-Bouchot, G. 40
Goldsmith, S. 80
Gold, Z. 60
Gollnisch, R. 47
Gomes,C. 39
Gomes, H. 81
Gomes Selman, J. 39
Gomez, A. 69, 76
Gómez, A. 88
Gomez-Andujar, N. 66, 67
Gomez Arias, S. 64
Gomez-Chiarri, M. 79
Gómez-Consarnau, L. 73
Gomez, J. 44
Gomez, M. 59
Gomez-Saez, G. 39, 71
Gon, C. 78
Gonçalves, A. 81
Gonçalves-Araujo, R. 37
Gong, J. 76
Goni, G. 86
Gonneea, M. 55
Gonsior, M. 44, 53
Gonzales, D. 54
Gonzalez, A. 38, 89
Gonzalez Colmenares, G. 52
Gonzalez, D. 48
González Díaz, P. 85
Gonzalez, F. 85
Gonzalez, G. 60
Gonzalez-Garcia, M. 77
Gonzalez Hernandez, N. 52
González Méndez, J. 85
Gonzalez-Pinzón, R. 63
Gonzalez, S. 35
Goodbody-Gringley, G. 63
Goode, A. 46
Goode, W. 49
Goodman, K. 56, 79
Goodwin, D. 78
Goodwin-Segal, T. 82
Gooseff, M. 83
Gopalakrishnan, K. 44, 68, 87
Gorbunov, M. 37
Gordon, E. 37
Gordon, R. 67
Gorgues, T. 42, 57
Gorham, C. 56, 88
Górski, K. 82
Gossiaux, D. 89
Gough, M. 78
Gould, R. 71
Govender, Y. 51, 80
Grabowski, R. 84
Graco, M. 42
Gradoville, M. 86
Graeve, M. 86
Grafals-Soto, R. 40, 81
Graff, j. 46
Graff, J. 45
Graham, E. 73
Graham, J. 37
Graham, N. 52
Granger, J. 42, 51, 78
Granzow, B. 69, 75
Grater, E. 66
Grattepanche, J. 45
Gravinese, P. 68
Gray, A. 62
Gray, D. 44, 47, 53, 56, 77, 85
Gray, M. 59
Grear, J. 42
Greaver, T. 70
Grébert, T. 60
Greeley, D. 49
Green, A. 82
Greenfield, D. 68, 88
Greengrove, C. 38
Greening, C. 69
Greenlee, S. 53
Greenslade, D. 67
Greenwood, S. 45
Greer, A. 47, 49
Grégori, G. 86
Gregory, A. 89
Gregory-Eaves, I. 46, 86, 87
Greif, J. 49
Griebler, C. 44
Grier, S. 76
Griffin, A. 59
Griffin, S. 71
Griffin, T. 87
Griffiths, K. 46
Griffiths, N. 43
Grigel, H. 90
Grigoropoulou, A. 76
Grimes, K. 36, 62, 66, 74, 89, 90
Grinham, A. 40, 59, 72
Griniene, E. 43
Grossart, H. 47, 65, 87
Groß, E. 47, 67
Grosso, O. 86
Groussman, R. 69, 73
Groves, S. 46
Grundle, D. 42, 49, 60
Grunert, B. 80
Guannel, M. 50, 64, 66, 74
Guéguen, C. 70, 85
Gugele, S. 67
Guidi, L. 83
Guilderson, T. 43, 44
Guilherme de Oliveira, D. 76
Guill, C. 46
Guillemette, F. 35, 66
Guillou, L. 61
Guimaraes, M. 76
Guimarão, L. 83
Guislain, A. 37
Guizien, K. 71
Gulick, A. 83
Gullström, M. 88
Gunnarsson, J. 81, 89, 90
Gunn, C. 70
Guo, C. 52
Guo, L. 52
Guo, X. 70
Gusmão, L. 78
Gustafsson, B. 68
Gutierrez, A. 36
Gutierrez, B. 50
Gutierrez-Fonseca, P. 72
Gutiérrez-Fonseca, P. 44
Gutner-Hoch, E. 55
Gutting, A. 54
Guzzo, M. 38

H

Haacker, R. 38
Haak, C. 62
Haas, S. 60
Habeck, E. 49
Habit, E. 82
Habtes, S. 36, 38, 48, 54, 74, 78, 80, 89
Hach, P. 35
Haëntjens, N. 46
Hagemeyer, A. 49
Haggett, E. 46
Hagley, C. 35
Hahn, M. 45, 54
Haig, H. 59, 66
Haig, J. 77
Hala, D. 40, 84
Hale, R. 87
Hales, B. 59
Halewood, E. 44, 52
Haley, S. 69
Hallaj, A. 66
Hall, C. 83
Hall, E. 39, 61
Hallen, C. 82
Hall, M. 48
Hallock, P. 65
Hall, R. 62, 67, 68
Hall, S. 83
Hallstrøm, S. 43
Halsey, K. 45
Hambright, K. 89
Hamersley, M. 81
Hamisi, M. 88
Hamman, C. 54
Hammerman, N. 76, 85
Hampel, J. 41, 67
Hancock, H. 41
Haney, J. 35
Hannigan, R. 49
Hansell, D. 52
Hansen, A. 51, 86
Hansen, C. 68
Hansen, J. 52

Hanson, A. 61, 77
Hanson, P. 43, 47
Hansson, T. 47
Han, Y. 47
Hanz, U. 65, 89
Happel, E. 61
Haraguchi, L. 64, 83
Harden, B. 50
Hardison, A. 40, 48
Hargan, K. 73, 75
Hargrave, K. 81
Harir, M. 44
Harley, C. 49
Harley, J. 68
Harms, T. 72
Harms-Tuohy, C. 52
Harris, E. 78
Harris, J. 78
Harris, L. 48, 64, 70
Harris, N. 74
Harrison, J. 73
Harrison, N. 56
Harris, T. 37, 68
Hartland, A. 49
Hartman, S. 62
Hartwell, I. 62
Harutyunyan, R. 80
Harvey, E. 45, 46, 51, 63, 70, 74, 79, 83
Harvey, H. 80
Harvey, J. 47
Harvey, R. 59
Häse, C. 61
Hashem, S. 52
Hassanzadeh, B. 73
Hassler, C. 69
Hastie, J. 36
Hastings, C. 86
Hatcher, P. 78
Hatje, V. 77, 79
Hatzianestis, I. 47
Hauke, A. 53
Hauss, H. 71
Hawco, N. 41, 64, 73
Hawkes, J. 63
Hayes, C. 37, 62
Hayes, N. 66, 86
Hay, M. 84
Hays, M. 79
Hazen, E. 60
Hazlett, M. 52
Head, M. 36
Heal, K. 69
Heathcote, A. 59, 68
Hébert-Houle, E. 82
Hébert, M. 38, 89
Hedberg, J. 81
Hedberg, S. 53
Hedley, J. 37
Hehemann, L. 71
Heidelberg, J. 73
Hein, C. 75
Heinle, M. 45, 84
Heiss, E. 67
Heithaus, M. 62
Held, N. 42, 64, 71
Helenius, L. 83
Heliás-Nunige, S. 86
Hellweger, F. 37
Helmer, D. 75
Helmuth, B. 36
Hemmler, D. 44
Hempel, P. 60
Hemsley, V. 43, 83
Hendrick, G. 76
Hendricks, S. 64
Hendriks, I. 62
Hendry, A. 38
Henke, B. 61
Hennessy, S. 67
Henrichs, D. 48
Henríquez, C. 42
Henry, M. 74
Henslee, H. 54, 79
Hentschel, I. 59
Heppell, S. 55
Herbig, J. 60
Herbold, C. 35
Herdman, L. 56
Herguera, J. 65
Herlihy, A. 86
Hermann, A. 39
Hernandez Ayon, M. 65
Hernández Carrero, I. 88
Hernandez-Delgado, E. 35, 48, 66, 67
Hernandez Figueroa, E. 81
Hernandez FIgueroa, E. 73
Hernandez, S. 38
Hernandez, W. 72, 76
Herndl, G. 35, 89
Hernes, P. 63
Heron, S. 69
Herreid, A. 81
Herren, C. 44
Herrera, M. 68
Herrera, S. 47
Herrera-Silveira, J. 88
Herrick, C. 85
Herrmann, T. 82
Hertkorn, N. 44
Hertler, H. 48
Herut, B. 76
Herzka, S. 52, 65
Herzschuh, U. 46
Herzsprung, P. 44
Hicks, T. 48
Hideshima, K. 51
Higgins, J. 57
Hikmawan, T. 45, 84
Hildebrand, M. 57, 84
Hill Cruz, M. 42
Hillebrand, H. 47
Hillhouse, J. 54, 76, 84
Hillis-Starr, Z. 88
Hill, R. 86
Hill, V. 70
Hilton, Z. 59
Hiltunen, M. 84
Hinchey, E. 69, 70
Hinchey-Malloy, E. 79
Hines, J. 73
Hinojosa, A. 62
Hinson, A. 46
Hintz, C. 59
Hjermann, D. 61, 89
Hobbs, J. 35
Hochberg, E. 69
Hoch, M. 54, 76, 79
Hockenberry, Z. 54
Hodder, J. 48
Hodder, K. 59
Hoegh-Guldberg, O. 85
Hoffman, D. 67, 89
Hoffman, E. 51
Hofmann, G. 38, 87
Hofmann, L. 59
Hogan, J. 48
Holbrook, S. 44, 68
Holen, D. 81
Holland, E. 87
Hollander, D. 65
Hollibaugh, J. 35
Hollingshead, K. 76
Hollis, B. 79
Hollister, A. 42
Hollister, K. 56, 85
Holmes, C. 41
Holohan, B. 87
Holsen, T. 43
Holstein, D. 89
Holt, J. 56
Honda, M. 51, 57
Hong, A. 75
Honisch, B. 46, 53
Hood, J. 79
Hood, K. 72
Hood, R. 70
Hooker, K. 89
Hook, T. 37, 70
Hooper, M. 48
Hopcroft, R. 45
Hopkinson, B. 85
Horgby, A. 59
Hormazabal Fritz, S. 42
Hormazábal, S. 57
Horodysky, A. 36, 53
Horváth, Z. 60, 84
Hosen, J. 48
Hoskins-Brown, D. 55
Hossler, K. 37
Host, G. 35
Ho, T. 76
Hotchkin, P. 38
Houde, E. 61
Hourigan, T. 55
Hovel, K. 39
Howard, E. 39, 67
Howard, H. 54
Howard, J. 83, 88
Howarth, R. 68, 75
Howden, S. 62
Howells, E. 87
Hrycik, A. 37
Hsieh, Y. 66
Hsu, A. 81
Hsu, T. 46
Huang, C. 39
Huang, H. 43
Huang, J. 37
Huang, M. 63
Hubbart, K. 42
Huber, A. 43
Huchaiah, V. 61
Hudak, C. 74
Hudak, G. 35
Hudson-Heck, E. 57
Hudson, J. 37
Hudson, K. 49
Huebner, J. 78
Huertas, E. 61, 77
Huesgen, P. 68
Huete-Stauffer, T. 89
Huggett, M. 76
Huggins, D. 81
Hughes, T. 69
Hug, J. 37
Huisman, J. 86
Hundey, B. 86
Hung, S. 83
Hunt, B. 67, 86
Hunter, R. 76
Hunt, R. 72
Hure, M. 53
Hur, J. 52
Hurst, T. 53
Hurtado, G. 54
Hurzeler, I. 74
Huss, M. 84
Hutchins, D. 41, 46, 84
Hutchins, R. 40
Hutley, L. 65
Hutley, N. 72
Huwer, B. 53
Hu, X. 42, 48
Huynh, M. 47
Huynh, N. 52, 70
Hwan, Y. 80
Hyatt, C. 41
Hylander, S. 43, 83

I

Iabichella-Armas, M. 71
Ianiri, H. 44
Iannino, A. 66
Ibanez, A. 54, 74
Ibelings, B. 36, 84
Idaghdour, Y. 87
Ignacio Espinoza, J. 36, 53, 61
Ignoffo, T. 76, 83
Iiames, J. 86
Ilgen, A. 85
Ilia, A. 79
Ilic, M. 47
Imai, A. 75
Imamura, M. 51
Imfeld, A. 80
Imhof, K. 54
Inamdar, S. 72
Indivero, J. 70
Ingalls, A. 60, 69
Inglett, P. 43
Inoue, K. 73
Inoue, R. 57
Irizarry, Y. 64
Irwin, A. 84
Isemonger, E. 35
Isensee, K. 86
Ishaque, A. 77
Islam, K. 70
Isles, P. 47
Israelsson, S. 69
Iturbe, M. 47
Ivanković, M. 65, 81
Ivanova, M. 36
Iversen, M. 71
Iverson, E. 48
Ives, A. 61, 78
Ivetic, S. 68
Iwahana, G. 85

J

Jabanoski, K. 48
Jabbur, L. 37
Jaccard, P. 89
Jacksin, R. 51
Jackson, C. 72
Jackson, M. 60
Jackson Mojica, A. 75
Jacoby, C. 57
Jacox, M. 60
Jaegler, T. 46
Jaffe, M. 41
Jaffery, R. 37, 54, 55, 60
Jagoe, C. 73, 77, 87
Jahn, O. 49, 89
Jaillon, O. 89
Jaimes, R. 81
Jakobsen, H. 64, 83
Jakuba, M. 49, 73
James, A. 61, 70
James, M. 40
Jarczak, J. 45, 47, 75
Jarrett, J. 76
Jarrold, M. 59
Jarvis, L. 44
Jarvis, S. 84
Jauzein, C. 63
Jearld, A. 50
Jeffrey, K. 67
Jenkins, A. 54
Jenkins, B. 41, 75
Jensen, A. 62
Jensen, B. 56
Jensen, L. 40, 42
Jeppesen, R. 77
Jerris, K. 56, 67, 76
Jesien, R. 77
Jesus, R. 77
Jewett, L. 86
Jeyasingh, P. 90
Jiang, H. 45
Jiang, M. 65
Jiang, X. 81
Jiang, Y. 40
Jiao, Y. 80
Jia, X. 47
Ji, C. 51
Jickling, N. 81
Jilbert, T. 68
Jiménez, F. 36
Jimenez, M. 66, 70
Jimenez-Reyes, D. 57
Jiménez-Rosenberg, S. 52, 65
Jinks, K. 67
Jin, P. 46
Jin, Y. 52
Ji, Q. 42, 60
Ji, R. 74
Jobsis, P. 62, 64, 79

Joehnk, K. 47
Johengen, T. 89
John, J. 70
John, S. 41
Johnson, B. 45, 52, 55
Johnson, C. 37, 74, 75, 83
Johnson, G. 53
Johnson, K. 50
Johnson, L. 35
Johnson, M. 53
Johnson, N. 68
Johnson, R. 35, 49, 73, 76, 78, 80, 83, 84, 86
Johnson, S. 79
Johnson, W. 69
Johnson, Z. 63
Johnston, S. 85
Joli, N. 73
Jones, A. 76
Jones, B. 48, 60, 85
Jones-Cervantes, S. 82
Jones, I. 47
Jones, J. 37, 57
Jones, M. 48
Jones, N. 50, 66
Jones, R. 54
Jones, S. 49
Jordan, T. 67
Jose, Y. 42
Jovanovic, B. 79, 87
Joye, S. 65, 81
Juescher, E. 79
Juhl, A. 54, 73
Juliussen, C. 77
Jungbluth, M. 76
Jung, M. 55
Juravel, K. 61
Jury, J. 59
Justic, D. 40, 43, 47

K

Kadjeski, M. 63
Kadko, D. 42
Kahilainen, K. 84
Kainz, M. 40
Kaiser, K. 40, 71
Kaldy, J. 40, 59, 62
Kallajoki, L. 37
Kalogerakis, N. 47
Kamalanathan, M. 40, 75, 76, 84
Kamjunke, N. 44
Kanagawa, M. 71
Kanaoka, K. 81
Kana, T. 86
Kandori, H. 73
Kanevskiy, M. 85
Kang, S. 75
Kankaala, P. 84
Kao, S. 36, 46
Kaplan, E. 53
Karatayev, A. 69, 70
Karatayev, V. 70
Kärcher, O. 84
Karl, D. 69
Karlen, J. 52
Karlsson, C. 61
Karp-Boss, L. 46
Karsh, K. 85
Karthäuser, C. 71
Karty, J. 60
Kashian, D. 44, 68, 87
Kaste, Ø. 40, 61, 63
Kath, N. 46
Katin, A. 47
Katkov, E. 44
Katsev, S. 74, 83
Kattan, Y. 60
Katz, L. 45
Kaufmann, P. 86
Kaup, T. 53
Kavanaugh, M. 36
Kawaguchi, S. 71
Kawai, K. 52
Kawakami, H. 51
Kay, M. 62, 67
Kazmiruk, Z. 70
Kearney, K. 39
Keeling, R. 62
Keep, M. 55
Keffer, J. 60
Kehoe, D. 60
Keil, R. 71
Keith, D. 73
Keller, A. 44, 65
Keller, J. 60
Kellerman, A. 60
Keller, P. 59
Kelleway, J. 88
Kelley, D. 38, 86
Kellogg, L. 60
Kellogg, M. 84
Kelly, D. 56
Kelly, E. 69
Kelly, L. 79
Kelly, M. 72
Kelly, R. 41
Kelly, T. 62, 69
Kelpsiene, E. 47
Kempnich, M. 63
Kendall, C. 67
Kennedy, B. 73
Kennedy, E. 68
Kenney, R. 74
Kenney, W. 73
Kenworthy, J. 79
Kessler, A. 69
Kessler, J. 51
Kessouri, F. 67, 89
Keyes, P. 53
Khandelwal, A. 63
Kidane, A. 35
Kido-Soule, M. 69, 86
Kienast, S. 42
Kiffney, P. 55
Kiko, R. 71
Killam, D. 36
Kim, D. 76
Kimera, S. 38
Kim, J. 52, 75
Kimmerer, W. 76, 77, 80, 83
Kimpe, L. 83
Kim, S. 77, 80
Kimura, S. 45
Kinard, S. 48
Kincaid, D. 72
King, A. 54, 63, 89
King, B. 48
King, C. 74
King, L. 81
King, N. 59
King, R. 55, 71
Kirchhoff, S. 73
Kirchman, D. 61
Kirk, R. 75
Kirsch, R. 48
Kirshen, P. 49
Kisand, V. 61
Kislik, C. 72
Kiszka, J. 62, 87
Kitahashi, T. 78
Kitamura, M. 51
Kitzinger, K. 35
Klauschies, T. 46
Kleiber, R. 67
Klein, E. 36
Klein, F. 89
Klein, S. 83
Klemt, W. 62
Klepikow, R. 37
Klinger, T. 39
Kling, J. 84
Klose, M. 83
Knap, A. 40, 43
Knapp, A. 36, 51, 78
Knight, R. 37
Knorr, K. 52
Knose, L. 37
Knowles, B. 46
Knowlton, N. 41
Koch, B. 41, 44
Koch, F. 41
Koch, M. 59
Kocovsky, P. 70
Koerting, C. 51
Koestner, D. 85
Kogovšek, T. 53
Kohlbach, D. 86
Köhler, J. 37
Kohut, J. 49
Kohzu, K. 75
Kolasa, K. 83
Komatsu, K. 75
Kominoski, J. 48, 80, 81
Koontz, E. 38
Koren, I. 88
Köring, M. 47, 67
Kortelainen, P. 79
Körtzinger, A. 42, 70
Koschorreck, M. 59, 63
Koski, M. 87
Kotabova, E. 45
Kothawala, D. 44
Kotwicki, L. 67
Kovalenko, K. 86
Kraal, P. 68
Krause, D. 37
Krause, J. 57, 62
Kraus, S. 74
Krauss, R. 56
Kraus, T. 51, 52, 67, 86
Krediet, C. 77
Krehlik, A. 72
Kremer, C. 46, 84
Krick, K. 49
Kriest, I. 42
Krinos, A. 40
Krishna, S. 61
Kristiansen, T. 89
Kritzberg, E. 72
Krock, B. 41
Kromkamp, J. 45
Krueger, S. 35
Krüger, S. 43
Krupp, K. 41
Kruse, S. 46
Kubanek, J. 79
Kubicek, A. 85
Kudela, R. 68
Kuehne, L. 38
Kuhn, A. 48
Kuhn, M. 40, 52
Kuiper, J. 35
Kujawa, E. 74
Kujawinski, E. 63, 69, 86
Kulinski, K. 42
Kulis, D. 46
Kumagai, J. 88
Kumar, V. 47
Kumler, W. 80
Kunzmann, A. 44
Kuo, L. 40, 59
Kurek, M. 75
Kurtay, G. 41, 56
Kurt, P. 43
Kustka, A. 57, 84
Kuypers, M. 35, 71
Kuzminov, F. 37
Kuzyk, Z. 70, 85
Kwietniewski, E. 86

L

Labarre, A. 76
Labiosa, R. 59
LaBone, E. 43
Labonté, J. 40
LaBrie, R. 35
Lacharité, M. 73
Lachaussee, N. 62
Lachkar, Z. 89
Ladah, L 62
LaDeau, S. 47
Lafrancois, B. 68
LaGarenne, R. 79
Lajaunie-Salla, K. 42
LaJeunesse, T. 37
Lamborg, C. 64
Lambrecht, N. 83
Lam, P. 35, 42, 43, 71, 83
Lamptey, D. 52
Landing, W. 42
Landry, B. 63
Landry, Z. 69
Lane, B. 79
Lane, C. 88
Laney, S. 85
Langdon, C. 49, 59
Lange, B. 86
Lange, J. 68
Langenheder, S. 47
Lannér, J. 40
Lannuzel, R. 40
Lanpher, K. 69
Lanphier, K. 68
Lantry, B. 45, 60
Lapid, G. 57
Lapierre, J. 43, 49, 65, 66, 86
LaRoche, J. 60
Laroche, R. 78
Larson, E. 39, 48
Larson, J. 68
Larson, R. 40, 41, 57, 69
Lascelles, N. 55, 87
Lasdin, K. 54
Lasi, M. 43
Lasseigne, D. 87
Lassiter, M. 70
Lassudrie, M. 46
Latour, D. 56
Latta, A. 77
Latz, M. 63, 88
Laubach, A. 51
Laudon, H. 47
Lauer, M. 44
Lau, M. 44
Laundry, J. 54
Laurenceau, R. 69
Laurent, A. 70
Lauterman, F. 73
Lau, Y. 51
Lavaleye, M. 65
Lavaud, J. 76
Lavender Law, K. 78
Laverdure, J. 74
Laverock, B. 49
Lavery, P. 56, 76, 88
Lavik, G. 71
Lavrentyev, P. 83
Lawlor, K. 88
Lawrence, J. 46, 54
Lawrence, S. 74
Layton, J. 36
Leach, T. 38, 55, 71, 87
Leary, D. 69
Leavitt, P. 59, 63, 66, 86
Lebreton, B. 86
Lecher, A. 36, 78
Lechtenfeld, O. 44
Leduc, M. 75
Lee, C. 49
Leefmann, T. 44
Lee, H. 77
Lee, J. 55, 77, 80
Lee, K. 38, 70, 83
Lee, P. 77
Lee-Patterson, D. 43
Leermakers, M. 90
Lee, S. 35, 37, 76
Lee, T. 77, 81
Lee, W. 51, 68
Lee, Y. 52
Lefcheck, J. 63
Le Fouest, V. 62, 76
Le Gall, S. 69
Le Gal, M. 80
Legendre, L. 83
Leggat, B. 85
Le Goff, M. 83
Legrand, B. 56
Legrand, C. 69, 83

Lehahn, Y. 88
Le Hir, P. 71
Lehrter, J. 49
Leibowitz, S. 86
Leichter, J. 40, 88
Leitao de Oliveira Junior, E 83
Leiva Dueñas, C. 88
Leiva-Dueñas, C. 88
Leles, S. 64
Le Moine, O. 40
Lenaker, P. 68
Leng, B. 37, 54, 55, 60
León, D. 88
Leone, F. 80
Leon Zayas, R. 52, 55
Leon-Zayas, R. 73
Leopo, D. 77
Lepak, J. 60
Lesht, B. 37, 69
Leslie, J. 78
Lester, S. 44
Letelier, R. 86
Lethaby, P. 80
Letourneau, M. 80
Letscher, R. 43
Leung, L. 63
Leung, T. 68
Lévesque, E. 82
Levin, L. 55
Levin, P. 49
Lévy, M. 89
Lewallen, E. 37
Lewis, D. 71
Lewis, S. 62
Liachko, I. 36
Li, B. 51
Liberti, K. 59
Li, C. 41
Li, D. 78, 87
Liefer, J. 84
Liess, A. 40
Liew, Y. 87
Li, H. 63
Liight, T. 63
Li, J. 57, 70
Liles, G. 50
Lilly, L. 42
Lilyestrom, C. 72
Li, M. 39, 40
Lima, F. 36, 62
Lima, I. 46
Lim, S. 61
Lindehoff, E. 83
Lindell, D. 71
Linde, M. 40
Lindemann, C. 40
Lindh, M. 69
Lindo Atichati, D. 80
Lindo, R. 60
Lindsay, D. 78
Link, J. 63
Lin, M. 60
Lin, S. 68, 81, 84, 85
Lin, X. 48, 71
Lisboa, J. 52
Lise, C. 38
Liskow, I. 43
Li, T. 45
Litchman, E. 84
Littmann, S. 35
Litvin, S. 37
Liu, C. 51
Liu, G. 43, 51, 69, 77
Liu, H. 41, 43
Liu, S. 52
Liu, X. 51
Liu, Y. 40, 43, 54
Liu, Z. 48
Li, X. 84
Li, Y. 36, 44, 56, 63
Llerandi-Roman, I. 44
Llopiz, J. 55
Loboschefsky, E. 77
Locke, S. 39
Lockwood, L. 49
Lockwood, S. 89
Lo, E. 81
Loftin, K. 68
Lofton, M. 71
Loftus, S. 63
Logan, C. 41
Logares, R. 89
Logozzo, L. 80
Loick-Wilde, N. 43
Lomas, D. 86
Lomas, M. 73, 78, 86
Lombard, F. 89
Lombardi, A. 45
Londoño, E. 36
Long, A. 53, 54, 61
Long, D. 57
Long, M. 46, 53, 62, 73, 77, 85
Longnecker, K. 69, 85, 86
Long, R. 79
Lonhart, S. 55
Lopes, C. 83, 88
Lopes, R. 57, 71
Lopez, A. 55
Lopez, C. 52
Lopez-Fernandez, M. 61
López, G. 64
Lopez-Gasca, M. 71
López-Lloreda, C. 56
Lopez-Merino, L. 56
Lopez, O. 72, 81
Lopez, P. 73, 78
Lopez Sandoval, D. 61
Lorda, J. 43
Lorenson, T. 85
Lorenzoni, L. 71
Los Huertos, M. 73, 75
Lott, S. 89
Louda, W. 68
Lougheed, V. 38
Louime, C. 75
Lovejoy, C. 73
Lovelock, C. 88
Lovko, V. 68
Low-Décarie, É. 44
Lowder, K. 59
Lowe, A. 89
Lubelczyk, L. 53, 78
Lucas, C. 83
Lucas, M. 76
Lucena, Z. 75
Lučić, D. 53
Luckey, F. 45
Ludlam, J. 37
Ludsin, S. 37
Lueker, T. 62
Lugo, A. 80
Lugo, K. 55
Lukeš, M. 45
Lukić, D. 84
Lumpi, T. 60
Lumpkin, R. 86
Lunden, J. 65
Lundgren, M. 47
Lundin, D. 61, 69
Luo, H. 42, 68, 81
Luo, J. 56
Luo, Z. 53
Lurling, M. 35
Lürling, M. 35
Lu, Y. 71
Luyando-Flusa, S. 64, 75
Lyimo, L. 88
Lyimo, T. 88
Lykkebo Petersen, K. 36
Lynch, J. 55
Lyons, W. 83

M

Maas, A. 53, 74
Maberly, S. 47
Mabry, L. 78
Macaya, E. 36
MacCready, P. 39, 70
Macdonald, H. 48
Macdonald, R. 85
MacFarland, A. 53
Machain-Castillo, M. 65
MacIntyre, H. 80
Mackay, E. 47
Mackin-McLaughlin, J. 60
Macknight, N. 61
MacMahan, J. 62, 78
MacMillan, G. 82, 85
MacMillan, J. 78
Macovei, V. 62
MacVittie, S. 74, 77
Maddox, T. 48
Madera, A. 62
Madin, J. 41
Madkour, F. 58
Magiopoulos, I. 47
Magnezi, L. 84
Magnusson, J. 74
Maguire, T. 67
Mahardika, N. 36
Maier, M. 70
Maire, O. 61
Maisonneuve, P. 66
Maiti, K. 41
Majaneva, S. 83
Makovec, T. 53
Maldonado-Benítez, N. 51
Maldonado, P. 48, 64
Malej, A. 53
Malik, M. 45
Malkin, S. 79
Manck, L. 41, 57
Maneval, P. 61
Mangolte, I. 89
Mangoni, O. 77
Mangora, M. 88
Manh Nguyen, H. 62
Manosalva, A. 82
Manswell-Butty, J. 38
Mantzouki, E. 36
Manzello, D. 49
Mapes, S. 75, 84
Maranda, L. 79
Maranger, R. 35, 49
Mara, P. 89
Marazzi, L. 39
Marbà, N. 62
Marcarelli, A. 59, 60, 72
Marcé, R. 47, 63
Marchant, H. 35, 56
Marchese, C. 46
Marchetti, A. 84
Maresca, J. 60
Margolin, A. 62, 74
Mariani-Rios, A. 52
Mariappan, K. 87
Markager, S. 37
Markovic, D. 84
Marliave, J. 49
Marques, A. 36
Marques, R. 39
Marquez, M. 77
Marroquim, A. 60
Marro, S. 45
Marsay, C. 42
Mars Brisbin, M. 44, 81
Marshak, T. 63
Marshall, A. 49
Marshall, T. 43
Marsh, B. 69
Martell, H. 41
Martens, I. 49
Marti de Ocampo, C. 56
Martin Bras, M. 63, 80, 88
Martin, C. 87
Martin-Creuzburg, D. 45, 75, 76
Martínez-Colón, M. 43, 65, 73
Martinez-Cortizas, A. 88
Martínez-Cortizas, A. 88
Martinez, E. 52
Martinez, F. 63
Martinez Fernandez, A. 36
Martinez, G. 35
Martínez, G. 81
Martinez Garcia, S. 61
Martinez, J. 46, 66
Martinez, K. 80
Martinez, L. 52
Martínez, M. 52, 65
Martinez Martinez, J. 36
Martínez Martínez, J. 46
Martinez, N. 38
Martinez, R. 80
Martinez-Suarez, A. 57
Martín, M. 69
Martin, R. 61, 77
Martins, E. 85
Martiny, A. 86
Martone, P. 49
Marty, S. 89
Martz, T. 57
Marx, C. 78
Marzocchi, U. 90
Marzolf, N. 72
Ma, S. 53
Mash, H. 37
Mason, E. 52
Mason, O. 69
Masque, P. 56, 77, 88
Mas, S. 84
Massamba-N'SIala, G. 38
Massana, R. 76
Massey, E. 52
Mast, M. 72
Masuda, T. 45
Masura, J. 38, 87
Maszczyk, P. 67
Matassa, C. 44, 55
Mateo, M. 88
Mathers, A. 54
Mathes, C. 60
Mathew Valayil, J. 84
Mathieu, L. 42
Mathuri, M. 51
Matiasek, S. 51
Matlaga, D. 56
Matrai, P. 46, 53, 54, 79
Matson, C. 38
Matsumoto, G. 74
Matsuzaki, S. 75, 90
Maurer-Alcalá, X. 45
Ma, X. 42
Maximov, T. 75
Mayali, X. 73
Mayer, C. 37
Mayer, L. 59
May, H. 76
Mayo, C. 74
Mayor, E. 45
Mayorga, C. 55
Mayot, N. 46, 54
Mays, J. 79
Mazzotta, M. 43
Mazzuco, A. 36
Mcadoo, B. 73
McAdoo, B. 75
McAmis, A. 41
McAskill, S. 48, 56
McBride, E. 57
McCann, K. 38
McCanty, S. 55
McCarthy, J. 36
McCarthy, M. 37, 38, 41, 43, 44, 60, 65, 67, 75, 78, 89
McCary, M. 78
McClain-Counts, J. 65
McClatchie, S. 39
McClelland, J. 37, 40, 85
McCorkle, D. 53, 85
McCormick, A. 61, 78
McCormick, L. 55
McCoy, C. 52, 55
McCrackin, M. 49, 68, 86
McCrow, J. 64
McCutcheon, M. 42
McDaris, J. 48
McDermith, E. 78
McDonald, C. 43

McDonald, K. 76, 77
McDonnell, J. 62
McDowell, J. 55
McDowell, N. 70
McDowell, W.
35, 40, 56, 67, 72, 79, 81
McFarland, M. 49, 71
McGlathery, K. 88
McGuinn, R. 55
McGuire, S. 73
McHenry, M. 84
Mcilvin, M. 49
McIlvin, M.
42, 43, 57, 64, 71, 73, 84
McKay, S. 39
Mckee, K. 81
McKenna, A. 36
McKenzie, T. 41, 65
McKnight, D. 44, 72
McLaughlin, K. 67
McLean, C. 69
McLean, J. 38
McLeod, K. 79
McLuckie, S. 69
McMahon, K. 76
McMahon, R. 80
McManus, G. 45, 53
McManus, M. 79
McMeans, B. 38, 44
McMinn, A. 85
McNab, M. 54, 76, 79
McNair, H. 70
McNally, S. 79
McNicholl, C. 59
McWilliams, J. 67, 89
Medeiros, P. 80
Medero, L. 64
Medina, L. 71, 78
Medina, M. 50, 61, 66, 84
Medina-Muñiz, J. 67
Meeder, J. 80, 81
Megan, C. 70
Mehler, K. 69, 70
Mehlrose, M. 86
Mejias-Rivera, C. 79
Melanson, J. 45
Méléder, V. 76
Melendez Diaz, J. 39, 79
Meléndez-Díaz, J. 39
Melendez, M. 49
Meléndez, M. 58
Mello-Athayde, M. 85
Mena, C. 69
Menden-Deuer, S.
45, 46, 53, 84
Mendoza, G. 55
Mendoza, J. 88
Mendoza, R. 88
Menendez, A. 80
Menezes, L. 39
Mengeot, C. 89
Menze, M. 52
Mercado, A. 35, 64
Mercado-Bettín, D. 47
Mercado-Molina, A. 84, 87
Merchant, C. 89
Merrifield, M. 62
Mertens, K. 46
Merz, J. 52
Mescioglu, E. 36, 76
Messager, M. 38, 49
Messer, P. 49
Messing, C. 65
Methot, R. 39
Metian, M. 59
Mette, M. 86
Metz, A. 40
Metz-Estrella, T. 35
Meunier, C. 47, 67
Meyer, C. 36, 78
Meyer, D. 43
Meyer, J. 61
Meyer-Jacob, C. 61, 67
Meyer, L. 37, 89
Meyer, R. 60
Miccoli, A. 53
Michel, A. 53, 78
Michel, C. 86
Michell, C. 87
Michisaki, R. 42
Middelboe, M. 61
Mienis, F. 65, 89
Miglietta, F. 59
Miles, T. 49
Milford, C. 36
Miller, A. 39, 56, 69
Miller, L. 85
Miller, M. 87
Miller, N. 61
Miller, R. 60
Miller, T. 63
Miller, W. 44
Mills, K. 73
Milošević, D. 79, 87
Miloslavich, P. 36
Milton, I. 55, 85
Miner, P. 83
Min, M. 36
Minnigh, H. 48
Minor, E. 67
Mino, Y. 51, 57
Miranda, A. 36
Mirand, C. 36
Miras, Y. 56
Mirhakak Esfahani, M. 37
Mitarai, S. 44, 80, 81
Mitchell, C. 54
Mitra, A. 64
Mize, Z. 77
Mizrahi, D. 57
Mladinich, K. 87
Mocaer, P. 69
Mochon Collura, T. 40, 59, 62
Moffat, C. 42
Moffett, J. 84
Moffett, K. 40
Mohl, J. 47
Moiti, N. 36
Molina, J. 52
Molina, M. 35
Molina, O. 43
Moline, M. 49
Monahan, D. 56
Moncrief-Cox, H. 54
Monfette, M. 82
Monismith, S. 40
Monks, L. 81
Monnier, B. 88
Monroe, A. 59
Monrose Mills, N. 64, 74
Monrose, N. 50
Montagna, P. 48, 56
Montalvo-Rodriguez, P. 54
Montana, C. 55
Montanez, A. 81
Montes, E. 36
Montes-Pérez, J. 63
Montgomery, Q. 76
Montoya, J. 43
Montressor, M. 76
Moody, E. 48, 90
Mooij, W. 35
Moore, B. 75
Moore, C. 62
Moore, G. 82
Moore, K. 79
Moore, M. 90
Moore, T. 61, 71
Morag, E. 55
Morales, C. 42
Morales-Nunez, A. 77
Morales, R. 73
Morales, S. 88, 89
Morales-Williams, A. 59, 60
Moran, D. 49, 64, 71, 73, 84
Morando, M. 73
Moran, M. 69, 76
Morán, X. 44, 89
Morelle, J. 61
Morell, J. 49, 56, 58, 70, 79
Moreno, D. 88
Moreno-Ostos, E. 63
Morey, S. 42
Morgan, S. 62
Moriarty, J. 63
Mori, C. 57
Moriceau, B. 83
Morison, F. 45, 46
Morley, D. 60
Morley, S. 84
Morrison, A. 78
Morrison, C. 65
Morrison, F. 46
Morrison, J. 42
Morrissette, H. 70
Morris, V. 38
Moser, F. 48, 63
Moser, K. 76
Motyka, J. 42
Moulton, T. 39
Moura, J. 74
Moutin, T. 86
Mtolera, M. 88
Mucci, M. 35
Mühle, J. 73
Mukherjee, S. 48, 80, 84
Mullane, K. 82
Muller, E. 54
Müller, J. 46
Muller-Karger, F. 36, 71
Muller-Karulis, B. 68
Muller-Parker, G. 66
Munday, P. 59
Mundy, S. 54
Muniz Baretto, M. 85
Munksgaard, N. 65
Muñoz, T. 57
Munroe, D. 77
Munroe, M. 88
Murarka, M. 45
Murdoch, W. 67
Murgulet, D. 41, 57
Murillo, A. 89
Murphy, C. 45
Murphy, D. 71
Murphy, S. 72
Murry, B. 44, 72
Mushet, G. 67
Musielewicz, S. 49
Mustaffa, N. 37
Muthukrishnan, R. 62
Muthusamy, S. 69
Muylaert, K. 75
Mydlarz, L. 61
Myers, E. 54, 73
Myers, J. 75, 89
Myers-Pigg, A. 43
Myint, S. 81
Myneni, S. 70, 80

N

Nadathur, G. 60
Nagarkar, M. 76
Nagel, J. 56
Nagy, E. 38
Naish, K. 41
Nakajima, R. 78
Nakamura, Y. 51, 52
Nalepa, T. 69
Nanayakkara, L. 63
Nardin, W. 79
Nascimento, F. 89, 90
Naski Keffer, L. 74
Natsuike, M. 70
Naumann, M. 43
Naustvoll, L. 61
Nava, M. 83
Navarro, B. 55, 85
Nawala, J. 67
Nayak, A. 49, 65, 71
Naziri-Saeed, M. 43
Ndlovu, O. 40
Neal, B. 53
Neale, P. 37
Neary, L. 62, 67
Needham, D. 73
Neilson, D. 39
Nejstgaard, J. 47, 65
Nelson, A. 41
Nelson, C. 61, 67, 76, 79
Nelson, H. 78
Nemeth, R. 89
Neres-Lima, V. 39
Nesbit, K. 82
Netburn, A. 45
Neuer, S. 79, 81, 89
Nevarez, J. 80
Nevers, M. 90
Neves, F. 71
Nevorski, K. 60, 72
Newell, S.
37, 38, 41, 60, 67, 75, 89
Newton, J. 39, 70, 86
Newton, R. 82
Nezlin, N. 59
Nguyen, A. 60
Nguyen, N. 87
Nguyen, T. 59
Nichols, J. 63
Nicholson, D. 46, 53
Nicholson, M. 76
Nichols, S. 35
Nick, S. 62, 89, 90
Nielsen, T. 87
Niemisto, M. 45, 53
Niewinski, D. 60
Nifong, R. 90
Nigrelli, C. 70
Nikonova, E. 37
Nilsson, E. 61
Niviere, M. 62
Noell, S. 61
Noffke, N. 78
Noisette, F. 38
Norf, H. 47
Norli, M. 63, 89
Northcott, D. 42
North, R. 37, 43
Norton, L. 38
Norton, S. 35
Novotny, A. 44, 83
Nuangsaeng, B. 73, 75
Null, K. 53
Nunn, B. 79
Nunoura, T. 69
Nuss, S. 74
Nuwer, M. 38
Nyboer, E. 38
Nybom, I. 90
Nydahl, A. 63

O

Obenour, D. 47
Oberhaensli, F. 59
Obiol, A. 76
Obrador, B. 59, 63
Obrecht, D. 37
O'Brien-Clayton, K. 42
Ocasio Torres, M. 64
Ocean, T. 89
Ochoa, J. 65
Ochsenkühn, M. 63
O'Connell, K. 48
O'Connor, A. 37
Oczkowski, A. 61, 77
O'Donnell, J. 40, 73, 79
O'Donnell, K. 52
Ogata, H. 36
Ogden, R. 35
Ogurcak, D. 38, 56, 80, 81
O'Hare, M. 77
Ohman, M. 42, 53, 71
Ohtsuki, H. 45
Okada, T. 51
Okazaki, R. 49
Okkonen, S. 85
Okotie-Oyekan, A. 35
Olascoaga, M. 86
Olden, J. 38
Oldham, V. 57

Olds, A. 67
O'Leary, J. 62, 78
Olefeldt, D. 40, 52, 72
Oleheiser, K. 43
Olive, D. 54, 55
Oliver, M. 49
Oliver, S. 86
Olivo Delgado, C. 48
Olker, J. 86
Olmeda, M. 44, 72
Olmeda-Saldana, M. 76
Olsen, D. 71
Olsen, Y. 35
Olson, J. 48
Olson, R. 53
O'Malley, B. 65, 80
O'Neil, J. 45, 82
O'Neill, S. 51, 60
Onofri, V. 53
Opalk, K. 44, 52
Opdal, A. 40
Oramadike, C. 52
Orcutt, B. 54
Orellana, M. 52
Oreska, M. 88
Ormand, C. 48
Orphan, V. 55
Ortega, A. 88
Ortega Pieck, A. 81
Orth, R. 63
Ortiz, D. 68, 90
Ortiz, G. 57
Ortiz, J. 79
Ortiz Zayas, J. 39
Ortiz-Zayas, J. 39, 67, 75, 78, 79
Orvain, F. 61, 62, 71
Osantowski, E. 70
Osborne, E. 51
Osborne, T. 73
Osborn, J. 71
Osburn, C. 52
Osburn, F. 45, 68
Oschlies, A. 39, 42
Osenberg, C. 44
Ossai, S. 61
Ossolinski, J. 80
Otake, Y. 45
Otero, E. 80
Otero, M. 88
Otero Morales, E. 72
Otis, D. 36
Oursel, B. 87
Ousley, A. 55
Overholt, E. 37
Overlinge, D. 43
Overton, E. 54, 63
Owca, T. 67
Owen, P. 74
Owens, M. 60
Oziolor, E. 38
Öztürk, D. 79, 87

P

Pacella, S. 40, 59, 62
Pace, M. 40, 68, 71
Pacheco, C. 72
Pacheco, F. 80
Pachiadaki, M. 61, 71
Packard, A. 56, 76
Padilla, C. 35
Padilla-Gamino, J. 41
Padilla Maldonado, B. 52
Padilla Plaza, G. 80
Paerl, R. 43, 52
Pagès Fauria, J. 88
Pagliosa, P. 88
Pahlow, M. 39
Pahnke, K. 57
Pajala, G. 59
Palenik, B. 76
Palevsky, H. 46, 89
Palkovacs, E. 52
Palladino, D. 89
Palmer, K. 42
Palomo, M. 36
Palter, J. 81
Palumbi, S. 41
Pan, C. 65
Papacek, J. 43
Papakyriakou, T. 70, 85
Papiol Nieves, V. 71
Papotto, J. 80
Paquette, C. 86
Pardis, W. 53, 57
Parin, D. 84
Parinos, C. 47
Paris, C. 62
Parkefelt, L. 40
Parker, A. 53, 86
Parker, S. 56
Parker, V. 62
Park, G. 38
Park, J. 37, 39, 77
Parrish, C. 83
Parrish, J. 48
Parry, D. 53
Parsons, C. 68
Parsons, R. 52, 79
Partensky, F. 42, 60
Parveen, S. 61
Passow, U. 45, 51, 56
Patceva, S. 84
Patel, C. 76
Pathak, A. 77
Pathare, M. 41, 56
Patin, N. 79
Patiño, R. 75
Patrick, C. 48
Pattengill-Semmens, C. 52, 55
Paulenich, C. 48
Paulsen, S. 86
Paul, V. 61
Pavan, D. 39
Pavia, A. 40, 54
Pavia, F. 71
Pavloudi, C. 47
Pawlak, G. 51, 62, 80
Pawlowicz, R. 60
Payet, J. 66
Paytan, A. 36, 54, 67, 76, 81
Pazanini, C. 83
Peach, C. 48
Peacock, E. 76
Peacock, M. 66, 70, 76
Pearman, J. 60
Pearson, R. 67, 77
Peart, S. 79
Pebbles, V. 35
Pech, E. 88
Pech, M. 88
Pedersen, A. 44, 87
Pedersen, J. 43
Pedler, B. 54
Peeken, I. 86
Peeler, M. 56
Peeters, F. 47, 59, 75
Peets-Allamby, O. 74
Pegion, K. 39
Peixoto, R. 85
Pellerin, B. 86
Peltier, S. 69
Penaluna, B. 48
Peñalver, P. 88
Pena, S. 55
Pendleton, D. 74
Peng, G. 78
Peng, S. 80
Penna, A. 80
Pennings, S. 48
Pennington, S. 70
Penske, A. 47
Penta, B. 49
Penta, W. 45
Péquin, B. 35
Pérez-Alegría, L. 35
Perez Brunius, P. 65
Perez, C. 38
Perez-Cruz, I. 64
Perez Ibanez, D. 70
Perez, L. 78
Pérez, O. 88
Perez-Reyes, O. 39, 72
Perez Rivera, K. 39
Peri, F. 48, 80
Perkins, R. 47, 83
Perkins, Z. 79
Perle, C. 60
Perron, S. 53
Perrtree, R. 78
Perry, R. 60
Pershing, A. 73, 74
Persons, M. 56
Pespeni, M. 38
Pess, G. 84
Pessoa, I. 51
Peter, E. 47
Petersen, L. 36
Peterson, D. 35
Peterson, P. 70
Peterson, R. 41, 43
Petkuviene, J. 43
Petrik, C. 39
Petrova, M. 36
Petty, E. 37, 43
Peura, S. 61, 73
Peychers, C. 59
Pfreundt, U. 71
Phan, M. 84
Philippe, C. 39
Phillips, J. 61, 78
Phlips, E. 43
Pichlo, C. 68
Pidgeon, E. 88
Pierce, R. 68
Pierella, J. 89
Pierson, D. 47
Pierson, J. 48, 64, 83
Pike, S. 57
Pilcher, D. 39
Pilditch, C. 49
Pilla, R. 37
Pillay, K. 42
Pimenta, A. 42
Pina, A. 38
Pinard, Z. 54
Pinazo, C. 42
Pinceel, T. 39
Pinckney, J. 88
Pineau, P. 62
Piñeiro-Juncal, N. 88
Pinhassi, J. 61, 69
Pinsonneault, A. 70
Pinto Pacheco, S. 73, 81
Pitarch, J. 37
Pitta, P. 47
Pitt, K. 83
Pitz, K. 42
Plaisted, H. 83
Planquette, H. 42
Plante, S. 49
Platz, M. 59
Plueddemann, A. 86
Plyley, C. 50
Póda, C. 84
Podgorski, D. 35, 85
Poisot, T. 49
Poje, A. 45
Pokhrel, S. 60
Polerecky, L. 45
Polidoro, B. 79
Polimene, L. 64
Politi, T. 77
Pollara, S. 63, 79
Pollard, A. 86
Pollock, J. 38
Polsenaere, P. 40, 62, 76
Polson, S. 60
Pomati, F 47
Pombrol, M. 45
Pompey, A. 74
Pondaven, P. 45
Pontiller, B. 61
Ponton, D. 43
Popendorf, K. 69
Popiel, S. 67
Popp, B. 35
Portilla, K. 75
Portrais, K. 54
Porubsky, W. 81
Posman, K. 53, 54
Posselt, J. 48
Post, D. 38
Pott, C. 54, 89
Potter, H. 40, 43
Potter, J. 56, 72
Potvin, M. 73
Poulson, S. 72
Poulton, N. 53, 74, 78
Powell, E. 88
Powers, L. 44, 53
Poynton, H. 49
Prada, M. 72
Prasanna, A. 85, 87
Prasil, O. 45
Preiler, C. 60
Prellwitz, S. 43
Premo, Z. 84
Presley, R. 51, 60
Prestegaard, K. 43, 83
Pretty, T. 77
Price, N. 46, 53
Price, R. 56
Prien, R. 43
Prieur, L. 83
Prihartato, P. 45, 84
Princiotta, S. 64
Pringle, C. 39, 72
Prokopenko, M. 45
Propson, B. 55
Protsenko, E. 89
Prouty, N. 65
Ptacnik, R. 60, 65, 81, 84
Pugh, E. 72
Pushkarev, A. 73
Putnam, H. 87

Q

Qin, W. 60
Quattrini, A. 65
Quay, P. 89
Quigg, A. 40, 41, 54, 75, 76, 84
Quiñones-Rivera, Z. 63, 66
Quintric, L. 46
Quiquempois, V. 62
Qurban, M. 45, 60, 84

R

Rabalais, N. 40
Rabouille, S. 45
Rabsatt, T. 55
Rachid, C. 85
Ragg, N. 59
Rahav, E. 76
Rahev, O. 76
Rahman, Z. 52
Raigoza, S. 77
Rainville, E. 62
Rakotomalala, C. 71
Rakshit, S. 71
Ramírez, A. 44, 51, 52, 57, 72, 75, 82
Ramírez-Flandes, S. 71
Ramirez-Irizarry, N. 52
Ramírez, J. 88
Ramirez-Lugo, J. 38, 82
Ramirez, S. 39
Ramírez Toro, G. 48
Ramirez, W. 77
Rämö, R. 90
Ramos, J. 67, 80, 88
Ramos-Scharron, C. 39, 40
Ramos Soler, I. 80
Randla, M. 89
Rangel-Parra, L. 54
Ranieri, J. 41
Rankinen, J. 40
Rao, D. 57, 85

Rapp, H. 89
Rasher, D. 54
Rasina, B. 82
Rassweiler, A. 44
Rathwell, C. 36, 79
Rauschenberg, C. 79
Rauschenberg, S. 54
Raut, Y. 73
Ravasi, T. 59
Raven, J. 84, 86
Rawat, A. 54
Rawlins, M. 85
Raymond, C. 89
Raymond, P. 66, 80
Raymore, M. 56
Razavi, R. 86
Reale, J. 63
Reale-Munroe, K. 88
Reascos, D. 75
Reavie, E. 61, 67, 69, 76, 86
Record, N. 74
Reddy, C. 54
Reeder-Myers, L. 75
Reed, J. 65
Reed, M. 38, 41, 75
Reed, V. 51
Rees, A. 35
Reese, B. 48, 73
Regaudie de Gioux, A. 81
Regier, P. 55, 63
Reglero, P. 69
Reichart, G. 89
Reidhaar, P. 88
Reid, N. 38
Reinart, A. 89
Reinl, K. 68
Reinthaler, T. 89
Relyea, R. 83
Rember, R. 42
Remmer, C. 62, 67
Remón, J. 88
Renaud, P. 86
Renault, L. 89
Rengefors, K. 47
Reniers, A. 62
Renman, G. 81
Rentería, M. 43
Repeta, D. 41, 42, 57, 69, 75
Resplandy, L. 57
Retelletti Brogi, S. 52
Reusser, B. 77
Reyes-Maldonado, R. 82
Reygondeau, G. 83
Rezende, C. 39
Rheuban, J. 53, 85
Rhew, R. 80
Rhude, K. 84
Ricciardi, A. 87
Rice, A. 49
Richard, A. 61
Richardson, C. 51, 67, 81
Richardson, J. 79
Richardson, K. 53
Richardson, M. 85
Richards, R. 67
Rich, J. 51, 60
Richmond, B. 85
Richmond, R. 41
Richon, C. 41
Rick, T. 75
Ridal, J. 43
Ridley, C. 35
Rielly, E. 77
Riemann, L. 43, 61
Riemer, S. 44
Rigonato, J. 89
Riise, G. 56
Rijos, C. 52
Riley, K. 44
Rilov, G. 36
Rines, J. 79
Ríos-Guzmán, H. 54
Ríos, J. 80
Rios Morales, A. 64
Rios, O. 55
Ripken, C. 44
Risley, S. 53
Rissanen, N. 84
Ritchie, K. 84
Riul, P. 88
Rivas Rivera, Y. 64
Rivas-Ubach, A. 59
Rivera, C. 76
Rivera-Carrasquillo, Y. 64
Rivera Casillas, P. 56
Rivera-Cruz, J. 80, 81
Rivera García, L. 79
Rivera, J. 86
Rivera, M. 52
Rivera Oquendo, E. 80
Rivera, R. 64
Rivera-Vicéns, R. 61, 79
Rivero, U. 86
Roberson, D. 38
Roberson, L. 53, 54, 55
Robert, M. 42
Roberts, B. 76
Roberts, E. 89
Robertson, C. 56
Robertson, D. 43
Robertson, G. 83
Roberts, Q. 35, 90
Roberts, S. 87
Robey, N. 53
Robinson, K. 41, 56
Robinson, L. 45
Robles-Flores, J. 65
Roca Marti, M. 57
Rocap, G. 36, 71, 79
Rocha, C. 41
Rocha de Souza, M. 85
Rodkey, T. 54
Rodrigues, L. 41, 80
Rodriguez-Abudo, S. 56
Rodríguez, C. 52, 80
Rodríguez-Cardona, B. 79
Rodriguez-Casariego, J. 87
Rodríguez Curet, T. 48
Rodriguez, E. 78
Rodríguez, J. 80
Rodriguez, L. 72
Rodríguez-Léon, E. 71
Rodriguez-Matos, P. 60
Rodríguez-Ortiz, N. 82
Rodriguez, S. 40, 70
Roethler, M. 59
Rogers, C. 48
Rogers, S. 37, 53, 60, 77
Rogge, A. 71
Rogosch, J. 38
Rohlfs, R. 38
Rohrlack, T. 56
Rojas, D. 70
Roldan Ayala, Z. 55
Roldán, Z. 80
Roley, S. 81
Rolle, S. 79
Rollinson, V. 51, 78
Rollwagen-Bollens, G. 66, 90
Roman, M. 63
Romano-Young, K. 79
Rom, E. 55
Romney, C. 62
Roper, Z. 56, 74, 78
Rörig, L. 76
Rosado Borrero, G. 87
Rosado, P. 85
Rosa-Marin, A. 73
Rosa, P. 38
Rosario, M. 79
Rosati, A. 39
Rosato, A. 47
Rose, K. 37, 39, 43, 63
Rosenfeld, S. 76
Rosengard, S. 70, 74
Rosen, J. 62
Rosenwasser, S. 46
Rose, V. 66
Ros, M. 85
Ross, A. 40
Ross, B. 65, 82
Rossi, E. 80
Rossi, R. 76
Ross, M. 45, 80, 81
Ross, S. 65
Rothenberger, M. 48
Roth, F. 68
Rothhaupt, K. 44, 67, 75, 76
Roth, V. 63
Rotjan, R. 87
Rovai, A. 70, 88
Rowell, J. 82
Rowe, M. 69
Rowley, D. 79
Ruacho, A. 57
Rubalcava, K. 44
Rubenson, E. 38
Ruben, Z. 57
Rubinoff, B. 37
Rudberg, D. 59
Rudstam, L. 37, 45, 60, 69, 79
Rueda, D. 71
Rue, G. 44
Rugh, W. 40
Ruginis, T. 77
Ruhi, A. 39
Ruhl, H. 70
Ruiz, C. 64, 67, 77
Ruiz-Cooley, R. 68
Ruiz Diaz, C. 38
Ruiz Díaz, C. 82
Ruiz, G. 73
Ruiz, H. 72
Ruiz-Jones, L. 85
Ruiz, P. 71
Ruiz-Ramos, D. 60
Runge, J. 74
Ruppel, C. 65
Rusak, J. 86
Russell, B. 65
Russell, J. 81
Rust, F. 59
Rutherford, E. 37
Rutherford, K. 70
Ryan, K. 72
Rykaczewski, R. 44, 78
Rynearson, T. 46, 76, 84
Rysgaard, S. 70
Ryu, J. 52

S

Saavedra-Flores, A. 65
Saba, G. 49
Sabal, M. 52
Sabat, A. 87
Sabate, K. 53
Sabater, J. 72
Sabel, M. 75
Sablan, L. 77
Sadofsky, E. 70
Saenger, C. 59
Sáez-Uribe, R. 39, 75
Sah, J. 39
Saito, M. 42, 49, 57, 64, 71, 73, 85
Saiz, E. 76
Sakuma, K. 53
Salazar, K. 70
Salem Jubran, Y. 64
Salgado-Hernanz, P. 81
Salinas, C. 56, 88
Salinas-de-León, P. 88
Salisbury, J. 49, 58, 65
Salk, K. 49
Salls, W. 86
Saltarelli, W. 39
Salter, I. 71
Saltvedt, M. 46
Samaniego, S. 47
Samarkin, V. 81
Sameoto, J. 73
Samuiloviene, A. 43
San Antonio, C. 49
Sánchez-Garcia, M. 54
Sanchez, J. 64
Sanchez, P. 85
Sanchez-Rios, A. 55
Sancho, G. 60
Sandaa, R. 46
Sanders, A. 52
Sanderson, M. 35
Sanders, R. 62
Sanfilippo, J. 60
Sanford, L. 53
Sans, A. 88
Santana-Cruz, A. 81
Santana, J. 83
Santana, N. 67
Santano, J. 55, 85
Santiago, J. 67
Santiago-Vazquez, L. 48
Santiago-Vera, J. 52
Santinelli, C. 76
Santoferrara, L. 45
Santora, J. 60
Santoro, A. 71
Santos, A. 77
Santos Andrade, M. 79
Santos, C. 52, 60, 81, 82
Santos-Crespo, M. 39, 80, 81
Santos, E. 61, 77
Santos Flores, C. 77
Santos-Flores, C. 52
Santos, H. 78
Santos, M. 76
Santos Mercado, H. 52
Santos, R. 39
Sañudo-Wilhelmy, S. 73
Sanzone, D. 67
Sapir, Y. 62
Saraceno, J. 72
Sarmiento, J. 39
Sarno, D. 76
Sasai, Y. 67, 80
Sasaki, H. 37
Sasaki, M. 38, 53
Sasaki, T. 90
Saso, E. 68
Sastre, M. 80
Sastre Wirshing, M. 55
Sato, K. 44
Satterthwaite, E. 60
Saunders, J. 64, 71
Sautter, L. 65
Savalia, P. 55
Savatier, M. 41
Savelli, R. 62, 76
Savic, A. 79
Savić, D. 87
Savic-Zdravkovic, D. 79
Savoca, M. 60
Savoie, A. 37, 62
Savva, I. 62
Sawakuchi, H. 59
Sawall, Y. 69
Saw, J. 69
Sawyer, D. 40, 41
Schaafsma, F. 86
Schaeffer, B. 70, 86
Schafer, T. 73
Schalk, C. 55
Schalles, J. 89
Schanke, N. 77
Scharer, M. 72
Scharnweber, K. 47, 56
Schattenhofer, M. 73
Schatz, D. 46
Schechner, A. 72
Scheef, L. 41
Scheib, A. 67
Schelker, J. 40
Schenk, J. 59
Schiebel, H. 48, 53, 88
Schiebelhut, L. 38
Schillaci, C. 79
Schizas, N. 36, 52, 60, 76, 79, 85, 87
Schlacher, T. 67
Schlenker, A. 37
Schlesinger, D. 70, 80

Schleyer, G. 46
Schlink, C. 78
Schluchter, W. 60
Schmidt, N. 87
Schmidt-Roach, S. 83, 85, 87
Schmidt, S. 46
Schmidt, W. 66, 70
Schmitt, J. 77
Schmitt-Kopplin, P. 44
Schmitt, R. 44, 68
Schneider, B. 42
Schneider, G. 76
Schneider, T. 39
Schnetger, B. 39, 56, 57
Schnetzer, A. 41
Schockman, K. 51
Schofield, K. 35
Schofield, O. 49, 85
Scholl, M. 72
Scholz, K. 59
Schomberg, J. 35
Schonhoff, B. 38
Schopflin, L. 55, 73
Schott, E. 45, 64, 70
Schreier, J. 76
Schreiner, K. 74
Schroeder, H. 59
Schroeder, I. 60
Schroth, A. 54, 56, 72, 75
Schubotz, F. 89
Schultz, J. 49
Schulz-Bull, D. 43
Schulz, M. 70
Schuster, U. 62
Schwartz, J. 80
Schwarz, M. 53
Schweizer, T. 60
Schwieterman, G. 54
Schwing, F. 63
Schwing, P. 41, 57, 65
Scofield, A. 37
Scott, J. 73
Scott, N. 77
Scott-Price, O. 50
Scott, T. 37, 38, 45, 54, 55, 60, 68, 90
Scranton, M. 71
Scriven, V. 76
Searles, A. 54
Sebestyen, S. 43
Sebourn, W. 53
Sed'a, J. 47
Sediva, B. 45
Sedlak, J. 75
Sedwick, P. 73
Seelen, L. 35
Seeley, M. 87
Seelmann, K. 42
Segarra, K. 81
Seger, A. 85
Segui, L. 76
Sehein, T. 61
Seidel, L. 43
Seidel, M. 39
Selbie, D. 46
Selig, G. 45
Seltzer, P. 55
Semmens, B. 52, 55
Sempéré, R. 87
Send, U. 57
Sengupta, A. 70, 71
Sensors, S. 57
Serafy, J. 54
Serrano, A. 51
Serrano, O. 56, 88
Setegn, S. 39
Sethi, S. 39
Setta, S. 40
Setterfield, S. 65
Severin, T. 46, 63
Seybold, E. 72, 75
Shahabinia, A. 86
Shah Esmaeili, Y. 36
Shakil, S. 85
Shalapyonok, A. 76
Shaltout, N. 58
Shama, L. 87
Shamberger, K. 48
Shanks, A. 62
Shanley, J. 72
Shapiro, J. 38
Shapiro, S. 60
Sharma, S. 76
Sharoni, S. 88
Sharp, J. 59
Sharpley, A. 68
Shattuck, M. 72
Shaul, N. 54
Shaw, A. 60
Shaw, J. 38
Shearman, R. 55
Sheik, C. 60, 67
Sheinbaum, J. 65
Shein, K. 76
Shen, S. 39
Shen, Y. 43
Sherenco, F. 54
Sherman, C. 52
Sheward, R. 84
Shibl, A. 63
Shields, A. 37
Shiller, A. 42
Shilling, E. 57
Shingai, Q. 68
Shin, K. 75
Shinohara, R. 75, 90
Shire, D. 84
Shirey, P. 86
Shirshin, E. 37
Shogren, A. 48
Shotorban, K. 78
Shrestha, R. 57, 84
Shropshire, T. 69, 80
Shukle, J. 75
Shulman, I. 49, 71
Shumway, S. 87
Siano, R. 46
Sickler, S. 48
Siebens Morales, Y. 80
Siebers, B. 43
Sieczko, A. 59
Siedlecki, S. 39, 42, 70
Sierra, P. 88
Sievers, M. 77
Sikkel, P. 76
Silliman, B. 44
Silsbe, G. 80
Silva, A. 76
Silva, D. 85
Silva, L. 68
Silveira, R. 39
Simier, M. 84
Simis, S. 89
Simmons, K. 48
Simmons, M. 70
Simmons, R. 66
Simonetti, S. 53
Simon, H. 56
Simon, M. 69
Simpson, G. 59
Simpson, L. 73
Singh, A. 60
Singh Yadav, N. 62
Sinninghe-Damste, J. 71, 73
Sintes, E. 69
Sipler, R. 35, 90
Siriwong, S. 73, 75
Siron, R. 49
Sison-Mangus, M. 63
Sivonen, K. 61
Siwe-Noundou, X. 35
Sjöstedt, J. 69
Skerlep, M. 72
Skerrit, C. 64, 74
Skinner, A. 43, 83
Skipp, P. 35
Skirving, W. 69
Slade, S. 49
Slater, M. 56
Slaughter, A. 76, 77, 80, 83
Slavin, E. 47
Slimani, A. 61
Sloan, V. 38
Slomp, C. 68
Smith, B. 65
Smith, C. 57, 76
Smith, E. 62, 79
Smith, G. 71
Smithheart, J. 47
Smith, J. 77
Smith, K. 69
Smith, N. 55
Smith, R. 37, 63, 70
Smith, S. 45, 59, 60, 67, 83, 89
Smith, T. 41, 48, 54, 71, 84, 85
Smoak, J. 40, 80, 81
Smol, J. 61, 83
Smoot, C. 45
Smulders, F. 83
Smyth, R. 37, 47
Snook, H. 35
Snowball, H. 82
Snyder, B. 35
Snyder, L. 72
Soares Quinete, N. 38
Sobarzo, M. 42
Sobek, A. 90
Soler-Figueroa, B. 73
Soler, M. 88
Soletchnik, P. 40
Solow, A. 46
Sommaruga, R. 59, 65
Sommer, C. 89
Sommer, S. 53
Song, B. 43, 60, 87, 90
Song, Q. 70
Sonnerup, R. 51
Son, Y. 77
Soomets, T. 89
Søreide, J. 86
Sørensen, K. 63, 89
Soriano, A. 85
Sorokopud-Jones, M. 78
Sosa, O. 69
Sosik, H. 53
Sotomayor-Ramírez, D. 35, 52
Souflis, C. 76
Soumya, S. 71
Souter, J. 72
Souza, J. 83
Soviadan, D. 89
Sowah, R. 35
Sowers, D. 65
Spalding, H. 85
Spanbauer, T. 74
Spehn, N. 43
Spencer, L. 87
Spencer, R. 35, 43, 85
Spielman, L. 89
Spitz, M. 48
Spitz, Y. 57
Spivak, A. 55
Spore, P. 37, 54, 55, 60
Sprecher, B. 84
Stahl, D. 60
Ståhle, M. 43
Stalker, J. 70
Stamieszkin, K. 53, 74
Stancioff, E. 42
Stanev, E. 37
Stanford, A. 62, 79
Stanganelli, J. 77
Stanislauskiene, R. 43
Stankovic, J. 79
Stanković, J. 87
Stanley, B. 35, 90
Stanley, E. 86
Starke, C. 45
Starr, B. 72
Staryk, C. 42
Stauffer, B. 41, 56, 78
Stead, Z. 79
Stecher, H. 40, 59
Stecher, J. 62
Steele, C. 87
Stegen, J. 70
Steichen, J. 40, 41
Steinberg, D. 53, 85
Steinhoff, T. 42
Steinman, B. 74
Steinmetz, H. 54
Stein, R. 46
Steinschneider, S. 39
Stelmark, K. 57
Stelzer, E. 68
Stemmann, L. 89
Stepanauskas, R. 69
Stephens, B. 44
Stephens, D. 36
Sterling, A. 41, 75
Sterner, R. 68, 74, 84
Steury, T. 38
Stevens, B. 61
Stevens, M. 54
Stevenson, A. 49
Stewart, F. 35, 79, 84
Stewart, J. 67
Stewart, T. 37
Stibor, H. 45, 47
Stiling, R. 38
Stillman, A. 47
St. Laurent, K. 75
Stock, C. 39, 40
Stockenreiter, M. 45, 47
Stocker, R. 71
Stockley, N. 49, 71
Stockwell, J. 37, 56, 80
Stoer, A. 80
Stojilovic, O. 37, 53
Stokowski, M. 42
Stoll, J. 68
Stoltenberg, L. 59
Stomp, M. 86
Stoof-Leichsenring, K. 46
Stout, J. 50, 74
Stovall, A. 54
Stow, C. 67
Strader, M. 55, 87
Straight, B. 72
Straile, D. 44, 47, 75
Stramski, D. 85
Strandberg, U. 84
Strand, E. 63
Strass, V. 42, 71
Strauss, S. 43
Streibel, M. 37
Strickler, J. 71
Striebel, M. 37, 47, 57
Striegl, R. 51, 65, 85
Strnat, J. 60
Strock, J. 75
Strock, K. 86
Strong, A. 42, 81
Strope, E. 75
Stuart, C. 54
Stubbins, A. 66, 70
Studivan, M. 77, 85
Stuhrenburg, J. 52
Stukel, M. 57, 69, 80
Sturaro, N. 66
Sturm, A. 85
Sturm, K. 59
Stutzman, N. 69
Subramaniam, A. 43
Subramanian, A. 56
Suca, J. 55
Sudek, M. 69
Suggett, D. 45, 85
Suggs, E. 76
Sugimoto, A. 75
Sukigara, C. 51, 57
Suleiman-Ramos, S. 35, 48, 66, 67
Sullivan, J. 71
Sullivan, M. 89
Sullivan, S. 36, 37
Summers, A. 38
Sunday, J. 39

Sundgren, I. 59
Sunstone, C. 55
Suominen, S. 71
Suskiewicz, T. 54
Suter, E. 71
Sutherland, K. 78
Sutton, A. 49, 62
Sutula, M. 67, 89
Suzuki, K. 85
Svanback, R. 61
Svanbäck, R. 76, 84
Svec, R. 49, 68
S. Vezhapparambu, V. 35
Swaney, D. 68, 75
Swanner, E. 68, 83
Swarbrick, V. 66
Swarr, G. 69, 86
Swarzenski, C. 70
Swarzenski, P. 59
Sweet, S. 40
Swoboda, S. 86
Sylvan, J. 48, 89
Syverson, V. 65
Szafraniec, M. 73
Szekely, A. 73
Szetela, J. 54
Szewczyk, C. 54
Szymczycha, B. 42, 61

T

Tagliabue, A. 41, 42
Taipale, S. 84
Takeda, S. 51
Takekawa, J. 77
Takizawa, Y. 75
Talbot, J. 49
Talmy, D. 46
Tamburri, M. 73
Tan, C. 81
Tan, E. 46
Tank, S. 40, 85
Tan, Z. 63
Tapia, F. 42
Tappa, E. 51
Tasevska, O. 84
Taş, S. 71
Tassawad, P. 73, 75
Taub, F. 75
Taveras, Z. 80
Taylor, A. 59
Taylor, B. 83
Taylor, G. 36, 67, 71, 78
Taylor, J. 59, 77, 90
Taylor, M. 74
Tebben, J. 41
Techtmann, S. 59
Teikari, J. 61
Teillet, M. 77
Templeton, T. 88
Tenkouano, G. 68
Terry, L. 54
Tesfa, T. 63
Testa, J. 54, 59
Teurlincx, S. 35
Teutli, C. 88
Teutonico, R. 38
Tfaily, M. 43
Thamatrakoln, K. 57, 80
Theriot, E. 74
Thewes, D. 37
Thibault, C. 38
Thillman, T. 81
Thomas, A. 73
Thomas, B. 45
Thomas, F. 69
Thomas, R. 47, 51
Thomas, S. 41
Thompson, C. 74
Thompson, L. 89
Thoré, E. 39
Thouvenin, B. 71
Thrift, E. 81
Thunell, R. 51, 71
Tian, L. 46
Tiedemann, R. 46
Tierney, C. 69
Tietjen, B. 37
Tilbrook, B. 86
Tilney, C. 42
Titocci, J. 47
Titus, B. 78
Tlusty, M. 49, 79
Tobias, C. 51, 78
Tobin, C. 78
Tobin, T. 56
Togami, H. 77
Toha, H. 36
Tolar, B. 86, 89
Toledo-Hernandez, C. 38, 64, 67, 82
Toledo-Hernández, D. 38
Tolic, N. 40
Toline, A. 62
Toline, C. 62
Tomas, C. 88
Tomasetti, S. 42
Tomelleri, E. 59
Tominack, S. 68
Tommasi, D. 39
Tonon, G. 59
Tonti, N. 78
Topor, Z. 41, 56
Torfstein, A. 36, 52, 57
Torrens, C. 83
Torres, A. 72
Torres, D. 67
Torres Díaz, M. 82
Torres-Figueroa, A. 64
Torres Garcia, M. 86
Torres, M. 80
Torres, P. 72
Torres-Perez, J. 64, 72, 76
Torres-Pérez, W. 39
Torres-Pulliza, D. 41
Torres Rivera, A. 64
Torres-Valdes, S. 62
Tortell, P. 70
Tossas Deida, T. 53
Toullec, J. 83
Townley, E. 82
Townsend, H. 60
Townsend, J. 56, 84
Townsend, T. 53
Trainer, V. 77
Tran, A. 76, 79
Tran, C. 54
Trannum, H. 61
Tran, T. 51
Tranvik, L. 47, 56
Trask, C. 54
Travik, L. 47
Traving, S. 61
Travis, C. 80
Tremblay, A. 43, 59
Tremblay, L. 35
Tremblay, N. 47, 67
Trench, S. 37
Trexler, J. 60, 72, 84
Trick, C. 40, 84
Trimborn, S. 41
Trimmer, M. 43, 83
Trombetta, T. 84
Tromboni, F. 72
Trowbridge, J. 86
True, A. 71
Truong Phan, P. 55
Tsai, S. 55
Tsuchiya, K. 75
Tsuchiya, M. 78
Tucru, A. 56
Tulloch, V. 77
Tully, B. 55, 61, 73
Tully, E. 75
Turley, B. 44
Turner, B. 45, 88
Turner, C. 36, 53
Turner, E. 42
Turner, R. 40, 70
Turner, S. 61
Turner, T. 62, 67, 79
Turschwell, M. 77
Tussenbroek, B. 79
Twardoski, M. 65
Twardowski, M. 71
Twilley, R. 70, 88
Twining, B. 54
Twiss, M. 37, 43, 66
Tyler, K. 65
Tyrell, A. 45
Tzadik, O. 86
Tzortziou, M. 80
Tzubari, Y. 84

U

Uchida, K. 90
Ullman, D. 79
Ulloa, O. 42, 71
Ulseth, A. 59
Umhau, B. 57
Urabe, J. 45
Urban, M. 38
Urban-Rich, J. 78, 87
Urban-RIch, J. 78
Uribe, M. 75
Urrutia Cordero, P. 47
Urvoy, M. 69
Ushijima, B. 61
Usman, A. 52
Uudeberg, K. 89
Uye, S. 89

V

Vad, C. 60, 65, 84
Vadstein, O. 35
Valdes, D. 64
Valente, R. 60
Vancova, M. 45
van den Engh, G. 42
Van Der Hout, J. 35
Vanderploeg, H. 37
van der Zande, R. 85
van Dorst, R. 84
van Grinsven, S. 73
Van Haren, H. 65
van Helmond, N. 68
Van Horn, D. 63
Van Meter, K. 68
Van Mooy, B. 80
Van Oostende, N. 69, 79
Van Parijs, S. 53
van Sebille, E. 37
Van Wambeke, F. 86
VanZwieten, J. 65
Vardar, D. 36
Vardi, A. 46, 88
Vardi, T. 41
Varela, D. 49
Varela, R. 71
Vargas, C. 71, 89
Varg, J. 61, 76
Varner, R. 81
Vasquez-Loarte, A. 88
Vaudrey, J. 51
Vaughan, M. 72
Vaughn, J. 44
Vaverka, S. 75
Vebrová, L. 79, 87
Vecchi, G. 39
Veglia, A. 36, 76, 79
Veitch, J. 42
Veitch, M. 65
Velarde, E. 75
Velez Gonzalez, I. 38
Vélez Negrón, V. 54
Vélez, S. 80
Velez Villamil, S. 55
Venkatachalam, S. 35
Venn, C. 82
Vermandele, F. 38
Vermilyea, A. 52, 72
Vernet, M. 69
VerPloeg, S. 38
VerWey, B. 45
Vesterinen, J. 84
Vicente Ramos, K. 64
Vidal Geraldino, P. 64
Vidussi, F. 84
Viggiano Beltrocco, M. 56
Viggiano, M. 56
Villa, E. 76
Villalobos, R. 60
Villalon, B. 54
Villanueva, L. 71, 73
Villareal, T. 43, 49
Villela, h. 85
Vincent, D. 83
Viqueira, R. 72
Vivancos, S. 71
Vivas, S. 88
Vohsen, S. 45, 54, 65
Vokhshoori, N. 65, 78
Volaric, M. 61
Vollmer, S. 61
Volpitta, A. 45
von Appen, W. 42, 71
von Elert, E. 45, 68
Vonk, J. 85
von Stackelberg, K. 41
von Tümpling, W. 44
Voolstra, C. 63
Vopel, K. 49
Vora, G. 69
Vortkamp, M. 86
Vosshage, A. 66
Voss, J. 57, 77
Voss, M. 43
Vucic, J. 53, 85
Vybernaite-Lubiene, I. 43

W

Waajen, G. 35
Wachnicka, A. 68
Wade, T. 40
Waggoner, D. 78
Wagner, D. 65
Wagner, E. 80
Wagner, J. 78
Wagner, M. 35
Wagner, N. 45, 68
Wagner, S. 66, 70
Wagner, Z. 53
Wahle, R. 45
Wain, D. 47
Wainger, L. 63
Waite, A. 71
Wakita, M. 51, 57
Waldbusser, G. 59
Waldo, S. 59
Walker, B. 71
Walker, C. 52
Walker, J. 71
Walker, L. 48, 56
Wallace, D. 60
Wallace, Z. 57
Wall, C. 53
Waller, J. 45
Waller, M. 43
Walles, T. 47
Wallin, M. 63
Wallsgrove, N. 35
Walpert, J. 40
Walsh, D. 46
Walsh, E. 38, 47, 53, 54
Walsh, G. 40
Walter, J. 40, 90
Walter, K. 60
Walteros, J. 51, 75
Walter, R. 57, 62, 78
Walters, T. 77
Walve, J. 81, 90
Walz, A. 84
Walz, K. 42
Wang, D. 71
Wang, F. 70
Wang, J. 45, 68, 69

Wang, L. 43, 47
Wang, P. 66
Wang, R. 51
Wang, S. 51
Wang, W. 70
Wang, X. 35, 78, 85, 87
Wang, Y. 77
Waniek, J. 43
Wanninkhof, R. 65
Waples, J. 57
Ward, A. 53
Ward, B. 69, 79
Ward, J. 87
Ward, K. 36
Ward, N. 59, 70
Wares, J. 44
Warren, R. 55, 85
Warren, T. 62
Waska, H. 56
Wasmund, N. 53
Wasson, K. 77
Waterbury, J. 42
Waterfield, H. 86
Waterhouse, L. 52
Waters, J. 46
Waters, M. 40, 73
Waterworth, S. 35
Watkins, J. 45, 60, 69, 79
Watson, E. 56, 61, 62, 77, 88
Wear, E. 89
Weathers, K. 47
Weaver, C. 48
Webb, A. 35
Webb, E. 42, 71
Webber, E. 39
Webb, J. 59
Webb, L. 37
Webb, S. 55
Weber, F. 67
Weber, M. 86
Weber, P. 73
Weber, T. 51, 59
Webster, A. 48, 49, 72
Webster, D. 71
Wee, S. 89
Wegley Kelly, L. 76
Wehrli, B. 70
Weidel, B. 37, 45, 60
Weigelhofer, G. 40
Wei, H. 40
Weil, E. 77
Weinke, T. 73
Weinnig, A. 66
Weinstock, J. 51
Weisberg, S. 67
Weisend, R. 73
Weissburg, M. 71
Weiss, S. 54
Weitere, M. 44, 47, 66
Wei, Y. 51
Wekerle, C. 71
Welch, H. 60
Welch, K. 83
Wells, B. 86
Wells, R. 87
Welsh, D. 37
Wenker, R. 61
Wenley, J. 89
Werne, J. 61
Wesselmann, M. 62
West, A. 63
Westberry, T. 46
West, D. 83
West, K. 52
Weston, N. 81
Wetz, M. 48, 68
Wewerka, S. 57, 78
Weyhenmeyer, G. 63
Whalen, K. 63, 79
Whilden, K. 40, 43
White, A. 86
White, D. 64
White, H. 78
Whitehead, A. 38
Whitehead, C. 68
White, K. 59
White, T. 64, 74
White, W. 53
Whitfield, C. 59
Whitman, E. 62
Whitmore, B. 71
Whitmore, L. 42
Whitmore, T. 73
Whitney, L. 78
Wiberg, P. 88
Wickland, K. 72
Widner, B. 69, 86
Wiegand, M. 78
Wiegner, T. 67, 77
Wierenga, J. 84
Wigand, C. 61, 77, 88
Wiggert, J. 51
Wiggin, K. 87
Wijesekera, H. 71
Wiklund, J. 67
Wikström, J. 81, 90
Wilbanks, E. 70
Wilburn, P. 84
Wile, A. 60
Wiley, D. 55
Wilhelm, F. 68
Wilkerson, F. 84, 86
Wilkes, A. 56, 72, 75
Wilkins, K. 37
Wilkinson, G. 40, 68, 90
Willette, D. 62, 79
Williams, B. 73, 75
Williams, C. 44
Williams, J. 74
Williams-McLeod, S. 36, 53
Williams, N. 63
Williamson, C. 37
Williams, S. 41, 65, 76
Wilson, A. 38
Wilson, C. 49
Wilson Grimes, K. 50, 78
Wilson, J. 52, 73, 77
Wilson, K. 87
Wilson, P. 79
Wilson, R. 43
Wilson, S. 48, 60, 83
Wincker, P. 89
Winder, M. 44, 83
Windham, R. 40, 41
Windle, M. 43
Wingard, L. 68
Winogradow, A. 42
Winter, C. 70
Winters, G. 62
Wise, D. 40
Wiseman, N. 59
Wissel, B. 63
Wissinger, S. 83
Wittkop, C. 83
Wöerheide, G. 61
Wohlfahrt, G. 59
Wolfe, B. 62, 67
Wolfe, W. 57, 82
Wollrab, S. 47
Wollschlaeger, J. 37
Wolthers, M. 68
Womack, N. 77
Wong, M. 53
Won, H. 75
Wood-Charlson, E. 86
Woodland, R. 61, 70
Woodley, C. 73
Woolway, I. 47
Worden, A. 46
Workman, M. 52
Wright, H. 78
Wu, C. 87
Wuerz, M. 78
Wüest, A. 47
Wu, J. 41, 53, 60, 77
Wu, M. 76
Wunsch, C. 76
Wurl, O. 37
Wu, Z. 43, 68
Wyatt, A. 57
Wyatt, S. 49
Wyllie-Echeverria, S. 62, 79
Wymore, A. 35, 40, 79, 81
Wynn, J. 65

X

Xenopoulos, M. 63, 70, 86
Xiao, X. 89
Xiao, Y. 56
Ximenes, D. 85
Xiong, X. 87
Xu, C. 38
Xue, P. 39
Xu, H. 67, 70, 81
Xu, J. 46, 78
Xu, K. 45
Xu, M. 36
Xu, X. 40, 71
Xu, Y. 57

Y

Yabusaki, S. 70
Yackulic, C. 86
Yair, S. 38
Yakubovskaya, E. 36
Yalcin, K. 55
Yalçın, G. 79, 87
Yamada, K. 45
Yanez Zapata, T. 38
Yang, E. 37, 75
Yang, S. 40
Yang, W. 70
Yang, X. 39
Yang, Y. 70
Yan, M. 84
Yan, Y. 45
Yao, H. 42, 86
Yao, X. 35
Yargeau, V. 38
Yazzie, T. 74
Yeager, L. 48
Yeh, Y. 36
Yelton, R. 80
Yen, J. 71, 80
Ye, X. 39
Yilmaz, I. 36
Yin, H. 69
Yin, J. 84
Yin, Z. 70
Ylitalo, G. 78
Yohannes, E. 44, 67, 75
Yokota, K. 86
Yoo, S. 77
Yoshida, K. 85
Yoshida, T. 45
Yoshikawa, C. 67
Yoshimura, C. 70
Yoshimura, M. 66
Younan, L. 68
Young, C. 59
Young, J. 70
Young, M. 62
Young, R. 37
Yousef, F. 89
Youtsey, B. 52
Yoza, B. 53
Yuksek, A. 36
Yu, L. 68, 85
Yu, M. 75, 80
Yvon-Lewis, S. 40, 54

Z

Zahran, N. 87
Zaldei, A. 59
Zaliznyak, T. 36, 67, 78
Zamora-Terol, S. 44, 83
Zanaroli, G. 47
Zandonà, E. 39
Zapfe, G. 45, 48
Zaremba, N. 57
Zavrel, T. 45
Zayas, B. 87
Zayas-Santiago, C. 59
Zehr, J. 86
Zeki, S. 36
Zemenick, A. 49
Zemlys, P. 43
Zerecero Marin, E. 55
Zhai, L. 39
Zhang, F. 78
Zhang, H. 84, 85
Zhang, K. 56
Zhang, L. 51, 67, 73
Zhang, P. 70, 71
Zhang, R. 35
Zhang, S. 44
Zhang, Y. 89
Zhao, F. 60
Zhao, J. 39
Zhao, M. 64
Zhao, S. 78, 87
Zhao, Y. 70
Zhao, Z. 35, 76
Zheng, Y. 60
Zheng, Z. 46
Zhou, H. 43
Zhou, L. 84
Zhou, Q. 40
Zhu, G. 67
Zhu, L. 78
Zhu, Q. 63, 69
Zieger, S. 67
Ziegler, S. 43, 83
Zielinski, O. 37
Zilius, M. 43, 77
Zimberlin, M. 88
Zimmer-Faust, A. 40
Zimmerman, J. 66
Zimmermann, H. 46
Zimmerman, R. 41, 59, 70
Zink, I. 54
Zong, H. 52
Zou, W. 46
Zwick, M. 38

OCEAN SCIENCES MEETING

16–21 FEBRUARY 2020
SAN DIEGO, CA, USA

AGU100 ADVANCING EARTH AND SPACE SCIENCE • ASLO Association for the Sciences of Limnology and Oceanography • THE OCEANOGRAPHY SOCIETY

Join the largest international gathering of the ocean science disciplines.

Mark Your Calendar: 16-22 February 2020

For more information on the 2019 Aquatic Sciences Meeting, please address all correspondence and questions regarding registration, conference logistics, and hotel accommodations to:

ASLO Business Office
1105 Wooded Acres, Suite 260
Waco, Texas 76710
business@aslo.org E-mail
254-776-3550 Phone
https://www.aslo.org/sanjuan2019/main Web

FUTURE ASLO MEETINGS

2020 Ocean Sciences Meeting
16–21 February 2020
San Diego Convention Center
San Diego, California, USA

2020 ASLO Summer Meeting
A joint meeting with ASLO and SFS
7–12 June 2020
Monona Terrace Community and Convention Center
Madison, Wisconsin, USA

2021 Aquatic Sciences Meeting
28 February–March 5, 2021
Palau de Congresos Palma de Mallorca
Palma de Mallorca, Spain

2022 Ocean Sciences Meeting
A joint meeting with ASLO, AGU and TOS
27 February–4 March 2022
Hawaii Convention Center
Honolulu, Hawaii, USA

2022 Joint Aquatic Sciences Meeting
A joint meeting of societies representing the Consortium of Aquatic Scientific Societies (CASS): American Fisheries Society (AFS), Association for the Sciences of Limnology and Oceanography (ASLO), Coastal and Estuarine Research Federation (CERF), Phycological Society of America (PSA), Society for Freshwater Science (SFS), Society of Wetland Scientists (SWS)
Dates and Location TBD

2024 Ocean Sciences Meeting
A joint meeting with ASLO, AGU and TOS
18–23 February 2024
New Orleans, Louisiana, USA

ASLO 2019 AQUATIC SCIENCES MEETING
PROGRAM ADDENDUM

Changes to Printed Scientific Program Schedule as of 2/27/19

Revision to ASLO Board of Directors Officers/Executive Committee listing on **Page 2.**
Please note correction: Dianne Greenfield, Secretary

MONDAY ORAL SESSIONS

CS013: Dissolved organic matter - DOC, DON, DOP, fDOM, cDOM
Room 202 > 2/25/2019 2:45 PM
DON CYCLING IN THE SURFACE SOUTH CHINA SEA AS INFERRED FROM ISOTOPE SIGNATURES
Oral Presentation Withdrawn: Zhang, Run

SS010: Exploring What Makes Undergraduate Research Experiences Work: Evidence From Students And Ment
Room 204 > 2/25/2019 9:45:00 AM
INCREASING DIVERSITY IN ENVIRONMENTAL STUDIES AND EVOLUTIONARY DEVELOPMENTAL BIOLOGY
Oral Presentation Withdrawn: Cohen, Sarah

SS010: Exploring what makes undergraduate research experiences work: evidence from students and mentors
Room 204 > 2/25/2019 9:45 AM
CO-OPERATIVE EDUCATION AS A VEHICLE FOR IMMERSIVE RESEARCH EXPERIENCES FOR UNDERGRADUATE STUDENTS – *Presenting Author: Watson, Elizabeth B.*
Presentation Revised from Poster to Oral; Withdrawn as Poster on Tuesday in Session SS010. Now an Oral on Monday.

SS010: Exploring what makes undergraduate research experiences work: evidence from students and mentors
Room 204 > 2/25/2019 10:15:00 AM
WHAT IS AND ISN'T WORKING IN REU DIVERSITY EFFORTS: TIPS FOR THE INTREPID
Presenting Author Revised: Batchelor, Rebecca

SS013 ADAPTATION OF AQUATIC BIODIVERSITY TO GLOBAL CHANGE
Room 102 > 2/25/2019 5:15 PM
Calosi, P.; Thibault, C.; Massamba-N'SIala, G.; Noisette, F.; Vermandele, F.; Babin, M.: RARITY, TRANSGENERATIONAL RESPONSES AND THE GLOBAL CHANGE
Time Slot Revised from 14:00 to 17:15

SS040: Radionuclides in Aqueous Systems
Room 208 A/B > 2/25/2019 9:15:00 AM
DETECTION OF SHORT-TERM SEDIMENTATION USING RADIOISOTOPES: FROM THE COAST TO THE DEEP SEA
Oral Presentation Withdrawn: Larson, Rebekka

SS050: Improving Coral Reef Resilience with Transformational Science
Room 103 B > 2/25/2019 4:45:00 PM
THE CORAL RESTORATION CONSORTIUM: EFFORTS TO FUND, COORDINATE, AND TRANSFORM REEF CONSERVATION
Oral Presentation Withdrawn: Vardi, Tali

TUESDAY ORAL SESSIONS

CS006: Trace metals
Room 102 A/B/C > 2/26/2019 5:15:00 PM
Mazzotta, M; McIlvin, M.R.; Moran, D.; Saito, M.A.: CHARACTERIZATION OF THE METALLOPROTEOME OF THE MARINE BACTERIUM, *ALTEROMONAS*
Abstract Title Revised; Author List Revised to include Co-Authors

(New 02-27-19)
CS007: Hypoxia
FORMATION OF A METALIMNETIC OXYGEN MINIMUM IN A DEEP LAKE
Room 201 => Tue, Feb 26, 2019 (10:00 AM)
Presentation Withdrawn: Cory McDonald

CS020: Fish and fisheries
Room 209 C > 2/26/2019 10:15:00 AM
A WATERSCAPE APPROACH TO FRESHWATER ARTISANAL FISHERIES
Oral Presentation Withdrawn: Garcia, Miguel

CS023: Community Ecology
Room 201 > 2/26/2019 4:15:00 PM
Ares, A.; Sato, K.; Mars Brisbin, M.; Diaz, J.; Ripken, C.; Iinura, Y.; Mitarai, S.: PLANKTON COMMUNITY STRUCTURE DYNAMICS UNDER RED SOIL POLLUTION WITHIN CORAL REEF ECOSYSTEMS
Author List Revised to include 6th Author

SS043: Beyond the Numbers: Strategies for Inclusive Practices Across the Aquatic Sciences
Room 208 C > 2/26/2019 2:30:00 PM
LESSONS LEARNED IN ENGAGING UNDERSERVED AUDIENCES IN NOAA EDUCATION PROGRAMS
Oral Presentation Withdrawn: Jones, Marissa

SS060: The advent of sampling biological processes in aquatic systems using autonomous platforms
Room 209 A/B > 2/26/2019 4:30:00 PM
LONG-DURATION SAMPLING OF THE PACIFIC GYRE USING HOLOGRAPHY AND ACTIVE FLUORESCENCE ON A WAVE GLIDER SV2
Oral Presentation Withdrawn: Villareal, Tracy

TUESDAY POSTER SESSION

(New 02-27-19)
CS005: Acidification
AMELIORATING OCEAN ACIDIFICATION: TOWARDS A MODEL RELATING PCO2, IRRADIANCE AND LEAF AREA INDEX OF ZOSTERA MARINA (EELGRASS) IN PADILLA BAY, WA
Poster #78 => Tue, Feb 26, 2019 (05:30 PM)
Poster Presentation Withdrawn – Tyler Tran

CS009: Gas Fluxes
Exhibit Hall B > 2/26/2019 5:30:00 PM
Poster #91
INTERIOR ALASKA LAKES – NET SOURCES OR SINKS OF CARBON GASES?
Poster Withdrawn: Dornblaser, Mark

CS013: Dissolved organic matter - DOC, DON, DOP, fDOM, cDOM
Exhibit Hall B > 2/26/2019 5:30:00 PM
Poster #106
CHALLENGES OF FLUORESCENT DISSOLVED ORGANIC MATTER CHEMICAL INTERPRETATIONS: A CASE STUDY OF POLAR ICE CORES
Poster Withdrawn: D'Andrilli, Juliana

SS004: Undergraduate Research in Aquatic Sciences Posters
Exhibit Hall B > 2/26/2019 5:30:00 PM
Poster #240
ACOUSTICAL ANALYSIS OF TWO SHALLOW WATER OCEAN NOISE REFERENCE STATION NETWORK SITES: AMERICAN SAMOA AND BUCK ISLAND
Poster Withdrawn: Cosca-Baresh, Allison

SS004: Undergraduate Research in Aquatic Sciences Posters
Exhibit Hall B > 2/26/2019 5:30:00 PM
Poster #262
EXAMINING THE EFFECTS OF FLOOD MITIGATION INFRASTRUCTURE ON THERMAL PERFORMANCE OF *P. VULGARIS* IN THE RUMNEY MARSH (REVERE, MASSACHUSETTS}
Abstract Title Revised

SS010: Exploring what makes undergraduate research experiences work: evidence from students and mentors
Exhibit Hall B > 2/26/2019 5:30:00 PM
Poster #340
CO-OPERATIVE EDUCATION AS A VEHICLE FOR IMMERSIVE RESEARCH EXPERIENCES FOR UNDERGRADUATE STUDENTS – Presenting Author: Watson, Elizabeth B.
Presentation Revised from Poster to Oral; Withdrawn as Poster on Tuesday in Session SS010. Now an Oral on Monday.

WEDNESDAY ORAL SESSIONS

CS010: Nitrogen biogeochemistry and cycling
Room 208 A/B => Wed, Feb 27, 2019 10:00 AM
EVALUATING SEDIMENTS AS AN ECOSYSTEM SERVICE IN WESTERN LAKE ERIE THROUGH QUANTIFICATION OF NITROGEN CYCLING PATHWAYS
Presenting Author Revised: McCarthy, Mark

CS022: Biodiversity
Room 103 B > 2/27/2019 3:00:00 PM
ENVIRONMENTAL BASELINE SURVEY DATA IN PERDIDO AND BAY OF CAMPECHE (SOUTHERN GULF OF MEXICO)
Oral Presentation Withdrawn: Perry, Ruth

CS022: Biodiversity
Room 103 B > 2/27/2019 3:15:00 PM
BASELINE CHARACTERIZATION OF SOFT-BOTTOM BENTHIC COMMUNITIES IN SUPPORT OF REGULATORY PERMITTING FOR DRILLING IN THE DEEPWATER MEXICAN GULF OF MEXICO
Oral Presentation Withdrawn: Valente, Raymond

SS023: Carbon Cycling Across Gradients in the Land-Ocean-Continuum
Room 208 C > 2/27/2019 9:30 AM
D'Andrilli, J.; Hawkes, J.; QUALITY CONTROL IN DISSOLVED ORGANIC MATTER COMPOSITION ASSESSMENT BY HIGH RESOLTUION MASS SPECTROMETRY, AN INTERNATIONAL LABORATORY
COMPARISON
Presenting Author Revised

SS041 THE NEXT GENERATION: UNDERGRADUATE RESEARCH IN PUERTO RICO AND THE US VIRGIN ISLANDS
Room 204 > 2/27/2019 2:15:00 PM
Berberena, K. Rivera, R.; Diaz, K.; Barberena-Arias, M.: ABUNDANCE OF DIFFERENT SIZE CLASSES OF PLASTICS IN HIGH AND LOW ENERGY BEACHES
First author/presenting author omitted from listing in the program book.

SS065: Turning the Lights on for Deep-Sea Ecosystems in the Caribbean, Gulf of Mexico, and US SE Atlantic
Room 103 A > 2/27/2019 10:00:00 AM
FISH LARVAE DIVERSITY IN THE DEEP WATER REGION OF THE GULF OF MEXICO
Oral Presentation Withdrawn: Jiménez Rosenberg, S.

SS065: Turning the Lights on for Deep-Sea Ecosystems in the Caribbean, Gulf of Mexico, and US SE Atl
Room 103 A > 2/27/2019 3:00:00 PM
OBSERVATIONS OF AN UNDESCRIBED CTENOPHORE FROM A 3,910 M DEPTH OFF PUERTO RICO
Oral Presentation Withdrawn: Ford, Mike

SS066: Large rivers of the world as pipes, chimneys and reactors
Room 202 > 2/27/2019 2:00:00 PM
CARBON DIOXIDE AND METHANE EXCHANGE WITH RIVERS AND STREAMS IN THE UPPER MISSISSIPPI RIVER NETWORK
Oral Presentation Withdrawn: Striegl, Robert

THURSDAY ORAL SESSION

AS001 Success Through Science: Using Limnology and Oceanography to Tackle Difficult Management Questions
Please note revised presentation order for the following:
Location: Room 204
09:00 McCarthy, M.; Boedecker, A.; Chaffin, J.; Gardner, W.; Hampel, J.; Hoffman, D.; Xu, H.; Zhang, L.; Zhu, G.; **Newell, S.:** CHEMICALLY REDUCED NITROGEN FORMS AS DRIVERS OF NON-N-FIXING CYANOBACTERIA BLOOMS IN LARGE LAKES ERIE AND TAIHU: MANAGEMENT IMPLICATIONS (T)
Presenting author revised

09:30 **Barouillet, C.**; Meyer-Jacob, C.; Mushet, G.; Hennessy, S.; Bertin, A.; Cumming, B.: DISSOLVED ORGANIC CARBON CONCENTRATIONS EXERT A STRONGER CONTROL ON THE CLADOCERAN COMMUNITY COMPOSITION OF BOREAL LAKES THAN WARMING

AS005: Extreme Events
Room 209 A/B > 2/28/2019 3:15:00 PM
EXTREME WARMING DECREASES CONTINENTAL SHELF NITROGEN REMOVAL CAPACITY
Oral Presentation Withdrawn: Fulweiler, Robinson

CS017: Harmful blooms
Room 102 A/B/C > 2/28/2019 3:15:00 PM
NUTRIENT AND TRACE METAL CO-LIMITATION OF CYANOBACTERIAL BLOOMS IN THE GREAT LAKES
Oral Presentation Withdrawn: Larson, James

SS023: Carbon Cycling Across Gradients in the Land-Ocean-Continuum
Room 104 > 2/28/2019 9:45:00 AM
DIAGNOSIS OF CO2 FLUXES IN THE COASTAL OCEAN
Oral Presentation Withdrawn: Dai, Minhan

SS023: Carbon Cycling Across Gradients in the Land-Ocean-Continuum
Room 104 > 2/28/2019 9:45 AM
THE COUPLED CYCLING OF DISSOLVED IRON AND DISSOLVED ORGANIC MATTER IN THE CONNECTICUT RIVER
Presentation Revised from Poster to Oral; Withdrawn as Poster on Thursday in Session SS023. Now an Oral on Thursday.
Presenting author: Laura Ann Logozzo

(New 2-27-19)
SS023: Carbon Cycling Across Gradients in the Land-Ocean-Continuum
CARBON CYCLING AND TRACE GAS DYNAMICS IN DISTURBED, RESTORED, AND REFERENCE WETLANDS ALONG A SALINITY GRADIENT IN OREGON, USA
Room 104 => Thu, Feb 28, 2019 (10:15 AM)
Presentation Withdrawn: Scott D. Bridgham

(New 02-27-19)
SS042: Long-term ecological studies of freshwater ecosystems in the Greater Caribbean Biogeographic Region
INTRA-COMMUNITY DIVERSITY OF INVASIVE SPECIES IMPACTS IN SPACE AND TIME: SCALING UP TO ECOSYSTEM FUNCTION
Room 103 A => Thu, Feb 28, 2019 (03:00 PM)
Presentation Withdrawn: Joel Trexler

THURSDAY POSTER SESSION

CS021: Aquatic food webs
Exhibit Hall B > 2/28/2019 5:30:00 PM
Poster #145
Patel, C.; Lee, C.; Ignoffo, T.; Slaughter, A.; Jungbluth, M.; Kimmerer, W.: INVESTIGATING COPEPOD CONSUMPTION OF PHYTOPLANKTON IN SAN FRANCISCO ESTUARY USING QPCR
Abstract Title Revised; Author List Revised

CS021: Aquatic food webs
Exhibit Hall B > 2/28/2019 5:30:00 PM
Poster #146
DNA ANALYSIS REVEALS ONTOGENETIC DIET SHIFTS FROM LARVAE TO ADULTS OF A PREDATORY COPEPOD
Poster Withdrawn: Cohen, Sarah

SS023: Carbon Cycling Across Gradients in the Land-Ocean-Continuum
Exhibit Hall B > 2/28/2019 5:30:00 PM
Poster #364
THE COUPLED CYCLING OF DISSOLVED IRON AND DISSOLVED ORGANIC MATTER IN THE CONNECTICUT RIVER
Presentation Revised from Poster to Oral; Withdrawn as Poster on Thursday. Now an Oral on Thursday.

SS030 BIOLUMINESCENT BAYS OF THE CARIBBEAN: SCIENCE, MANAGEMENT, OUTREACH, AND RECOVERY
Exhibit Hall B > 2/28/2019 5:30:00 PM
Poster #390
Flores, J.; Siebens Morales, Y.; Govender, Y.: PATTERNS OF SEDIMENTATION AND DINOFLAGELLATE DENSITY IN THE LAGUNA GRANDE, FAJARDO, PUERTO RICO
Author listing revised

FRIDAY ORAL SESSIONS

(New 2-25-19)
CS012: Carbon fluxes in FW and marine environment
101 AB > 3/1/2019 2:45 PM
SPATIOTEMPORAL ANALYSIS OF CARBON FLUXES IN GLACIAL MELTWATER STREAMS, ANTARCTICA
Presentation Withdrawn: Christa Torrens

CS033: Coral Reef Ecosystems
Room 104 > 3/1/2019 4:45:00 PM
Sturm, A.; Gonzalez Mendez, J.; Eckert, R.; Chei, E.; Dodge, D.; Studivan, M.; Gonzalez Diaz, P.; Voss, J.D.: ASSESSING POPULATION CONNECTIVITY OF THE CORAL SPECIES, *MONTASTRAEA CAVERNOSA*, ACROSS LOCAL AND REGIONAL SCALES IN CUBA AND THE TROPICAL WESTERN ATLANTIC
Author List Revised to include 8th Author

CS043: Global oceanography and limnology
Room 208 C > 3/1/2019 9:30:00 AM
THE OCEAN OBSERVATORIES INITIATIVE (OOI)
Presenting Author Revised: Dever, Ed

SPOT001: Publications Spotlight Session
Room 209 A/B > 3/1/2019 2:45:00 PM
USING OPTICAL PROPERTIES TO LOOK BACKWARDS AND FORWARDS IN TIME: DISSOLVED ORGANIC MATTER SOURCE, PROCESSING, REACTIVITY AND FATE
Oral Presentation Withdrawn: Kraus, Tamara

SPOT001: Publications Spotlight Session
Room 209 A/B > 3/1/2019 4:00:00 PM
CRITICAL IMPORTANCE OF SEA ICE-DERIVED CARBON IN ARCTIC AND ANTARCTIC FOOD WEB PROCESSES
Oral Presentation Withdrawn: Kohlbach, Doreen

SS054: Remote Sensing of Water Quality and Quantity
Room 202 > 3/1/2019 5:00:00 PM
USE OF FERRYBOX SHIPS OF OPPORTUNITY SYSTEMS FOR SATELLITE PRODUCT VALIDATION
Presenting Author Revised: *Deininger, Anne*

SS058: Brave new world: the ecology of highly impacted waterbodies
Room 208 A/B > 3/1/2019 3:45:00 PM
DISCONTINUITY BETWEEN PATTERN AND PROCESS IN SIMULATED AGRICULTURAL DITCHES YIELD ECOLOGICAL AND METHODOLOGICAL QUESTIONS REGARDING DIEL DENITRIFICATION MODELS
Oral Presentation Withdrawn: Taylor, Jason